Politecnico di Torino

Master of Science in Computer Engineering

Department of Control and Computer Engineering (DAUIN)

Database and Data Mining Group (DBDMG)

Large Language Models for Software Engineering

Theoretical foundations, methodologies, and practical architectures

Comprehensive master's course study guide

Authored for graduate students, researchers, and practitioners

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Preface

Pedagogical vision

The emergence of Large Language Models (LLMs) represents a fundamental paradigm shift in computer science and software engineering. Moving beyond rigid syntax-directed rules, modern software engineering increasingly leverages deep probabilistic models capable of synthesizing code, repairing complex bugs, formalizing requirements, and orchestrating multi-agent systems.

This study guide accompanies the Master of Science course *Large Language Models for Software Engineering* at Politecnico di Torino, taught by Prof. Flavio Giobergia and Prof. Riccardo Coppola. The curriculum is intentionally structured into two complementary pillars:

1. **Part I: Foundations of large language models** — Traces the mathematical and architectural lineage of language modeling, beginning with n-gram Markov chains, proceeding through feedforward networks, distributed word embeddings (Word2Vec, FastText), recurrent networks and vanishing gradient proofs, and culminating in the self-attention Transformer architecture, pre-training scaling laws, instruction tuning (RLHF, DPO), and parameter-efficient fine-tuning (LoRA, QLoRA).

2. **Part II: Large language models for software engineering** — Bridges foundational model capabilities with the Software Development Life Cycle (SDLC), exploring prompt engineering patterns, autonomous agent frameworks (ReAct, multi-agent collaboration), automated test generation, software maintenance and refactoring, rigorous evaluation methodologies (Pass@k, CodeBLEU), and security, intellectual property, and ethical implications.

Visual callout structure

To maximize learning effectiveness and exam readiness, the guide embeds three specialized pedagogical environments:

Exam insights: Recurrent exam patterns and pitfalls

Highlighted in terracotta/amber, these boxes isolate specific questions frequently asked in oral and written examinations, point out common mathematical misconceptions (e.g., mixing logarithm bases when computing perplexity), and underscore key grading criteria.

Intuition & analogy: Geometric and conceptual analogies

Highlighted in royal purple, these boxes provide conceptual, visual, and intuitive interpretations of complex mathematical machinery (e.g., attention as a soft differentiable lookup table, high-dimensional vector space geometry).

External deep dive: Seminal research and authoritative multimedia

Highlighted in pine teal, these boxes contextualize textbook theory with seminal academic papers [1–3] and breakdowns by leading researchers and educators.

Examination framework and course organization

As detailed by the instructors in the introductory lecture, assessment in this course is divided into two distinct components:

1. **Written Examination (18 Points)**:
    - Duration: 75 minutes.
    - Format: Typically around 12 questions (combining closed-ended multiple-choice items and open-ended technical/mathematical derivations).
    - Scope: Comprehensive coverage across all foundational and applied software engineering topics.
    - Threshold: A minimum score of 10/18 is strictly required to achieve sufficiency.
2. **Practical Course Project (14 Points)**:
    - Implementation: Development of an applied LLM-for-software-engineering system using Python and PyTorch.
    - Modality: Undertaken individually or in small teams.
    - Scoring: Total points across the written test (18) and project (14) sum to 32, with scores exceeding 30 earning *30 e Lode* (honors).
3. **Lecture and Lab Schedule**:
    - Monday: 08:30 – 11:30 (Room 8) — Core theoretical foundations.
    - Wednesday: 11:30 – 13:00 (Room R1) — Practical workshops, interactive coding, and PyTorch laboratories.

How to use this study guide

Each chapter is fully self-contained and begins with precise learning objectives, followed by exhaustive mathematical derivations, algorithmic walkthroughs, and practical software engineering implications. Students are encouraged to independently verify all mathematical derivations and implement the core algorithms to develop deep mechanical empathy for large language models.

Notation and conventions

Throughout this study guide, we adopt consistent mathematical, algorithmic, and typographic conventions aligned with standard deep learning and software engineering literature.

Linear algebra and tensor operations

Symbol	Type	Description
x, y, z	Scalar	Real-valued scalars or scalar components ($x \in \mathbb{R}$).
$\mathbf{x}, \mathbf{y}, \mathbf{h}$	Vector	Column vectors ($\mathbf{x} \in \mathbb{R}^d$). Lowercase bold font.
$\mathbf{x}^\top$	Transpose	Transpose of vector $\mathbf{x}$ (row vector $\mathbf{x}^\top \in \mathbb{R}^{1\times d}$).
$\mathbf{A}, \mathbf{W}, \mathbf{X}$	Matrix	Real matrices ($\mathbf{A} \in \mathbb{R}^{m\times n}$). Uppercase bold font.
$\mathbf{A}_{i,j}$ or a_{ij}	Scalar	Entry at row i and column j of matrix $\mathbf{A}$.
$\mathbf{I}$ or $\mathbf{I}_d$	Matrix	Identity matrix of dimension $d \times d$.
$\|\mathbf{x}\|_2$ or $\|\mathbf{x}\|$	Scalar	Euclidean (L_2) norm of vector $\mathbf{x}$, $\sqrt{\sum_i x_i^2}$.
$\|\mathbf{A}\|_F$	Scalar	Frobenius norm of matrix $\mathbf{A}$, $\sqrt{\sum_{i,j} A_{i,j}^2}$.
$\langle \mathbf{u}, \mathbf{v} \rangle$ or $\mathbf{u}^\top \mathbf{v}$	Scalar	Standard dot (inner) product between vectors $\mathbf{u}$ and $\mathbf{v}$.
$\odot$	Operator	Element-wise (Hadamard) product between tensors of identical shape.
$\otimes$	Operator	Outer product or Kronecker tensor product.

Table 1: Linear algebra and tensor notation summary.

Probability and information theory

Neural networks, transformers, and software engineering

Symbol	Type	Description
$\mathbb{P}(A)$	Scalar	Probability of event A ($\mathbb{P} : \mathcal{F} \to [0, 1]$).
$\mathbb{P}(X = x)$	Scalar	Probability mass function for discrete random variable X.
$\mathbb{P}(Y \mid X)$	Distribution	Conditional probability distribution of Y given X.
$\mathbb{E}_{x\sim P}[f(x)]$	Scalar	Expected value of function $f(x)$ under distribution P.
$\mathbb{V}\text{ar}(X)$	Scalar	Variance of random variable X, $\mathbb{E}[(X - \mathbb{E}[X])^2]$.
$\mathcal{H}(P)$	Scalar	Shannon entropy of distribution P, $-\sum_x P(x) \log P(x)$.
$\mathcal{H}(P, Q)$	Scalar	Cross-entropy between target distribution P and model Q.
$D_{\mathrm{KL}}(P \parallel Q)$	Scalar	Kullback–Leibler (KL) divergence from Q to P.
$\mathrm{PPL}(W)$	Scalar	Perplexity of word sequence W under a language model.

Table 2: Probability and information theoretic notation.

Symbol	Type	Description
$\mathcal{V}$	Set	Discrete vocabulary of tokens; $\lvert\mathcal{V}\rvert = V$ denotes vocabulary size.
T or T	Integer	Sequence length (number of input or output tokens).
d_{model}	Integer	Hidden embedding dimensionality of the Transformer model.
d_{ff}	Integer	Dimensionality of the intermediate Feed-Forward Network layer.
d_k, d_v	Integer	Dimensionality of query/key and value projections per head.
h	Integer	Number of parallel attention heads ($h \cdot d_k = d_{\text{model}}$).
$\sigma(z)$	Function	Standard logistic sigmoid function: $\sigma(z) = \frac{1}{1+e^{-z}}$.
$\mathrm{softmax}(\mathbf{z})$	Function	Softmax normalized exponential vector operator.
$\mathrm{ReLU}(z)$	Function	Rectified Linear Unit activation: $\max(0, z)$.
$\mathrm{GELU}(z)$	Function	Gaussian Error Linear Unit activation.
$\mathcal{L}$ or $\mathcal{L}$	Scalar	Scalar objective / loss function minimized during optimization.
Pass@k	Metric	Probability that at least one of k generated code samples passes unit tests.
CodeBLEU	Metric	Multi-aspect code evaluation metric combining BLEU, AST, and data flow.

Table 3: Model architecture parameters and SE metrics.

Part I

Foundations of large language models

CHAPTER 1

Introduction to language models

1.1 The sequence prediction problem and probabilistic language modeling

At its core, a language model formalizes the intuitive mechanism by which humans evaluate the fluency, plausibility, and likelihood of word sequences. In Natural Language Processing (NLP) and modern AI-assisted Software Engineering, we model human language and source code not through rigid, brittle rule systems, but as a discrete stochastic process governed by probability distributions.

Let $\mathcal{V} = \{w_1, w_2, \ldots, w_V\}$ denote a finite vocabulary of $V = |\mathcal{V}|$ discrete tokens (e.g., words, subwords, or punctuation symbols). An arbitrary natural language sentence, code snippet, or document of length N is represented as an ordered sequence of discrete random variables $W = (w_1, w_2, \ldots, w_N) \in \mathcal{V}^N$.

Definition 1.1: *Statistical language model*

A *statistical language model* is a formal probability distribution $\mathbb{P}$ defined over the set of all possible discrete sequences $\mathcal{S} = \bigcup_{N=1}^{\infty} \mathcal{V}^N$. For any specific sequence $W = (w_1, \ldots, w_N)$, the language model assigns a scalar probability $\mathbb{P}(W) \in [0, 1]$ satisfying the Kolmogorov probability axioms:

$$\mathbb{P}(W) \geq 0 \quad \forall W \in \mathcal{S}, \quad \text{and} \quad \sum_{S \in \mathcal{S}} \mathbb{P}(S) = 1. \tag{1.1}$$

The model maps the syntactic and semantic conventions of human language into real values on the unit interval:

- Fluent, grammatically coherent sentences receive high probability mass:

$$\mathbb{P}(\text{“this”}, \text{“is”}, \text{“a”}, \text{“reasonable”}, \text{“sentence”}) = 0.1$$

- Syntactically valid but semantically unusual or rare sentences receive noticeably lower probability:

$$\mathbb{P}(\text{“this”}, \text{“is”}, \text{“a”}, \text{“purple”}, \text{“sentence”}) = 0.01$$

- Ungrammatical, corrupted, or disjoint sequences receive negligible probability:

$$\mathbb{P}(\text{“this”}, \text{“are”}, \text{“a”}, \text{“reasonable”}, \text{“sentence”}) = 0.001$$

Evaluating the joint probability $\mathbb{P}(w_1, \ldots, w_N)$ directly over the combinatorial space $\mathcal{V}^N$ is intractable because observing every conceivable multi-word sequence in a training dataset

is impossible. To structure this estimation problem, we invoke the fundamental *chain rule of probability*.

Definition 1.2: *Chain rule decomposition of language models*

For any sequence of random variables $(w_1, w_2, \ldots, w_N)$, the joint probability factors identically and unconditionally into an ordered product of conditional next-token probabilities:

$$\begin{aligned}\mathbb{P}(w_1, w_2, \ldots, w_N) &= \mathbb{P}(w_1) \cdot \mathbb{P}(w_2 \mid w_1) \cdot \mathbb{P}(w_3 \mid w_1, w_2) \cdots \mathbb{P}(w_N \mid w_1, \ldots, w_{N-1}) \\ &= \prod_{t=1}^{N} \mathbb{P}(w_t \mid w_1, \ldots, w_{t-1}) = \prod_{t=1}^{N} \mathbb{P}(w_t \mid w_{<t}),\end{aligned} \tag{1.2}$$

where $w_{<t} = (w_1, \ldots, w_{t-1})$ represents the prefix history preceding timestep t. For the initial token $t = 1$, the conditioning prefix is the empty sequence $\emptyset$, giving $\mathbb{P}(w_1 \mid w_{<1}) = \mathbb{P}(w_1)$.

Applying Equation (1.2) to our running example yields the concrete factorization:

$$\begin{aligned}\mathbb{P}(\text{this, is, a, reasonable, sentence}) = \mathbb{P}(&\text{this}) \\ &\cdot \mathbb{P}(\text{is} \mid \text{this}) \\ &\cdot \mathbb{P}(\text{a} \mid \text{this, is}) \\ &\cdot \mathbb{P}(\text{reasonable} \mid \text{this, is, a}) \\ &\cdot \mathbb{P}(\text{sentence} \mid \text{this, is, a, reasonable}).\end{aligned}$$

Reframing language modeling as conditional prediction. This mathematical decomposition marks a pivotal conceptual shift in Natural Language Processing. Instead of attempting to score complete, static documents simultaneously, the modeling focus shifts entirely to sequential, conditional next-word estimation:

Given history $w_{<t}$, estimate the categorical probability distribution over all words in $\mathcal{V}$: $\mathbb{P}(w_t = w \mid w_{<t})$.

This formulation aligns directly with the classic psychological *cloze test* (fill-in-the-blank):

"This is a reasonable _____*"* $\implies$ $\mathbb{P}(\text{sentence} \mid \text{This is a reasonable}) = ?$

Every modern autoregressive language model, including GPT-4 and Claude, relies fundamentally on this conditional next-token formulation.

Intuition & analogy: The core thesis: next-word prediction as full language modeling

In the opening lecture of the Politecnico di Torino course, the fundamental essence of modern language modeling was crystallized:

> *"If we can predict the next word, then we have estimated this conditional probability. And then we can plug this into the chain rule, and estimate the probability of any arbitrary sentence. This is what a language model does: nothing more, nothing less. If you understand this idea, you understand the core mechanism driving the entire course, from 1980s Markov chains to modern trillion-parameter LLMs."*

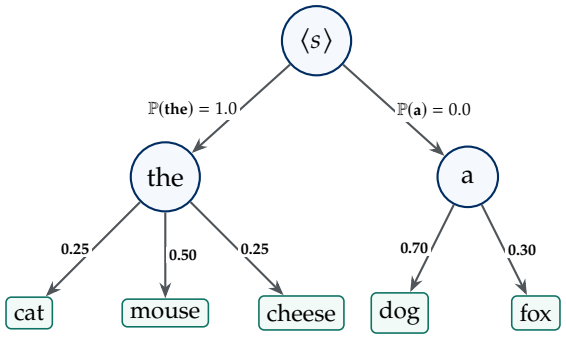


Figure 1.1: Visualizing autoregressive sequence generation as path traversal across a probability branching tree. At the root symbol $\langle s \rangle$, outgoing edges partition total probability mass across vocabulary tokens. At each node $w_{<t}$, children represent the conditional distribution $\mathbb{P}(w_t \mid w_{<t})$. The law of total probability guarantees that branch probabilities sum to unity at each decision point.

Intuition & analogy: The branching probability tree

Sequence generation can be visualized geometrically as walking down a directed decision tree (Figure 1.1). At the root node $\langle s \rangle$, the model confronts the entire vocabulary $\mathcal{V}$. Choosing a token w_1 traverses an edge with probability $\mathbb{P}(w_1)$ and lands at a new conditioning state. From this state, another set of branches opens, weighted by $\mathbb{P}(w_2 \mid w_1)$. The joint probability of an entire sentence is simply the product of edge weights along that unique trajectory. Generating text consists of sampling a path from the root to a designated terminal token $\langle /s \rangle$.

1.2 Markov assumptions and n-gram language models

Although Equation (1.2) is mathematically exact and preserves all syntactic and semantic dependencies, computing $\mathbb{P}(w_t \mid w_{<t})$ directly in empirical applications presents an insurmountable combinatorial bottleneck. As the sequence index t grows, the conditioning history $w_{<t} = (w_1, \ldots, w_{t-1})$ expands monotonically. Because each constituent token is drawn from a vocabulary $\mathcal{V}$ of size $V = |\mathcal{V}|$, the space of potential prefix histories preceding token t is the Cartesian product space $\mathcal{V}^{t-1}$, whose cardinality is:

$$|\mathcal{H}_t| = |\mathcal{V}|^{t-1} = V^{t-1}. \tag{1.3}$$

Even for a modest vocabulary of $V = 10{,}000$ words, tracking histories of length $t = 4$ requires estimating probabilities over $10{,}000^3 = 10^{12}$ distinct conditioning states. In any finite training corpus, almost every sequence of length greater than three occurs exactly zero times.

To make estimation mathematically tractable, the principle of conditional independence formalizes that future tokens depend only on a bounded historical window. In the landmark 1948 treatise on information theory [4], it was demonstrated that modeling natural language as a discrete Markov process produces increasingly coherent synthetic English as the conditioning order increases. Modern pedagogical treatments [5] formalize this simplification as the foundation of classical statistical natural language processing.

Definition 1.3: *Markov assumption of order $n-1$*

An n-gram language model assumes that the conditional probability of the current

word w_t depends exclusively on the immediately preceding $n-1$ words, rendering it conditionally independent of all earlier discourse history:

$$\mathbb{P}(w_t \mid w_1, w_2, \ldots, w_{t-1}) \approx \mathbb{P}(w_t \mid w_{t-n+1}, \ldots, w_{t-1}). \tag{1.4}$$

The integer $n \geq 1$ dictates the order of the model:

- **Unigram Model ($n = 1$, Order 0)**: Completely memoryless. Each word is treated as statistically independent of all past and future tokens:

$$\mathbb{P}(w_1, \ldots, w_N) \approx \prod_{t=1}^{N} \mathbb{P}(w_t).$$

- **Bigram Model ($n = 2$, Order 1)**: First-order Markov chain. Conditioning depends exclusively on the single preceding token w_{t-1}:

$$\mathbb{P}(w_1, \ldots, w_N) \approx \prod_{t=1}^{N} \mathbb{P}(w_t \mid w_{t-1}).$$

- **Trigram Model ($n = 3$, Order 2)**: Second-order Markov chain. Conditioning depends on the preceding pair of tokens (w_{t-2}, w_{t-1}):

$$\mathbb{P}(w_1, \ldots, w_N) \approx \prod_{t=1}^{N} \mathbb{P}(w_t \mid w_{t-2}, w_{t-1}).$$

1.2.1 Combinatorial state-space reduction

The primary mathematical virtue of the Markov assumption is that it truncates an exponentially growing state space into a fixed, stationary cardinality. Under an n-gram model, the conditioning context is always restricted to $n-1$ words, regardless of the absolute sequence position t:

$$|\mathcal{H}^{(n)}| = |\mathcal{V}|^{n-1} = V^{n-1}. \tag{1.5}$$

Consequently, specifying the complete transition probability distribution across the entire vocabulary requires an $(n-1)$-dimensional conditioning tensor containing:

$$P_{\text{params}} = V^{n-1} \times (V - 1) \approx V^n \tag{1.6}$$

free parameters. Table 1.1 contrasts the unconstrained historical state space with fixed n-gram models for realistic vocabulary sizes.

While the unconstrained state space exceeds the total number of atoms in the observable universe ($\approx 10^{80}$) by step $t = 20$, a bigram model maintains a manageable state space of 50,000 rows. However, this drastic dimensional reduction comes at the cost of complete amnesia for any context preceding the immediate $n-1$ tokens.

1.2.2 Maximum likelihood estimation (MLE)

Under Maximum Likelihood Estimation, we estimate the conditional transition probabilities directly by counting occurrences of exact token sequences in a training corpus [5]. For a generic n-gram:

$$\mathbb{P}_{\text{MLE}}(w_t \mid w_{t-n+1}, \ldots, w_{t-1}) = \frac{C(w_{t-n+1}, \ldots, w_{t-1}, w_t)}{C(w_{t-n+1}, \ldots, w_{t-1})}, \tag{1.7}$$

Table 1.1: State-space cardinality $|\mathcal{H}|$ as a function of sequence position t versus fixed n-gram Markov orders for vocabularies $V = 10{,}000$ and $V = 50{,}000$.

Model / Order	**Effective Context**	**Formula**	**States ($V = 10^4$)**	**States ($V = 5 \times 10^4$)**
Unconstrained ($t = 2$)	1 previous token	V^1	1.0×10^4	5.0×10^4
Unconstrained ($t = 3$)	2 previous tokens	V^2	1.0×10^8	2.5×10^9
Unconstrained ($t = 5$)	4 previous tokens	V^4	1.0×10^{16}	6.25×10^{18}
Unconstrained ($t = 10$)	9 previous tokens	V^9	1.0×10^{36}	1.95×10^{42}
Unconstrained ($t = 20$)	19 previous tokens	V^{19}	1.0×10^{76}	1.90×10^{89}
Unigram ($n = 1$)	0 tokens (memoryless)	$V^0 = 1$	1	1
Bigram ($n = 2$)	1 previous token	V^1	1.0×10^4	5.0×10^4
Trigram ($n = 3$)	2 previous tokens	V^2	1.0×10^8	2.5×10^9
4-gram ($n = 4$)	3 previous tokens	V^3	1.0×10^{12}	1.25×10^{14}

where $C(\cdot)$ denotes the raw frequency of that exact sequence in the training text. For a bigram model ($n = 2$), Equation (1.7) specializes to:

$$\mathbb{P}_{\text{MLE}}(w_t \mid w_{t-1}) = \frac{C(w_{t-1}, w_t)}{C(w_{t-1})} = \frac{C(w_{t-1}, w_t)}{\sum_{w \in \mathcal{V}} C(w_{t-1}, w)}. \tag{1.8}$$

Normalizing each row by the unigram count of the conditioning prefix ensures that $\sum_{w \in \mathcal{V}} \mathbb{P}_{\text{MLE}}(w \mid w_{t-1}) = 1$.

1.3 Transition count matrix construction walkthrough

To understand exactly how statistical n-gram language models operate, we trace the step-by-step extraction of bigram frequencies and transition probabilities from the authoritative lecture corpus:

1. Sentence 1: *"The cat chased the mouse happily"*
2. Sentence 2: *"The mouse ate the cheese"*

Vocabulary extraction. Extracting all unique tokens (case-normalized with standard initial capitalization for clarity) yields a vocabulary of size $|V| = 7$:

$$\mathcal{V} = \{\text{Ate, Cat, Chased, Cheese, Happily, Mouse, The}\}.$$

Sequential bigram decomposition. We iterate through both training sentences token by token, recording each observed transition $(w_{t-1} \to w_t)$:

- From Sentence 1 (*"The cat chased the mouse happily"*):
  1. $(\text{The, cat}) \implies C(\text{The, Cat}) \leftarrow 1$
  2. $(\text{cat, chased}) \implies C(\text{Cat, Chased}) \leftarrow 1$
  3. $(\text{chased, the}) \implies C(\text{Chased, The}) \leftarrow 1$
  4. $(\text{the, mouse}) \implies C(\text{The, Mouse}) \leftarrow 1$
  5. $(\text{mouse, happily}) \implies C(\text{Mouse, Happily}) \leftarrow 1$
- From Sentence 2 (*"The mouse ate the cheese"*):

6. $(\text{The}, \text{mouse}) \implies C(\text{The}, \text{Mouse}) \leftarrow 2$
7. $(\text{mouse}, \text{ate}) \implies C(\text{Mouse}, \text{Ate}) \leftarrow 1$
8. $(\text{ate}, \text{the}) \implies C(\text{Ate}, \text{The}) \leftarrow 1$
9. $(\text{the}, \text{cheese}) \implies C(\text{The}, \text{Cheese}) \leftarrow 1$

Unigram normalization counts. Summing across rows provides the total unigram occurrences $C(w_{t-1})$:

$$\begin{aligned} C(\text{The}) &= 4 \quad (\text{cat} : 1, \text{mouse} : 2, \text{cheese} : 1) \\ C(\text{Cat}) &= 1 \quad (\text{chased} : 1) \\ C(\text{Chased}) &= 1 \quad (\text{the} : 1) \\ C(\text{Mouse}) &= 2 \quad (\text{happily} : 1, \text{ate} : 1) \\ C(\text{Ate}) &= 1 \quad (\text{the} : 1) \\ C(\text{Cheese}) &= 0 \quad (\text{terminal word, no outgoing transitions}) \\ C(\text{Happily}) &= 0 \quad (\text{terminal word, no outgoing transitions}) \end{aligned}$$

Full empirical transition probability matrix. Applying Equation (1.8) yields the complete 7×7 transition matrix $\mathbf{M} \in \mathbb{R}^{7\times7}$ where row i represents condition w_{t-1} and column j represents next token w_t. Table 1.2 presents both raw counts and normalized MLE probabilities.

Table 1.2: Empirical transition count matrix and normalized bigram conditional probabilities $\mathbb{P}(w_t \mid w_{t-1})$ for vocabulary $\mathcal{V} = \{\text{Ate, Cat, Chased, Cheese, Happily, Mouse, The}\}$.

Condition (w_{t-1})	Ate	Cat	Chased	Cheese	Happily	Mouse	The	Row Total
Ate	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	1 (1.00)	1
Cat	0 (0.0)	0 (0.0)	1 (1.00)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	1
Chased	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	1 (1.00)	1
Cheese	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0
Happily	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0 (0.0)	0
Mouse	1 (0.50)	0 (0.0)	0 (0.0)	0 (0.0)	1 (0.50)	0 (0.0)	0 (0.0)	2
The	0 (0.0)	1 (0.25)	0 (0.0)	1 (0.25)	0 (0.0)	2 (0.50)	0 (0.0)	4

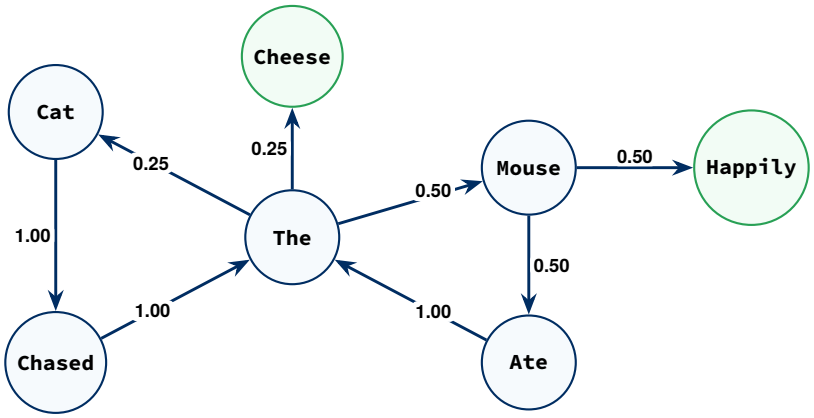


Figure 1.2: Markov state transition graph for the 7-token cat/mouse corpus. Directed edges indicate bigram transition probabilities $\mathbb{P}(w_t \mid w_{t-1})$. States `Cheese` and `Happily` (green) are terminal absorbing states with zero outgoing probability mass.

1.4 Autoregressive estimation and generation

With the transition matrix **M** constructed (Table 1.2) and its state graph mapped (Figure 1.2), the language model can execute two distinct operational tasks:

1. **Sequence Probability Estimation**: Assign a likelihood score to any proposed test sentence.
2. **Autoregressive Generation**: Incrementally sample tokens to produce novel synthetic text.

1.4.1 Worked sentence probability calculation

Consider evaluating the joint probability of the sentence:

"the cat chased the cheese"

Under the first-order Markov assumption, the joint probability factors as:

$$\mathbb{P}(\text{the, cat, chased, the, cheese}) = \mathbb{P}(\text{the})\cdot\mathbb{P}(\text{cat} \mid \text{the})\cdot\mathbb{P}(\text{chased} \mid \text{cat})\cdot\mathbb{P}(\text{the} \mid \text{chased})\cdot\mathbb{P}(\text{cheese} \mid \text{the}). \tag{1.9}$$

Assuming our generation begins deterministically with the initial token $\mathbb{P}(\text{the}) = 1.0$, we substitute the corresponding conditional probabilities from Table 1.2:

$$\begin{aligned}\mathbb{P}(\text{cat} \mid \text{the}) &= \frac{1}{4} = 0.25\\ \mathbb{P}(\text{chased} \mid \text{cat}) &= 1.0\\ \mathbb{P}(\text{the} \mid \text{chased}) &= 1.0\\ \mathbb{P}(\text{cheese} \mid \text{the}) &= \frac{1}{4} = 0.25\end{aligned}$$

Multiplying these values step by step:

$$\mathbb{P}(\text{the, cat, chased, the, cheese}) = 1.0 \times \frac{1}{4} \times 1 \times 1 \times \frac{1}{4} = \frac{1}{16} = 0.0625. \tag{1.10}$$

Despite the sequence *"the cat chased the cheese"* never occurring as an entire sentence in the training corpus, the bigram model assigns it a strictly positive probability of 6.25%. By decomposing text into local constituent pairs, n-gram models exhibit rudimentary combinatorial generalization.

1.4.2 5-step autoregressive generation trace

Now consider using the bigram model for open-ended text generation. Autoregressive sampling begins with a seed prefix, samples the next token from the resulting conditional probability distribution, appends the sampled token to the history, and repeats.

We trace the exact 5-step generation sequence recorded in the lecture slides, starting from the seed token *"the"*:

Step 1: Condition on prefix: history = ("the").

- Active row in transition matrix: Row "The".
- Candidate distribution: $\mathbb{P}(\text{cat} \mid \text{the}) = 0.25$, $\mathbb{P}(\text{cheese} \mid \text{the}) = 0.25$, $\mathbb{P}(\text{mouse} \mid \text{the}) = 0.50$.
- Random sample: **"mouse"**.
- Generated prefix: *"the mouse"*.

Step 2: Condition on prefix: history = (. . . , "mouse").

- Active row in transition matrix: Row "Mouse".
- Candidate distribution: $\mathbb{P}(\text{ate} \mid \text{mouse}) = 0.50$, $\mathbb{P}(\text{happily} \mid \text{mouse}) = 0.50$.
- Random sample: **"ate"**.
- Generated prefix: *"the mouse ate"*.

Step 3: Condition on prefix: history = (. . . , "ate").

- Active row in transition matrix: Row "Ate".
- Candidate distribution: $\mathbb{P}(\text{the} \mid \text{ate}) = 1.00$ (deterministic transition).
- Emitted token: **"the"**.
- Generated prefix: *"the mouse ate the"*.

Step 4: Condition on prefix: history = (. . . , "the").

- Active row in transition matrix: Row "The".
- Candidate distribution: $\mathbb{P}(\text{cat} \mid \text{the}) = 0.25$, $\mathbb{P}(\text{cheese} \mid \text{the}) = 0.25$, $\mathbb{P}(\text{mouse} \mid \text{the}) = 0.50$.
- Random sample: **"mouse"**.
- *Slide Note*: **:(** (A repetitive structural loop has emerged because the model cannot remember that it already emitted "mouse" two words ago).
- Generated prefix: *"the mouse ate the mouse"*.

Step 5: Condition on prefix: history = (. . . , "mouse").

- Active row in transition matrix: Row "Mouse".
- Candidate distribution: $\mathbb{P}(\text{ate} \mid \text{mouse}) = 0.50$, $\mathbb{P}(\text{happily} \mid \text{mouse}) = 0.50$.
- Random sample: **"happily"**.
- *Slide Note*: **(shocked face)** (A syntactically valid but bizarre, cannibalistic sentence is generated).
- Generated sequence: ***"The mouse ate the mouse happily"***.

Because the word *"happily"* has zero outgoing transitions in Table 1.2, the process reaches an absorbing terminal state and generation halts naturally.

Analysis of absorbing terminal states and infinite loops. During the sequence generation analysis, two fundamental edge cases emerge:

1. **Absorbing terminal states after sentence-ending tokens**: In Table 1.2, terminal sentence-ending words have a row count of zero ($C(\text{Happily}) = 0$), making the conditional probability division $0/0$ mathematically undefined. In standard production language modeling, this is formalized by appending an explicit end-of-sequence token $\langle /s \rangle$ (or [EOS]) to every training sentence, guaranteeing that every word possesses a non-zero outgoing transition to either another word or $\langle /s \rangle$.

2. **Infinite cyclic loops under greedy decoding**: Because bigram models condition exclusively on the single preceding token w_{t-1}, once the model transitions from *"ate"* to *"the"*, it retains zero memory that it already described the mouse eating cheese two steps earlier. If greedy decoding is used (arg max), the transition path becomes fully deterministic and falls into a stationary limit cycle: *"the* $\rightarrow$ *mouse* $\rightarrow$ *ate* $\rightarrow$ *the* $\rightarrow$ *mouse* $\rightarrow$ . . . *"*. Modern LLMs resolve this through long-context self-attention horizons, temperature-controlled stochastic sampling, and repetition penalties.

1.5 Fundamental limitations of statistical n-gram models: sparsity and zero-frequency collapse

While n-gram models provided the foundation for speech recognition, statistical machine translation, and spelling correction throughout the 1980s and 1990s, they suffer from fatal structural limitations that ultimately motivated the deep learning revolution.

1.5.1 Catastrophic data sparsity and parameter explosion

To condition on $n-1$ words, an n-gram model requires storing a discrete probability table of dimensionality $|\mathcal{V}|^n$. For realistic natural language and software applications:

- Vocabulary size: $|\mathcal{V}| \approx 50{,}000$ to $100{,}000$ unique words or code identifiers.
- Bigram table ($n = 2$): $|\mathcal{V}|^2 = 50{,}000^2 = 2.5 \times 10^9$ parameters ($\approx$ 10 GB at 32-bit floating point).
- Trigram table ($n = 3$): $|\mathcal{V}|^3 = 50{,}000^3 = 1.25 \times 10^{14}$ parameters (over 500 Terabytes).
- 4-gram table ($n = 4$): $|\mathcal{V}|^4 = 50{,}000^4 = 6.25 \times 10^{18}$ parameters (Exabyte scale).

Even if storage were free, the number of distinct 4-grams in all digitized human literature is dwarfed by 10^{18}. Consequently, the overwhelming majority of valid phrases never appear in training corpora.

1.5.2 The zero-frequency catastrophe: probability and perplexity collapse

The most lethal consequence of combinatorial sparsity is the *zero-frequency catastrophe*. Under Maximum Likelihood Estimation, any n-gram that never appeared verbatim in the training text receives count $C = 0$, yielding:

$$\mathbb{P}_{\text{MLE}}(w_t \mid w_{t-n+1}^{t-1}) = \frac{C(w_{t-n+1}^{t})}{C(w_{t-n+1}^{t-1})} = 0. \tag{1.11}$$

By the probabilistic chain rule (Equation (1.2)), evaluating a test sequence $W_{\text{test}} = (w_1, \ldots, w_N)$ requires taking the product across all constituent conditional transitions:

$$\mathbb{P}(W_{\text{test}}) = \prod_{t=1}^{N} \mathbb{P}(w_t \mid w_{<t}) = \mathbb{P}(w_1) \cdots 0 \cdots \mathbb{P}(w_N \mid w_{<N}) = 0. \tag{1.12}$$

A single unobserved token transition collapses the estimated joint probability of an entire document, article, or codebase to absolute zero.

Furthermore, when evaluating the model via Empirical Cross-Entropy $\mathcal{H}(W_{\text{test}})$ or Perplexity $\text{PPL}(W_{\text{test}})$:

$$\mathcal{H}(W_{\text{test}}) = -\frac{1}{N} \sum_{t=1}^{N} \log_2 \mathbb{P}(w_t \mid w_{<t}) \longrightarrow \infty, \quad \text{PPL}(W_{\text{test}}) = 2^{\mathcal{H}(W_{\text{test}})} \longrightarrow \infty. \tag{1.13}$$

An unobserved test n-gram results in infinite perplexity ($\text{PPL} = \infty$), rendering the entire language model statistically useless for evaluation or comparison.

1.5.3 Finite context horizon limitations

An n-gram model is strictly blind to any token that occurred more than $n-1$ steps in the past:

$$\mathbb{P}(w_t \mid w_1, w_2, \ldots, w_{t-1}) \approx \mathbb{P}(w_t \mid w_{t-n+1}, \ldots, w_{t-1}). \tag{1.14}$$

Because the conditioning window is rigidly restricted to the preceding $n-1$ tokens, any semantic constraint, topic coherence, or syntactic agreement established earlier in the discourse is discarded. This finite context horizon causes severe memory amnesia across two critical domains:

Linguistic agreement breakdown in natural language. Consider long-range syntactic agreements where an arbitrary number of modifiers, prepositional phrases, or subordinate relative clauses separate the grammatical subject from its governing predicate:

*"The **engineers** who designed the complex distributed compiler infrastructure [...] **were** vs **was** awarded the patent."*

If $n = 4$ (a 4-gram model), the model predicts the verb by conditioning exclusively on the three preceding tokens:

$$\mathbb{P}(\text{verb} \mid \text{compiler}, \text{infrastructure}, [\ldots]).$$

Because the singular noun *"infrastructure"* is immediately adjacent to the verb position, the local 4-gram transition statistics strongly favor the singular verb *"was"*. The model has completely forgotten the true plural antecedent subject *"engineers"*, producing an ungrammatical agreement violation.

Structural and scoping breakdown in software engineering code. In programming languages, code validity requires strict, long-range determinism governed by context-free grammars and scope resolution rules:

1. **Lexical Scoping and Variable Lifecycles**: A variable declared in a function header:

   ```
   void process_batch(std::vector<TransactionRequest>& transactions) { ...}
   ```

   must be invoked using valid `std::vector` member methods (e.g., `.push_back()`, `.size()`) thirty lines later. An n-gram model sees only `transactions.` without access to the declaration, leading to syntactically invalid completions.

2. **Bracket and Block Matching**: Program code requires exact hierarchical pairing of opening and closing delimiters (`{ ...}`, `[ ...]`, `( ...)`) and control flow constructs (`try ...catch ...finally`). Because n-gram models lack internal state stacks, they frequently emit mismatched braces or premature block terminations.

1.5.4 Discrete orthogonal symbols and lack of semantic generalization

In classical statistical NLP, words are treated as discrete, orthogonal atomic symbols. If the model observes:

*"The **cat** chased the mouse happily"*

it gains zero statistical confidence regarding the identical sentence containing synonyms:

*"The **feline** chased the mouse happily"*

Because *"cat"* and *"feline"* are completely separate row indices in Table 1.2, observations of one do not inform the other. There is no concept of semantic distance, synonymy, or conceptual overlap.

The causal imperative. While long-range memory amnesia and semantic blindness demand continuous neural architectures (introduced in Chapters 2 and 4), the immediate, fatal crisis in statistical language modeling is the **zero-frequency catastrophe** ($C = 0 \implies P = 0 \implies \text{PPL} = \infty$). Before we can even evaluate an n-gram model, we must establish rigorous mathematical algorithms that steal probability mass from frequent events and reallocate it to unobserved events: this is the domain of **statistical smoothing**.

1.6 Statistical smoothing and discounting algorithms

To prevent probability collapse ($P = 0$) and infinite perplexity ($\text{PPL} = \infty$) on unseen test sequences, statistical language modeling relies on *smoothing* (or discounting). The governing principle of smoothing is conservative probability reallocation: we shave off a fraction of probability mass from observed events and redistribute it across unseen events, ensuring that every valid sequence receives a strictly positive probability:

$$\forall w \in \mathcal{V}, \quad \mathbb{P}(w \mid \text{context}) > 0.$$

1.6.1 Laplace add-one and add-α smoothing

The simplest smoothing formulation, originated by Pierre-Simon Laplace, adds a uniform pseudo-count $\alpha > 0$ to every possible transition.

Mathematical formulation. For an unconditioned unigram model, Laplace add-α smoothing assigns:

$$\mathbb{P}_{\text{Laplace}}(w) = \frac{C(w) + \alpha}{N + \alpha|\mathcal{V}|}, \tag{1.15}$$

where $N = \sum_{w \in \mathcal{V}} C(w)$ is the total token count. For a bigram model conditioned on w_{t-1}, the estimator modifies Equation (1.8) as:

$$\mathbb{P}_{\text{Laplace}}(w_t \mid w_{t-1}) = \frac{C(w_{t-1}, w_t) + \alpha}{C(w_{t-1}) + \alpha|\mathcal{V}|}. \tag{1.16}$$

When $\alpha = 1$, the technique is known as *Laplace smoothing* (or Add-1); when $0 < \alpha < 1$, it is termed *Lidstone smoothing*.

Proof of proper probability distribution. We verify that Equation (1.16) satisfies the Kolmogorov axioms by summing over the entire vocabulary $\mathcal{V}$:

$$\begin{aligned}
\sum_{w_t \in \mathcal{V}} \mathbb{P}_{\text{Laplace}}(w_t \mid w_{t-1}) &= \sum_{w_t \in \mathcal{V}} \frac{C(w_{t-1}, w_t) + \alpha}{C(w_{t-1}) + \alpha|\mathcal{V}|} \\
&= \frac{\sum_{w_t \in \mathcal{V}} C(w_{t-1}, w_t) + \sum_{w_t \in \mathcal{V}} \alpha}{C(w_{t-1}) + \alpha|\mathcal{V}|} \\
&= \frac{C(w_{t-1}) + \alpha|\mathcal{V}|}{C(w_{t-1}) + \alpha|\mathcal{V}|} = 1.
\end{aligned} \tag{1.17}$$

Why Add-1 fails in practice. Although mathematically clean, Add-1 smoothing performs terribly in natural language processing. In realistic vocabularies where $|\mathcal{V}| = 50{,}000$, if word w_{t-1} occurred $C(w_{t-1}) = 100$ times, adding $\alpha|\mathcal{V}| = 50{,}000$ inflates the denominator from 100 to 50,100. A frequent observed transition with $C(w_{t-1}, w_t) = 50$ is diluted from an empirical probability of 0.50 to:

$$\frac{50 + 1}{100 + 50{,}000} = \frac{51}{50{,}100} \approx 0.00102.$$

Add-1 smoothing steals more than 99.8% of the probability mass from observed events and reallocates it uniformly to unobserved transitions. Even Lidstone smoothing ($\alpha = 0.05$) assigns the same probability to all unseen words, failing to recognize that some unseen words (e.g., common nouns) are inherently more probable than rare technical jargon.

1.6.2 Good-Turing discounting

Originally formulated in [6] during cryptographic research at Bletchley Park, *Good-Turing discounting* derives the probability of events seen r times directly from the count of events seen $r + 1$ times.

Frequency of frequencies. Let $r = C(w_{t-n+1}^{t}) \geq 0$ denote the empirical frequency of an n-gram. We define the *frequency of frequencies* N_r as the total number of distinct n-grams that appeared exactly r times in the training corpus:

$$N_r = |\{x : C(x) = r\}|. \tag{1.18}$$

The total number of tokens in the corpus is $N = \sum_{r=1}^{\infty} rN_r$, and N_0 represents the enormous number of valid n-grams that never appeared ($r = 0$).

Adjusted count derivation. Good-Turing replaces each raw count r with an effective discounted count r^*:

$$r^* = (r + 1)\frac{N_{r+1}}{N_r}. \tag{1.19}$$

For any observed n-gram with count $r > 0$, its Good-Turing smoothed probability is:

$$\mathbb{P}_{\text{GT}}(w_t \mid w_{t-n+1}^{t-1}) = \frac{r^*}{N} = \frac{(r + 1)N_{r+1}}{N \cdot N_r}. \tag{1.20}$$

The total probability mass reserved for all unobserved n-grams ($r = 0$) is determined by the proportion of *singletons* (N_1, words seen exactly once):

$$P_0 = \frac{N_1}{N}. \tag{1.21}$$

Because there are N_0 unobserved n-grams, each individual unseen n-gram receives probability:

$$\mathbb{P}_{\text{GT}}(\text{unseen}) = \frac{P_0}{N_0} = \frac{N_1}{N \cdot N_0}. \tag{1.22}$$

Smoothing N_r for large r. For large values of r, empirical counts become sparse (N_r fluctuates wildly or drops to zero, making N_{r+1}/N_r undefined). In production systems, *Simple Good-Turing* [7] replaces raw N_r values with smoothed estimates derived from a linear regression fit in log space ($\log N_r = a + b \log r$) for $r > k$.

1.6.3 Katz backoff with α normalization

The recursive framework introduced in [8] extended Good-Turing discounting through a hierarchical backoff mechanism: if an n-gram has been observed, use its discounted probability; if it has never been observed, back off to the $(n - 1)$-gram distribution, scaled by a normalizing factor.

Mathematical formulation. For a bigram model, the Katz backoff estimator is:

$$\mathbb{P}_{\text{Katz}}(w_t \mid w_{t-1}) = \begin{cases} \mathbb{P}^*(w_t \mid w_{t-1}) & \text{if } C(w_{t-1}, w_t) > 0, \\ \alpha(w_{t-1})\mathbb{P}_{\text{Katz}}(w_t) & \text{if } C(w_{t-1}, w_t) = 0, \end{cases} \tag{1.23}$$

where $\mathbb{P}^*(w_t \mid w_{t-1}) = \frac{C^*(w_{t-1}, w_t)}{C(w_{t-1})}$ represents the discounted probability calculated using Good-Turing adjusted counts for low frequencies ($r \leq k$, typically $k = 5$) and MLE for high counts ($r > k$):

$$C^*(w_{t-1}, w_t) = \frac{\frac{r^*}{r} - \frac{(k+1)N_{k+1}}{N_1}}{1 - \frac{(k+1)N_{k+1}}{N_1}} \cdot r \quad \text{for } 1 \leq r \leq k. \tag{1.24}$$

Derivation of the normalizing backoff weight $\alpha(w_{t-1})$. The leftover probability mass shaved off from observed bigrams is:

$$\beta(w_{t-1}) = 1 - \sum_{w_t : C(w_{t-1}, w_t) > 0} \mathbb{P}^*(w_t \mid w_{t-1}). \tag{1.25}$$

To ensure that $\sum_{w_t \in \mathcal{V}} \mathbb{P}_{\text{Katz}}(w_t \mid w_{t-1}) = 1$, the backoff weight $\alpha(w_{t-1})$ scales the lower-order unigram distribution over unobserved words:

$$\alpha(w_{t-1}) = \frac{\beta(w_{t-1})}{\sum_{w_t : C(w_{t-1}, w_t) = 0} \mathbb{P}_{\text{Katz}}(w_t)}. \tag{1.26}$$

Summing across the entire vocabulary confirms proper normalization:

$$\begin{aligned} \sum_{w_t \in \mathcal{V}} \mathbb{P}_{\text{Katz}}(w_t \mid w_{t-1}) &= \sum_{C>0} \mathbb{P}^*(w_t \mid w_{t-1}) + \sum_{C=0} \alpha(w_{t-1})\mathbb{P}_{\text{Katz}}(w_t) \\ &= \big(1 - \beta(w_{t-1})\big) + \alpha(w_{t-1}) \sum_{C=0} \mathbb{P}_{\text{Katz}}(w_t) \\ &= \big(1 - \beta(w_{t-1})\big) + \beta(w_{t-1}) = 1. \end{aligned} \tag{1.27}$$

1.6.4 Kneser-Ney smoothing and continuation probabilities

The pinnacle of statistical language modeling is *Kneser-Ney smoothing* [9]. Kneser-Ney introduces two transformative innovations: absolute discounting and the continuation probability.

Absolute discounting. Rather than scaling counts proportionally, empirical analysis of corpus frequencies demonstrates that subtracting a fixed constant $d \in (0, 1)$ (typically $d \approx 0.75$) from all positive counts accurately models the true discounted frequency:

$$C_{\text{discounted}} = \max\left(C(w_{t-n+1}^t) - d,\, 0\right).$$

The continuation probability failure mode of standard backoff. Consider predicting the missing word in the sentence:

"I can't see without my reading _____"

The word *"glasses"* should receive a high conditional probability. Now consider the word *"Francisco"*. In any standard text corpus, *"Francisco"* has a high unigram frequency because the proper noun *"San Francisco"* appears frequently. Under Katz backoff, because the bigram *"reading Francisco"* has zero count, the model backs off to the unigram frequency of *"Francisco"*, assigning it a dangerously high probability of completing the sentence!

The breakthrough insight in [9] recognized that the unigram backoff distribution should not reflect *how often* a word appears overall, but *how versatile* the word is across diverse preceding contexts.

Definition 1.4: *Continuation probability*

The *continuation probability* $\mathbb{P}_{\text{cont}}(w_t)$ is the ratio of the number of unique preceding word types that precede w_t to the total number of unique bigram types in the vocabulary:

$$\mathbb{P}_{\text{cont}}(w_t) = \frac{|\{w_{t-1} : C(w_{t-1}, w_t) > 0\}|}{\sum_{w'} |\{w_{t-1} : C(w_{t-1}, w') > 0\}|} = \frac{|\{w_{t-1} : C(w_{t-1}, w_t) > 0\}|}{|\{(w_{t-1}, w_t) : C(w_{t-1}, w_t) > 0\}|}. \tag{1.28}$$

Because *"Francisco"* almost exclusively appears preceded by the single word *"San"*, $|\{w_{t-1} : C(w_{t-1}, \text{Francisco}) > 0\}| \approx 1$, resulting in a tiny continuation probability. In contrast, words like *"glasses"* or *"book"* appear in dozens of distinct contexts, yielding high continuation probabilities.

Interpolated Kneser-Ney formulation. Combining absolute discounting with continuation probabilities yields the *Interpolated Kneser-Ney* estimator:

$$\mathbb{P}_{\text{KN}}(w_t \mid w_{t-n+1}^{t-1}) = \frac{\max\left(C(w_{t-n+1}^{t}) - d,\, 0\right)}{C(w_{t-n+1}^{t-1})} + \lambda(w_{t-n+1}^{t-1})\mathbb{P}_{\text{KN}}(w_t \mid w_{t-n+2}^{t-1}), \tag{1.29}$$

where $\lambda(w_{t-n+1}^{t-1})$ is an explicit normalizing interpolation weight:

$$\lambda(w_{t-n+1}^{t-1}) = \frac{d}{C(w_{t-n+1}^{t-1})} \cdot \left|\{w : C(w_{t-n+1}^{t-1}, w) > 0\}\right|. \tag{1.30}$$

For a bigram model ($n = 2$), the base recursion terminates at the unigram continuation probability:

$$\mathbb{P}_{\text{KN}}(w_t \mid w_{t-1}) = \frac{\max\left(C(w_{t-1}, w_t) - d,\, 0\right)}{C(w_{t-1})} + \frac{d}{C(w_{t-1})} \cdot |\{w : C(w_{t-1}, w) > 0\}| \cdot \mathbb{P}_{\text{cont}}(w_t). \tag{1.31}$$

Proof of proper normalization. Summing Equation (1.31) over all $w_t \in \mathcal{V}$:

$$\begin{aligned}
\sum_{w_t \in \mathcal{V}} \mathbb{P}_{\text{KN}}(w_t \mid w_{t-1}) &= \sum_{w_t : C>0} \frac{C(w_{t-1}, w_t) - d}{C(w_{t-1})} + \lambda(w_{t-1}) \sum_{w_t \in \mathcal{V}} \mathbb{P}_{\text{cont}}(w_t) \\
&= \frac{\sum_{C>0} C(w_{t-1}, w_t) - d \cdot |\{w : C(w_{t-1}, w) > 0\}|}{C(w_{t-1})} + \lambda(w_{t-1}) \cdot 1 \\
&= \frac{C(w_{t-1}) - d|\{w : C(w_{t-1}, w) > 0\}|}{C(w_{t-1})} + \frac{d|\{w : C(w_{t-1}, w) > 0\}|}{C(w_{t-1})} = 1.
\end{aligned} \tag{1.32}$$

Interpolated Kneser-Ney consistently outperforms all other statistical smoothing techniques, serving as the industry standard until the advent of deep learning.

The causal imperative. Now that we have established competing smoothing algorithms (Laplace, Good-Turing, Katz, Kneser-Ney), each reallocating probability mass under different mathematical assumptions, we face a crucial practical question: *How do we quantitatively benchmark and compare these models on held-out evaluation text without running costly downstream applications?* This brings us directly to formal evaluation metrics.

1.7 Evaluation metrics: cross-entropy and perplexity

To benchmark and compare different language models quantitatively without executing costly downstream applications, natural language processing relies on intrinsic, information-theoretic metrics: *Cross-Entropy* and *Perplexity* [4, 5].

1.7.1 Information-theoretic foundations: entropy and KL divergence

Let $\mathcal{X}$ denote a discrete sample space (the vocabulary $\mathcal{V}$), and let $P(x)$ represent the true, underlying, but inaccessible probability distribution of human language.

Definition 1.5: *Shannon entropy*

The *Shannon entropy* $H(P)$ measures the fundamental, irreducible uncertainty or average information content inherent in distribution P [4]:

$$H(P) = -\sum_{x \in \mathcal{X}} P(x) \log_2 P(x) = \mathop{\mathbb{E}}_{x \sim P} \left[\log_2 \frac{1}{P(x)} \right]. \tag{1.33}$$

By Shannon's source coding theorem, $H(P)$ represents the absolute theoretical lower bound on the expected number of bits required to encode a symbol emitted by P without loss.

When constructing a language model, we approximate the true distribution P with an estimated parameterized distribution Q_θ. To quantify the statistical discrepancy between the true distribution P and our approximating model Q_θ, we employ the *Kullback-Leibler (KL) divergence*.

Definition 1.6: *Kullback-Leibler (KL) divergence*

The *KL divergence* (or relative entropy) from model distribution Q to target distribution P is:

$$D_{\mathrm{KL}}(P \parallel Q) = \sum_{x \in \mathcal{X}} P(x) \log_2 \left(\frac{P(x)}{Q(x)} \right) = \mathop{\mathbb{E}}_{x \sim P} \left[\log_2 \frac{P(x)}{Q(x)} \right]. \tag{1.34}$$

Theorem 1.1: *Gibbs' inequality*

For any two discrete probability distributions P and Q, the KL divergence is non-negative:

$$D_{\mathrm{KL}}(P \parallel Q) \geq 0, \tag{1.35}$$

with equality $D_{\mathrm{KL}}(P \parallel Q) = 0$ if and only if $P(x) = Q(x)$ for all $x \in \mathcal{X}$.

Proof. Applying the natural logarithm and utilizing the standard inequality $\ln(u) \leq u - 1$ for all $u > 0$ (with equality if and only if $u = 1$):

$$\begin{aligned} -D_{\mathrm{KL}}(P \parallel Q) &= \sum_{x \in \mathcal{X}} P(x) \log_2 \left(\frac{Q(x)}{P(x)} \right) = \frac{1}{\ln 2} \sum_{x \in \mathcal{X}} P(x) \ln \left(\frac{Q(x)}{P(x)} \right) \\ &\leq \frac{1}{\ln 2} \sum_{x \in \mathcal{X}} P(x) \left(\frac{Q(x)}{P(x)} - 1 \right) \\ &= \frac{1}{\ln 2} \left(\sum_{x \in \mathcal{X}} Q(x) - \sum_{x \in \mathcal{X}} P(x) \right) = \frac{1}{\ln 2} (1 - 1) = 0. \end{aligned}$$

Multiplying both sides by -1 reverses the inequality, yielding $D_{\mathrm{KL}}(P \parallel Q) \geq 0$. ■

1.7.2 Cross-entropy decomposition

Expanding the logarithm in the definition of KL divergence demonstrates the fundamental relationship connecting Entropy, KL Divergence, and Cross-Entropy:

$$\begin{aligned} D_{\mathrm{KL}}(P \parallel Q) &= \sum_{x \in \mathcal{X}} P(x) \log_2 P(x) - \sum_{x \in \mathcal{X}} P(x) \log_2 Q(x) \\ &= -H(P) + \left(- \sum_{x \in \mathcal{X}} P(x) \log_2 Q(x) \right). \end{aligned} \tag{1.36}$$

Rearranging terms defines the *Cross-Entropy* $H(P, Q)$:

$$H(P, Q) \triangleq - \sum_{x \in \mathcal{X}} P(x) \log_2 Q(x) = H(P) + D_{\mathrm{KL}}(P \parallel Q). \tag{1.37}$$

Why optimizing cross-entropy optimizes KL divergence. Equation (1.37) reveals the theoretical justification for cross-entropy training:

- The true language distribution P is determined by human linguistic behavior; therefore, the true entropy $H(P)$ is a fixed, immutable physical constant.
- When we minimize cross-entropy with respect to model parameters θ:

$$\arg\min_{\theta} H(P, Q_\theta) = \arg\min_{\theta} \left(H(P) + D_{\mathrm{KL}}(P \parallel Q_\theta) \right) = \arg\min_{\theta} D_{\mathrm{KL}}(P \parallel Q_\theta). \tag{1.38}$$

Minimizing cross-entropy is mathematically identical to driving the KL divergence between model Q_θ and the true language distribution P to zero.

1.7.3 Empirical sequence cross-entropy

Because we do not have closed-form access to the true distribution $P(x)$, we cannot compute the expectation $\mathbb{E}_{x \sim P}[\cdot]$ directly. Instead, by the Shannon-McMillan-Breiman theorem (the ergodic theorem for stationary information sources), the expected value is estimated empirically by sampling a long test sequence $W = (w_1, w_2, \ldots, w_T) \sim P$:

$$\mathcal{H}(W) = -\frac{1}{T} \log_2 Q(w_1, w_2, \ldots, w_T) = -\frac{1}{T} \sum_{t=1}^{T} \log_2 Q(w_t \mid w_{<t}). \tag{1.39}$$

Empirical cross-entropy measures the average number of bits required to compress each token of test text using our model Q.

1.7.4 Perplexity (PPL)

While cross-entropy evaluates performance on a logarithmic scale, researchers and practitioners overwhelmingly report *Perplexity* (PPL), defined as the exponentiated cross-entropy loss:

$$\mathrm{PPL}(W) = 2^{\mathcal{H}(W)} = 2^{-\frac{1}{T}\sum_{t=1}^{T}\log_2 Q(w_t|w_{<t})} \tag{1.40}$$

$$= \left(\prod_{t=1}^{T}\frac{1}{Q(w_t \mid w_{<t})}\right)^{\frac{1}{T}}. \tag{1.41}$$

Perplexity represents the reciprocal geometric mean of the token probabilities assigned by the model across the test corpus.

1.7.5 Rigorous proof: perplexity as the effective branching factor

The fundamental intuition taught across NLP literature is that perplexity quantifies the *effective branching factor* of a language model. We now formalize this property through a rigorous proof.

Theorem 1.2: *Perplexity of a uniform K-ary distribution*

Let a language model Q predict next tokens such that at each timestep t, the probability mass is distributed uniformly across exactly K equally likely candidate tokens ($Q(w_t \mid w_{<t}) = \frac{1}{K}$ for each candidate). Then the perplexity on any generated sequence W of length T is identically K:

$$\mathrm{PPL}(W) = K. \tag{1.42}$$

Proof. Substituting the uniform prediction probability $Q(w_t \mid w_{<t}) = \frac{1}{K}$ into the empirical cross-entropy formula:

$$\mathcal{H}(W) = -\frac{1}{T}\sum_{t=1}^{T}\log_2\left(\frac{1}{K}\right) = -\frac{1}{T}\sum_{t=1}^{T}\left(-\log_2 K\right)$$
$$= \frac{1}{T}\left(T\log_2 K\right) = \log_2 K.$$

Exponentiating the cross-entropy:

$$\mathrm{PPL}(W) = 2^{\mathcal{H}(W)} = 2^{\log_2 K} = K. \quad \blacksquare$$

Physical interpretation. When a model evaluates a text corpus and achieves PPL = 42, it indicates that, on average, the model's predictive uncertainty at each token is mathematically equivalent to choosing uniformly at random from a shortlist of 42 equally plausible words:

- A uniform random guesser across vocabulary $\mathcal{V}$ achieves PPL = $|\mathcal{V}|$ (e.g., PPL = 50,000).
- An optimal oracle that predicts every token with probability 1.0 achieves the theoretical lower bound PPL = 1.0.
- State-of-the-art LLMs on natural language benchmarks typically achieve perplexities between 8 and 15.

1.7.6 Base-2 vs. base-e logarithmic conversions and perplexity computation

A critical mathematical nuance arises when connecting information-theoretic formulations (which traditionally employ base-2 logarithms) with deep learning frameworks such as PyTorch (which exclusively compute natural base-e cross-entropy loss):

Property	Information Theory (Bits)	Deep Learning / PyTorch (Nats)
Loss Formula	$\mathcal{H}_2 = -\frac{1}{T}\sum \log_2 Q(w_t \mid w_{<t})$	$\mathcal{L}_{\text{CE}} = -\frac{1}{T}\sum \ln Q(w_t \mid w_{<t})$
Unit	Bits per token	Nats per token
Perplexity Formula	$\text{PPL} = 2^{\mathcal{H}_2}$	$\text{PPL} = \exp(\mathcal{L}_{\text{CE}}) = e^{\mathcal{L}_{\text{CE}}}$
Conversion Identity	$\mathcal{H}_2 = \frac{\mathcal{L}_{\text{CE}}}{\ln 2} \approx \frac{\mathcal{L}_{\text{CE}}}{0.693147}$,	$\mathcal{L}_{\text{CE}} = \mathcal{H}_2 \cdot \ln 2 \approx 0.693147 \cdot \mathcal{H}_2$

Worked numerical calculation. Suppose a model trained on code completion reports an evaluation cross-entropy loss of $\mathcal{L}_{\text{CE}} = 2.7726$ nats:

1. **Direct Base-e Perplexity**:
$$\text{PPL} = e^{2.7726} = \exp(2.7726) \approx 16.00.$$
2. **Conversion to Information Bits**:
$$\mathcal{H}_2 = \frac{2.7726}{\ln 2} = \frac{2.7726}{0.693147} = 4.00 \text{ bits/token}.$$
3. **Base-2 Perplexity Verification**:
$$\text{PPL} = 2^{\mathcal{H}_2} = 2^{4.00} = 16.00.$$

Notice that evaluating $2^{\mathcal{L}_{\text{CE}}} = 2^{2.7726} \approx 6.83$ would severely underestimate the true branching factor by more than 57%. One must always ensure consistent logarithmic bases when deriving perplexity from loss.

1.7.7 The glass ceiling of smoothed n-gram models

While Kneser-Ney smoothing dramatically reduces test perplexity by eliminating zero-frequency collapses, smoothed n-gram models remain fundamentally handicapped:

1. **Horizon Truncation**: Conditioning remains locked to $n-1$ tokens; increasing $n \geq 5$ yields negligible gains because higher-order frequencies are almost entirely unobserved, causing the model to constantly back off to lower orders.
2. **Zero Parameter Sharing**: Words remain orthogonal indices. The model cannot generalize that *"The programmer wrote a test"* informs *"The developer authored a test"*.
3. **Combinatorial Storage**: Counting tables for large corpora require gigabytes to terabytes of RAM.

These structural limits motivated researchers to explore new paradigms that replace discrete counting tables with continuous, parameterized functions.

1.8 Historical evolution of language models

Understanding modern large language models requires appreciating the 75-year evolutionary trajectory that progressively dismantled the mathematical and computational barriers of discrete symbol counting. Figure 1.3 synthesizes the macro-historical progression of language modeling from Claude Shannon's foundational 1948 information theory to modern autonomous software engineering systems.

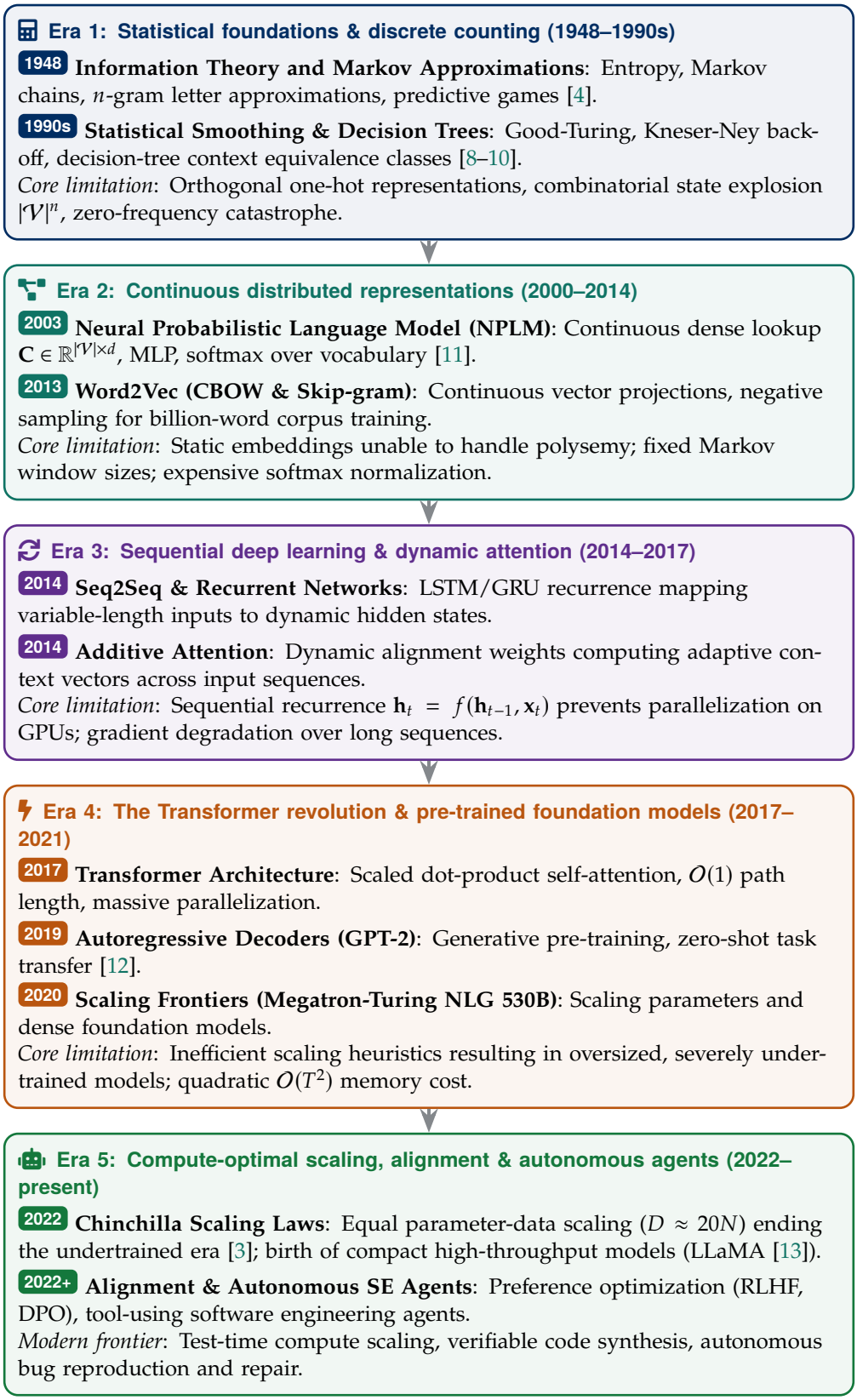


Figure 1.3: Chronological timeline of language model evolution from 1948 to the present.

1.8.1 1948–1980s: Shannon's information theory and Markov n-grams

The conceptual origin of language modeling stems from the landmark 1948 paper *"A Mathematical Theory of Communication"* [4]. In that work, written human language was treated as a discrete stochastic process generated by an underlying stationary Markov source over a finite alphabet.

Shannon introduced the *Shannon game*—a pedagogical thought experiment where an observer attempts to predict the next letter in an unknown sequence of text given the preceding letters. Shannon demonstrated that as the conditioning context length increases from zero-order (independent uniform character selection) to unigram (individual character frequencies), bigram (pairs), and trigram (triplets) models, the generated text rapidly converges toward natural English orthography and syntax.

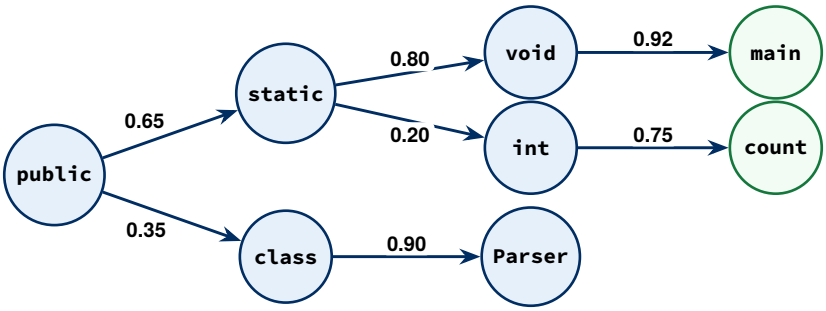


Figure 1.4: Markov chain state transition graph for software tokens.

In this framework, the conditional probability of token w_t given context $w_1, \ldots, w_{t-1}$ is simplified via an order-$(n-1)$ Markov assumption, as illustrated in Figure 1.4 for code syntax.

1.8.2 1990s: statistical decision trees and context partitioning

Throughout the 1990s, as speech recognition and machine translation systems scaled, researchers recognized that flat n-gram tables could not condition on long histories without exponential parameter explosion.

To overcome this limitation in the course progression, *decision-tree language models* were introduced in [10] (as examined on Slide 19 of the course lecture deck). Rather than conditioning probability on a rigid, exact sequence of $n-1$ tokens, a binary decision tree asks a sequence of syntactic and lexical questions about the historical context $\mathcal{H} = (w_{t-1}, w_{t-2}, \ldots)$.

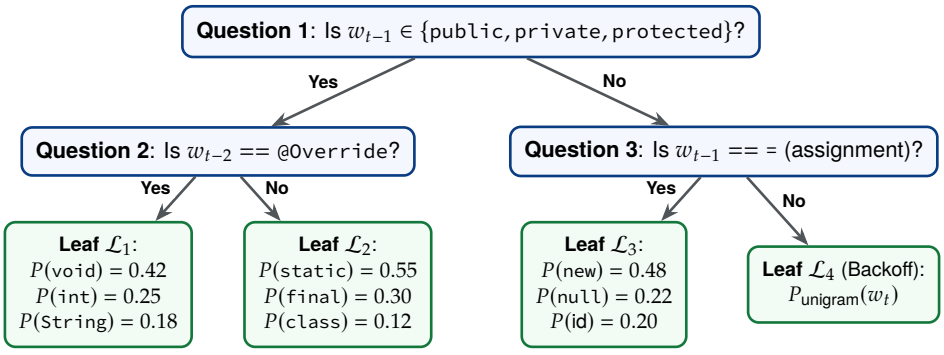


Figure 1.5: Decision tree context partitioning for conditional language modeling [10].

As illustrated in Figure 1.5, context histories that share identical predictive characteristics are clustered into the same leaf node $\mathcal{L}_k$, yielding conditional distributions $\mathbb{P}(w_t \mid \mathcal{L}_k)$. While this reduced the number of free parameters compared to full tables, the model remained fundamentally bound to discrete categorical splitting without true semantic interpolation.

1.8.3 2000–2003: The neural probabilistic language model

In the landmark 2003 paper *"A Neural Probabilistic Language Model"* [11], natural language processing was revolutionized by introducing continuous distributed representations to defeat the curse of dimensionality (as highlighted on Slide 20 of the course lectures).

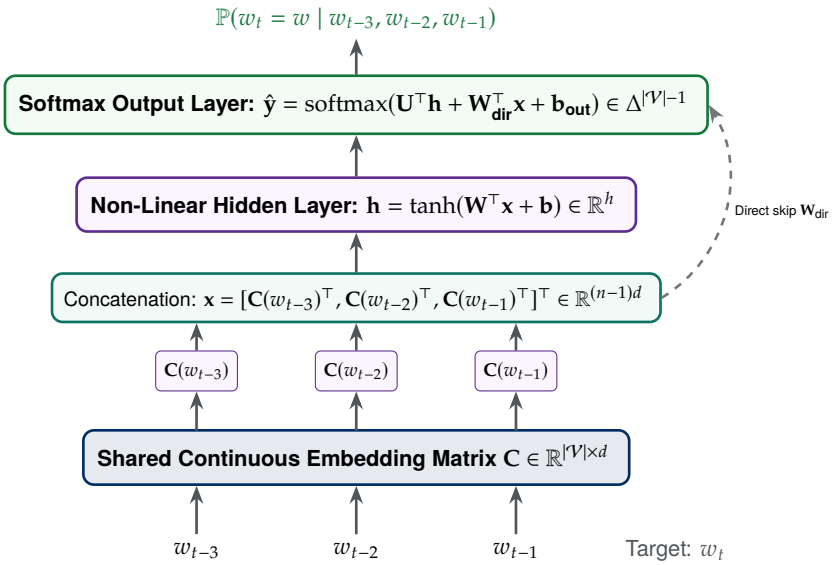


Figure 1.6: The Neural Probabilistic Language Model (NPLM) architecture [11]. Discrete context words are mapped into continuous dense feature vectors via shared matrix $\mathbf{C} \in \mathbb{R}^{|\mathcal{V}| \times d}$, concatenated, and processed through a non-linear hidden layer to predict next-token probabilities via softmax.

The neural probabilistic formulation in [11] replaced discrete count tables with a continuous parameterized architecture (Figure 1.6):

1. **Continuous Embedding Projection**: Each word $w \in \mathcal{V}$ is associated with a real-valued distributed feature vector $\mathbf{c}_w \in \mathbb{R}^d$ ($d \ll |\mathcal{V}|$, e.g., $d = 100$). The vocabulary lookup matrix $\mathbf{C} \in \mathbb{R}^{|\mathcal{V}| \times d}$ is learned jointly via backpropagation.

2. **Semantic Smoothing via Continuous Space**: If *"cat"* and *"feline"* have similar embedding vectors ($\mathbf{c}_{\text{cat}} \approx \mathbf{c}_{\text{feline}}$), observing *"the cat chased the mouse"* automatically increases the model's confidence in *"the feline chased the mouse"*. Continuous geometry solves the orthogonal symbol limitation.

3. **Softmax Output Normalization**: The network projects hidden activations to $|\mathcal{V}|$ unnormalized logits and applies the softmax function to guarantee a valid probability distribution:

$$\mathbb{P}(w_t = i \mid w_{<t}) = \frac{\exp(z_i)}{\sum_{j=1}^{|\mathcal{V}|} \exp(z_j)}. \tag{1.43}$$

1.8.4 2018–2019: Generative pre-training and GPT-2

While continuous representations were established in [11], models throughout the 2000s and 2010s remained restricted in scale and context length. In 2018–2019, the GPT-2 architecture was introduced in [12] (*"Language Models are Unsupervised Multitask Learners"*, examined on Slide 21 of the course slides).

GPT-2 demonstrated that massive, autoregressive decoder-only models trained on hundreds of gigabytes of uncurated internet text (WebText) could perform diverse natural language

and software tasks without task-specific supervised training. By predicting next tokens over context windows of 1,024 tokens, the model exhibited emergent zero-shot transfer across reading comprehension, machine translation, and Python code generation.

1.8.5 2020–2022: Megatron-Turing NLG 530B and compute-optimal scaling

Following GPT-2 and GPT-3, the deep learning industry initiated an aggressive scaling arms race, culminating in monolithic models such as the Megatron-Turing NLG 530B (Slide 22). However, early scaling heuristics [14] recommended scaling model parameters faster than training data. Consequently, models like Megatron-Turing NLG 530B and GPT-3 (175B parameters trained on 300B tokens) were severely **undertrained** relative to their parameter capacity.

In 2022, the landmark study *"Training Compute-Optimal Large Language Models"* [3] established the **Chinchilla** scaling laws:

$$D \approx 20 \times N, \tag{1.44}$$

where N is parameter count and D is training tokens. For compute optimality, parameter count and training tokens must scale in equal proportion. This insight prompted the modern era of compact, heavily trained open-weight models (e.g., LLaMA [13] trained on trillions of tokens), dramatically reducing deployment and inference overhead in software engineering environments.

1.8.6 From discrete counting tables to continuous parametric learners: bridge to deep learning

Our exploration of statistical language models in this chapter has revealed a fundamental evolutionary trajectory:

- **The Discrete Counting Limit**: Exact probability factorization via the chain rule (Equation (1.2)) is theoretically exact, but empirical n-gram counting tables suffer from exponential state-space explosion (V^n), context horizon amnesia ($n-1$), and orthogonal semantic blindness.
- **The Smoothing Ceiling**: While smoothing algorithms (Good-Turing, Katz, Kneser-Ney) rescue models from zero-frequency collapse ($P = 0 \implies \text{PPL} = \infty$), they merely redistribute probability mass across discrete table bins. They cannot learn high-dimensional feature interactions or share statistical strength between synonymous phrases.

To transcend the discrete counting paradigm, natural language processing abandoned frequency tables in favor of **continuous, parameterized neural networks**:

$$f(\mathbf{x}; \boldsymbol{\theta}) = \mathbf{W}^\top \mathbf{x} + \mathbf{b}.$$

Instead of counting co-occurrences in text, we parameterize language predictors as continuous functions whose weights $\boldsymbol{\theta}$ are optimized iteratively through gradient descent and backpropagation.

This transition requires establishing the mathematical and computational foundations of deep learning: How does an artificial neuron compute? Why is stacking linear layers without non-linear activations mathematically futile? What allows multi-layer perceptrons to approximate arbitrary complex functions? And how does backpropagation automatically compute gradients across computational graphs?

We address each of these foundational questions in Chapter 2: *Deep Learning Foundations for NLP*.

CHAPTER 2

Deep learning foundations for NLP

2.1 From biological neurons to the artificial perceptron

Having established the probabilistic formulation of language modeling and the combinatorial limitations of discrete symbol counting in Chapter 1, we now develop the neural computational foundations that allow language models to scale. In contemporary Natural Language Processing and AI-assisted Software Engineering, next-token prediction is reformulated as a high-dimensional, non-linear classification problem over continuous vector spaces. Rather than storing astronomical tables of discrete counts, we parameterize language predictors as continuous functions whose weights are optimized via gradient descent.

The foundational building block of modern deep learning is the *perceptron*, originally inspired by biological neurons in the human nervous system. A biological neuron receives electrochemical signals from upstream cells through branching *dendrites*, integrates these incoming potentials in its central cell body (*soma*), and, when the accumulated electrical voltage exceeds a physiological threshold (the action potential), fires a spike through its *axon* to downstream synapses.

In computational neural networks, this biological mechanism was first formalized in 1943 [15] as a threshold logic unit and subsequently refined in 1958 [16] into the trainable *Perceptron* (examined on Slide 7 of the course deep learning foundations).

Definition 2.1: *The artificial perceptron*

Let $\mathbf{x} = [x_1, x_2, \ldots, x_n]^\top \in \mathbb{R}^n$ represent an input feature vector, and let $\mathbf{w} = [w_1, w_2, \ldots, w_n]^\top \in \mathbb{R}^n$ represent the corresponding parameter weights, with scalar bias $w_0 = b \in \mathbb{R}$. The artificial perceptron computes:

$$y = f\left(w_0 + \sum_{i=1}^{n} w_i x_i\right) = f\left(w_0 + w_1 x_1 + w_2 x_2 + \cdots + w_n x_n\right), \tag{2.1}$$

where $f : \mathbb{R} \to \mathbb{R}$ is the activation function. By adopting the standard convention of augmenting the input vector with a constant dummy feature $x_0 = 1$ and defining the parameter vector as $\mathbf{w} = [w_0, w_1, \ldots, w_n]^\top \in \mathbb{R}^{n+1}$, Equation (2.1) is expressed in compact vector inner-product notation:

$$y = f\left(\sum_{i=0}^{n} w_i x_i\right) = f\left(\mathbf{w}^\top \mathbf{x}\right) \quad \text{with } x_0 = 1,\ w_0 = \text{bias}. \tag{2.2}$$

Figure 2.1 illustrates the structural mechanics of the artificial perceptron: input signals are weighted by synaptic connection strengths, summed at a central junction, and passed through an activation function to generate the output signal.

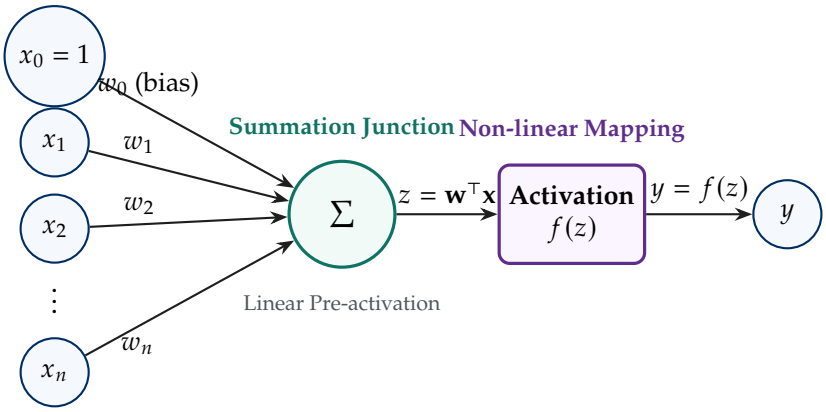


Figure 2.1: Architectural schematic of the artificial perceptron. Inputs x_i are modulated by synaptic weights w_i, accumulated with bias w_0 at the summation junction Σ to yield pre-activation $z = \mathbf{w}^\top \mathbf{x}$, and mapped through activation function $f(z)$ to produce scalar output y.

2.1.1 The perceptron in two dimensions: linear decision hyperplane

To establish geometric intuition, consider a two-dimensional feature space $\mathbf{x} = [x_1, x_2]^\top \in \mathbb{R}^2$ with weights $\mathbf{w} = [w_1, w_2]^\top$ and bias w_0. The pre-activation value is:

$$z(\mathbf{x}) = w_1 x_1 + w_2 x_2 + w_0. \tag{2.3}$$

When the perceptron acts as a binary classifier using a step or sign activation ($y = \text{sgn}(z)$), the decision boundary separating the two output classes is precisely the set of points where the pre-activation vanishes:

$$\mathcal{H} = \left\{ \mathbf{x} \in \mathbb{R}^2 \mid w_1 x_1 + w_2 x_2 + w_0 = 0 \right\}. \tag{2.4}$$

Geometric interpretation of weights and bias. In two dimensions, Equation (2.4) defines a straight line in the (x_1, x_2) Cartesian plane:

- **Normal Vector**: The weight vector $\mathbf{w} = [w_1, w_2]^\top$ is strictly perpendicular (orthogonal) to the separating line $\mathcal{H}$. For any two points $\mathbf{x}_a, \mathbf{x}_b \in \mathcal{H}$:

  $$\mathbf{w}^\top \mathbf{x}_a + w_0 = 0 \quad \text{and} \quad \mathbf{w}^\top \mathbf{x}_b + w_0 = 0 \implies \mathbf{w}^\top (\mathbf{x}_a - \mathbf{x}_b) = 0.$$

  The vector $\mathbf{w}$ points directly into the positive decision region ($z > 0$).

- **Orthogonal Distance to Origin**: The shortest Euclidean distance from the origin $\mathbf{0} = [0, 0]^\top$ to the separating hyperplane $\mathcal{H}$ is:

  $$d = \frac{|w_0|}{\|\mathbf{w}\|_2} = \frac{|w_0|}{\sqrt{w_1^2 + w_2^2}}. \tag{2.5}$$

  The bias w_0 shifts the hyperplane parallel to itself: if $w_0 = 0$, the decision boundary passes exactly through the origin; adjusting w_0 translates the boundary without altering its orientation.

Figure 2.2 diagrams this geometric relationship in $\mathbb{R}^2$.

2.1.2 Connection to classical linear and logistic regression

The artificial perceptron unifies two foundational statistical models studied in classical data analysis:

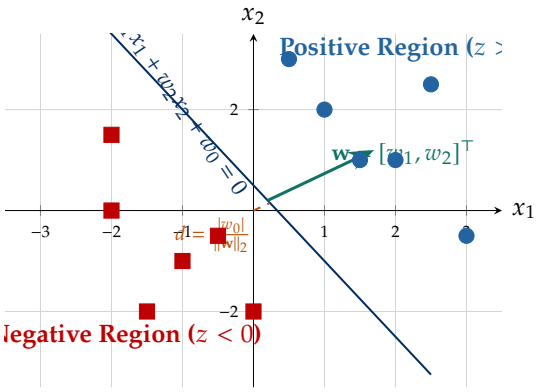


Figure 2.2: Geometric representation of the perceptron decision boundary in $\mathbb{R}^2$. The line $w_1 x_1 + w_2 x_2 + w_0 = 0$ partitions feature space into positive ($\mathcal{H}^+$) and negative ($\mathcal{H}^-$) half-spaces. The weight vector $\mathbf{w}$ defines the normal vector to the separating hyperplane, and the scalar bias w_0 determines its distance from the origin.

- **Linear Regression**: If the activation function is the linear identity $f(z) = z$, Equation (2.2) computes:
$$y = \mathbf{w}^\top \mathbf{x},$$
predicting an unbounded real-valued continuous response variable.
- **Logistic Regression**: If the activation function is the logistic sigmoid $f(z) = \sigma(z) = \frac{1}{1+e^{-z}}$, Equation (2.2) computes:
$$y = \sigma(\mathbf{w}^\top \mathbf{x}) = \mathbb{P}(Y = 1 \mid \mathbf{x}),$$
predicting the posterior Bernoulli probability of belonging to the positive class.

2.1.3 The fatal barrier of single-layer perceptrons: XOR linear inseparability

Despite its mathematical elegance, the single-layer perceptron suffers from a fatal representational barrier that brought artificial intelligence research to a virtual standstill for over a decade. In their seminal 1969 monograph *"Perceptrons"*, Marvin Minsky and Seymour Papert demonstrated that a single-layer perceptron cannot compute the basic *exclusive-or* (XOR) logical function.

The XOR truth table and linear system. Consider the XOR logic function over binary inputs $x_1, x_2 \in \{0, 1\}$:

x_1	x_2	Target $y = x_1 \oplus x_2$
0	0	0
0	1	1
1	0	1
1	1	0

For a single-layer perceptron with weights w_1, w_2 and bias w_0 to compute this function, there must exist parameters such that the linear boundary $w_1 x_1 + w_2 x_2 + w_0$ outputs positive values ($z \geq 0$) for target 1 and negative values ($z < 0$) for target 0. Substituting the four input pairs yields the system of linear inequalities:

$$(x_1 = 0, x_2 = 0) \implies w_0 < 0, \tag{2.6}$$
$$(x_1 = 1, x_2 = 0) \implies w_1 + w_0 \geq 0, \tag{2.7}$$
$$(x_1 = 0, x_2 = 1) \implies w_2 + w_0 \geq 0, \tag{2.8}$$
$$(x_1 = 1, x_2 = 1) \implies w_1 + w_2 + w_0 < 0. \tag{2.9}$$

Proof by mathematical contradiction. Summing inequalities (2.7) and (2.8):

$$(w_1 + w_0) + (w_2 + w_0) \geq 0 \implies w_1 + w_2 + 2w_0 \geq 0 \implies w_1 + w_2 \geq -2w_0. \tag{2.10}$$

Now examine inequality (2.9):

$$w_1 + w_2 + w_0 < 0 \implies w_1 + w_2 < -w_0. \tag{2.11}$$

Combining inequalities (2.10) and (2.11) yields:

$$-2w_0 \leq w_1 + w_2 < -w_0 \implies -2w_0 < -w_0.$$

Adding $2w_0$ to both sides yields:

$$0 < w_0 \implies w_0 > 0.$$

However, inequality (2.6) strictly demands that $w_0 < 0$. The assumption that a single linear hyperplane can compute XOR produces a direct mathematical contradiction ($w_0 > 0$ and $w_0 < 0$). ■

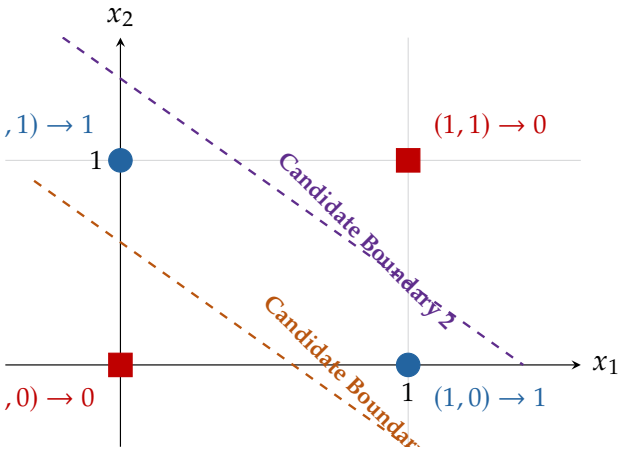


Figure 2.3: The XOR linear inseparability barrier (Minsky & Papert 1969). Class 1 points $(0, 1)$ and $(1, 0)$ (blue circles) cannot be separated from Class 0 points $(0, 0)$ and $(1, 1)$ (red squares) by any single linear hyperplane $w_1x_1 + w_2x_2 + w_0 = 0$. Any single line either isolates $(0, 0)$ or $(1, 1)$, but cannot simultaneously group both off-diagonal points.

As visualized in Figure 2.3, no straight line can separate the diagonal pairs. Because natural language and software engineering phenomena are riddled with non-linear relationships (e.g., mutually dependent programming idioms, nested conditional scopes), single-layer linear perceptrons are fundamentally inadequate.

The causal question. How can we overcome this linear inseparability barrier? An immediate intuitive engineering hypothesis presents itself: *If one linear perceptron can draw one line, can we solve non-linear problems simply by cascading multiple linear perceptrons into a multi-layer network?* We test this hypothesis next.

2.2 Stacking linear layers and the proof of linear stacking collapse

2.2.1 Fully connected linear layers (dense layers)

In multi-layer architectures, individual perceptrons do not operate in isolation; instead, multiple neurons run concurrently across a shared input vector $\mathbf{x} \in \mathbb{R}^{d_{\text{in}}}$, forming a *dense layer* (or fully connected layer).

Let d_{in} denote the input feature dimensionality and d_{out} denote the number of hidden or output units. Stacking the individual weight vectors $\mathbf{w}_j \in \mathbb{R}^{d_{\text{in}}}$ as columns of a weight matrix $\mathbf{W} \in \mathbb{R}^{d_{\text{in}} \times d_{\text{out}}}$ and collecting the scalar biases b_j into a bias vector $\mathbf{b} \in \mathbb{R}^{d_{\text{out}}}$, the layer computation expresses compactly in matrix notation:

$$\mathbf{y} = f\left(\mathbf{W}^\top \mathbf{x} + \mathbf{b}\right), \tag{2.12}$$

where $f(\cdot)$ is applied elementwise. Figure 2.4 diagrams this fully connected bipartite structure with explicit tensor dimensions and connection weights.

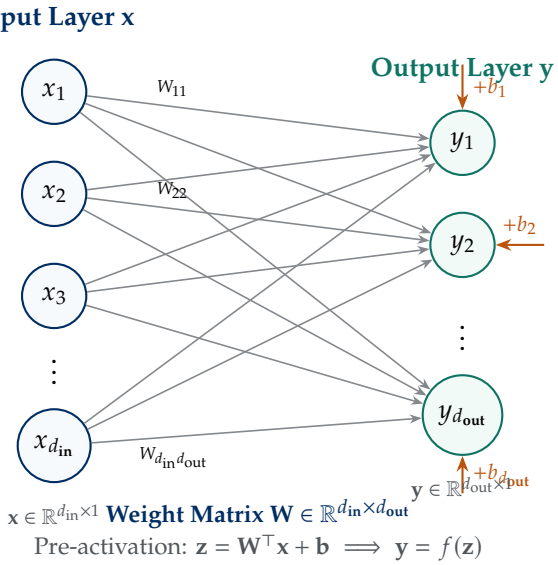


Figure 2.4: Architecture of a fully connected dense layer mapping an input vector $\mathbf{x} \in \mathbb{R}^{d_{\text{in}}}$ to output $\mathbf{y} \in \mathbb{R}^{d_{\text{out}}}$. Each directed edge carries an individual scalar weight W_{ij}, and each output neuron incorporates an additive bias b_j.

Exact parameter counting across stacked dense layers. For an affine transformation mapping $\mathbb{R}^{d_{\text{in}}} \longrightarrow \mathbb{R}^{d_{\text{out}}}$:

$$N_{\text{params}} = \underbrace{d_{\text{in}} \times d_{\text{out}}}_{\text{weight parameters}} + \underbrace{d_{\text{out}}}_{\text{bias parameters}} = d_{\text{out}}(d_{\text{in}} + 1). \tag{2.13}$$

Example Calculation: Consider a two-layer Multi-Layer Perceptron (MLP) mapping an embedding of dimension $d_{\text{in}} = 768$ through a hidden layer $d_{\text{hidden}} = 3072$ to an output projection $d_{\text{out}} = 768$:

1. Layer 1: $3072 \times (768 + 1) = 2{,}359{,}296 + 3072 = 2{,}362{,}368$ parameters.
2. Layer 2: $768 \times (3072 + 1) = 2{,}359{,}296 + 768 = 2{,}360{,}064$ parameters.
3. Total Parameters: $2{,}362{,}368 + 2{,}360{,}064 = 4{,}722{,}432$ parameters.

It is essential to account for the additive bias parameters (d_{out}) alongside the multiplicative weight matrix ($d_{\text{in}} \times d_{\text{out}}$) to obtain exact parameter counts.

2.2.2 The mathematical proof of linear stacking collapse

Now we return to our hypothesis: *Can we solve XOR and non-linear problems simply by cascading multiple linear layers without activation functions?*

The following theorem demonstrates that without non-linear activations, stacking arbitrary numbers of linear layers provides zero additional representational power over a single-layer linear model.

Theorem 2.1: *Linear stacking collapse*

Let $f_1 : \mathbb{R}^n \to \mathbb{R}^m$ and $f_2 : \mathbb{R}^m \to \mathbb{R}^k$ denote two consecutive fully connected layers operating without non-linear activations ($f(x) = x$). Let $\mathbf{W}_1 \in \mathbb{R}^{n \times m}$ and $\mathbf{b}_1 \in \mathbb{R}^m$ be the parameters of the first layer, and let $\mathbf{W}_2 \in \mathbb{R}^{m \times k}$ and $\mathbf{b}_2 \in \mathbb{R}^k$ be the parameters of the second layer:

$$\mathbf{h} = f_1(\mathbf{x}) = \mathbf{W}_1^\top \mathbf{x} + \mathbf{b}_1, \tag{2.14}$$

$$\mathbf{z} = f_2(\mathbf{h}) = \mathbf{W}_2^\top \mathbf{h} + \mathbf{b}_2. \tag{2.15}$$

The composite function $\mathbf{z} = f_2(f_1(\mathbf{x}))$ collapses identically to a single affine transformation:

$$\begin{aligned} \mathbf{z} &= \mathbf{W}_2^\top \left(\mathbf{W}_1^\top \mathbf{x} + \mathbf{b}_1\right) + \mathbf{b}_2 \\ &= \mathbf{W}_2^\top \mathbf{W}_1^\top \mathbf{x} + \mathbf{W}_2^\top \mathbf{b}_1 + \mathbf{b}_2 \\ &= (\mathbf{W}_1 \mathbf{W}_2)^\top \mathbf{x} + \left(\mathbf{W}_2^\top \mathbf{b}_1 + \mathbf{b}_2\right) \\ &= \mathbf{W}_{\text{net}}^\top \mathbf{x} + \mathbf{b}_{\text{net}}, \end{aligned} \tag{2.16}$$

where $\mathbf{W}_{\text{net}} = \mathbf{W}_1 \mathbf{W}_2 \in \mathbb{R}^{n \times k}$ and $\mathbf{b}_{\text{net}} = \mathbf{W}_2^\top \mathbf{b}_1 + \mathbf{b}_2 \in \mathbb{R}^k$.

Proof by mathematical induction. Matrix multiplication is associative and closed under composition. The product of two matrices $\mathbf{W}_1 \in \mathbb{R}^{n \times m}$ and $\mathbf{W}_2 \in \mathbb{R}^{m \times k}$ is simply a single matrix $\mathbf{W}_{\text{net}} \in \mathbb{R}^{n \times k}$. Likewise, the affine transformation of bias vector $\mathbf{b}_1$ by $\mathbf{W}_2^\top$ added to $\mathbf{b}_2$ is simply a single constant vector $\mathbf{b}_{\text{net}} \in \mathbb{R}^k$.

By mathematical induction, for any cascade of L linear layers:

$$\mathbf{z} = \mathbf{W}_L^\top \left(\dots \left(\mathbf{W}_2^\top \left(\mathbf{W}_1^\top \mathbf{x} + \mathbf{b}_1\right) + \mathbf{b}_2\right) \dots\right) + \mathbf{b}_L = \mathbf{W}_{\text{net}}^\top \mathbf{x} + \mathbf{b}_{\text{net}}, \tag{2.17}$$

where:

$$\mathbf{W}_{\text{net}} = \prod_{l=1}^{L} \mathbf{W}_l \in \mathbb{R}^{d_{\text{in}} \times d_{\text{out}}}, \quad \mathbf{b}_{\text{net}} = \mathbf{b}_L + \sum_{l=1}^{L-1} \left(\prod_{j=l+1}^{L} \mathbf{W}_j \right)^\top \mathbf{b}_l. \tag{2.18}$$

The composite mapping is strictly linear. Stacking 100 purely linear layers introduces millions of redundant matrix multiplications while learning nothing beyond standard linear regression. It strictly cannot compute XOR or any non-linear function. ■

The fundamental takeaway. Non-linear activation functions are strictly and mathematically mandatory. Without non-linearity between layers, deep networks collapse into trivial shallow affine mappings. Non-linear activations warp and fold coordinate space, transforming linearly inseparable data manifolds into representations where classes can be cleanly separated.

2.2.3 Universal approximation theorem and bump-function synthesis

Given that non-linear activation functions prevent linear collapse, what family of functions can a non-linear neural network represent? The theoretical foundation validating neural networks as general-purpose function approximators is the *Universal Approximation Theorem*.

Theorem 2.2: *Universal approximation theorem*

Let $\sigma : \mathbb{R} \to \mathbb{R}$ be any continuous, non-constant, bounded, and monotonically increasing activation function (e.g., sigmoid) [17]. Let $I_n \subset \mathbb{R}^n$ denote a compact (closed and bounded) subset of n-dimensional Euclidean space, and let $C(I_n)$ denote the space of continuous real-valued functions on I_n. Then, for any target continuous function $g \in C(I_n)$ and any arbitrary error tolerance $\epsilon > 0$, there exist a finite integer $M \in \mathbb{N}$, parameter weights $\mathbf{w}_j \in \mathbb{R}^n$, outer weights $\alpha_j \in \mathbb{R}$, and biases $b_j \in \mathbb{R}$ such that the single-hidden-layer feedforward network:

$$F(\mathbf{x}) = \sum_{j=1}^{M} \alpha_j \sigma\left(\mathbf{w}_j^\top \mathbf{x} + b_j\right) \tag{2.19}$$

satisfies uniform approximation error:

$$\sup_{\mathbf{x} \in I_n} \left|F(\mathbf{x}) - g(\mathbf{x})\right| < \epsilon. \tag{2.20}$$

Originally proved in [17] for sigmoidal activations and generalized in [18] and [19] to arbitrary continuous non-polynomial activations, this theorem guarantees that a feedforward network with a single hidden layer has the mathematical capacity to approximate any continuous function (as discussed in the course theoretical foundations).

Intuitive geometric derivation: bump-function synthesis. To understand intuitively why a single hidden layer can approximate any continuous curve, consider a 1D function $g(x) : [a, b] \to \mathbb{R}$.

Step 1: Constructing a localized rectangular bump. Consider two sigmoid neurons with very high slope $w \gg 0$ located at spatial offsets x_1 and x_2 (with $x_1 < x_2$). As $w \to \infty$, the sigmoid $\sigma(w(x - x_1))$ converges to a Heaviside step function $H(x - x_1)$.

Subtracting the second sigmoid from the first produces a localized *bump function* (or rectangular wavelet):

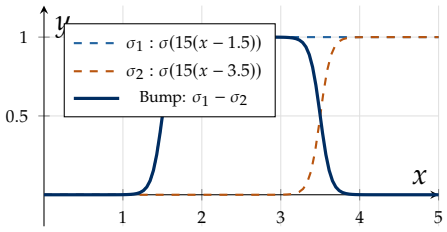
$$B(x) = \sigma(w(x - x_1)) - \sigma(w(x - x_2)) \approx \begin{cases} 1 & \text{if } x_1 \le x \le x_2 \\ 0 & \text{otherwise} \end{cases}. \tag{2.21}$$

Figure 2.5(a) illustrates how two opposing sigmoids subtract to create a localized rectangular pulse.

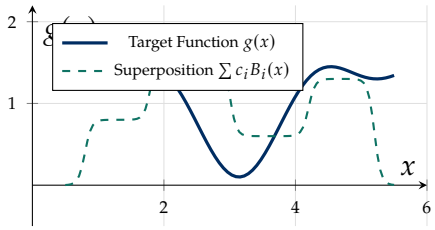
Step 2: Riemann sum function reconstruction. By the definition of continuous functions on compact intervals, $g(x)$ can be approximated to arbitrary accuracy ϵ by a piecewise-constant step function consisting of M uniform sub-intervals $[x_k, x_{k+1}]$ with heights $c_k = g(x_k)$, exactly as in standard Riemann integration:

$$g(x) \approx \sum_{k=1}^{M} c_k B_k(x) = \sum_{k=1}^{M} c_k \left[\sigma(w(x - x_k)) - \sigma(w(x - x_{k+1}))\right]. \tag{2.22}$$

Each bump $B_k(x)$ requires exactly two hidden sigmoid neurons. Consequently, a single hidden layer containing $2M$ neurons can synthesize any continuous function to arbitrary precision, as visualized in Figure 2.5(b).



(a) Bump Synthesis via Opposing Sigmoids



(b) Riemann-style Function Reconstruction

Figure 2.5: Visual derivation of the Universal Approximation Theorem. (a) Subtracting two shifted steep sigmoids produces a localized rectangular bump function $B(x)$; (b) Superimposing multiple scaled bump functions reconstructs any continuous target function to arbitrary accuracy ϵ.

2.2.4 Why deep networks if shallow networks are universal approximators?

If a single hidden layer can approximate any function, why does modern deep learning exclusively use deep architectures with dozens or hundreds of stacked layers (e.g., 96 layers in GPT-3)?

1. **Non-Constructive Existence**: The theorem guarantees that parameters exist, but provides no gradient-based algorithm to discover them and establishes no finite bound on the width M.

2. **Exponential Parameter Inefficiency**: Approximating highly compositional functions in n dimensions with a single hidden layer requires a width M that scales exponentially with input dimensionality ($M \sim O(2^n)$).

3. **Hierarchical Compositionality**: Deep networks factorize functions into hierarchical abstractions: early layers learn low-level primitives (subwords, syntax), intermediate layers learn grammatical structure, and deep layers capture long-range semantic discourse. Deep networks achieve equal approximation power with exponentially fewer parameters than shallow networks.

2.3 Taxonomy of neural activation functions

Modern neural NLP architectures deploy diverse activation functions tailored to hidden representations and output heads. Figure 2.6 presents exact mathematical plots and derivatives for the primary activation functions.

1. The logistic sigmoid (σ).

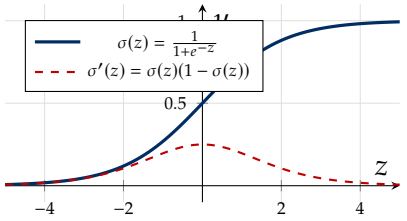
$$\sigma(z) = \frac{1}{1+e^{-z}}, \quad \text{Range: } (0,1), \quad \sigma'(z) = \sigma(z)(1-\sigma(z)). \tag{2.23}$$

The derivative attains its global maximum of $\sigma'(0) = 0.25$ at the origin. For large positive or negative inputs ($|z| \geq 4$), $\sigma'(z) \to 0$, causing severe *vanishing gradients* in deep networks.

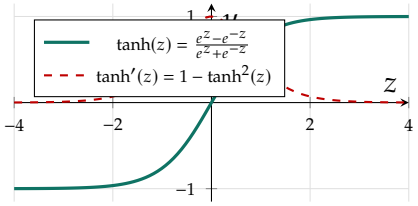
2. Hyperbolic tangent (tanh**).**

$$\tanh(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}} = 2\,\sigma(2z) - 1, \quad \text{Range: } (-1,1), \quad \tanh'(z) = 1 - \tanh^2(z). \tag{2.24}$$

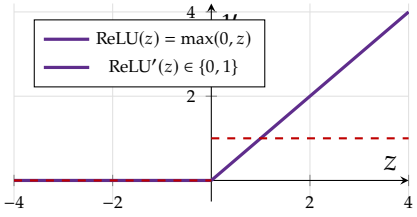
The hyperbolic tangent is strictly zero-centered, possessing average activation values near zero, which accelerates gradient descent compared to non-negative sigmoids. However, it still saturates for $|z| \geq 3$, yielding vanishing gradients.



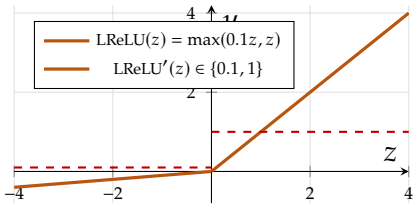
(a) Sigmoid (Logistic)



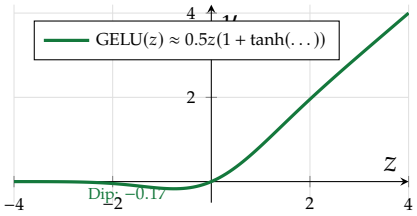
(b) Hyperbolic Tangent (tanh)



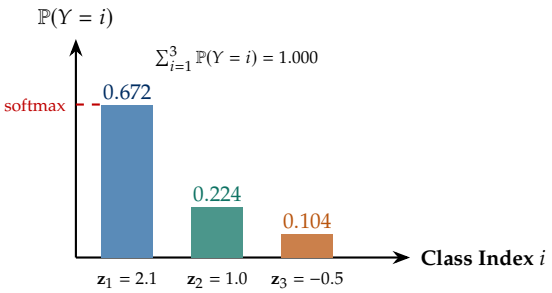
(c) Rectified Linear Unit (ReLU)



(d) Leaky ReLU ($\alpha = 0.1$)



(e) Gaussian Error Linear Unit (GELU)



(f) Softmax Simplex Mapping ($\mathbb{R}^3 \to \Delta^2$)

Figure 2.6: Taxonomy of primary neural activation functions plotted across their operating ranges: (a) Sigmoid $\sigma(z)$ with peak derivative 0.25; (b) Zero-centered $\tanh(z)$ with peak derivative 1.0; (c) Piecewise linear $\text{ReLU}(z)$; (d) Leaky ReLU preventing dead units; (e) Non-monotonic $\text{GELU}(z)$ used in modern Transformers; (f) Multi-class Softmax mapping logits to the probability simplex.

3. Rectified linear unit (ReLU**).**

$$\text{ReLU}(z) = \max(0, z), \quad \text{Range: } [0, \infty), \quad \text{ReLU}'(z) = \begin{cases} 1 & \text{if } z > 0 \\ 0 & \text{if } z < 0 \end{cases}. \tag{2.25}$$

ReLU completely eliminates gradient vanishing for positive pre-activations ($z > 0$). However, if a large gradient update shifts a neuron's weights such that $z \leq 0$ across all training samples, its activation and derivative permanently collapse to zero ($\text{ReLU}'(z) = 0$), a pathology known as the *dying ReLU problem*.

4. Leaky ReLU (LReLU).

$$\text{LReLU}(z) = \max(\alpha z, z) \quad \text{with } 0 < \alpha \ll 1, \quad \text{Range: } (-\infty, \infty), \quad \text{LReLU}'(z) = \begin{cases} 1 & \text{if } z > 0 \\ \alpha & \text{if } z \leq 0 \end{cases}. \tag{2.26}$$

By maintaining a small constant slope α (typically $\alpha = 0.01$) in the negative half-plane, Leaky ReLU guarantees that a non-zero gradient always backpropagates, preventing dead neurons.

5. Gaussian error linear unit (GELU**).** Introduced by Dan Hendrycks and Kevin Gimpel, GELU scales the input by its cumulative standard normal probability $\Phi(z)$:

$$\text{GELU}(z) = z \cdot \Phi(z) = z \cdot \frac{1}{2}\left[1 + \text{erf}\left(\frac{z}{\sqrt{2}}\right)\right] \approx 0.5z\left(1 + \tanh\left(\sqrt{\frac{2}{\pi}}\left(z + 0.044715z^3\right)\right)\right). \tag{2.27}$$

GELU is smooth, differentiable everywhere, and non-monotonic, dipping to a global minimum of ≈ -0.17 at $z \approx -0.75$. This smoothness provides curvature information during optimization, making GELU the default activation in Transformer language models (GPT-2, GPT-3, BERT, RoBERTa).

6. Softmax. For categorical multi-class predictions over C mutually exclusive classes, Softmax normalizes a real-valued logit vector $\mathbf{z} = [z_1, \ldots, z_C]^\top \in \mathbb{R}^C$ onto the unit probability simplex Δ^{C-1}:

$$\hat{y}_i = \text{softmax}(\mathbf{z})_i = \frac{e^{z_i}}{\sum_{j=1}^{C} e^{z_j}} \quad \text{such that } \hat{y}_i \in (0, 1) \quad \text{and} \quad \sum_{i=1}^{C} \hat{y}_i = 1. \tag{2.28}$$

Softmax forms the standard vocabulary output head of modern autoregressive language models, computing next-token probability distributions across vocabulary $\mathcal{V}$.

Table 2.1 provides a comparative reference summarizing the mathematical properties of each activation function.

2.4 Architectural task formulations and software engineering case studies

In machine learning applications for software engineering, task formulations map specific software artifacts (commits, ASTs, bug reports, source code tokens) into continuous predictions. Three primary architectural heads dominate:

Table 2.1: Comprehensive taxonomy of neural activation functions, mathematical formulations, operating bounds, and architectural domains.

Activation	Domain	Range	Peak Gradient	Primary Use Case in Language Modeling
Sigmoid (σ)	$\mathbb{R}$	$(0, 1)$	0.25 ($z = 0$)	Binary classification output heads; gating mechanisms in LSTM/GRU highways.
Tanh	$\mathbb{R}$	$(-1, 1)$	1.00 ($z = 0$)	Recurrent hidden state candidate generation; zero-centered coordinate representations.
ReLU	$\mathbb{R}$	$[0, \infty)$	1.00 ($z > 0$)	Classical deep feedforward networks; compute-efficient linear regions.
Leaky ReLU	$\mathbb{R}$	$\mathbb{R}$	1.00 ($z > 0$), α ($z \leq 0$)	Prevents dying neuron syndrome; robust discriminator networks.
GELU	$\mathbb{R}$	$[-0.17, \infty)$	≈ 1.08	Canonical Transformer activation (GPT, BERT); smooth, probabilistic gating.
Softmax	$\mathbb{R}^C$	Δ^{C-1}	Matrix Jacobian	Next-token prediction vocabulary head; maps logits to discrete probability simplex.
Identity	$\mathbb{R}$	$\mathbb{R}$	1.00 ($\forall z$)	Regression output head; unbounded linear prediction.

2.4.1 1. Binary classification: automated vulnerability detection

Consider predicting whether a proposed code commit contains a security vulnerability (e.g., buffer overflow, SQL injection).

- **Input Vector**: $\mathbf{x} \in \mathbb{R}^d$ (extracted code embeddings or static analysis metrics).
- **Output Layer**: A single linear neuron producing logit $z = \mathbf{w}^\top \mathbf{h} + b \in \mathbb{R}$.
- **Activation**: Logistic Sigmoid $\hat{y} = \sigma(z) = \mathbb{P}(\text{Vulnerable} = 1 \mid \mathbf{x}) \in [0, 1]$.
- **Decision Rule**: Flag for manual audit if $\hat{y} \geq \tau$ (where $\tau = 0.5$ by default, tuned to optimize recall).

Figure 2.7(a) illustrates the resulting separating hyperplane in feature space.

2.4.2 2. Multi-class classification: automated AST node type prediction

In program repair and neural compiler optimization, models classify Abstract Syntax Tree (AST) nodes into C discrete syntactic types (e.g., `MethodDeclaration`, `VariableDeclaration`, `IfStatement`, `WhileStatement`).

- **Output Layer**: C parallel linear neurons producing logit vector $\mathbf{z} = \mathbf{W}^\top \mathbf{h} + \mathbf{b} \in \mathbb{R}^C$.
- **Activation**: Softmax operator $\hat{\mathbf{y}} = \text{softmax}(\mathbf{z}) \in \Delta^{C-1}$.
- **Decision Rule**: Predicted AST type $\hat{c} = \arg\max_i \hat{y}_i$.

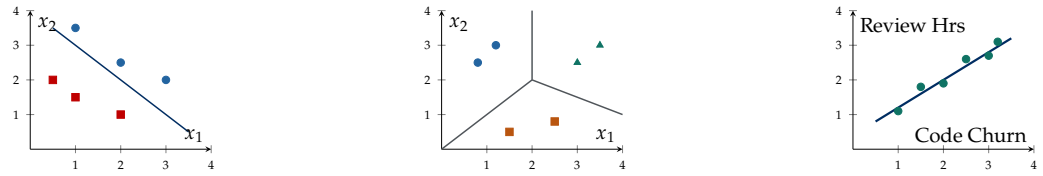
Figure 2.7(b) diagrams the resulting Voronoi-like partitioned multi-class decision space.

2.4.3 3. Regression: pull request review effort estimation

Engineering managers must predict continuous quantities, such as the estimated review time (in hours) required for an incoming pull request based on code churn, changed files, and author experience.

- **Output Layer**: A single linear neuron producing scalar $z = \mathbf{w}^\top \mathbf{h} + b$.
- **Activation**: Linear Identity function $f(z) = z \in \mathbb{R}$.
- **Prediction**: $\hat{y} = z \in \mathbb{R}$ (continuous unbounded prediction).

Figure 2.7(c) illustrates continuous line fitting through software engineering observation points.



(a) Binary: Defect Detection (b) Multi-class: AST Type (c) Regression: PR Review Effort

Figure 2.7: Task formulations across Software Engineering workflows: (a) Binary vulnerability detection with separating hyperplane; (b) Multi-class defect categorization with partitioned decision regions; (c) Continuous PR review effort regression with residual errors.

2.5 Loss function formulations and optimization landscapes

2.5.1 Loss function formulations

Training a neural network requires an objective function $\mathcal{L}(\theta)$ that measures the discrepancy between model predictions $\hat{y}$ and ground-truth targets y.

1. Mean squared error (MSE). For continuous regression targets:

$$\mathcal{L}_{\text{MSE}}(\theta) = \frac{1}{N} \sum_{i=1}^{N} \left(y_i - \hat{y}_i\right)^2. \tag{2.29}$$

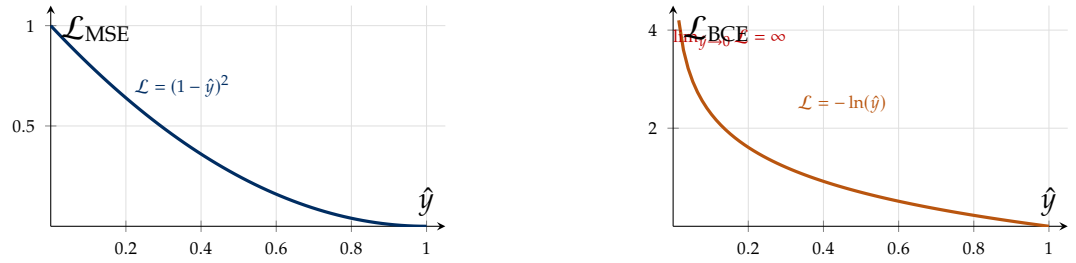
2. Binary cross-entropy (BCE). For binary classification with target $y \in \{0, 1\}$ and predicted probability $\hat{y} = \sigma(z) \in [0, 1]$:

$$\mathcal{L}_{\text{BCE}}(\theta) = -y \ln(\hat{y}) - (1 - y) \ln(1 - \hat{y}). \tag{2.30}$$

3. Categorical cross-entropy (CCE). For multi-class classification over C mutually exclusive classes with one-hot target vector $\mathbf{y} \in \{0, 1\}^C$:

$$\mathcal{L}_{\text{CE}}(\theta) = -\sum_{i=1}^{C} y_i \ln(\hat{y}_i) = -\ln\left(\hat{y}_{\text{target}}\right) = -\ln\left(\frac{e^{z_{\text{target}}}}{\sum_{j=1}^{C} e^{z_j}}\right). \tag{2.31}$$

Figure 2.8 compares the mathematical profiles of MSE and BCE for a positive target $y = 1$.



(a) MSE Loss for $y = 1$ (Bounded: $\mathcal{L} \in [0, 1]$)

(b) BCE Loss for $y = 1$ (Asymptotic Penalty)

Figure 2.8: Comparison of loss function behaviors for target $y = 1$. While MSE is bounded quadratically at 1.0, BCE imposes an asymptotic infinite penalty ($\mathcal{L} \to \infty$) as predicted probability $\hat{y} \to 0$, providing steep, non-vanishing corrective gradients.

2.5.2 Rigorous proof: why MSE fails for classification

A fundamental theoretical question is: *Why do we never use Mean Squared Error with sigmoid activation for classification?* The answer lies in gradient saturation and non-convexity.

Theorem 2.3: *Gradient saturation of MSE under sigmoid activation*

Let a binary classification model predict $\hat{y} = \sigma(z) = \frac{1}{1+e^{-z}}$ with ground-truth target $y \in \{0, 1\}$. Under the quadratic loss $\mathcal{L}_{\text{MSE}} = \frac{1}{2}(y - \hat{y})^2$, the gradient with respect to the pre-activation logit z is:

$$\frac{\partial \mathcal{L}_{\text{MSE}}}{\partial z} = -(y - \hat{y}) \cdot \hat{y}(1 - \hat{y}). \tag{2.32}$$

Conversely, under Binary Cross-Entropy $\mathcal{L}_{\text{BCE}} = -y \ln(\hat{y}) - (1 - y) \ln(1 - \hat{y})$, the gradient is:

$$\frac{\partial \mathcal{L}_{\text{BCE}}}{\partial z} = \hat{y} - y. \tag{2.33}$$

Proof. 1. For MSE, applying the calculus chain rule:

$$\frac{\partial \mathcal{L}_{\text{MSE}}}{\partial z} = \frac{\partial \mathcal{L}_{\text{MSE}}}{\partial \hat{y}} \frac{\partial \hat{y}}{\partial z}. \tag{2.34}$$

The derivative of the loss with respect to $\hat{y}$ is:

$$\frac{\partial \mathcal{L}_{\text{MSE}}}{\partial \hat{y}} = \frac{\partial}{\partial \hat{y}} \left[\frac{1}{2}(y - \hat{y})^2 \right] = -(y - \hat{y}).$$

The derivative of the sigmoid is $\sigma'(z) = \sigma(z)(1 - \sigma(z)) = \hat{y}(1 - \hat{y})$. Substituting:

$$\frac{\partial \mathcal{L}_{\text{MSE}}}{\partial z} = -(y - \hat{y})\hat{y}(1 - \hat{y}). \tag{2.35}$$

2. For BCE, expanding the loss:

$$\mathcal{L}_{\text{BCE}} = -y \ln(\sigma(z)) - (1 - y) \ln(1 - \sigma(z)). \tag{2.36}$$

Differentiating directly with respect to logit z:

$$\begin{aligned}\frac{\partial \mathcal{L}_{\text{BCE}}}{\partial z} &= -y\frac{\sigma'(z)}{\sigma(z)} - (1-y)\frac{-\sigma'(z)}{1-\sigma(z)} \\ &= -y\frac{\sigma(z)(1-\sigma(z))}{\sigma(z)} + (1-y)\frac{\sigma(z)(1-\sigma(z))}{1-\sigma(z)} \\ &= -y(1-\sigma(z)) + (1-y)\sigma(z) \\ &= -y + y\sigma(z) + \sigma(z) - y\sigma(z) \\ &= \sigma(z) - y = \hat{y} - y. \quad \blacksquare\end{aligned}$$

Analytical comparison of optimization behavior. The fundamental architectural flaws of MSE for classification are immediately apparent from Equations (2.35) and (2.33):

- **The Worst-Case MSE Pathology (Confident and Wrong)**: Suppose the true label is $y = 1$, but the network is completely mistaken, predicting $z = -10 \implies \hat{y} \approx 0.000045$. The prediction error is maximal: $(y - \hat{y}) \approx 1.0$. However, evaluating the MSE gradient:

  $$\frac{\partial \mathcal{L}_{\text{MSE}}}{\partial z} = -(1.0) \times (0.000045) \times (0.999955) \approx -0.000045 \approx 0!$$

  The gradient completely vanishes! Even though the model is entirely wrong, parameter updates halt because the sigmoid derivative $\hat{y}(1-\hat{y})$ saturates to zero. Optimization gets trapped on a flat plateau.

- **The BCE Remedy**: Under BCE, for the identical scenario ($y = 1, \hat{y} \approx 0$):

  $$\frac{\partial \mathcal{L}_{\text{BCE}}}{\partial z} = \hat{y} - y = 0 - 1 = -1.0.$$

  The sigmoid derivative in the numerator cancels perfectly with the denominator in the derivative of the logarithm! The gradient is strictly linear with respect to the error $(y - \hat{y})$ and provides maximum learning signal precisely when the model is most wrong.

- **Non-Convexity**: While MSE with linear regression is strictly convex, combining MSE with non-linear sigmoid activations creates a non-convex loss surface with numerous local plateaus. In contrast, BCE is strictly convex with respect to logit z.

2.5.3 Maximum likelihood estimation derivation of cross-entropy

Cross-entropy is not an empirical heuristic; it is the mathematically canonical loss derived from Maximum Likelihood Estimation (MLE).

Bernoulli likelihood derivation (BCE). For binary classification, target $y_i \in \{0, 1\}$ follows a Bernoulli distribution parameterized by probability $p_i = \hat{y}_i = \sigma(z_i)$:

$$\mathbb{P}(Y = y_i \mid \mathbf{x}_i) = \hat{y}_i^{y_i}(1-\hat{y}_i)^{1-y_i}. \tag{2.37}$$

For an independent dataset of N samples, the total likelihood is $\mathcal{L}_{\text{data}}(\theta) = \prod_{i=1}^{N} \hat{y}_i^{y_i}(1-\hat{y}_i)^{1-y_i}$. Taking the negative log-likelihood (NLL):

$$\mathcal{L}(\theta) = -\ln \mathcal{L}_{\text{data}}(\theta) = -\sum_{i=1}^{N} \left[y_i \ln \hat{y}_i + (1-y_i)\ln(1-\hat{y}_i) \right] = \sum_{i=1}^{N} \mathcal{L}_{\text{BCE},i}. \tag{2.38}$$

Minimizing BCE is mathematically identical to maximizing the data likelihood under a Bernoulli distribution.

Categorical likelihood derivation (CCE). For multi-class classification over C classes with one-hot targets $\mathbf{y}_i$, the likelihood follows a Categorical distribution:

$$\mathbb{P}(\mathbf{y}_i \mid \mathbf{x}_i) = \prod_{c=1}^{C} \hat{y}_{ic}^{\,y_{ic}} \implies -\ln \mathbb{P}(\mathbf{y}_i \mid \mathbf{x}_i) = -\sum_{c=1}^{C} y_{ic} \ln \hat{y}_{ic} = \mathcal{L}_{\text{CE}}. \tag{2.39}$$

2.5.4 Gradient descent and adaptive optimizers

Parameters θ are updated iteratively along the negative gradient:

$$\theta^{(t+1)} := \theta^{(t)} - \alpha \nabla_\theta \mathcal{L}(\theta^{(t)}), \tag{2.40}$$

where $\alpha > 0$ is the learning rate.

Limitations of vanilla gradient descent. In deep neural architectures, vanilla Gradient Descent struggles with:

1. **Ravines and Ill-Conditioned Curvature**: Valleys where gradient magnitude is large across the ravine walls but tiny along the base, causing erratic cross-valley oscillation.
2. **Saddle Points**: Points where $\nabla\mathcal{L} = 0$ while the Hessian possesses both positive and negative eigenvalues.

Adaptive remedies: Momentum and Adam.

- **Momentum**: Incorporates an exponentially decaying velocity vector $\mathbf{v}_t = \beta \mathbf{v}_{t-1} + (1 - \beta)\nabla\mathcal{L}(\theta_t)$, accelerating descent through saddle regions.
- **Adam (Adaptive Moment Estimation)** [20]: Tracks running averages of both gradients (first moment $\mathbf{m}_t$) and squared gradients (second uncentered moment $\mathbf{v}_t$), with bias corrections:

$$\mathbf{m}_t = \beta_1 \mathbf{m}_{t-1} + (1 - \beta_1)\mathbf{g}_t, \quad \hat{\mathbf{m}}_t = \frac{\mathbf{m}_t}{1 - \beta_1^t}, \tag{2.41}$$

$$\mathbf{v}_t = \beta_2 \mathbf{v}_{t-1} + (1 - \beta_2)\mathbf{g}_t^2, \quad \hat{\mathbf{v}}_t = \frac{\mathbf{v}_t}{1 - \beta_2^t}, \tag{2.42}$$

$$\theta_{t+1} = \theta_t - \frac{\alpha}{\sqrt{\hat{\mathbf{v}}_t} + \epsilon} \hat{\mathbf{m}}_t. \tag{2.43}$$

  Adam scales learning rates adaptively per parameter, making it the standard optimizer across LLM pre-training and fine-tuning pipelines.

2.6 Computational graphs and exact adjoint backpropagation

To compute gradients efficiently across billions of parameters, deep learning frameworks (PyTorch, JAX) represent mathematical workflows as a *Directed Acyclic Graph (DAG)*, known as a *computational graph*.

Definition 2.2: *Computational graph*

A computational graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ is a directed acyclic graph where:

- Each node $v \in \mathcal{V}$ represents an input variable, parameter, or primitive mathematical

operation $(+, \times, -, \exp)$.

- Each directed edge $e = (u, v) \in \mathcal{E}$ represents intermediate data flow from variable u into operation v.

2.6.1 Exact backpropagation walkthrough from slide 24

We reproduce and derive the exact pedagogical computational graph and adjoint trace from Slide 24 of the course materials.

Problem setup. Consider a model with two scalar parameters $\theta_1, \theta_2 \in \mathbb{R}$ evaluated on training pair $(x, y) \in \mathbb{R}^2$:

$$\text{Model Prediction:} \quad \hat{y} = \theta_1 \theta_2 x, \tag{2.44}$$

$$\text{Objective Loss:} \quad \mathcal{L} = (\theta_1 \theta_2 x - y)^2. \tag{2.45}$$

Forward decomposition. The graph decomposes Equation (2.45) into four primitive operations:

$$a = \theta_1 \theta_2 \quad \text{(parameter interaction product)}, \tag{2.46}$$

$$b = ax = \theta_1 \theta_2 x \quad \text{(prediction } \hat{y}\text{)}, \tag{2.47}$$

$$c = b - y = \theta_1 \theta_2 x - y \quad \text{(residual error)}, \tag{2.48}$$

$$\mathcal{L} = c^2 = (\theta_1 \theta_2 x - y)^2 \quad \text{(scalar squared loss)}. \tag{2.49}$$

Backward adjoint chain rule derivation. Backpropagation computes the derivative of scalar loss $\mathcal{L}$ with respect to every intermediate variable and parameter by traversing the graph in reverse topological order:

$$\text{Step 1:} \quad \frac{\partial \mathcal{L}}{\partial c} = \frac{\partial (c^2)}{\partial c} = 2c. \tag{2.50}$$

$$\text{Step 2:} \quad \frac{\partial \mathcal{L}}{\partial b} = \frac{\partial \mathcal{L}}{\partial c}\frac{\partial c}{\partial b} = 2c \cdot \frac{\partial (b - y)}{\partial b} = 2c \cdot 1 = 2c. \tag{2.51}$$

$$\text{Step 3:} \quad \frac{\partial \mathcal{L}}{\partial a} = \frac{\partial \mathcal{L}}{\partial b}\frac{\partial b}{\partial a} = 2c \cdot \frac{\partial (ax)}{\partial a} = 2c \cdot x = 2cx. \tag{2.52}$$

$$\text{Step 4:} \quad \frac{\partial \mathcal{L}}{\partial \theta_1} = \frac{\partial \mathcal{L}}{\partial a}\frac{\partial a}{\partial \theta_1} = (2cx) \cdot \frac{\partial (\theta_1 \theta_2)}{\partial \theta_1} = 2cx \cdot \theta_2 = 2(\theta_1 \theta_2 x - y) x \theta_2. \tag{2.53}$$

$$\text{Step 5:} \quad \frac{\partial \mathcal{L}}{\partial \theta_2} = \frac{\partial \mathcal{L}}{\partial a}\frac{\partial a}{\partial \theta_2} = (2cx) \cdot \frac{\partial (\theta_1 \theta_2)}{\partial \theta_2} = 2cx \cdot \theta_1 = 2(\theta_1 \theta_2 x - y) x \theta_1. \tag{2.54}$$

Figure 2.9 illustrates this complete forward and backward adjoint execution.

Intuition & analogy: 3Blue1Brown's geometric view of backpropagation

Grant Sanderson (3Blue1Brown) articulates backpropagation not as abstract calculus algebra, but as a systematic tracking of sensitivity *nudges*. Imagine wiggling parameter θ_1 by an infinitesimal perturbation ϵ. That nudge ripples through the product node to scale a by $\epsilon\theta_2$. The change in a wiggles b by $\epsilon\theta_2 x$, which directly shifts error c and finally scales loss $\mathcal{L}$ by $2c\epsilon\theta_2 x$. Backpropagation simply reads these connected multiplicative sensitivity tracks in reverse, gathering the composite derivative at each fork in the graph.

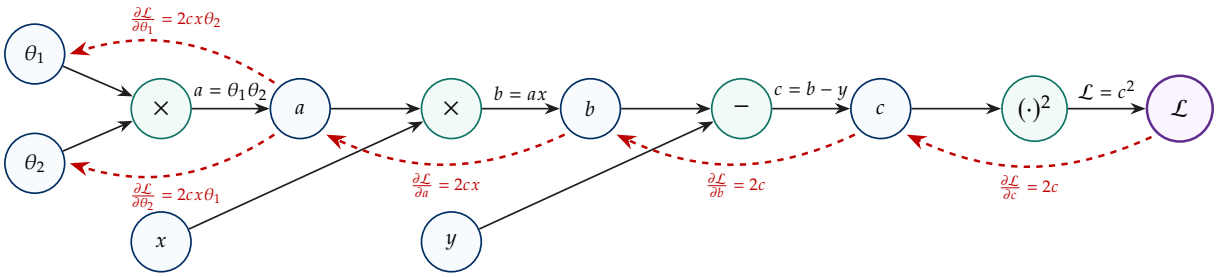


Figure 2.9: Directed Acyclic Computational Graph (DAG) for model $\hat{y} = \theta_1\theta_2 x$ with quadratic loss $\mathcal{L} = (\theta_1\theta_2 x - y)^2$ (reproduced from Slide 24). Solid black arrows trace forward execution; dashed red arrows trace reverse-mode adjoint backpropagation via the chain rule [21].

External deep dive: Building autograd from scratch: Andrej Karpathy's Micrograd

Andrej Karpathy's open-source library *Micrograd* implements reverse-mode automatic differentiation in fewer than 100 lines of pure Python, demonstrating the exact computational graph mechanics shown in Figure 2.9.
Each scalar variable is wrapped inside a `Value` class containing:

- `.data`: the floating-point scalar computed during the forward pass.
- `.grad`: the accumulated adjoint derivative $\partial\mathcal{L}/\partial\cdot$, initialized to 0.0.
- `._prev`: a set storing pointers to operand children nodes.
- `._backward`: a closure storing the local multivariate chain rule.

For example, multiplication is implemented as:

```
def __mul__(self, other):
    other = other if isinstance(other, Value) else Value(other)
    out = Value(self.data * other.data, (self, other), '*')
    def _backward():
        self.grad += other.data * out.grad
        other.grad += self.data * out.grad
    out._backward = _backward
    return out
```

When calling `loss.backward()`, the engine performs a **topological sort** on the DAG to establish reverse dependency ordering, sets `loss.grad = 1.0`, and executes `node._backward()` across all parent nodes in reverse topological order.

2.6.2 Vectorized matrix adjoint backpropagation for dense layers

While scalar computational graphs illustrate the chain rule clearly, production deep learning libraries execute backpropagation via *vectorized matrix calculus* across high-dimensional tensors. We now derive the exact vectorized matrix adjoint equations for fully connected layers with explicit tensor dimensions.

Layer formulation and forward variables. Consider a fully connected dense layer mapping an input vector $\mathbf{x} \in \mathbb{R}^{d_{\text{in}}}$ through weight matrix $\mathbf{W} \in \mathbb{R}^{d_{\text{in}} \times d_{\text{out}}}$ and bias $\mathbf{b} \in \mathbb{R}^{d_{\text{out}}}$ to pre-activation

$\mathbf{z} \in \mathbb{R}^{d_{\text{out}}}$ and activation $\mathbf{y} \in \mathbb{R}^{d_{\text{out}}}$:

$$\mathbf{z} = \mathbf{W}^\top \mathbf{x} + \mathbf{b} \in \mathbb{R}^{d_{\text{out}}}, \tag{2.55}$$

$$\mathbf{y} = f(\mathbf{z}) \in \mathbb{R}^{d_{\text{out}}}, \tag{2.56}$$

where scalar component z_j expands as:

$$z_j = \sum_{i=1}^{d_{\text{in}}} W_{ij} x_i + b_j. \tag{2.57}$$

The layer output $\mathbf{y}$ propagates downstream to contribute to a final scalar objective loss $\mathcal{L} \in \mathbb{R}$.

Adjoint definition. Assume downstream backpropagation has computed the loss gradient with respect to layer pre-activation $\mathbf{z}$. We define the adjoint error vector $\boldsymbol{\delta} \in \mathbb{R}^{d_{\text{out}}}$:

$$\boldsymbol{\delta} \triangleq \frac{\partial \mathcal{L}}{\partial \mathbf{z}} = \frac{\partial \mathcal{L}}{\partial \mathbf{y}} \odot f'(\mathbf{z}) \in \mathbb{R}^{d_{\text{out}}}, \tag{2.58}$$

where $\odot$ denotes the elementwise Hadamard product.

1. Weight matrix adjoint derivation ($\frac{\partial \mathcal{L}}{\partial \mathbf{W}}$). To compute the derivative of scalar loss $\mathcal{L}$ with respect to individual matrix element W_{ij}, we apply the multivariate chain rule across all output pre-activations z_k:

$$\frac{\partial \mathcal{L}}{\partial W_{ij}} = \sum_{k=1}^{d_{\text{out}}} \frac{\partial \mathcal{L}}{\partial z_k} \frac{\partial z_k}{\partial W_{ij}}. \tag{2.59}$$

Because $z_k = \sum_{l=1}^{d_{\text{in}}} W_{lk} x_l + b_k$, the partial derivative is non-zero only when $k = j$:

$$\frac{\partial z_k}{\partial W_{ij}} = \begin{cases} x_i & \text{if } k = j \\ 0 & \text{if } k \neq j \end{cases}. \tag{2.60}$$

Substituting this back:

$$\frac{\partial \mathcal{L}}{\partial W_{ij}} = \frac{\partial \mathcal{L}}{\partial z_j} x_i = \delta_j x_i = x_i \delta_j. \tag{2.61}$$

Collecting all elements W_{ij} into matrix form reveals that the weight gradient is identically the **outer product** of the upstream input vector $\mathbf{x} \in \mathbb{R}^{d_{\text{in}}}$ and the adjoint error vector $\boldsymbol{\delta} \in \mathbb{R}^{d_{\text{out}}}$:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{W}} = \mathbf{x}\boldsymbol{\delta}^\top \in \mathbb{R}^{d_{\text{in}} \times d_{\text{out}}}. \tag{2.62}$$

2. Bias vector adjoint derivation ($\frac{\partial \mathcal{L}}{\partial \mathbf{b}}$). For each scalar bias component b_j:

$$\frac{\partial \mathcal{L}}{\partial b_j} = \sum_{k=1}^{d_{\text{out}}} \frac{\partial \mathcal{L}}{\partial z_k} \frac{\partial z_k}{\partial b_j} = \delta_j \cdot 1 = \delta_j. \tag{2.63}$$

In vector notation:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{b}} = \boldsymbol{\delta} \in \mathbb{R}^{d_{\text{out}}}. \tag{2.64}$$

3. Upstream input adjoint derivation ($\frac{\partial \mathcal{L}}{\partial \mathbf{x}}$). To propagate error signals back to previous layers in a deep neural network, we must evaluate the gradient of loss $\mathcal{L}$ with respect to the input feature vector $\mathbf{x} \in \mathbb{R}^{d_{\text{in}}}$. Applying the chain rule to input component x_i:

$$\frac{\partial \mathcal{L}}{\partial x_i} = \sum_{k=1}^{d_{\text{out}}} \frac{\partial \mathcal{L}}{\partial z_k} \frac{\partial z_k}{\partial x_i} = \sum_{k=1}^{d_{\text{out}}} \delta_k W_{ik} = \sum_{k=1}^{d_{\text{out}}} W_{ik} \delta_k. \tag{2.65}$$

Recognizing this sum as the i-th element of the matrix-vector multiplication between weight matrix $\mathbf{W} \in \mathbb{R}^{d_{\text{in}} \times d_{\text{out}}}$ and adjoint vector $\boldsymbol{\delta} \in \mathbb{R}^{d_{\text{out}}}$:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{x}} = \mathbf{W}\boldsymbol{\delta} \in \mathbb{R}^{d_{\text{in}}}. \tag{2.66}$$

Figure 2.10 diagrams this vectorized adjoint error propagation across the dense layer.

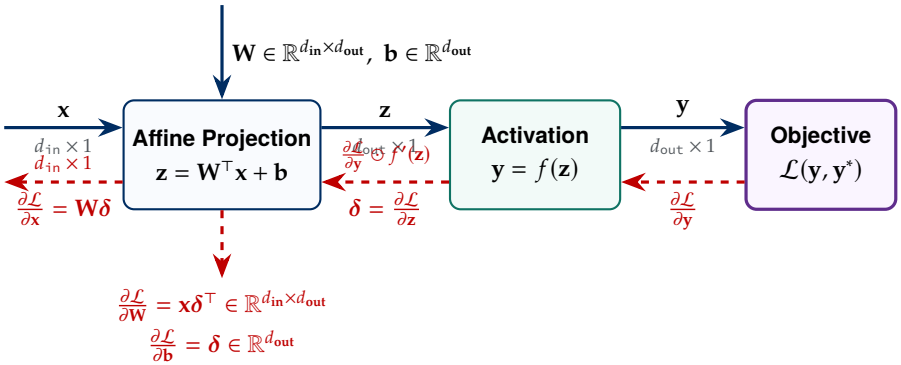


Figure 2.10: Vectorized matrix adjoint backpropagation across a fully connected dense layer. Solid navy arrows trace forward execution mapping input $\mathbf{x} \in \mathbb{R}^{d_{\text{in}}}$ to output $\mathbf{y} \in \mathbb{R}^{d_{\text{out}}}$; dashed red arrows trace reverse-mode adjoint error propagation ($\boldsymbol{\delta} \in \mathbb{R}^{d_{\text{out}}}$), yielding parameter updates $\frac{\partial \mathcal{L}}{\partial \mathbf{W}} = \mathbf{x}\boldsymbol{\delta}^\top$, $\frac{\partial \mathcal{L}}{\partial \mathbf{b}} = \boldsymbol{\delta}$, and upstream gradient $\frac{\partial \mathcal{L}}{\partial \mathbf{x}} = \mathbf{W}\boldsymbol{\delta}$.

Mini-batch matrix generalization. In GPU implementations, B samples are grouped into a batch data matrix $\mathbf{X} \in \mathbb{R}^{B \times d_{\text{in}}}$. The layer computes:

$$\mathbf{Z} = \mathbf{X}\mathbf{W} + \mathbf{1}_B \mathbf{b}^\top \in \mathbb{R}^{B \times d_{\text{out}}}, \quad \mathbf{Y} = f(\mathbf{Z}) \in \mathbb{R}^{B \times d_{\text{out}}}. \tag{2.67}$$

Defining batch adjoint matrix $\boldsymbol{\Delta} = \frac{\partial \mathcal{L}}{\partial \mathbf{Z}} \in \mathbb{R}^{B \times d_{\text{out}}}$, the gradient equations generalize to highly efficient Basic Linear Algebra Subprograms (BLAS) matrix multiplications:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{W}} = \mathbf{X}^\top \boldsymbol{\Delta} \in \mathbb{R}^{d_{\text{in}} \times d_{\text{out}}}, \tag{2.68}$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{b}} = \boldsymbol{\Delta}^\top \mathbf{1}_B \in \mathbb{R}^{d_{\text{out}}}, \tag{2.69}$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{X}} = \boldsymbol{\Delta}\mathbf{W}^\top \in \mathbb{R}^{B \times d_{\text{in}}}. \tag{2.70}$$

These matrix equations form the computational workhorse executed across billions of parameters in modern GPU training clusters.

2.7 The sequential computation impasse: bridge to recurrent neural networks

In this chapter, we have formulated the mathematical and architectural foundations of deep feedforward neural networks (FCNNs / Multi-Layer Perceptrons):

1. **Continuous Units**: The artificial perceptron computes weighted inner products over real-valued vectors.
2. **Non-Linearity**: Non-linear activations (σ, tanh, ReLU, GELU) prevent multi-layer linear stacking collapse, folding space to solve linearly inseparable problems like XOR and enabling universal function approximation.
3. **Principled Optimization**: Cross-entropy objective functions provide linear error signals without gradient saturation, and backpropagation calculates matrix adjoints across computational DAGs via vectorized matrix calculus.

The feedforward assumption and its sequential failure. Throughout these formulations, feedforward networks operate under a fundamental structural assumption: the input is a static, fixed-dimensional Euclidean vector $\mathbf{x} \in \mathbb{R}^{d_{\text{in}}}$, and the network maps this vector through a cascade of static linear projections $\mathbf{W}_l \in \mathbb{R}^{d_l \times d_{l-1}}$ to produce a fixed-dimensional output $\mathbf{y} \in \mathbb{R}^{d_{\text{out}}}$:

$$\mathbf{y} = \mathcal{F}(\mathbf{x}; \Theta) = f_L(\mathbf{W}_L \cdots f_1(\mathbf{W}_1\mathbf{x} + \mathbf{b}_1) \cdots + \mathbf{b}_L). \tag{2.71}$$

While this architecture is highly effective for tabular data and fixed-attribute classification, it immediately encounters a computational impasse when confronted with sequential data—such as human natural language, software source code tokens, execution traces, and temporal signals. Sequential data violates the core assumptions of feedforward processing in three fundamental ways:

1. **Rigid Input Dimensionality**: Natural language sentences and source code blocks have elastic, variable lengths T. An FCNN strictly expects an input of dimension d_{in}, forcing artificial truncation or massive zero-padding that wastes computational throughput and memory.
2. **Zero Internal State and Temporal Memory**: Feedforward networks are memoryless. Each forward pass $\mathcal{F}(\mathbf{x})$ is independent and decoupled from prior inputs. They cannot maintain temporal context or resolve long-range grammatical and syntactic dependencies across multiple time steps.
3. **Absence of Parameter Sharing Across Positions**: In an FCNN, each position in the input is governed by separate weight parameters. If a syntactic construct (such as a variable declaration `"let x = 5;"` or a noun phrase *"a large language model"*) appears at token position 1 versus token position 20, the FCNN must learn separate weight matrices for each position independently, destroying sample efficiency.

The path forward. To process variable-length sequences, maintain evolving context over time, and achieve translational invariance across sequence positions, deep learning requires a computational paradigm with internal recurrence and temporal weight sharing.

This architectural necessity leads directly to Chapter 3: *Recurrent Neural Networks and Sequence Modeling*.

CHAPTER 3

Recurrent neural networks and sequence modeling

3.1 Sequential data and constraints of feed-forward neural networks

In Chapter 2, we formulated the mathematical and architectural foundations of deep feedforward neural networks (FCNNs) and backpropagation across computational DAGs. As established at the conclusion of that chapter, however, feedforward architectures operate under a rigid, memoryless paradigm: they assume fixed-dimensional Euclidean inputs $\mathbf{x} \in \mathbb{R}^{d_{\text{in}}}$, possess no internal recurrence or persistent temporal state across sequential time steps, and cannot enforce grammatical agreement across temporal horizons.

In standard deep learning, a Fully-Connected Neural Network (FCNN) or Multi-Layer Perceptron is represented as a cascade of weight matrices interleaved with non-linear activation functions:

$$f(\mathbf{x}) = \sigma\left(\mathbf{W}_{\text{out}}\,\text{ReLU}\left(\mathbf{W}_2\,\text{ReLU}\left(\mathbf{W}_1\mathbf{x} + \mathbf{b}_1\right) + \mathbf{b}_2\right) + \mathbf{b}_{\text{out}}\right). \tag{3.1}$$

While FCNNs excel on tabular datasets with fixed, predefined attributes, they exhibit three critical architectural constraints when applied to natural language and temporal sequences (Slides 2–5):

1. The input is expected to be fixed in size. The input dimensionality is rigidly fixed by the number of columns in the first weight matrix $\mathbf{W}_1 \in \mathbb{R}^{d_1 \times d_{\text{in}}}$. For structured tabular data, this constraint is natural because every sample contains the same predefined attributes. However, for unstructured sequential data, fixed input sizing is deeply problematic:

- Natural language sentences come in all lengths: from 2 words (*"Run now"*) to 80 words in complex legal contracts.
- Audio recordings exhibit variable sampling frequencies and durations.
- Images come in varying resolutions and aspect ratios.

Flattening a variable-length text sequence into a rigid vector $\mathbf{x} \in \mathbb{R}^{d_{\text{in}}}$ requires setting an arbitrary maximum length $T_{\max}$ and applying zero-padding to shorter sentences while truncating longer ones. This wastes vast memory on empty pad tokens and destroys the elastic temporal nature of human language.

2. Lack of parameter sharing and translation invariance. In an FCNN, each position in the flattened input vector is associated with its own distinct set of weight parameters. Consequently, if the same linguistic pattern occurs in different portions of the input, the network must learn that pattern multiple times from scratch:

- In computer vision, a dog is a dog regardless of whether it appears on the left or the right side of an image. An FCNN must learn a separate "dog-detector" for the left pixels and the right pixels.
- In NLP, the noun phrase *"a dog"* possesses the exact same grammatical and semantic structure whether it appears at token position 1 (*"A dog barked loudly"*) or token position 10 (*"After a long walk in the park, a dog barked"*).

Because an FCNN does not share weights across sequence positions, it is wildly inefficient and requires massive datasets to observe every linguistic pattern at every conceivable spatial index.

3. The output is expected to be fixed in size. The output dimensionality is rigidly determined by the number of rows in the final projection matrix $\mathbf{W}_{\text{out}}$. In many sequential tasks (such as machine translation, text summarization, and dialogue generation), the output sequence length is unknown a priori and varies dynamically based on the input:

English (5 words): *"It is indeed sunny today"* $\implies$ Italian (4 words): *"Effettivamente oggi è soleggiato"*.

An FCNN cannot dynamically adjust its output layer to emit variable-length token sequences.

3.2 The recurrent neural network cell and weight sharing

To overcome the constraints of FCNNs, **Recurrent Neural Networks (RNNs)**—first formalized as the Simple Recurrent Network in [22] (as presented on Slides 6–7 of the course lecture deck)—process sequential data element by element, maintaining an internal continuous *hidden state* $\mathbf{h}_t \in \mathbb{R}^{d_h}$ that acts as a recurrent memory summarizing the sequence history observed up to time step t.

3.2.1 Recurrent cell architecture

At each discrete time step $t \in \{1, 2, \ldots, T\}$, the recurrent cell is characterized by two inputs and two outputs:

- **Input 1**: The current sequence element $\mathbf{x}_t \in \mathbb{R}^{d_{\text{in}}}$ (e.g., a one-hot vector or continuous embedding).
- **Input 2**: The hidden state produced in the previous time step $\mathbf{h}_{t-1} \in \mathbb{R}^{d_h}$.
- **Output 1**: The model's prediction for the current time step $\hat{\mathbf{y}}_t \in \mathbb{R}^{d_{\text{out}}}$.
- **Output 2**: The updated hidden state $\mathbf{h}_t \in \mathbb{R}^{d_h}$, which is recursively forwarded to time step $t + 1$.

3.2.2 Mathematical formulation with explicit tensor dimensions

Internally, the vanilla RNN cell comprises an input projection layer, a recurrent state transition layer, and an output projection layer (Slide 12):

$$\underbrace{\mathbf{h}_t}_{[d_h\times 1]} = \tanh\left(\underbrace{\mathbf{W}_{hh}}_{[d_h\times d_h]}\underbrace{\mathbf{h}_{t-1}}_{[d_h\times 1]} + \underbrace{\mathbf{W}_{xh}}_{[d_h\times d_{\text{in}}]}\underbrace{\mathbf{x}_t}_{[d_{\text{in}}\times 1]} + \underbrace{\mathbf{b}_h}_{[d_h\times 1]}\right), \tag{3.2}$$

$$\underbrace{\hat{\mathbf{y}}_t}_{[V\times 1]} = \text{softmax}\left(\underbrace{\mathbf{W}_{hy}}_{[V\times d_h]}\underbrace{\mathbf{h}_t}_{[d_h\times 1]} + \underbrace{\mathbf{b}_y}_{[V\times 1]}\right). \tag{3.3}$$

3.2.3 Weight sharing across time

The defining feature of RNNs is **temporal weight sharing** (Slide 13). The parameter matrices $\mathbf{W}_{xh}$, $\mathbf{W}_{hh}$, and $\mathbf{W}_{hy}$ are *strictly identical* across all time steps $t \in \{1, \ldots, T\}$. Because the same weights are applied regardless of time step, the model develops **translation invariance**: if the same context occurs at step $t = 2$ or step $t = 50$, the recurrent cell behaves identically. The network learns grammatical rules (such as capitalizing the first letter after a quotation mark) once, and applies that rule everywhere in the text.

Intuition & analogy: Recurrent dynamics as a trajectory through high-dimensional phase space

As visualized in geometric visual analyses [23], the hidden state $\mathbf{h}_t \in \mathbb{R}^{d_h}$ can be understood geometrically as the position of a particle traveling through a continuous d_h-dimensional *phase space*:

- At rest ($t = 0$), the system sits at initial coordinate $\mathbf{h}_{-1} = \mathbf{0}$.
- When each discrete input token $\mathbf{x}_t$ arrives, the linear projection $\mathbf{W}_{xh}\mathbf{x}_t$ applies an external directional impulse or "vector nudge" to the particle.
- Simultaneously, the recurrent transformation $\mathbf{W}_{hh}\mathbf{h}_{t-1}$ rotates, scales, and folds the previous position, while the non-linear squashing activation tanh confines the entire state space within the hypercube $(-1, 1)^{d_h}$.
- The unfolding sequence of text traces a smooth continuous trajectory through state space. Regions of this phase space function as **attractor basins**: for instance, reading an opening quote mark (") knocks the state into a specific sub-manifold where the emission projection $\mathbf{W}_{hy}$ naturally maps coordinates into uppercase character distributions. The state remains trapped in this basin until an exit trigger (such as a whitespace or subsequent character) forces the trajectory out into the lowercase manifold.

3.2.4 Unrolling the recurrent cell across time

To trace how an RNN processes a sequence, we *unroll* the recursive feedback loop across discrete time steps $t = 0, 1, \ldots, T$ (Slides 8–11).

Consider the character casing task:

- Input sequence: `"you are very kind, mr.  holmes, but i cannot do that"`
- Target sequence: `"You are very kind, Mr.  Holmes, but I cannot do that"`

Each ASCII character is represented as a 256-dimensional one-hot vector $\mathbf{x}_t \in \{0,1\}^{256}$:

- **Step** $t = 0$: Receives input s_0 = `"` with initial state $\mathbf{h}_{-1} = \mathbf{0}$. The cell produces state $\mathbf{h}_0$. The network realizes: *"We just encountered an opening quote (`"`), which signals the beginning of dialogue; the following character should be capitalized."*
- **Step** $t = 1$: Receives input s_1 = `y` and prior state $\mathbf{h}_0$. The network emits output $\hat{y}_1$ = `Y` and produces updated state $\mathbf{h}_1$. The state records: *"We just emitted a capitalized letter at the start of dialogue; following letters in this word should probably be lowercase."*
- **Step** $t = 2$: Receives input s_2 = `o` and prior state $\mathbf{h}_1$. The network emits output $\hat{y}_2$ = `o` and produces state $\mathbf{h}_2$, continuing lowercase word progression.

Figure 3.1 illustrates this unrolled computational structure and the backward flow of gradients.

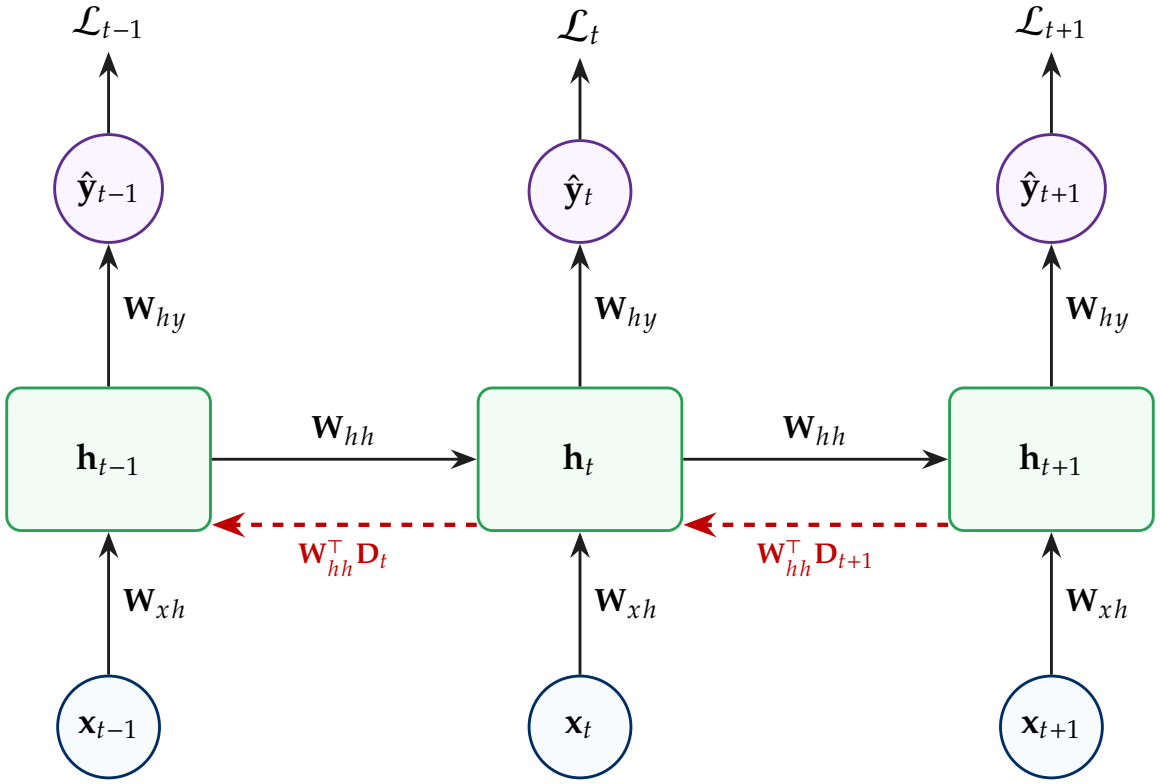


Figure 3.1: Unrolling an RNN cell across time steps $t-1, t, t+1$. Solid black arrows show the forward propagation of inputs $\mathbf{x}_t$ and recurrent states $\mathbf{h}_t$. Dashed red arrows trace Backpropagation Through Time (BPTT), where gradients accumulate across time via repeated multiplications by the Jacobian $\mathbf{W}_{hh}^\top \mathbf{D}_t$.

3.3 Backpropagation through time and the proof of vanishing gradients

When an RNN processes a sequence of length T, the computational graph is unrolled T times. The total sequence loss is the sum of cross-entropy losses incurred at each time step (Slides

14–17):

$$\mathcal{L} = \sum_{t=1}^{T} \mathcal{L}_t. \tag{3.4}$$

To optimize the shared recurrent weight matrix $\mathbf{W}_{hh}$ via Gradient Descent, we apply **Backpropagation Through Time (BPTT)**, first formalized for recurrent structures in [24]. The fundamental difficulty of learning long-term dependencies through gradient descent was rigorously analyzed in [25] and [26] (and emphasized on Slides 14–17 of the course lecture deck). The gradient of the total loss with respect to $\mathbf{W}_{hh}$ accumulates the contributions across all time steps:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{W}_{hh}} = \sum_{t=1}^{T} \sum_{k=1}^{t} \frac{\partial \mathcal{L}_t}{\partial \mathbf{h}_t} \frac{\partial \mathbf{h}_t}{\partial \mathbf{h}_k} \frac{\partial^+ \mathbf{h}_k}{\partial \mathbf{W}_{hh}}, \tag{3.5}$$

where $\frac{\partial^+ \mathbf{h}_k}{\partial \mathbf{W}_{hh}}$ denotes the direct derivative of $\mathbf{h}_k$ with respect to $\mathbf{W}_{hh}$ treating $\mathbf{h}_{k-1}$ as constant.

The temporal transmission term $\frac{\partial \mathbf{h}_t}{\partial \mathbf{h}_k}$ represents the sensitivity of state $\mathbf{h}_t$ to an earlier state $\mathbf{h}_k$ ($t > k$). By the multivariate chain rule, this sensitivity decomposes into a product of single-step Jacobian matrices:

$$\frac{\partial \mathbf{h}_t}{\partial \mathbf{h}_k} = \prod_{j=k+1}^{t} \frac{\partial \mathbf{h}_j}{\partial \mathbf{h}_{j-1}}. \tag{3.6}$$

3.3.1 Rigorous mathematical proof of gradient pathology

We now derive the explicit single-step Jacobian and prove why vanilla RNNs suffer from vanishing and exploding gradients.

Theorem 3.1: *Jacobian product and spectral radius gradient decay*

Let the recurrent state transition be $\mathbf{h}_j = \tanh(\mathbf{z}_j)$ where $\mathbf{z}_j = \mathbf{W}_{hh}\mathbf{h}_{j-1} + \mathbf{W}_{xh}\mathbf{x}_j + \mathbf{b}_h$ with $\mathbf{h}_j \in \mathbb{R}^{d_h}$ and $\mathbf{W}_{hh} \in \mathbb{R}^{d_h \times d_h}$. Under standard column-vector convention, the single-step forward temporal Jacobian $\mathbf{J}_j = \frac{\partial \mathbf{h}_j}{\partial \mathbf{h}_{j-1}} \in \mathbb{R}^{d_h \times d_h}$ is:

$$\mathbf{J}_j = \frac{\partial \mathbf{h}_j}{\partial \mathbf{h}_{j-1}} = \operatorname{diag}\left(1 - \mathbf{h}_j^2\right) \mathbf{W}_{hh} = \mathbf{D}_j \mathbf{W}_{hh}, \tag{3.7}$$

where $\mathbf{D}_j = \operatorname{diag}(1 - \tanh^2(\mathbf{z}_j)) \in \mathbb{R}^{d_h \times d_h}$ is a diagonal matrix containing the local derivatives of the hyperbolic tangent function. Consequently, the multi-step temporal transmission evaluates to:

$$\frac{\partial \mathbf{h}_t}{\partial \mathbf{h}_k} = \prod_{j=k+1}^{t} \mathbf{D}_j \mathbf{W}_{hh}. \tag{3.8}$$

When transmitting the gradient of a scalar loss $\mathcal{L}_t$ backward through time, the column-vector gradient $\nabla_{\mathbf{h}_{j-1}} \mathcal{L}_t \in \mathbb{R}^{d_h}$ satisfies the adjoint recurrence $\nabla_{\mathbf{h}_{j-1}} \mathcal{L}_t = \mathbf{W}_{hh}^\top \mathbf{D}_j \nabla_{\mathbf{h}_j} \mathcal{L}_t$. Let $\lambda_{\max} = \|\mathbf{W}_{hh}\|_2 = \sigma_{\max}(\mathbf{W}_{hh})$ denote the spectral norm (maximum singular value) of $\mathbf{W}_{hh}$. The Euclidean norm of the transmitted gradient is bounded by:

$$\left\| \frac{\partial \mathcal{L}_t}{\partial \mathbf{h}_k} \right\| \leq \left\| \frac{\partial \mathcal{L}_t}{\partial \mathbf{h}_t} \right\| (\lambda_{\max})^{t-k}. \tag{3.9}$$

Proof. Differentiating the i-th component of $\mathbf{h}_j$ with respect to the m-th component of $\mathbf{h}_{j-1}$:

$$\frac{\partial h_{j,i}}{\partial h_{j-1,m}} = \frac{\partial \tanh(z_{j,i})}{\partial z_{j,i}} \frac{\partial z_{j,i}}{\partial h_{j-1,m}} = (1 - h_{j,i}^2) W_{hh,i,m}.$$

In matrix notation, the entry in row i and column m of $\frac{\partial \mathbf{h}_j}{\partial \mathbf{h}_{j-1}}$ is $(1 - h_{j,i}^2) W_{hh,i,m} = (\mathbf{D}_j)_{ii} W_{hh,i,m}$, which corresponds exactly to the matrix product $\mathbf{D}_j \mathbf{W}_{hh}$. Taking the transpose for backward column-vector gradient propagation yields $(\mathbf{D}_j \mathbf{W}_{hh})^\top = \mathbf{W}_{hh}^\top \mathbf{D}_j$, matching the backward adjoint arrows in Figure 3.1. Because $|\tanh'(z)| = 1 - \tanh^2(z) \in [0, 1]$, the operator 2-norm of each diagonal matrix satisfies $\|\mathbf{D}_j\|_2 \leq 1$. Applying the sub-multiplicative property of matrix norms to Equation (3.8):

$$\begin{aligned} \|\frac{\partial \mathbf{h}_t}{\partial \mathbf{h}_k}\|_2 &= \|\prod_{j=k+1}^{t} \mathbf{D}_j \mathbf{W}_{hh}\|_2 \leq \prod_{j=k+1}^{t} \|\mathbf{D}_j\|_2 \|\mathbf{W}_{hh}\|_2 \\ &\leq \prod_{j=k+1}^{t} 1 \cdot \lambda_{\max} = (\lambda_{\max})^{t-k}. \end{aligned}$$

Multiplying by $\|\frac{\partial \mathcal{L}_t}{\partial \mathbf{h}_t}\|_2$ yields Equation (3.9). □

Two pathological regimes (slides 21–23):

1. **Vanishing Gradient ($\lambda_{\max} < 1$)**: If the singular values of $\mathbf{W}_{hh}$ are slightly less than 1 (e.g., $\lambda_{\max} = 0.9$), over a temporal distance of 50 steps:

   $$0.9^{50} \approx 0.00515 \quad \text{(over 99.5\% of the gradient signal is extinguished).}$$

   Over 100 steps, $0.9^{100} \approx 0.000026$. The gradient decays exponentially to zero. Parameter updates from distal tokens vanish, meaning the network cannot learn long-range temporal dependencies. Saturated activations (such as $\sigma'(z) \leq 0.25$) accelerate this decay exponentially.

2. **Exploding Gradient ($\lambda_{\max} > 1$)**: If the singular values exceed 1 (e.g., $\lambda_{\max} = 1.1$), over 50 steps:

   $$1.1^{50} \approx 117.39.$$

   Over 100 steps, $1.1^{100} \approx 13{,}780$. Gradients grow exponentially, destabilizing training, inducing severe numerical overflow, and producing `NaN` parameter corruptions. (Practitioners counter exploding gradients using heuristic *gradient clipping*: $\mathbf{g} \leftarrow \mathbf{g} \cdot \min(1, \frac{\text{threshold}}{\|\mathbf{g}\|})$).

The long-term dependency degradation (slides 24–25). Because hidden states are rewritten at every time step without persistent memory protection, vanilla RNNs suffer from catastrophic recency bias: they remember only the most recent tokens. When tested on long-range word prediction:

*"The **mouse** was getting chased by the cat through the big house. After a long chase, the cat finally managed to catch the _____"*

the target word is unambiguously *"mouse"*. However, vanilla RNNs fail completely because the hidden state has overwritten the subject *"mouse"* across the intervening 20 tokens.

3.4 Case correction demonstration: Sherlock Holmes training dynamics

To demonstrate how an RNN learns sequential syntax in practice, Slides 18–20 document an empirical case study trained on Arthur Conan Doyle's *The Adventures of Sherlock Holmes*:

Experimental configuration.

- **Corpus**: Raw text from Arthur Conan Doyle's *The Adventures of Sherlock Holmes*.
- **Sequence Length**: Subsequences sliced between 10 and 100 ASCII characters.
- **Token Representation**: 256-dimensional one-hot vectors ($d_{\text{in}} = 256$).
- **Architecture**: Single-layer vanilla RNN with hidden dimension $d_h = 32$.
- **Optimization**: Adam optimizer (lr = 0.001), batch size 32, trained for 10 epochs.
- **Padding Mechanism**: Null bytes (`0x00`, rendered visually as `X`) used to pad variable-length sequences to 100 characters.

Fixed evaluation test sentence. The model is evaluated after each epoch on a fixed unseen test string:

- **Input Test Sequence**: `upon his pale face.  "it may be so, or it may not, mr. holmes," said he, "but if you are so very sharp`
- **Target Test Sequence**: `upon his pale face.  "It may be so, or it may not, Mr. Holmes," said he, "but if you are so very sharp`

Qualitative phase transitions across 10 epochs. Table 3.1 records the loss trajectory and qualitative phase transitions across training:

3.4.1 Character-level modeling, temperature sampling, and hidden state interpretability

The Sherlock Holmes casing experiment exemplifies a broader paradigm analyzed in the landmark study *"The Unreasonable Effectiveness of Recurrent Neural Networks"* [27]: **character-level sequence modeling**.

1. The elegance of character-level modeling. Unlike word-level models that require complex tokenizers, morphological analyzers, and massive vocabularies ($V \geq 50{,}000$), character-level RNNs operate over an extremely compact alphabet (e.g., standard 256-dimensional ASCII).

- There are **zero Out-of-Vocabulary (OOV) tokens**: any arbitrary string, code snippet, mathematical formula, or neologism can be represented as an sequence of bytes.
- The model is forced to discover the hierarchical structure of language entirely from scratch: it must learn how characters form words, that words are separated by spaces, how punctuation delimits clauses, and how grammatical agreement spans across clauses.

Table 3.1: Epoch-by-epoch loss progression and qualitative phase transitions on the Sherlock Holmes casing task.

Epoch	Loss	Observed Generation & Qualitative Linguistic Transition
01	3.1681	`XXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX` *Phase 1 (Null-byte frequency)*: The model discovers that null padding bytes (`0x00` / `X`) appear with high statistical frequency due to padding imbalance, predicting all `X`'s.
02	2.3334	`eXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXXX` *Phase 2 (Vowel emergence)*: The model predicts `e` (the most frequent character in English orthography) alongside padding characters.
03	1.7436	`tst X s te e eee t e e e te se ee se e e ee eee XXXee` *Phase 3 (Whitespace and unigrams)*: The model learns that whitespace characters occur regularly, predicting spaces, `e`, `t`, and `s`.
04	0.9519	`upon his pale face.  cit may be sos or it may nots mr.  holresst` *Phase 4 (The Identity Breakthrough)*: A massive drop in loss occurs when the network learns to forward lowercase letters directly from input to output (`W.grad` captures the identity diagonal).
05–06	0.3579	`upon his pale face.  "it may be so, or it may not, mr.  holmes,"` *Phase 5 (Punctuation and Syntax)*: The model captures quotation marks (`"`), commas, and spaces accurately.
07–08	0.1009	`upon his pale face.  "It may be so, or it may not, mr.  holmes,"` *Phase 6 (Quote Capitalization)*: The model learns that characters immediately following opening quotes (`"`) must be capitalized (`"It`).
09–10	0.0515	`upon his pale face.  "It may be so, or it may not, mr.  Holmes,"` *Phase 7 (Period Capitalization and Causal Blindness)*: The model capitalizes `Holmes` because it follows a period (`.`). Crucially, the model **fails to capitalize** `mr.` because it cannot see into the future! It does not know that `m` begins the title `Mr.` until the `r.` arrives.

2. Autoregressive sampling and temperature dynamics. During text generation, the RNN emits an unnormalized logit vector $\mathbf{z}_t = \mathbf{W}_{hy}\mathbf{h}_t + \mathbf{b}_y \in \mathbb{R}^V$. Rather than taking the deterministic arg max, we convert logits into a sampling distribution modulated by a scalar hyperparameter: the **temperature** $T > 0$:

$$\mathbb{P}(y_t = c \mid \mathbf{h}_t) = \frac{\exp\left(\frac{z_{t,c}}{T}\right)}{\sum_{j=1}^{V} \exp\left(\frac{z_{t,j}}{T}\right)}. \tag{3.10}$$

The temperature controls the entropy and randomness of the emitted sequence:

- **Low Temperature ($T \to 0$ / Greedy Decoding)**: Exaggerates differences between logits. The largest logit absorbs almost all probability mass ($p_{\max} \to 1$). The model makes confident, conservative predictions, but frequently falls into pathological mode collapse—getting trapped in infinite repetitive loops (e.g., `"the the the the"` or repeating identical sentence fragments).
- **Balanced Temperature ($T \in [0.6, 0.8]$)**: Preserves the syntactic and morphological structure discovered by the network while introducing sufficient stochastic diversity to produce coherent, rich generations. In empirical experiments demonstrated in [27], character-level LSTMs trained at $T = 0.7$ generated structurally sound Shakespearean plays (including character names, stage directions, and poetic meter), syntactically compilable C code from the Linux kernel, and valid algebraic geometry LaTeX documents.
- **High Temperature ($T > 1.2$)**: Flattens the probability distribution toward a uniform distribution ($p_c \to 1/V$). The network makes chaotic, exploratory predictions, generating misspelled words, hallucinated punctuation, and unpronounceable character noise.

3. Interpretable hidden state neurons. By inspecting the internal cell activations $\mathbf{h}_t$ of a trained LSTM as it reads character sequences, the empirical investigation in [27] revealed that individual hidden units act as interpretable linguistic finite-state machines:

1. **The Quote Detection Neuron**: A single scalar unit in $\mathbf{h}_t$ that fires strongly positive (+1) the moment an opening quotation mark (`"`) is parsed, remains persistently active throughout the enclosed dialogue, and resets immediately to -1 when the matching closing quote mark is encountered.
2. **The Code Indentation / Scope Depth Neuron**: In code generation models, a specific neuron's activation ramps up linearly with every tab or whitespace indentation inside nested loops or conditionals, tracking variable scope depth and resetting when closing brackets (`}`) appear.
3. **The Line Length Tracker**: A unit whose activation monotonically increases across 80 characters, anticipating when to emit the carriage return (`\n`) character to conform to standard formatting conventions.

4. Classroom insights: causal lookahead blindness and overfitting dynamics. In the Politecnico di Torino course lecture, two essential practical takeaways are highlighted from the Sherlock Holmes casing demonstration:

- **Causal Lookahead Blindness**: Why did the one-to-one RNN fail to capitalize `"mr."` into `"Mr."` even after 10 epochs? The model operates strictly causally from left to right. When processing the character `'m'`, the model only sees past history. It cannot see into the future to know whether `'m'` begins the proper title `"Mr."` or common words like `"me"`, `"my"`, or `"much"`. Without lookahead, capitalizing `'m'` would be an unsupported guess. Conversely, the model successfully capitalizes `"Holmes"` because it has already observed the preceding period and space (`". h"`), triggering capitalization.

- **Overfitting vs. Generalization Dynamics**: When addressing why training was stopped at 10 epochs during the course lecture demo, it was emphasized that extending training for hundreds of epochs on a finite corpus (such as Doyle's novel) causes the model to memorize specific character n-grams verbatim. While training loss continues to drop toward zero, the network loses its ability to generalize to novel sentence structures, reinforcing the necessity of early stopping and validation monitoring emphasized throughout the course.

3.5 Gated architectures: LSTM and GRU

To solve the vanishing gradient problem and allow neural networks to retain dependencies over hundreds of time steps, researchers introduced **Gated RNNs** (Slide 26–29):

3.5.1 Long short-term memory (LSTM)

Introduced in [28] and covered on Slides 18–25 of the course lecture deck, the **Long Short-Term Memory (LSTM)** network partitions memory into two separate states:

1. **Cell State $\mathbf{c}_t \in \mathbb{R}^{d_h}$**: The *long-term memory* highway, which runs straight down the entire chain with minimal linear interactions.
2. **Hidden State $\mathbf{h}_t \in \mathbb{R}^{d_h}$**: The *short-term memory*, which acts as the visible output emitted at each step.

Definition 3.1: *LSTM gating equations*

At time step t, given input $\mathbf{x}_t \in \mathbb{R}^{d_{\text{in}}}$ and previous hidden state $\mathbf{h}_{t-1} \in \mathbb{R}^{d_h}$, the LSTM evaluates three sigmoid gates and a candidate state:

$$\text{Forget Gate:} \quad \mathbf{f}_t = \sigma\left(\mathbf{W}_f[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_f\right) \in (0,1)^{d_h}, \tag{3.11}$$

$$\text{Input Gate:} \quad \mathbf{i}_t = \sigma\left(\mathbf{W}_i[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_i\right) \in (0,1)^{d_h}, \tag{3.12}$$

$$\text{Candidate Cell:} \quad \tilde{\mathbf{c}}_t = \tanh\left(\mathbf{W}_c[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_c\right) \in (-1,1)^{d_h}, \tag{3.13}$$

$$\text{Cell State Update:} \quad \mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tilde{\mathbf{c}}_t \in \mathbb{R}^{d_h}, \tag{3.14}$$

$$\text{Output Gate:} \quad \mathbf{o}_t = \sigma\left(\mathbf{W}_o[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_o\right) \in (0,1)^{d_h}, \tag{3.15}$$

$$\text{Hidden State Update:} \quad \mathbf{h}_t = \mathbf{o}_t \odot \tanh(\mathbf{c}_t) \in (-1,1)^{d_h}, \tag{3.16}$$

where $\odot$ denotes the element-wise Hadamard product, and $[\mathbf{h}_{t-1}, \mathbf{x}_t] \in \mathbb{R}^{d_h + d_{\text{in}}}$ denotes vector concatenation.

Semantic roles of the gates (slide 29): Each gate squashes pre-activations through a sigmoid function $\sigma \in [0,1]$, functioning as a continuous valve:

- Multiplying by $0 \implies$ drop/forget everything.
- Multiplying by $1 \implies$ preserve/pass everything.
- **Forget Gate ($\mathbf{f}_t$)**: Regulates what proportion of the past cell state $\mathbf{c}_{t-1}$ to discard. (e.g., when a closing quote or period arrives, the forget gate purges past sentence subjects).
- **Input Gate ($\mathbf{i}_t$)**: Regulates what proportion of the new candidate information $\tilde{\mathbf{c}}_t$ to write into the long-term cell state.

- **Output Gate ($\mathbf{o}_t$)**: Regulates which portions of the internal cell state $\mathbf{c}_t$ should be revealed as visible hidden state $\mathbf{h}_t$ for current output predictions.

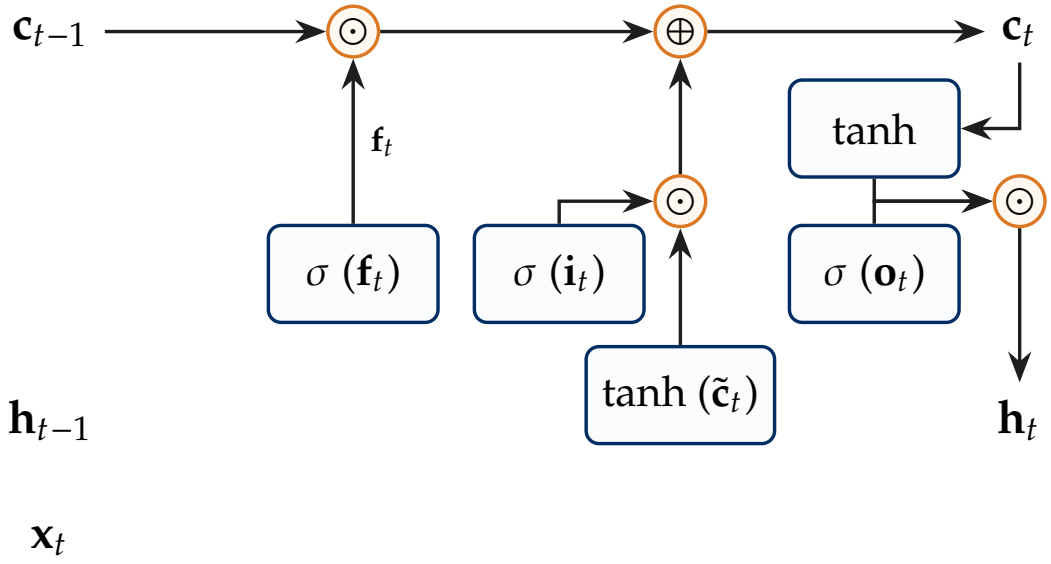


Figure 3.2: The LSTM memory cell architecture. The top horizontal path forms the uninterrupted additive cell state highway $\mathbf{c}_t$ (the Constant Error Carousel). Forget gate $\mathbf{f}_t$, input gate $\mathbf{i}_t$, and output gate $\mathbf{o}_t$ modulate information flow using multiplicative valves.

Theorem 3.2: *The Constant Error Carousel proof*

In an LSTM, the gradient of the loss with respect to past cell state $\mathbf{c}_{t-1}$ contains an additive direct path:

$$\frac{\partial \mathbf{c}_t}{\partial \mathbf{c}_{t-1}} = \mathbf{f}_t + \frac{\partial \mathbf{f}_t}{\partial \mathbf{c}_{t-1}}\mathbf{c}_{t-1} + \frac{\partial \mathbf{i}_t}{\partial \mathbf{c}_{t-1}}\tilde{\mathbf{c}}_t + \frac{\partial \tilde{\mathbf{c}}_t}{\partial \mathbf{c}_{t-1}}\mathbf{i}_t. \tag{3.17}$$

When the network sets forget gate $\mathbf{f}_t \approx \mathbf{1}$, the leading derivative collapses to $\frac{\partial \mathbf{c}_t}{\partial \mathbf{c}_{t-1}} \approx \mathbf{I}$. Consequently, over any arbitrary temporal horizon $T - k$:

$$\frac{\partial \mathbf{c}_T}{\partial \mathbf{c}_k} = \prod_{j=k+1}^{T} \frac{\partial \mathbf{c}_j}{\partial \mathbf{c}_{j-1}} \approx \prod_{j=k+1}^{T} \mathbf{I} = \mathbf{I}. \tag{3.18}$$

Gradients flow backward across hundreds of time steps without exponential decay. This linear additive gradient highway is the **Constant Error Carousel (CEC)**.

Intuition & analogy: The conveyor belt of memory

Think of the cell state $\mathbf{c}_t$ as a continuous factory conveyor belt running from the start of the sequence to the end. The standard recurrent transition in a vanilla RNN continually picks up the items, squashes them with weights and tanh, and puts a degraded version back on the belt. Over 50 time steps, the original items are unrecognizable.
In an LSTM, items on the conveyor belt are untouched by default. As the belt passes each station:

1. The forget gate $\mathbf{f}_t$ acts as a discard mechanism: if $\mathbf{f}_t \approx 1$, the past item rides past undisturbed; if $\mathbf{f}_t \approx 0$, it is swept off the belt.
2. The input gate $\mathbf{i}_t$ adds new relevant cargo $\tilde{\mathbf{c}}_t$ onto the belt via simple vector addition ($\oplus$).
3. The output gate $\mathbf{o}_t$ inspects the cargo on the belt, reads off a copy, and displays it to the outside world as hidden state $\mathbf{h}_t$, leaving the underlying cell state intact on the highway.

Because writing is additive rather than multiplicative, information can ride the conveyor belt across hundreds of words without experiencing exponential decay.

3.5.2 Gated recurrent unit (GRU)

Introduced in [29] and analyzed on Slides 28–32 of the course lecture deck, the **Gated Recurrent Unit (GRU)** simplifies the LSTM architecture by merging the cell state and hidden state into a single recurrent state $\mathbf{h}_t$ and coupling the forget and input gates into a single *update gate* $\mathbf{z}_t$:

Definition 3.2: *GRU equations*

At time step t, given input $\mathbf{x}_t$ and prior state $\mathbf{h}_{t-1}$:

$$\text{Update Gate:} \quad \mathbf{z}_t = \sigma\left(\mathbf{W}_z[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_z\right), \tag{3.19}$$

$$\text{Reset Gate:} \quad \mathbf{r}_t = \sigma\left(\mathbf{W}_r[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_r\right), \tag{3.20}$$

$$\text{Candidate Hidden State:} \quad \tilde{\mathbf{h}}_t = \tanh\left(\mathbf{W}_h[\mathbf{r}_t \odot \mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_h\right), \tag{3.21}$$

$$\text{Hidden State Interpolation:} \quad \mathbf{h}_t = (1 - \mathbf{z}_t) \odot \mathbf{h}_{t-1} + \mathbf{z}_t \odot \tilde{\mathbf{h}}_t. \tag{3.22}$$

Physical gate routing and dynamic memory modulation. The GRU cell operates through two coupled mechanisms illustrated in Figure 3.3:

1. **Reset Gate Modulation ($\mathbf{r}_t$)**: The reset gate vector $\mathbf{r}_t \in (0,1)^{d_h}$ acts as a dynamic filter applied to the previous hidden state $\mathbf{h}_{t-1}$ via Hadamard multiplication ($\mathbf{r}_t \odot \mathbf{h}_{t-1}$). If $\mathbf{r}_t \approx \mathbf{0}$, the candidate state $\tilde{\mathbf{h}}_t$ completely ignores past memory, resetting the representation to depend exclusively on the current token $\mathbf{x}_t$ (e.g., at sentence boundaries or clause resets). If $\mathbf{r}_t \approx \mathbf{1}$, the full history flows into the candidate calculation.
2. **Update Gate Convex Interpolation ($\mathbf{z}_t$)**: Rather than using distinct input and forget gates like the LSTM, the GRU couples memory retention and candidate addition into a single convex combination:
$$\mathbf{h}_t = (1 - \mathbf{z}_t) \odot \mathbf{h}_{t-1} + \mathbf{z}_t \odot \tilde{\mathbf{h}}_t.$$
When $\mathbf{z}_t \approx \mathbf{0}$, the unit passes the previous state $\mathbf{h}_{t-1}$ forward unchanged ($1 \odot \mathbf{h}_{t-1} + 0 \odot \tilde{\mathbf{h}}_t = \mathbf{h}_{t-1}$), establishing an unobstructed linear identity shortcut similar to the LSTM Constant Error Carousel. When $\mathbf{z}_t \approx \mathbf{1}$, the unit overwrites the state entirely with the newly proposed candidate $\tilde{\mathbf{h}}_t$.

Parameter count comparison: vanilla RNN vs. LSTM vs. GRU. Comparing the exact parameter counts of recurrent cells for hidden dimension d and input dimension m clarifies the architectural trade-offs across sequence models:

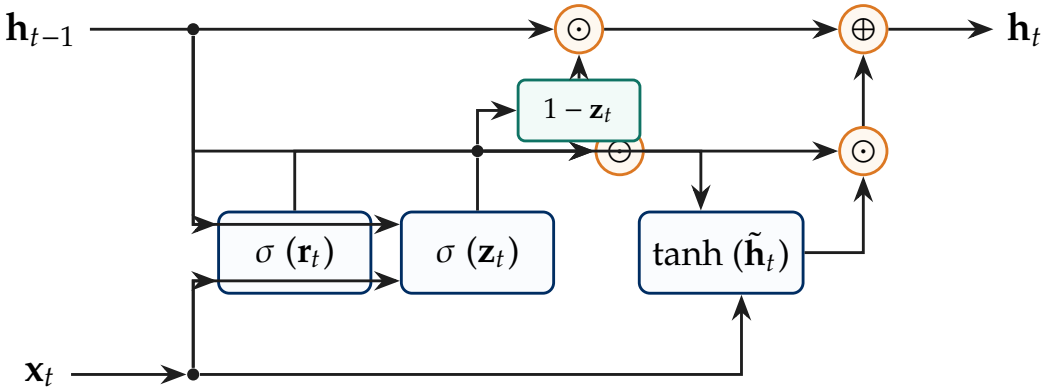


Figure 3.3: The Gated Recurrent Unit (GRU) cell architecture. Reset gate $\mathbf{r}_t$ dynamically filters previous hidden state $\mathbf{h}_{t-1}$ before constructing candidate state $\tilde{\mathbf{h}}_t$. Update gate $\mathbf{z}_t$ controls the convex interpolation $(1 - \mathbf{z}_t) \odot \mathbf{h}_{t-1} + \mathbf{z}_t \odot \tilde{\mathbf{h}}_t$, unifying long- and short-term memory into a single state vector $\mathbf{h}_t$.

- **Vanilla RNN**: Contains 1 transition set $(\mathbf{W}_{hh}, \mathbf{W}_{xh}, \mathbf{b}_h) \implies d^2 + dm + d = d(d + m + 1)$.
- **GRU**: Contains 3 gate sets $(\mathbf{z}_t, \mathbf{r}_t, \tilde{\mathbf{h}}_t) \implies 3 \times d(d + m + 1)$.
- **LSTM**: Contains 4 gate sets $(\mathbf{f}_t, \mathbf{i}_t, \tilde{\mathbf{c}}_t, \mathbf{o}_t) \implies 4 \times d(d + m + 1)$.

The GRU reduces parameters by 25% compared to the LSTM while exhibiting comparable empirical performance across diverse sequence modeling benchmarks.

3.6 Sequence-to-sequence (Seq2Seq) and encoder-decoder pipelines

3.6.1 Limitations of one-to-one sequence mapping

Thus far, we have considered models where each input step produces exactly one output step ($|X| = |Y|$), such as character-level case correction. However, the majority of core NLP tasks (machine translation, question answering, code generation) violate this assumption (Slides 30–32):

1. **Unequal Sequence Lengths**: In machine translation, source and target languages rarely match in word count:

   English (5 words): *[it] [is] [indeed] [sunny] [today]* $\implies$ Italian (4 words): *[effettivamente] [oggi] [è] [soleggiato].*

   A one-to-one recurrent model would process 5 tokens and be forced to produce 5 output tokens.

2. **Non-Monotonic Alignment and Causal Blindness**: Natural languages exhibit different word orders. In the example above, the Italian word *"effettivamente"* must be emitted at step 1, but its English counterpart *"indeed"* does not appear until step 3! In a standard one-to-one model, the cell cannot see the future, rendering early word generation impossible.

3.6.2 The encoder-decoder architecture

To overcome these limitations, the **Encoder-Decoder (Seq2Seq)** architecture was introduced in [30] and [29] (as explored on Slides 33–34 of the course lecture deck):

1. **Encoder**: An LSTM processes the entire variable-length source sequence $(x_1, \ldots, x_N)$ step by step, updating its internal state. The final hidden state $\mathbf{h}_N$ acts as a dense, compressed summary vector encoding the semantics of the whole input sentence.

2. **Decoder**: A separate LSTM initialized with state $\mathbf{h}_N$. The decoder autoregressively emits target tokens step by step. The initial input token is the special Beginning of Sequence symbol ⟨BOS⟩. At each subsequent step t, the input is the token emitted at step $t-1$. Generation continues until the model emits the special End of Sequence symbol ⟨EOS⟩.

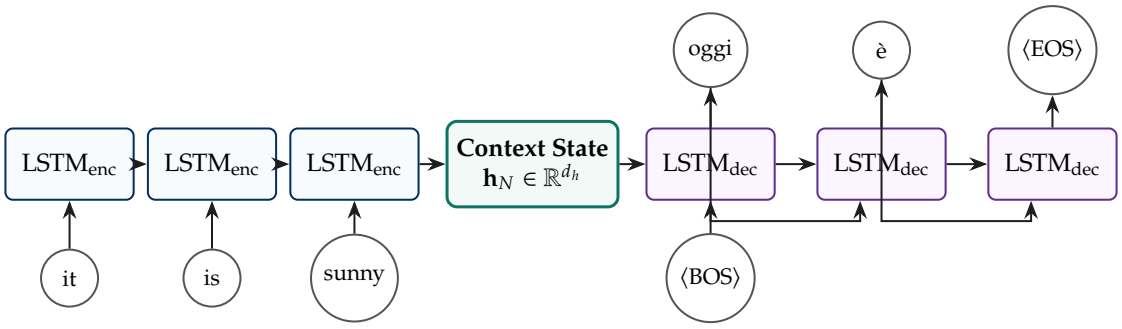


Figure 3.4: The Sequence-to-Sequence (Seq2Seq) Encoder-Decoder architecture. The Encoder LSTM compresses the variable-length source sentence into a single context state $\mathbf{h}_N$. The Decoder LSTM is initialized with $\mathbf{h}_N$ and autoregressively emits target tokens from ⟨BOS⟩ until emitting ⟨EOS⟩.

3.6.3 Teacher forcing vs. inference exposure bias

During training, the decoder's predicted token $\hat{y}_{t-1}$ is often incorrect in early epochs. If the model feeds its own faulty prediction into step t, errors compound catastrophically down the sequence.

To stabilize and accelerate training, practitioners use **Teacher Forcing** (Slide 34):

$$\mathbf{x}_{\text{dec},t} = y^*_{t-1} \quad \text{(pass true ground-truth target token instead of predicted } \hat{y}_{t-1}\text{)}. \tag{3.23}$$

Exposure Bias Warning: While Teacher Forcing provides rapid convergence during training, it creates a discrepancy at inference time when ground-truth tokens are unavailable, causing models to be fragile when recovering from early generation mistakes.

3.6.4 Sherlock Holmes case correction revised: one-to-one vs. encoder-decoder

To empirically contrast the representational power of one-to-one recurrent mapping against the Sequence-to-Sequence Encoder-Decoder architecture, Slides 35 and 36 evaluate both models on an identical, challenging character casing sentence:

Input Test Sequence: `"indeed i am sherlock holmes,"`
Target Test Sequence: `"Indeed I am Sherlock Holmes,"`

Theoretical expectations from architectural constraints (Slide 35). This evaluation sentence contains four distinct capitalization targets, each testing a specific causal and temporal capacity:

1. **`"I"` of `"Indeed"` (beginning of quote)**: Should be capitalized because it is immediately preceded by an opening quotation mark (`"`). A one-to-one model can easily learn this local transition, as the trigger character is already present in its past recurrent state $\mathbf{h}_{t-1}$.

2. **`"I"` of `"I"` (standalone subject pronoun)**: Must be capitalized because it functions as the first-person singular pronoun. A one-to-one model is structurally blind here: when reading `'i'`, the network only knows that a word is beginning with `'i'`. It cannot determine whether this character is an isolated pronoun or the initial letter of common words like `"in"`, `"it"`, or `"into"` until the subsequent whitespace arrives at step $t+1$. Without future lookahead, capitalizing `'i'` at step t is mathematically unsupported.

3. **`"S"` of `"Sherlock"` (proper noun first name)**: Must be capitalized because it begins the proper noun *Sherlock*. When the one-to-one cell receives `'s'`, it only observes an arbitrary lowercase word beginning with `'s'`. It cannot anticipate that the subsequent characters will form the name `"herlock"`.

4. **`"H"` of `"Holmes"` (proper noun last name)**: After the model parses the preceding token sequence `"sherlock "`, its recurrent state $\mathbf{h}_{t-1}$ retains the memory of the detective's first name. When an `'h'` arrives, the transition matrix $\mathbf{W}_{hh}$ leverages the learned bigram association $\mathbb{P}(\texttt{Holmes} \mid \texttt{Sherlock})$ to correctly capitalize the last name.

Empirical results across 1070 training epochs (Slide 36). Table 3.2 documents the verbatim loss trajectories and generated outputs across training checkpoints from Epoch 1 to Epoch 1070.

Table 3.2: Empirical comparison of character casing dynamics on the Sherlock Holmes corpus (Slide 36): One-to-One Recurrent Mapping vs. Sequence-to-Sequence Encoder-Decoder across training epochs 1 to 1070.

	One-to-One Mapping		Encoder-Decoder Architecture	
Epoch	**Loss**	**Generated Output**	**Loss**	**Generated Output**
0001	5.3212	xxxxxxxxxxxxxxxxxxxxxxxxxxxxxx	5.4591	wwXXXXXXXXXXXXXXXXXXXXXXXXXX
0002	3.7157	xxxxxxxxxxxxxxxxxxxxxxxxxxxxxx	4.0161	aXXXXXXXXXXXXXXXXXXXXXXXXXXX
0003	2.7964	xxxxxxxxxxxxxxxxxxxxxxxxxxxxxx	2.8268	xxxxxxxxxxxxxxxxxxxxxxxxxxxxx
0004	2.7981	xxxxxxxxxxxxxxxxxxxxxxxxxxxxxx	2.7671	xxxxxxxxxxxxxxxxxx
0030	0.2264	`"indeed i am sherlock holmes,"`	0.2315	`"indeed i am sherlock holmes,"`
0050	0.0802	`"indeed i am sherlock holmes,"`	0.0705	`"indeed i am sherlock holmes,"`
0160	0.0301	`"Indeed i am sherlock Holmes,"`	0.0259	`"Indeed i am sherlock Holmes,"`
0530	0.0224	`"Indeed i am sherlock Holmes,"`	0.0197	`"Indeed I am sherlock Holmes,"`
1070	0.0232	`"Indeed i am sherlock Holmes,"`	0.0156	`"Indeed I am Sherlock Holmes,"`

Analysis of qualitative phase transitions and convergence trade-offs. The comparative trajectories in Table 3.2 reveal three fundamental pedagogical principles:

1. **The Causal Barrier of One-to-One Recurrence**: The one-to-one model plateaus at Epoch 160 with loss 0.0301 and cannot improve further through Epoch 1070 (loss 0.0232). As predicted, it successfully capitalizes `"Indeed"` (from the quote mark) and `"Holmes"` (from the preceding `"sherlock"`), but remains permanently trapped producing lowercase `"i"` and `"sherlock"`. Because the lockstep recurrent cell must emit output $\hat{y}_t$ simultaneously with reading x_t, it suffers from irreversible causal lookahead blindness.

2. **Sequence-Level Information Integration in Seq2Seq**: The Encoder-Decoder model overcomes causal blindness by decoupling sequence ingestion from sequence emission. The encoder reads the entire string before generation begins. Consequently, the context vector $\mathbf{h}_N$ encodes global contextual features: it "remembers" that `"i"` is bounded by whitespace on both sides (making it the isolated pronoun `"I"`) and that `"s"` begins the full proper name `"sherlock"`. By Epoch 530, the decoder resolves `"Indeed I am sherlock`

`Holmes,"` (loss 0.0197), and by Epoch 1070 it fully resolves all four capitalization targets: `"Indeed I am Sherlock Holmes,"` with superior loss 0.0156.

3. **The Sample Complexity Trade-off (100× Training Epochs)**: Why does the Encoder-Decoder require 1070 epochs when the simple character model converged in 10 epochs? In one-to-one mapping, temporal alignment is strictly hardcoded: character t at the input maps directly to character t at the output. In Seq2Seq, the model is given zero alignment hints; it must simultaneously discover: (i) how to compress the entire sentence into vector $\mathbf{h}_N$, (ii) how to decode variable-length autoregressive tokens, and (iii) how to align output positions with latent source semantics. Learning this unconstrained non-monotonic mapping from scratch drastically increases optimization difficulty and sample complexity.

3.7 Persistent bottlenecks of gated RNNs and the road to transformers

While LSTMs and Encoder-Decoder architectures dominated sequence modeling from 2014 to 2017, Slide 37 identifies three irreducible physical and computational bottlenecks that ultimately necessitated a revolutionary paradigm shift:

1. The Shannon channel capacity and vector bottleneck [30]. In the canonical Seq2Seq framework, an input sequence $(x_1, \ldots, x_{T_x})$ of arbitrary length and linguistic entropy is forced to compress into a single continuous vector of fixed dimensionality:

$$\mathbf{h}_{T_x} = \text{Encoder}(x_1, \ldots, x_{T_x}) \in \mathbb{R}^{d_h}. \tag{3.24}$$

By Shannon's source coding and rate-distortion theory, a finite-dimensional vector in floating-point precision has bounded information capacity. As sequence length increases beyond $T_x \geq 30$ tokens, the recurrent state suffers from catastrophic information dilution: earlier tokens (often containing essential syntactic subjects and conditioning context) are progressively overwritten by later inputs.

2. The sequential computation wall ($O(T)$ training latency). Modern deep learning hardware accelerators (e.g., NVIDIA Hopper/Blackwell GPUs, Google TPUs) derive their performance from massive SIMD (Single Instruction, Multiple Data) parallelism across thousands of tensor cores. Recurrent architectures impose an unbreakable temporal recurrence dependency:

$$\mathbf{h}_t = f(\mathbf{h}_{t-1}, \mathbf{x}_t). \tag{3.25}$$

Computing hidden state $\mathbf{h}_t$ strictly requires the completion of state $\mathbf{h}_{t-1}$. Consequently, an RNN cannot parallelize training across the temporal dimension T. While convolutional and attention layers process the entire sequence in $O(1)$ parallel sequential steps via batched matrix multiplications, RNNs require $O(T)$ sequential operations, rendering pre-training on multi-trillion-token web-scale corpora computationally intractable.

3. Temporal path length ($O(T)$ vs. $O(1)$ direct routing). In an unrolled recurrent graph, the shortest path between token x_1 and prediction y_T scales linearly with sequence length: $O(T)$. Every additional step introduces an intermediate Jacobian multiplication $\frac{\partial \mathbf{h}_j}{\partial \mathbf{h}_{j-1}}$. Even with the additive Constant Error Carousel of LSTMs, error signals become noisy and diffuse over horizons exceeding 100 to 200 tokens.

The two-stage historical leap to foundation models: These structural ceilings precipitated a two-stage intellectual revolution:

1. **Stage 1: Soft Alignment [31]**: Additive Attention shattered the information vector bottleneck by maintaining direct memory access to all encoder annotations $\{\mathbf{h}_1, \dots, \mathbf{h}_{T_x}\}$. However, because the underlying backbone remained recurrent, training was still bound to the sequential execution wall.

2. **Stage 2: Attention Is All You Need [1]**: The landmark study demonstrated that if attention provides direct $O(1)$ information flow between any pair of tokens regardless of temporal distance, **recurrence is completely redundant**. Discarding recurrent cells entirely in favor of multi-head self-attention unlocked full GPU sequence parallelization and dynamic contextualization, establishing the modern Transformer architecture.

3.7.1 Breaking the vector bottleneck: Bahdanau additive attention

In the standard Seq2Seq architecture (Section 3.6), the encoder must squeeze all grammatical, lexical, and structural information of a source sentence of arbitrary length ($T_x = 40, 80$) into the single final state $\mathbf{h}_{T_x} \in \mathbb{R}^{d_h}$. As sentence length increases, empirical translation performance drops precipitously: the fixed-size hidden vector acts as an extreme information bottleneck.

To shatter this bottleneck, **Additive (Soft) Attention** was introduced in [31], allowing the decoder to dynamically search and focus across all encoder hidden states at every generation step.

1. Mathematical formulation with explicit tensor dimensions. Let the source sequence $(x_1, \dots, x_{T_x})$ be encoded by a Bidirectional RNN, producing a sequence of source annotation vectors:

$$\mathbf{h}_j = \begin{bmatrix} \overrightarrow{\mathbf{h}}_j \\ \overleftarrow{\mathbf{h}}_j \end{bmatrix} \in \mathbb{R}^{2d_h}, \quad \forall j \in \{1, 2, \dots, T_x\}. \tag{3.26}$$

At decoder generation step i, let $\mathbf{s}_{i-1} \in \mathbb{R}^{d_{\text{dec}}}$ denote the previous decoder hidden state.

1. **Alignment Energy Score**: The relevance of source token j to target emission i is evaluated by a small feed-forward network parameterized by $\mathbf{W}_a \in \mathbb{R}^{d_a \times d_{\text{dec}}}$, $\mathbf{U}_a \in \mathbb{R}^{d_a \times 2d_h}$, and $\mathbf{v}_a \in \mathbb{R}^{d_a}$:
$$e_{ij} = \mathbf{v}_a^\top \tanh\left(\mathbf{W}_a \mathbf{s}_{i-1} + \mathbf{U}_a \mathbf{h}_j\right) \in \mathbb{R}. \tag{3.27}$$

2. **Softmax Attention Weights**: Exponentiating and normalizing across all T_x source positions yields a valid categorical distribution over input tokens:
$$\alpha_{ij} = \frac{\exp(e_{ij})}{\sum_{k=1}^{T_x} \exp(e_{ik})}, \quad \text{where } \alpha_{ij} \in (0, 1) \text{ and } \sum_{j=1}^{T_x} \alpha_{ij} = 1. \tag{3.28}$$
The scalar α_{ij} represents the degree to which the decoder should attend to source word x_j when generating target word y_i.

3. **Dynamic Context Vector**: The context vector $\mathbf{c}_i$ is computed as the expected annotation under distribution $\boldsymbol{\alpha}_i$:
$$\mathbf{c}_i = \sum_{j=1}^{T_x} \alpha_{ij} \mathbf{h}_j \in \mathbb{R}^{2d_h}. \tag{3.29}$$
Unlike standard Seq2Seq where context $\mathbf{h}_{T_x}$ is static, $\mathbf{c}_i$ is dynamically recomputed at every decoder step i, steering the model's focus to the relevant source phrase.

4. **Decoder State and Emission Update**: The decoder hidden state updates by conditioning on the previous token y_{i-1} and dynamic context $\mathbf{c}_i$:

$$\mathbf{s}_i = \text{RNN}_{\text{dec}}\left(\mathbf{s}_{i-1},\ [\mathbf{y}_{i-1}, \mathbf{c}_i]\right), \tag{3.30}$$

and target token probabilities are emitted via $\hat{\mathbf{y}}_i = \text{softmax}(\mathbf{W}_s[\mathbf{s}_i, \mathbf{y}_{i-1}, \mathbf{c}_i] + \mathbf{b}_s)$.

Figure 3.5 illustrates the Bahdanau Additive Attention architecture.

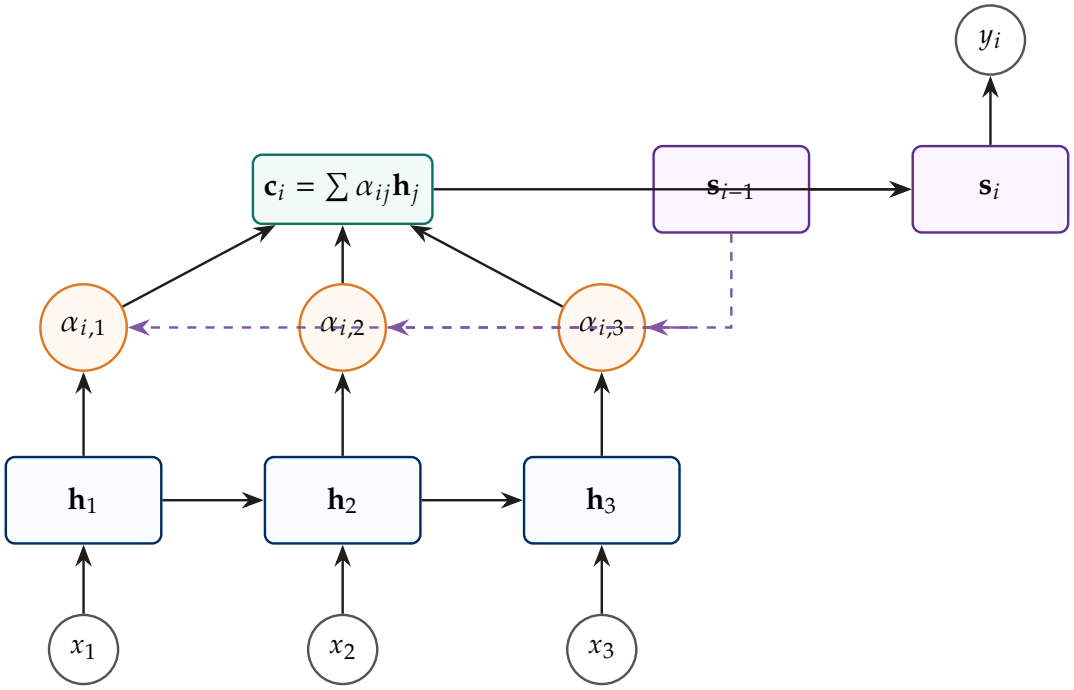


Figure 3.5: The Bahdanau Additive Attention mechanism. At decoder step i, previous decoder state $\mathbf{s}_{i-1}$ is compared against all encoder annotations $\mathbf{h}_j$ via Equation (3.27), producing attention distribution α_{ij}. The weighted sum synthesizes dynamic context vector $\mathbf{c}_i \in \mathbb{R}^{2d_h}$, directly feeding relevant source features into decoder state $\mathbf{s}_i$.

External deep dive: Historical milestone: from Schmidhuber's CEC to Sutskever's Seq2Seq

The lineage of sequence modeling represents one of the most intellectually cohesive arcs in artificial intelligence:

- In 1997, the milestone paper *"Long Short-Term Memory"* [28] solved the vanishing gradient problem via the Constant Error Carousel, proving that additive highways preserve error signals indefinitely.
- In 2014, the Gated Recurrent Unit (GRU) and RNN encoder-decoder architectures were formulated in [29] for statistical machine translation.
- Simultaneously in 2014, the landmark study *"Sequence to Sequence Learning with Neural Networks"* [30] demonstrated that multi-layer deep LSTMs could map sentences across languages end-to-end. Crucially, it was discovered that reversing the source sentence (*"A B C → C B A"*) dramatically improved performance by shortening the initial temporal gradient distance between source and target tokens.

3.7.2 Causal bridge to Chapter 4: the representation problem and continuous word embeddings

Throughout this chapter, we have traced the linear evolutionary arc of sequential neural architectures:

1. **Vanilla RNNs** introduced temporal recurrence and weight sharing to eliminate the fixed-dimensional constraints of feedforward networks, processing elastic sequences via an evolving hidden state $\mathbf{h}_t \in \mathbb{R}^{d_h}$.

2. **LSTMs and GRUs** resolved exponential vanishing gradients along the temporal chain via the Constant Error Carousel ($\frac{\partial \mathbf{c}_t}{\partial \mathbf{c}_{t-1}} \approx \mathbf{I}$) and multiplicative gating highways, enabling gradient propagation across extended sequences.

3. **Seq2Seq Encoder-Decoder Pipelines** overcame the causal lookahead blindness of one-to-one recurrent cells by mapping variable-length input sequences into dynamic variable-length output sequences.

4. **Bahdanau Additive Attention** shattered the single-vector bottleneck ($\mathbf{h}_N \in \mathbb{R}^{d_h}$) by granting the decoder dynamic soft addressing across all encoder annotations ($\mathbf{c}_i = \sum_j \alpha_{ij}\mathbf{h}_j$).

The unresolved input bottleneck: what is $\mathbf{x}_t$? Despite these mathematical triumphs in temporal modeling, every recurrent transition $\mathbf{h}_t = \tanh(\mathbf{W}_{hh}\mathbf{h}_{t-1} + \mathbf{W}_{xh}\mathbf{x}_t + \mathbf{b}_h)$ rests upon a fundamental, unaddressed prerequisite: *the input $\mathbf{x}_t$ must be a numeric vector in $\mathbb{R}^{d_{in}}$*. In our pedagogical character casing example, we used a toy ASCII one-hot vector $\mathbf{x}_t \in \{0,1\}^{256}$. However, scaling sequence models to real-world natural language or software engineering source code reveals three catastrophic failure modes of local one-hot representations:

1. **Dimensionality Explosion**: Realistic vocabularies contain $V = |\mathcal{V}| \geq 50{,}000$ to $1{,}000{,}000$ unique words. Representing tokens as one-hot vectors $\mathbf{e}_i \in \{0,1\}^V$ forces the recurrent input projection matrix $\mathbf{W}_{xh} \in \mathbb{R}^{d_h \times V}$ to balloon into millions of parameters, causing severe memory bloat and extreme sample inefficiency.

2. **Orthogonal Metric Collapse**: Standard basis vectors are strictly mutually orthogonal ($\mathbf{e}_i^\top \mathbf{e}_j = 0$ for all $i \neq j$) and uniformly equidistant ($d_E(\mathbf{e}_i, \mathbf{e}_j) = \sqrt{2}$). The recurrent cell receives zero geometric signal that *"dog"* and *"puppy"*, or keywords `"while"` and `"for"`, share syntactic or semantic roles.

3. **Zero Gradient Parameter Sharing**: During backpropagation through time (BPTT), updating weights for one-hot vector $\mathbf{e}_i$ provides absolutely zero learning transfer to semantically similar words $\mathbf{e}_j$, requiring astronomical training data to observe every word in every temporal context.

The path forward. While recurrent networks solved the *temporal dynamics* problem across sequences, they are crippled without a dense, low-dimensional, continuous metric space $\phi : \mathcal{V} \to \mathbb{R}^d$ ($d \ll V$) where geometric proximity directly encodes syntactic and semantic similarity.

This fundamental representational challenge leads directly to Chapter 4: *Distributed Word Representations and Word2Vec*.

Chapter 4

Distributed word representations and Word2Vec

4.1 The representation problem: why continuous embeddings are mandatory

Natural language and software engineering source code are fundamentally composed of discrete, symbolic tokens. In written English or in source files, programs and sentences present themselves as strings of alphanumeric characters: words such as `"compiler"`, `"optimizer"`, `"gradient"`, and keywords such as `"class"`, `"def"`, `"return"`.

Conversely, modern artificial neural networks are continuous, differentiable mathematical operators. An L-layer deep neural network $\mathcal{F}(\mathbf{x}; \Theta)$ computes outputs through compositions of parameterized linear projections and non-linear activation functions:

$$\mathcal{F}(\mathbf{x}; \Theta) = \sigma_L\Big(\mathbf{W}_L\, \sigma_{L-1}\big(\cdots \sigma_1(\mathbf{W}_1\mathbf{x} + \mathbf{b}_1)\cdots\big) + \mathbf{b}_L\Big), \tag{4.1}$$

where $\mathbf{W}_l \in \mathbb{R}^{d_l \times d_{l-1}}$ and $\mathbf{b}_l \in \mathbb{R}^{d_l}$. Such architectures rely fundamentally on the continuous mathematics of vector spaces $\mathbb{R}^d$, multivariable calculus, and gradient-based backpropagation. The optimization algorithm minimizes an empirical risk loss $\mathcal{L}(\Theta)$ by iteratively updating weight tensors along the negative gradient direction:

$$\Theta \leftarrow \Theta - \eta\, \nabla_\Theta \mathcal{L}(\Theta). \tag{4.2}$$

This mathematical reality creates an immediate, fundamental impasse: **discrete text strings cannot be differentiated**. The partial derivative of a scalar loss with respect to a raw string literal, $\frac{\partial \mathcal{L}}{\partial \texttt{"cat"}}$, is mathematically undefined. Discrete symbols possess no notion of infinitesimal perturbation, tangent spaces, or linear vector combinations. One cannot add `"cat"` to `"dog"`, nor compute an inner product between `"while"` and `"for"`.

To bridge this chasm between discrete symbolic communication and continuous gradient-based computation, we must establish a formal mapping from the discrete vocabulary into a continuous vector space.

Definition 4.1: *Continuous word embedding*

Let $\mathcal{V} = \{w_1, w_2, \ldots, w_V\}$ denote a finite vocabulary of $V = |\mathcal{V}|$ distinct discrete symbolic tokens. A **continuous word embedding** is an injective mapping:

$$\phi : \mathcal{V} \longrightarrow \mathbb{R}^d, \quad \text{with } d \ll V, \tag{4.3}$$

that maps each discrete symbol $w_i \in \mathcal{V}$ to a low-dimensional, dense continuous vector $\mathbf{v}_i = \phi(w_i) \in \mathbb{R}^d$. The target vector space $(\mathbb{R}^d, \|\cdot\|, \langle\cdot,\cdot\rangle)$ is a continuous metric space equipped with an inner product, where geometric distance $D(\mathbf{v}_i, \mathbf{v}_j)$ and angular orientation $\cos(\mathbf{v}_i, \mathbf{v}_j)$ preserve semantic and syntactic relationships.

In typical natural language and software engineering applications, vocabulary sizes range from $V \approx 30{,}000$ to upwards of $V = 1{,}000{,}000$ unique tokens, while the latent embedding dimension is chosen to be compact: $d \in [100, 1024]$ (e.g., $d = 300$ in classical Word2Vec and FastText; $d = 768$ in standard Transformer baselines). By mapping discrete words into $\mathbb{R}^d$, language tokens become continuous variables that participate directly in matrix algebra, linear transformations, and gradient backpropagation.

To understand why dense continuous representations are indispensable, we must first examine the standard categorical baseline: local one-hot encoding. While intuitive and trivial to construct, one-hot vectors suffer from severe geometric and computational pathologies that fundamentally incapacitate neural learning.

4.2 The baseline approach: local one-hot encodings and why they fail

The most straightforward, historical baseline for converting discrete categorical symbols into numeric vectors is **one-hot encoding**. Under this local representational scheme, every word $w_i \in \mathcal{V}$ is mapped to a standard basis coordinate vector $\mathbf{e}_i$ in a V-dimensional Euclidean space $\mathbb{R}^V$:

$$\mathbf{e}_i = [\underbrace{0, \dots, 0}_{i-1 \text{ zeros}}, 1, \underbrace{0, \dots, 0}_{V-i \text{ zeros}}]^\top \in \{0,1\}^V, \quad \text{where } (\mathbf{e}_i)_k = \delta_{ik} = \begin{cases} 1 & \text{if } k = i, \\ 0 & \text{if } k \neq i. \end{cases} \tag{4.4}$$

For example, consider a toy vocabulary of $V = 5$ tokens from the course lecture slides:

$$\mathcal{V} = \{\texttt{"dog"}, \texttt{"cat"}, \texttt{"fish"}, \texttt{"pen"}, \texttt{"pencil"}\}. \tag{4.5}$$

Under one-hot encoding, each word is assigned an isolated Cartesian coordinate axis:

$$\mathbf{e}_{\text{dog}} = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{e}_{\text{cat}} = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{e}_{\text{fish}} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{e}_{\text{pen}} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{e}_{\text{pencil}} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}. \tag{4.6}$$

4.2.1 Algebraic equivalence: one-hot matrix multiplication as embedding lookup

In the foundational guide *Build a Large Language Model from Scratch* [32], it is proven that feeding a one-hot vector into a dense linear layer is algebraically identical to retrieving a specific row or column from a weight lookup table.

Theorem 4.1: *Embedding lookup equivalence*

Let $\mathcal{V}$ be a vocabulary of size V, and let token $w_i \in \mathcal{V}$ be encoded as the standard basis vector $\mathbf{e}_i \in \{0,1\}^V$.

Case 1: Column-Vector Orientation (Lecture Slide Convention). Let $\mathbf{W}_{\text{in}} \in \mathbb{R}^{d \times V}$ be the embedding matrix whose k-th column corresponds to the d-dimensional embedding vector $\mathbf{w}_k = (\mathbf{W}_{\text{in}})_{:,k} \in \mathbb{R}^d$:

$$\mathbf{W}_{\text{in}} = \begin{bmatrix} | & | & & | \\ \mathbf{w}_1 & \mathbf{w}_2 & \cdots & \mathbf{w}_V \\ | & | & & | \end{bmatrix}. \tag{4.7}$$

Multiplying $\mathbf{W}_{\text{in}}$ by the one-hot column vector $\mathbf{e}_i$ extracts the i-th column of $\mathbf{W}_{\text{in}}$:

$$\mathbf{h} = \mathbf{W}_{\text{in}}\mathbf{e}_i = \sum_{k=1}^{V}(\mathbf{e}_i)_k(\mathbf{W}_{\text{in}})_{:,k} = 1 \cdot (\mathbf{W}_{\text{in}})_{:,i} + \sum_{k\neq i} 0 \cdot (\mathbf{W}_{\text{in}})_{:,k} = \mathbf{w}_i \in \mathbb{R}^d. \tag{4.8}$$

Case 2: Row-Vector Orientation (PyTorch and Modern Deep Learning Convention). Let $\mathbf{W} \in \mathbb{R}^{V\times d}$ be an embedding table whose k-th row contains the embedding vector $\mathbf{w}_k^\top = \mathbf{W}_{k,:} \in \mathbb{R}^{1\times d}$. Multiplying the one-hot row vector $\mathbf{e}_i^\top$ by $\mathbf{W}$ extracts the i-th row of $\mathbf{W}$:

$$\mathbf{e}_i^\top\mathbf{W} = \sum_{k=1}^{V}(\mathbf{e}_i)_k\mathbf{W}_{k,:} = 1 \cdot \mathbf{W}_{i,:} + \sum_{k\neq i} 0 \cdot \mathbf{W}_{k,:} = \mathbf{W}_{i,:} = \mathbf{w}_i^\top \in \mathbb{R}^{1\times d}. \tag{4.9}$$

Hence, matrix multiplication of an embedding weight matrix by a one-hot indicator vector is identical to an indexed memory table lookup:

$$\mathbf{W}_{\text{in}}\mathbf{e}_i \equiv \texttt{lookup}(\mathbf{W}_{\text{in}}, i). \tag{4.10}$$

4.2.2 Systems-level justification: why frameworks use dedicated embedding lookups

Although dense matrix multiplication $\mathbf{e}_i^\top\mathbf{W}$ and table indexing $\texttt{lookup}(\mathbf{W}, i)$ are algebraically equivalent, no modern deep learning framework (PyTorch, TensorFlow, JAX) ever implements word embeddings as a dense matrix multiplication. Deep learning frameworks provide specialized lookup modules, such as `torch.nn.Embedding(num_embeddings=V, embedding_dim=d)`, for three decisive systems engineering reasons:

1. **Elimination of $O(V \cdot d)$ Wasted FLOPs**: Evaluating the matrix multiplication $\mathbf{e}_i^\top\mathbf{W}$ requires executing $V \cdot d$ floating-point multiplications and $(V-1) \cdot d$ additions. In a standard language model with vocabulary $V = 50{,}257$ (GPT-2 tokenizer) and hidden dimension $d = 768$, computing the input projection for a single token via dense matrix multiplication requires:

   $$50{,}257 \times 768 \approx 3.86 \times 10^7 \text{ arithmetic operations per token.} \tag{4.11}$$

   Of these 38.6 million operations, exactly 99.998% evaluate $0 \times W_{k,j} = 0$. By contrast, `torch.nn.Embedding` accepts integer token IDs $i \in \{0, \ldots, V-1\}$ and performs direct $O(1)$ pointer arithmetic into physical GPU/CPU memory:

   $$\text{address}(\mathbf{w}_i) = \text{base_pointer} + i \times d \times \texttt{sizeof}(\text{float32}), \tag{4.12}$$

   retrieving the d-dimensional embedding slice in $O(1)$ time with exactly zero arithmetic FLOPs.

2. **Massive VRAM Savings (25,000× Memory Reduction)**: Consider a standard training batch with batch size $B = 64$ and sequence context length $T = 1{,}024$. If input sequences were materialized as dense one-hot tensors, the input tensor $\mathbf{X}_{\text{onehot}}$ would have shape $(B, T, V) = (64, 1{,}024, 50{,}257)$. Storing this tensor in single-precision floating point (`float32`, 4 bytes per element) would require:

   $$\text{VRAM}_{\text{one-hot}} = 64 \times 1{,}024 \times 50{,}257 \times 4 \text{ bytes} \approx 13{,}175{,}095{,}296 \text{ bytes} \approx 13.18\,\text{GB of VRAM}, \tag{4.13}$$

   simply to hold the input batch before computing a single activation. Conversely, storing sequences as integer token IDs produces an `int64` tensor of shape $(B, T) = (64, 1{,}024)$, requiring:

   $$\text{VRAM}_{\text{indices}} = 64 \times 1{,}024 \times 8 \text{ bytes} = 524{,}288 \text{ bytes} \approx 524\,\text{KB of VRAM}. \tag{4.14}$$

This constitutes a memory bandwidth reduction factor of $\frac{13.18\,\text{GB}}{524\,\text{KB}} \approx \mathbf{25{,}128}\times$, entirely preventing Out-Of-Memory (OOM) GPU crashes.

3. **Sparse Gradient Backpropagation**: Under dense matrix multiplication $\mathbf{y} = \mathbf{e}_i^\top \mathbf{W}$, the backpropagation adjoint with respect to the weight matrix is the outer product:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{W}} = \mathbf{e}_i \left(\frac{\partial \mathcal{L}}{\partial \mathbf{y}} \right) \in \mathbb{R}^{V \times d}. \tag{4.15}$$

This gradient matrix is massive ($V \times d$) and strictly sparse: only row i contains non-zero values, while the remaining $V - 1$ rows are populated entirely by zeros. Synchronizing such dense gradient tensors across multi-GPU distributed clusters saturates PCIe and NVLink bandwidth. `torch.nn.Embedding` executes sparse indexed gradient accumulation (`index_add_`), routing the upstream vector $\frac{\partial \mathcal{L}}{\partial \mathbf{y}} \in \mathbb{R}^d$ directly into the i-th row of $\mathbf{W}$.

4.2.3 The 5 structural failure modes of one-hot representations

While the lookup table implementation rescues computational throughput, local one-hot representations fail catastrophically as semantic representations. They suffer from five fatal pathologies:

1. Extreme Sparsity and Curse of Dimensionality. As vocabulary size scales to web scale ($V \geq 50{,}000$ to 10^6), one-hot vectors become astronomical sparse vectors where 99.998% of entries are zero. The dimensionality of the vector space equals the number of distinct words. Under this curse of dimensionality, the volume of the enclosing space grows exponentially, leaving data points isolated at the extreme vertices of independent axes.

2. Metric Collapse and Semantic Orthogonality ($\cos = 0, d_E = \sqrt{2}$). Because one-hot vectors form the standard Cartesian basis of $\mathbb{R}^V$, every pair of distinct words $w_i \neq w_j$ is mutually perpendicular and equidistant:

$$\cos(\mathbf{e}_i, \mathbf{e}_j) = \frac{\mathbf{e}_i^\top \mathbf{e}_j}{\|\mathbf{e}_i\|_2 \|\mathbf{e}_j\|_2} = \frac{0}{1 \cdot 1} = 0, \quad \forall i \neq j, \tag{4.16}$$

$$d_E(\mathbf{e}_i, \mathbf{e}_j) = \|\mathbf{e}_i - \mathbf{e}_j\|_2 = \sqrt{(1)^2 + (-1)^2 + \sum_{k \neq i,j} 0^2} = \sqrt{2} \approx 1.414, \quad \forall i \neq j. \tag{4.17}$$

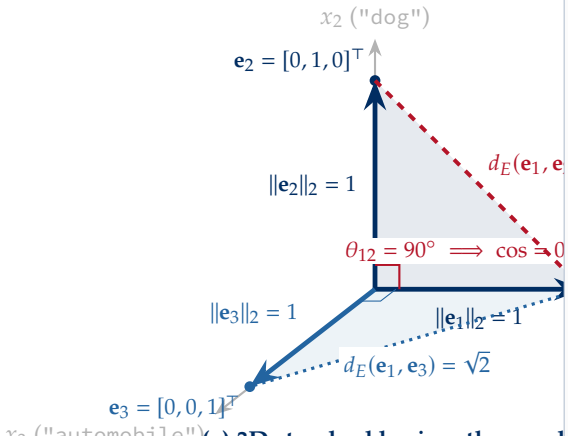
Under one-hot encoding, every word is orthogonal and equidistant to every other word:

$$d_E(\texttt{"pen"}, \texttt{"pencil"}) = d_E(\texttt{"pen"}, \texttt{"crocodile"}) = \sqrt{2}. \tag{4.18}$$

The underlying metric space collapses into a degenerate topology where open balls $B_r(\mathbf{e}_i) = \{\mathbf{x} \in \mathcal{V} : d_E(\mathbf{e}_i, \mathbf{x}) < r\}$ contain only $\{\mathbf{e}_i\}$ for $r \leq \sqrt{2}$, and immediately engulf the entire universe of vocabulary tokens for $r > \sqrt{2}$. There is zero geometric capacity to reflect semantic proximity, synonymy, or grammatical category.

3. Zero Gradient Parameter Sharing During Backpropagation. Because one-hot vectors activate strictly disjoint coordinates, training a neural network on an occurrence of token w_i produces gradients exclusively for the parameters associated with index i:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{W}_{k,:}} = \mathbf{0}, \quad \forall k \neq i. \tag{4.19}$$



(a) 3D standard basis orthogonal

Pairwise metric equality matrices:

• **Cosine similarity matrix** (Gramian $\mathbf{E}^\top\mathbf{E} = \mathbf{I}_V$):

$$\mathbf{S}_{\cos} = \begin{pmatrix} & \text{cat} & \text{dog} & \text{car} \\ \text{cat} & 1 & \mathbf{0} & \mathbf{0} \\ \text{dog} & \mathbf{0} & 1 & \mathbf{0} \\ \text{car} & \mathbf{0} & \mathbf{0} & 1 \end{pmatrix}$$

• **Euclidean distance matrix** ($\mathbf{D}_{ij} = \sqrt{2}(1 - \delta_{ij})$):

$$\mathbf{D}_{L_2} = \begin{pmatrix} & \text{cat} & \text{dog} & \text{car} \\ \text{cat} & 0 & \mathbf{\sqrt{2}} & \mathbf{\sqrt{2}} \\ \text{dog} & \mathbf{\sqrt{2}} & 0 & \mathbf{\sqrt{2}} \\ \text{car} & \mathbf{\sqrt{2}} & \mathbf{\sqrt{2}} & 0 \end{pmatrix}$$

The geometric pathology:
dist("cat", "dog") = dist("cat", "car") = $\sqrt{2}$
cos("cat", "dog") = cos("cat", "car") = 0

(b) Semantic metric collapse

All tokens are equidistant and mutually perpendicular, completely destroying semantic metric topology.

Figure 4.1: Visual proof of pairwise orthogonality and equidistance in one-hot vector representations. (a) In V-dimensional Euclidean space, each word is an orthogonal standard basis vector of unit norm ($\|\mathbf{e}_i\|_2 = 1$). By the Pythagorean theorem, every distinct pair forms a right isosceles triangle with origin $\mathbf{0}$, yielding an identical Euclidean distance of $\sqrt{1^2 + 1^2} = \sqrt{2}$ and an inner product of 0 ($\theta = 90°$). (b) The resulting Gramian matrix is the identity $\mathbf{I}_V$, treating related concepts like `"cat"` and `"dog"` with the exact same geometric distance as totally unrelated concepts like `"cat"` and `"automobile"`.

If a model updates its weights upon observing the sentence *"The feline climbed the tree"*, the parameters associated with `"feline"` receive gradient updates, but the parameters for `"cat"`, `"kitten"`, or `"leopard"` remain completely unchanged. The model cannot transfer statistical strength across semantically synonymous or related words; it must encounter every word hundreds of times independently in every grammatical context.

4. Simplex Confinement and Absence of Latent Manifolds. One-hot vectors are confined strictly to the V extreme vertices of the standard $(V-1)$-dimensional probability simplex:

$$\Delta^{V-1} = \left\{ \mathbf{x} \in \mathbb{R}^V : \sum_{i=1}^{V} x_i = 1,\ x_i \geq 0 \right\}. \tag{4.20}$$

The interior and edges of the simplex correspond to fractional probabilistic mixtures, but distinct words themselves inhabit isolated, non-overlapping boundary corners. Words cannot reside on smooth, lower-dimensional semantic submanifolds, nor can they support linear translation arithmetic ($\mathbf{w}_{\text{king}} - \mathbf{w}_{\text{man}} + \mathbf{w}_{\text{woman}} \approx \mathbf{w}_{\text{queen}}$).

5. Software Engineering Identifier Polymorphism. In software engineering source code, developers routinely express identical semantic intents through differing lexical notations, identifier casings, and API conventions:

- **Casing variants**: `getUserById` vs. `get_user_by_id` vs. `GetUserById`.
- **API synonyms**: `push()` vs. `append()` vs. `add()` vs. `enqueue()`.
- **Syntactic equivalents**: Type references such as `ArrayList` and `LinkedList` vs. unrelated punctuation symbols like `";"`.

Under one-hot encoding, each casing variation and API synonym is assigned a totally independent, mutually orthogonal basis vector. To a one-hot model:

$$\cos(\mathbf{e}_{\texttt{get_user_by_id}}, \mathbf{e}_{\texttt{getUserById}}) = 0, \quad d_E(\mathbf{e}_{\texttt{ArrayList}}, \mathbf{e}_{\texttt{LinkedList}}) = d_E(\mathbf{e}_{\texttt{ArrayList}}, \mathbf{e}_{\texttt{";"}}) = \sqrt{2}. \tag{4.21}$$

This lexical fragmentation completely prevents code-intelligence models from transferring refactoring patterns, bug-fix templates, or type semantics across codebases.

Because local one-hot encodings confine words to isolated, orthogonal coordinate axes where semantic similarity is geometrically undefinable, we require a fundamentally different representational paradigm. This paradigm shift rests on distributing linguistic meaning across continuous latent dimensions—guided by the foundational Distributional Hypothesis.

4.3 Distributed representations and the distributional hypothesis

To overcome the orthogonality and isolation of local representations, modern deep learning adopts **distributed representations**, a concept formalized in connectionist neural networks in 1986 [21].

Definition 4.2: *Local vs. distributed representations*

- **Local Representation**: Each concept or symbol is represented by a single dedicated computational unit (e.g., one coordinate axis in $\mathbb{R}^V$). If that unit is inactive, the concept is entirely absent. There is zero overlap between distinct symbols.

- **Distributed Representation**: Each concept is represented by a continuous pattern of activation across all d dimensions of a shared vector space $\mathbb{R}^d$ ($d \ll V$). Conversely, each individual dimension participates in representing many distinct concepts.

In a distributed representation, meaning is not stored in an isolated slot; rather, it emerges from the continuous coordinates along latent dimensions. For example, latent dimensions might implicitly reflect attributes such as animacy, size, sentiment, syntactic part of speech, or software typing constructs.

4.3.1 The distributional hypothesis: context as meaning

How can an optimization algorithm discover meaningful continuous coordinates $\mathbf{v}_w \in \mathbb{R}^d$ for every word purely from raw text, without human supervision? The answer lies in the **Distributional Hypothesis**, formulated in structural linguistics in 1954 [33] and popularized in 1957 [34]:

> *"You shall know a word by the company it keeps."*
> — Distributional Hypothesis [34]

The distributional hypothesis posits that words that appear in similar linguistic contexts tend to share similar semantic and pragmatic meanings. Consider two sentences:

Sentence 1: "I used a **pencil** to write the essay."
Sentence 2: "You used my **pen** to write a letter."

Even if a machine has never encountered a dictionary definition of `"pencil"` or `"pen"`, it observes that both tokens co-occur with identical context words: `"used"`, `"write"`, `"essay"`, `"letter"`. Conversely, an unrelated word such as `"crocodile"` or `"refrigerator"` almost never occurs in the context of `"used a [...] to write"`.

Therefore, statistical co-occurrence within local contextual windows serves as an empirical, computable proxy for semantic similarity. If a learning algorithm is tasked with predicting surrounding context words from a target word (or vice versa), the learned vector coordinates must position words with similar co-occurrence profiles close together in vector space.

While the Distributional Hypothesis asserts that context defines meaning, it does not specify an optimization algorithm. To translate linguistic co-occurrence into optimized vector coordinates, we must frame a self-supervised prediction task and establish an end-to-end differentiable training loop.

4.4 The embedding training loop and semantic geometry evolution

To convert the distributional hypothesis into concrete numerical representations, we formulate a **self-supervised prediction cloze task** from unannotated text corpora.

4.4.1 The 7-step self-supervised optimization cycle

Given a raw, unannotated corpus of T running words $C = (w_1, w_2, \ldots, w_T)$, we slide a context window of half-width C across the text. At each step t, the model solves a fill-in-the-blank prediction task: estimating the probability distribution over the vocabulary for a missing token given its surrounding context:

$$\mathbb{P}(w_t = w \mid w_{t-C}, \ldots, w_{t-1}, w_{t+1}, \ldots, w_{t+C}). \tag{4.22}$$

As presented in the course lecture slides, this prediction task is executed through an iterative **7-step optimization cycle**:

1. **Initialize Parameter Matrices**: Allocate two weight matrices with small random values: the input embedding matrix $\mathbf{W}_{\text{in}} \in \mathbb{R}^{d \times V}$ and the output embedding matrix $\mathbf{W}_{\text{out}} \in \mathbb{R}^{d \times V}$.

2. **Context Lookup and Aggregation**: For the active context window, retrieve the input embedding vectors $\mathbf{v}_c = (\mathbf{W}_{\text{in}})_{:,c}$ and aggregate them into a single hidden context vector $\mathbf{h} \in \mathbb{R}^d$ (e.g., via vector summation or averaging).

3. **Output Candidate Projection**: Compute the dot product between the hidden context state $\mathbf{h}$ and every candidate output word vector $\mathbf{u}_w = (\mathbf{W}_{\text{out}})_{:,w}$ in the vocabulary:

$$z_w = \mathbf{u}_w^\top \mathbf{h}, \quad \text{yielding logit vector } \mathbf{z} = \mathbf{W}_{\text{out}}^\top \mathbf{h} \in \mathbb{R}^V. \tag{4.23}$$

4. **Probability Normalization**: Pass the similarity logits through the softmax activation function to produce a valid probability distribution $\hat{\mathbf{y}} \in \Delta^{V-1}$:

$$\hat{y}_w = \mathbb{P}(w \mid \text{context}) = \frac{\exp(z_w)}{\sum_{k=1}^{V} \exp(z_k)} = \frac{\exp(\mathbf{u}_w^\top \mathbf{h})}{\sum_{k=1}^{V} \exp(\mathbf{u}_k^\top \mathbf{h})}. \tag{4.24}$$

5. **Loss Evaluation**: Evaluate the prediction error against the true ground-truth target token $w^* = w_t$ using Categorical Cross-Entropy:

$$\mathcal{L}_{\text{CE}} = -\log \hat{y}_{w^*} = -\log \left(\frac{\exp(\mathbf{u}_{w^*}^\top \mathbf{h})}{\sum_{k=1}^{V} \exp(\mathbf{u}_k^\top \mathbf{h})} \right). \tag{4.25}$$

6. **Backpropagation Error Adjoints**: Apply reverse-mode automatic differentiation to compute error adjoints. The gradient with respect to logit vector $\mathbf{z}$ is the prediction residual $\boldsymbol{\delta} = \hat{\mathbf{y}} - \mathbf{y} \in \mathbb{R}^V$, where $\mathbf{y} = \mathbf{e}_{w^*}$. Propagating this residual backward yields parameter gradients:

$$\nabla_{\mathbf{W}_{\text{out}}} \mathcal{L} = \mathbf{h}\, \boldsymbol{\delta}^\top \in \mathbb{R}^{d \times V}, \quad \nabla_{\mathbf{h}} \mathcal{L} = \mathbf{W}_{\text{out}} \boldsymbol{\delta} \in \mathbb{R}^d, \quad \nabla_{\mathbf{W}_{\text{in}}} \mathcal{L} = (\nabla_{\mathbf{h}} \mathcal{L})\, \mathbf{e}_{\text{context}}^\top \in \mathbb{R}^{d \times V}. \tag{4.26}$$

7. **Stochastic Gradient Descent Parameter Update**: Adjust the coordinates in both embedding matrices by stepping along the negative gradient direction:

$$\mathbf{W}_{\text{out}} \leftarrow \mathbf{W}_{\text{out}} - \eta \nabla_{\mathbf{W}_{\text{out}}} \mathcal{L}, \qquad \mathbf{W}_{\text{in}} \leftarrow \mathbf{W}_{\text{in}} - \eta \nabla_{\mathbf{W}_{\text{in}}} \mathcal{L}. \tag{4.27}$$

Figure 4.2 illustrates this complete learning mechanism, juxtaposing the forward/backward computational dataflow in Panel (a) with the resulting geometric evolution of the embedding space across training epochs in Panel (b).

4.4.2 Geometric phase transition of the embedding space

As shown in Panel (b) of Figure 4.2, the continuous embedding space undergoes a profound topological self-organization over the course of training:

- **Phase 1: Initial State ($t = 0$).** Vectors are initialized using small Gaussian noise, $\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_d)$ with $\sigma \approx 0.01$ or $\sigma \approx 1/\sqrt{d}$. In high dimensions ($d \geq 100$), randomly sampled vectors are almost exactly orthogonal ($\cos \approx 0$). Tokens from disparate semantic domains—`"dog"`, `"compiler"`, `"apple"`—are scattered isotropically across the unit sphere with zero structural order.

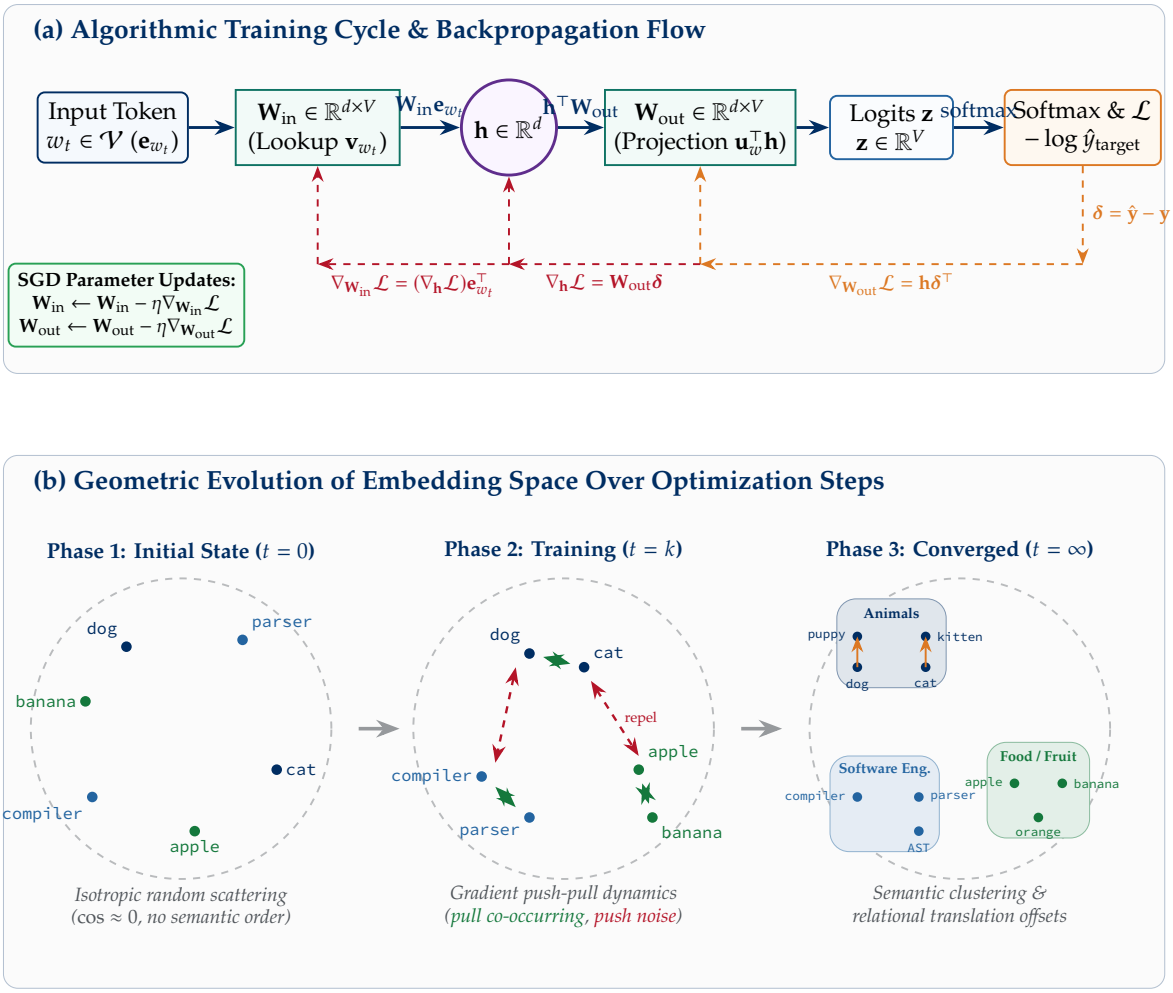


Figure 4.2: The embedding training loop and semantic space evolution. (a) Computational dataflow of the forward prediction pass and backward adjoint parameter updates. (b) Three-phase geometric evolution of embedding space from isotropic random scattering to clustered semantic topology with parallel relational offsets.

- **Phase 2: Training Dynamics ($t = k$).** Whenever the model processes an observed sentence (e.g., *"The compiler parsed the syntax tree"*), backpropagation produces an attractive gradient force pulling the vectors of co-occurring words closer together. Concurrently, negative gradients or normalizer updates exert repulsive forces, pushing non-co-occurring words apart.
- **Phase 3: Converged Semantic Topology ($t = \infty$).** Upon convergence across millions of sentences, words coalesce into dense semantic neighborhoods (e.g., domestic animals, compiler construction terms, fruits). Remarkably, linear vector differences between related concepts become parallel across distinct pairs, giving rise to relational vector arithmetic (e.g., $\mathbf{v}_{\text{dog}} - \mathbf{v}_{\text{puppy}} \approx \mathbf{v}_{\text{cat}} - \mathbf{v}_{\text{kitten}}$).

4.4.3 Initialization dynamics and the hockey-stick loss curve

In educational analyses of neural language models and lookup tables [35], it is demonstrated that the scale of initial weights in $\mathbf{W}_{\text{in}}$ and $\mathbf{W}_{\text{out}}$ dictates optimization stability. At initialization, before the network has learned any linguistic patterns, the model should assign an approximately uniform probability to every word in the vocabulary:

$$\hat{y}_w \approx \frac{1}{V}, \quad \forall w \in \mathcal{V}. \tag{4.28}$$

Consequently, the theoretical expected initial cross-entropy loss is strictly:

$$\mathcal{L}_{\text{init}} = -\log\left(\frac{1}{V}\right) = \log V. \tag{4.29}$$

For a vocabulary of $V = 50{,}000$, the initial loss must be $\mathcal{L}_{\text{init}} = \ln(50{,}000) \approx 10.82$. If $\mathbf{W}_{\text{in}}$ or $\mathbf{W}_{\text{out}}$ are initialized with excessively large variance, logits $z_w = \mathbf{u}_w^\top \mathbf{h}$ take extreme values; the softmax distribution becomes peaky and confident on arbitrary words, causing the initial loss to spike drastically ($\mathcal{L} \gg 20$). The optimizer must then spend thousands of iterations flattening these unearned confidences—producing a pathological "hockey-stick" loss curve. Proper variance scaling ($\mathcal{N}(0, 0.01^2)$ or $\mathcal{N}(0, 1/d)$) ensures stable, well-conditioned gradient flow from step zero.

With the general self-supervised training cycle established, we must concretize the network architecture. The first foundational architecture investigated in the course is the Continuous Bag-of-Words (CBOW) model, which aggregates surrounding context words to predict an omitted target center word.

4.5 Word2Vec continuous bag-of-words (CBOW)

In 2013, **Word2Vec** was introduced in [36, 37] (covered on Slides 9–12 of the course lecture deck) as a suite of log-linear models designed to scale distributed representation learning to web-scale corpora containing billions of tokens.

The first of these architectures is the **Continuous Bag-of-Words (CBOW)** model. In CBOW, the objective is to predict the omitted center target token w_t given the symmetric surrounding context window:

$$C_t = \{w_{t-C}, \ldots, w_{t-1}, w_{t+1}, \ldots, w_{t+C}\} \setminus \{w_t\}. \tag{4.30}$$

4.5.1 Matrix formulation and tensor plumbing

The network consists of two weight matrices:

- $\mathbf{W}_{\text{in}} \in \mathbb{R}^{d \times V}$: The input embedding matrix, where each column $(\mathbf{W}_{\text{in}})_{:,k} = \mathbf{v}_k \in \mathbb{R}^d$ represents the vector of token k when acting as context.

- $\mathbf{W}_{\text{out}} \in \mathbb{R}^{d \times V}$: The output embedding matrix, where each column $(\mathbf{W}_{\text{out}})_{:,k} = \mathbf{u}_k \in \mathbb{R}^d$ represents the vector of token k when acting as the target.

Let $\mathbf{e} \in \{0,1\}^V$ denote the multi-hot context vector indicating the presence of words in C_t:

$$e_k = \sum_{j \in \{-C,\dots,C\} \setminus \{0\}} \mathbf{1}_{\{w_{t+j} = w_k\}}. \tag{4.31}$$

The hidden state $\mathbf{h} \in \mathbb{R}^d$ is obtained by projecting the context vector through $\mathbf{W}_{\text{in}}$ and averaging over the $2C$ context positions:

$$\mathbf{h} = \frac{1}{2C} \mathbf{W}_{\text{in}} \mathbf{e} = \frac{1}{2C} \sum_{\substack{-C \le j \le C \\ j \neq 0}} \mathbf{v}_{w_{t+j}} \in \mathbb{R}^d. \tag{4.32}$$

Next, the model scores candidate target words by projecting $\mathbf{h}$ against the columns of $\mathbf{W}_{\text{out}}$:

$$\hat{\mathbf{p}} = \mathbf{h}^\top \mathbf{W}_{\text{out}} \in \mathbb{R}^{1 \times V}, \quad \text{where } \hat{p}_k = \mathbf{u}_k^\top \mathbf{h}. \tag{4.33}$$

As emphasized in geometric visual analyses [23], the inner product $\mathbf{u}_k^\top \mathbf{h} = \|\mathbf{u}_k\|_2 \|\mathbf{h}\|_2 \cos\theta$ acts as a directional alignment sensor: it evaluates how closely the target word's representation points in the same direction as the accumulated context vector.

The raw similarity logits $\hat{\mathbf{p}}$ are normalized via the softmax function to form the conditional probability distribution:

$$\mathbb{P}(w_t = w \mid C_t) = \frac{\exp(\mathbf{u}_w^\top \mathbf{h})}{\sum_{k=1}^{V} \exp(\mathbf{u}_k^\top \mathbf{h})}. \tag{4.34}$$

The model is trained by minimizing the negative log-likelihood of the ground-truth target word $w^* = w_t$:

$$\mathcal{L}_{\text{CBOW}} = -\log \mathbb{P}(w_t = w^* \mid C_t) = -\mathbf{u}_{w^*}^\top \mathbf{h} + \log \sum_{k=1}^{V} \exp(\mathbf{u}_k^\top \mathbf{h}). \tag{4.35}$$

Figure 4.3 illustrates this computational flow.

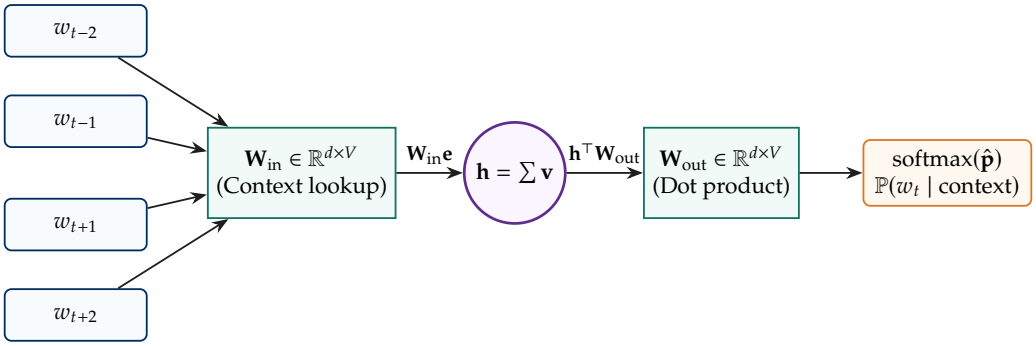


Figure 4.3: Continuous Bag-of-Words (CBOW) architecture. Context words $w_{t-c}, \dots, w_{t+c}$ are looked up in $\mathbf{W}_{\text{in}}$ and summed to form context state $\mathbf{h} \in \mathbb{R}^d$. Inner products with output matrix $\mathbf{W}_{\text{out}}$ yield similarity logits normalized via softmax to predict the center target token w_t.

4.5.2 Comparison with earlier neural language models: the log-linear breakthrough

A neural language model learning distributed embeddings had been introduced a decade prior in 2003 [11]. However, that architecture had two major differences:

1. **Strict Causal Prediction**: In [11], language was modeled autoregressively, predicting w_t strictly from past tokens $w_{t-n+1}, \ldots, w_{t-1}$. In Word2Vec, the window was opened symmetrically to both left and right context tokens ($w_{t-C}, \ldots, w_{t+C}$), doubling the statistical signal.

2. **Elimination of the Non-Linear Hidden Layer**: In [11], concatenated context embeddings were passed through a non-linear dense hidden layer $\tanh(\mathbf{W}_h\mathbf{x} + \mathbf{b})$. The matrix multiplication with $\mathbf{W}_h$ dominated computation time ($O(C \cdot d \cdot d_h)$). In Word2Vec, the non-linear activation was stripped away entirely, using a direct linear projection (vector averaging). This made Word2Vec a log-linear model, reducing compute costs by orders of magnitude and enabling training on billions of words in a few hours.

4.5.3 Strengths and representational bottlenecks of CBOW

Strengths:

- **Rapid Training Convergence**: Context averaging smooths the gradient signal, allowing CBOW to converge in fewer epochs over large corpora.
- **Robust Representations for Frequent Words**: For high-frequency tokens with abundant contextual observations, the smoothed averaging stabilizes coordinate estimates against noisy outliers.

Representational Bottlenecks:

- **Loss of Sequential Word Order (Syntactic Blindness)**: Vector summation in Equation (4.32) is strictly commutative: $\mathbf{v}_a + \mathbf{v}_b = \mathbf{v}_b + \mathbf{v}_a$. The model is incapable of distinguishing *"the dog bit the man"* from *"the man bit the dog"*.
- **Rare Word Dilution**: Because context words are averaged into a single vector $\mathbf{h}$, if a rare technical token appears alongside multiple high-frequency stopwords (e.g., *"the **polymorphism** in the"*), the rare token's embedding is diluted by the dominant stopwords: $\frac{1}{2C}\mathbf{v}_{\text{rare}} \approx \mathbf{0}$. The gradient update to the rare word is severely attenuated.

Although CBOW trains rapidly by averaging context representations, this very averaging creates a representational bottleneck: it washes out rare vocabulary tokens and discards syntactic word order. To learn high-fidelity representations for infrequent words, inverting the predictive formulation yields the Continuous Skip-Gram architecture.

4.6 Word2Vec continuous skip-gram and the softmax bottleneck

To preserve the distinct gradient signal of rare words, the **Continuous Skip-Gram** architecture was introduced in [36] (examined on Slide 11 of the course slides). Instead of aggregating context words to predict a single center token, Skip-Gram **inverts the predictive direction**: given a single center target word w_t, the model independently predicts each surrounding context word within window radius C.

Definition 4.3: *Skip-gram objective function*

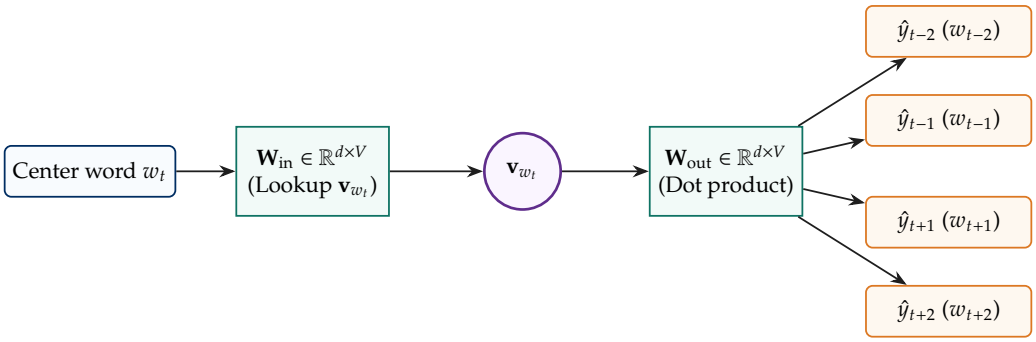


Figure 4.4: Continuous Skip-Gram architecture. A single center target word w_t is projected via $\mathbf{W}_{\text{in}}$ into vector $\mathbf{v}_{w_t} \in \mathbb{R}^d$, which predicts the probability distribution of context words across window radius c through output matrix $\mathbf{W}_{\text{out}}$.

Given a training sequence of T words $w_1, w_2, \ldots, w_T$ and a context window of half-width C, the Skip-gram model maximizes the average log-likelihood:

$$\mathcal{L}_{\text{SG}} = \frac{1}{T} \sum_{t=1}^{T} \sum_{\substack{-C \le j \le C \\ j \neq 0}} \log \mathbb{P}(w_{t+j} \mid w_t). \tag{4.36}$$

Under standard multinomial softmax, the conditional probability of context word w_O given center word w_I is:

$$\mathbb{P}(w_O \mid w_I) = \frac{\exp\left(\mathbf{u}_{w_O}^\top \mathbf{v}_{w_I}\right)}{\sum_{w=1}^{V} \exp\left(\mathbf{u}_w^\top \mathbf{v}_{w_I}\right)}, \tag{4.37}$$

where $\mathbf{v}_w \in \mathbb{R}^d$ denotes the column of $\mathbf{W}_{\text{in}}$ (input embedding) and $\mathbf{u}_w \in \mathbb{R}^d$ denotes the column of $\mathbf{W}_{\text{out}}$ (output context embedding).

4.6.1 Why Skip-gram captures rare words

In Skip-Gram, context words are never averaged together. Instead, every context-target co-occurrence (w_I, w_O) generates an independent, dedicated pairwise training instance. When an infrequent, domain-specific term (such as `"refactoring"` or `"polymorphism"`) appears in text, its vector $\mathbf{v}_{w_I}$ directly predicts each context word:

$$\nabla_{\mathbf{v}_{w_I}} \mathcal{L} = \sum_{w_O \in C} \left(\mathbf{u}_{w_O} - \sum_{k=1}^{V} \mathbb{P}(w_k \mid w_I) \mathbf{u}_k \right). \tag{4.38}$$

The gradient is never diluted by surrounding words. Consequently, Skip-Gram produces significantly higher-quality embeddings for rare words, specialized technical jargon, and nuanced semantic distinctions than CBOW.

4.6.2 The shared computational wall: the full softmax bottleneck

Despite its empirical superiority, the standard formulation of Skip-Gram crashes into an insurmountable computational wall: **the full Softmax partition function bottleneck**.

Examining the denominator of Equation (4.37):

$$Z(w_I) = \sum_{w=1}^{V} \exp\left(\mathbf{u}_w^\top \mathbf{v}_{w_I}\right). \tag{4.39}$$

Computing this normalizing denominator requires evaluating an inner product $\mathbf{u}_w^\top \mathbf{v}_{w_I}$ and an exponential $\exp(\cdot)$ for **every single token w in the entire vocabulary $\mathcal{V}$**.

- **Computational Complexity per Token**: Computing $Z(w_I)$ takes $O(V \cdot d)$ floating-point operations.
- **Total Corpus Cost**: For a window of radius $C = 5$, predicting $2C = 10$ context words per token over a corpus of $T = 10^9$ words with vocabulary $V = 10^5$ and dimension $d = 300$ requires:

$$10^9 \text{ tokens} \times 10 \text{ predictions} \times 10^5 \text{ words} \times 300 \text{ dims} = 3 \times 10^{17} \text{ FLOPs}. \tag{4.40}$$

- **Gradient Synchronization Bottleneck**: In backpropagation, the derivative $\frac{\partial \mathcal{L}}{\partial \mathbf{u}_w}$ touches every column $w \in \{1, \ldots, V\}$ of $\mathbf{W}_{\text{out}}$ on every single training step, forcing full-matrix memory writes.

This $O(V \cdot d)$ complexity renders standard multinomial softmax completely intractable for web-scale corpora.

While Continuous Skip-Gram excels at capturing nuanced semantic relationships and rare tokens, its standard formulation crashes into a severe computational barrier: normalizing over the full vocabulary requires an $O(V \cdot d)$ denominator calculation for every single token prediction. To scale to realistic vocabularies, we must circumvent this partition function bottleneck.

4.7 Scaling softmax: hierarchical softmax and negative sampling

To bypass the $O(V \cdot d)$ partition function barrier, two distinct scaling methods were formulated in [37]: **Hierarchical Softmax** (a structured tree-based approximation) and **Skip-Gram with Negative Sampling (SGNS)** (a sample-based binary discrimination formulation).

4.7.1 Hierarchical softmax ($O(\log_2 V)$)

First developed in 2005 [38] and adapted in [37], **Hierarchical Softmax** decomposes the flat multiclass probability distribution into a hierarchical sequence of binary routing decisions over a binary tree.

- **Binary Tree Topology**: The V vocabulary words form the leaves of a binary tree (constructed as a binary Huffman tree where frequent words have shorter code lengths).
- **Path Specification**: For any word $w \in \mathcal{V}$, let $L(w)$ denote the depth of the leaf from the root. There exists a unique path of nodes from the root node $n(w, 1) =$ root to the leaf node $n(w, L(w)) = w$.
- **Internal Node Parameters**: Each internal node $j \in \{1, \ldots, L(w)-1\}$ is assigned a continuous parameter vector $\mathbf{v}'_{n(w,j)} \in \mathbb{R}^d$.
- **Binary Branching Probability**: At each internal node, the probability of branching to the left child is parameterized by the logistic sigmoid function:

$$\mathbb{P}(\text{branch left} \mid \mathbf{v}_{w_I}) = \sigma\left(\mathbf{v}'_{n(w,j)}{}^\top \mathbf{v}_{w_I}\right) = \frac{1}{1 + \exp\left(-\mathbf{v}'_{n(w,j)}{}^\top \mathbf{v}_{w_I}\right)}. \tag{4.41}$$

The probability of branching to the right child is:

$$\mathbb{P}(\text{branch right} \mid \mathbf{v}_{w_I}) = 1 - \sigma\left(\mathbf{v}'_{n(w,j)}{}^{\top}\mathbf{v}_{w_I}\right) = \sigma\left(-\mathbf{v}'_{n(w,j)}{}^{\top}\mathbf{v}_{w_I}\right). \tag{4.42}$$

- **Path Probability Product**: The conditional probability of reaching leaf word w is the product of binary decision probabilities along its unique path:

$$\mathbb{P}(w \mid w_I) = \prod_{j=1}^{L(w)-1} \sigma\left(\text{sign}(w,j) \cdot \mathbf{v}'_{n(w,j)}{}^{\top}\mathbf{v}_{w_I}\right), \tag{4.43}$$

where $\text{sign}(w,j) = +1$ if the path branches left at node j, and -1 if it branches right.

Proof of Exact Normalization. Because probabilities sum to 1 at each binary branching step ($\sigma(z) + \sigma(-z) = 1$), telescoping products guarantee that the sum of probabilities over all leaves equals 1 identically:

$$\sum_{w \in \mathcal{V}} \mathbb{P}(w \mid w_I) \equiv 1. \tag{4.44}$$

No global partition function $Z(w_I)$ ever needs to be computed!

Complexity Reduction. Evaluating $\mathbb{P}(w \mid w_I)$ and its gradients requires traversing only the $L(w) - 1$ nodes along the path. In a balanced or Huffman binary tree, the average path length is $O(\log_2 V)$. For a vocabulary of $V = 100{,}000$:

$$\log_2(100{,}000) \approx 16.6 \implies \frac{100{,}000}{17} \approx \mathbf{5{,}880}\times \text{ acceleration}. \tag{4.45}$$

4.7.2 Skip-gram negative sampling (SGNS)

While Hierarchical Softmax scales to $O(\log_2 V)$, an even more efficient and empirically superior alternative was formulated in [37]: **Negative Sampling (SGNS)**, derived as a simplified approximation to Noise Contrastive Estimation (NCE).

Instead of treating context prediction as a massive multinomial classification task over V mutually exclusive categories, SGNS reformulates the problem as $1 + K$ **independent binary logistic classifications**:

1. **Positive Sample ($y = 1$)**: The true center-context pair (w_c, w_o) observed in the text corpus. The model maximizes the probability that this pair is a true co-occurrence.
2. **Negative Samples ($y = 0$)**: K noise words $\{w_{n_1}, w_{n_2}, \ldots, w_{n_K}\}$ drawn at random from a noise distribution $P_n(w)$. The model maximizes the probability that these randomly sampled pairs are synthetic noise.

Definition 4.4: *Skip-gram negative sampling objective*

For an observed center-context token pair (w_c, w_o) and K negative samples $\{w_{n_1}, \ldots, w_{n_K}\} \sim P_n(w)$, the SGNS loss is:

$$\mathcal{L}_{\text{SGNS}} = -\log\sigma\left(\mathbf{u}_{w_o}^{\top}\mathbf{v}_{w_c}\right) - \sum_{k=1}^{K} \log\sigma\left(-\mathbf{u}_{w_{n_k}}^{\top}\mathbf{v}_{w_c}\right), \tag{4.46}$$

where $\sigma(z) = \frac{1}{1+e^{-z}}$ denotes the standard logistic sigmoid function.

4.7.3 Step-by-step display math analytical gradient derivations

To understand how SGNS shapes the geometry of the embedding space, we derive the exact analytical gradients of $\mathcal{L}_{\text{SGNS}}$ with respect to all parameter vectors.

Recall the fundamental identities for the logistic sigmoid derivative:

$$\sigma'(z) = \sigma(z)(1 - \sigma(z)), \tag{4.47}$$

$$\frac{d}{dz}\log\sigma(z) = \frac{\sigma'(z)}{\sigma(z)} = 1 - \sigma(z), \tag{4.48}$$

$$\frac{d}{dz}\log\sigma(-z) = \frac{-\sigma'(-z)}{\sigma(-z)} = \frac{-\sigma(-z)(1-\sigma(-z))}{\sigma(-z)} = -(1-\sigma(-z)) = -\sigma(z). \tag{4.49}$$

1. Gradient with respect to Positive Target Vector $\mathbf{u}_{w_o}$. The positive target vector $\mathbf{u}_{w_o}$ appears exclusively in the first term of Equation (4.46):

$$\begin{aligned}\frac{\partial \mathcal{L}_{\text{SGNS}}}{\partial \mathbf{u}_{w_o}} &= -\frac{\partial}{\partial \mathbf{u}_{w_o}}\log\sigma(\mathbf{u}_{w_o}^\top \mathbf{v}_{w_c}) \\ &= -\left[1 - \sigma(\mathbf{u}_{w_o}^\top \mathbf{v}_{w_c})\right]\frac{\partial(\mathbf{u}_{w_o}^\top \mathbf{v}_{w_c})}{\partial \mathbf{u}_{w_o}} \\ &= \left[\sigma(\mathbf{u}_{w_o}^\top \mathbf{v}_{w_c}) - 1\right]\mathbf{v}_{w_c}.\end{aligned} \tag{4.50}$$

Defining the positive prediction error as $e_{\text{pos}} = \sigma(\mathbf{u}_{w_o}^\top \mathbf{v}_{w_c}) - 1 = \hat{y}_{\text{pos}} - 1$, this simplifies to:

$$\frac{\partial \mathcal{L}_{\text{SGNS}}}{\partial \mathbf{u}_{w_o}} = (\hat{y}_{\text{pos}} - 1)\mathbf{v}_{w_c}. \tag{4.51}$$

2. Gradient with respect to Negative Noise Vector $\mathbf{u}_{w_{n_k}}$. Each negative sample vector $\mathbf{u}_{w_{n_k}}$ appears in exactly one term of the summation in Equation (4.46):

$$\begin{aligned}\frac{\partial \mathcal{L}_{\text{SGNS}}}{\partial \mathbf{u}_{w_{n_k}}} &= -\frac{\partial}{\partial \mathbf{u}_{w_{n_k}}}\log\sigma(-\mathbf{u}_{w_{n_k}}^\top \mathbf{v}_{w_c}) \\ &= -\left[-\sigma(\mathbf{u}_{w_{n_k}}^\top \mathbf{v}_{w_c})\right]\frac{\partial(\mathbf{u}_{w_{n_k}}^\top \mathbf{v}_{w_c})}{\partial \mathbf{u}_{w_{n_k}}} \\ &= \sigma(\mathbf{u}_{w_{n_k}}^\top \mathbf{v}_{w_c})\,\mathbf{v}_{w_c}.\end{aligned} \tag{4.52}$$

Defining the negative prediction error as $e_{\text{neg},k} = \sigma(\mathbf{u}_{w_{n_k}}^\top \mathbf{v}_{w_c}) - 0 = \hat{y}_{\text{neg},k}$, this simplifies to:

$$\frac{\partial \mathcal{L}_{\text{SGNS}}}{\partial \mathbf{u}_{w_{n_k}}} = \hat{y}_{\text{neg},k}\,\mathbf{v}_{w_c}. \tag{4.53}$$

3. Gradient with respect to Center Word Vector $\mathbf{v}_{w_c}$. The center word vector $\mathbf{v}_{w_c}$ appears in both the positive term and all K negative terms. Applying the chain rule to the complete loss:

$$\begin{aligned}\frac{\partial \mathcal{L}_{\text{SGNS}}}{\partial \mathbf{v}_{w_c}} &= -\frac{\partial}{\partial \mathbf{v}_{w_c}}\log\sigma(\mathbf{u}_{w_o}^\top \mathbf{v}_{w_c}) - \sum_{k=1}^{K}\frac{\partial}{\partial \mathbf{v}_{w_c}}\log\sigma(-\mathbf{u}_{w_{n_k}}^\top \mathbf{v}_{w_c}) \\ &= \underbrace{\left[\sigma(\mathbf{u}_{w_o}^\top \mathbf{v}_{w_c}) - 1\right]\mathbf{u}_{w_o}}_{\text{Attractive Component (Pull Context)}} + \underbrace{\sum_{k=1}^{K}\sigma(\mathbf{u}_{w_{n_k}}^\top \mathbf{v}_{w_c})\,\mathbf{u}_{w_{n_k}}}_{\text{Repulsive Component (Push Noise)}}.\end{aligned} \tag{4.54}$$

4.7.4 Physical vector mechanics interpretation

Under gradient descent, parameter updates subtract the gradient: $\mathbf{v}_{w_c} \leftarrow \mathbf{v}_{w_c} - \eta \nabla_{\mathbf{v}_{w_c}} \mathcal{L}$:

$$\mathbf{v}_{w_c} \leftarrow \mathbf{v}_{w_c} + \eta \left(1 - \hat{y}_{\text{pos}}\right) \mathbf{u}_{w_o} - \eta \sum_{k=1}^{K} \hat{y}_{\text{neg},k} \, \mathbf{u}_{w_{n_k}}. \tag{4.55}$$

This update formula admits a clean, intuitive **physical vector mechanics** interpretation:

- **The Attractive Pull Force**: Since $\hat{y}_{\text{pos}} = \sigma(\mathbf{u}_{w_o}^\top \mathbf{v}_{w_c}) \in (0, 1)$, the coefficient $(1 - \hat{y}_{\text{pos}})$ is strictly positive. The update adds a vector component in the direction $+\mathbf{u}_{w_o}$, **pulling the center word vector toward the true context vector**. If the model is already confident ($\hat{y}_{\text{pos}} \approx 1$), the force vanishes; if the model is surprised ($\hat{y}_{\text{pos}} \approx 0$), the force reaches maximum strength $+1 \cdot \mathbf{u}_{w_o}$.
- **The Repulsive Push Force**: For every negative sample k, $\hat{y}_{\text{neg},k} = \sigma(\mathbf{u}_{w_{n_k}}^\top \mathbf{v}_{w_c}) \in (0, 1)$ is strictly positive. The update subtracts a vector component along each noise vector $-\mathbf{u}_{w_{n_k}}$, **pushing the center word vector away from noise words**. If the noise word is far away ($\hat{y}_{\text{neg},k} \approx 0$), no force is exerted; if the noise word is erroneously close ($\hat{y}_{\text{neg},k} \approx 1$), a strong repulsive kick $-\mathbf{u}_{w_{n_k}}$ repels the points.

Crucially, the computational complexity of evaluating Equation (4.55) is $O(K \cdot d)$, which is **entirely independent of vocabulary size** V. In practice, choosing $K \in [5, 20]$ for small corpora and $K \in [2, 5]$ for web-scale corpora provides sufficient negative contrast, reducing training time from days to hours.

4.7.5 Theoretical justification of the $3/4$ unigram noise exponent

How should negative samples be drawn? In [37], the noise distribution $P_n(w)$ is defined as a smoothed unigram distribution raised to the power of $\alpha = 3/4 = 0.75$:

$$P_n(w) = \frac{f(w)^{3/4}}{\sum_{w' \in \mathcal{V}} f(w')^{3/4}}, \tag{4.56}$$

where $f(w)$ denotes the raw unigram occurrence frequency of token w in the corpus.

Why $\alpha = 0.75$ instead of $\alpha = 1.0$ (Raw Frequency) or $\alpha = 0.0$ (Uniform)?

1. **Failure of Raw Frequency ($\alpha = 1.0$)**: Natural language word frequencies follow Zipf's law ($f(r) \propto 1/r$). A tiny fraction of stopwords (`"the"`, `"of"`, `"and"`, `"to"`) account for over 50% of all token occurrences in a typical corpus. If $\alpha = 1.0$, more than half of all negative samples drawn are uninformative stopwords. The model wastes its entire discriminative budget separating words from `"the"`, while rare content words (`"polymorphism"`, `"compiler"`) have near-zero probability of ever being selected as negative samples, receiving virtually zero negative gradient updates.
2. **Failure of Uniform Sampling ($\alpha = 0.0$)**: Under a uniform distribution $P_n(w) = 1/V$, every vocabulary word has equal probability. Because the vast majority of words in any natural vocabulary reside in the long tail of rare words, sampling uniformly selects almost exclusively rare words as negatives. The network rarely practices distinguishing moderately frequent content words from one another.
3. **The Sub-linear Compromise ($\alpha = 0.75$)**: The exponent 3/4 acts as a concave, sub-linear dampener. It compresses the gap between ultra-frequent stopwords and rare content words.

Numerical Demonstration of the Rare-Word Boost. Consider a frequent stopword w_{freq} with corpus frequency $f(w_{\text{freq}}) = 0.01$ (e.g., 10^6 occurrences in 10^8 words) and a rare technical term w_{rare} with corpus frequency $f(w_{\text{rare}}) = 10^{-6}$ (e.g., 100 occurrences in 10^8 words):

- Under raw unigram frequency ($\alpha = 1.0$), the sampling ratio is:

$$\frac{f(w_{\text{freq}})}{f(w_{\text{rare}})} = \frac{10^{-2}}{10^{-6}} = 10{,}000. \tag{4.57}$$

  The stopword is sampled 10,000 times more frequently than the rare word.

- Under the smoothed distribution ($\alpha = 0.75$), the ratio becomes:

$$\frac{f(w_{\text{freq}})^{0.75}}{f(w_{\text{rare}})^{0.75}} = \left(\frac{10^{-2}}{10^{-6}}\right)^{0.75} = (10{,}000)^{0.75} = (10^4)^{3/4} = 10^3 = 1{,}000. \tag{4.58}$$

By applying the 3/4 power, the relative probability of sampling the rare word is amplified by a factor of:

$$\frac{10{,}000}{1{,}000} = \mathbf{10}\times. \tag{4.59}$$

This 10× boost provides rare words with sufficient negative gradient pushes to carve out accurate decision boundaries, while still reflecting broad corpus statistics.

4.7.6 Global count matrix factorization: GloVe

Around the same time, **GloVe (Global Vectors for Word Representation)** was introduced in [39]. Whereas Word2Vec is a local window-based predictive model that processes streaming word pairs, GloVe demonstrated that distributed embeddings could be trained by directly factorizing a **global co-occurrence matrix** $\mathbf{X} \in \mathbb{R}^{V\times V}$, where X_{ij} counts the number of times word j appears in the context window of word i.

GloVe's core insight is that the semantic distinction between two words i and j relative to a probe word k is encoded not by raw probabilities $P_{ik} = X_{ik}/X_i$, but by the **ratio of co-occurrence probabilities**:

$$\frac{P_{ik}}{P_{jk}} = \frac{P(k \mid i)}{P(k \mid j)}. \tag{4.60}$$

Table 4.1 reproduces the seminal demonstration from [39] comparing i = `"ice"` and j = `"steam"` across probe words k.

Table 4.1: Co-occurrence probability ratios illustrating semantic discrimination in GloVe [39].

Probability & Ratio	k = `"solid"`	k = `"gas"`	k = `"water"`	k = `"fashion"`
$P(k \mid$ `"ice"`$)$	1.9×10^{-4}	6.6×10^{-5}	3.0×10^{-3}	1.7×10^{-5}
$P(k \mid$ `"steam"`$)$	2.2×10^{-5}	7.8×10^{-4}	2.2×10^{-3}	1.8×10^{-5}
$P(k \mid$ `"ice"`$)/P(k \mid$ `"steam"`$)$	**8.9**	**0.085**	1.36	0.96

When k = `"solid"` is strongly related to `"ice"` but not `"steam"`, the ratio is large ($8.9 \gg 1$). When k = `"gas"` is related to `"steam"` but not `"ice"`, the ratio is tiny ($0.085 \ll 1$). When k = `"water"` is equally related to both (or k = `"fashion"` is related to neither), the ratio hovers near 1.0.

GloVe enforces this relationship by fitting a log-bilinear model where dot products approximate log co-occurrence counts, minimized via weighted least squares:

$$J = \sum_{i=1}^{V}\sum_{j=1}^{V} f(X_{ij})\left(\mathbf{w}_i^\top \tilde{\mathbf{w}}_j + b_i + \tilde{b}_j - \log X_{ij}\right)^2, \tag{4.61}$$

where $f(X_{ij}) = \min(1, (X_{ij}/x_{\max})^{\alpha})$ is a sub-linear weighting function preventing frequent stopwords from dominating the loss. Both Word2Vec and GloVe construct semantically structured vector spaces, with SGNS operating on streaming text and GloVe operating on aggregated count statistics.

Skip-Gram Negative Sampling and GloVe resolve the computational bottleneck, making large-scale distributed representation learning feasible. However, Word2Vec operates under an implicit architectural assumption: that the word is the minimal, indivisible atomic unit of language. In real-world deployment, this assumption creates severe structural vulnerabilities.

4.8 Limitations of Word2Vec and the out-of-vocabulary (OOV) barrier

Despite their revolutionary impact, classical Word2Vec and GloVe architectures suffer from fundamental linguistic and software engineering limitations.

4.8.1 The atomic word token assumption and out-of-vocabulary (OOV) failure

Word2Vec models assume a closed, fixed vocabulary $\mathcal{V} = \{w_1, \ldots, w_V\}$ established during training. Each word is treated as an indivisible, atomic entity: its character composition is completely ignored. This creates the **Out-of-Vocabulary (OOV) catastrophe**:

- **Unseen Words at Test Time**: In open-domain text and evolving codebases, users constantly produce novel tokens: technical neologisms, composite identifiers (e.g., `AsyncHttpClientFactory`), misspelled words (e.g., `"recusive"`), and inflected stems.
- **Mapping to `<UNK>`**: When a trained Word2Vec model encounters an OOV word at inference, it has no entry in $\mathbf{W}_{\text{in}}$ or $\mathbf{W}_{\text{out}}$. Deep learning pipelines are forced to map all OOV tokens to a generic unknown token placeholder:

$$w_{\text{oov}} \longmapsto \texttt{<UNK>} \implies \mathbf{v}_{w_{\text{oov}}} = \mathbf{v}_{\texttt{<UNK>}}. \tag{4.62}$$

  This discards all semantic identity. To Word2Vec, a typo like `"algorithmm"` is as completely devoid of meaning as an alien string.

4.8.2 Morphological blindness

Word2Vec is entirely blind to internal word morphology. In natural languages with rich inflectional systems (e.g., Italian, German, Turkish) as well as programming languages, words share sub-morphemic stems, prefixes, and suffixes:

Natural language: `"develop"`, `"developer"`, `"developing"`, `"undeveloped"`,
Software identifiers: `"parseFile"`, `"parseTokens"`, `"parseTree"`.

Under Word2Vec, each of these variants is assigned a totally disjoint, independent column in $\mathbf{W}_{\text{in}}$. The model possesses zero architectural mechanism to share parameters across words containing the common root `"develop"` or `"parse"`. If `"undeveloped"` occurs only twice in the corpus, its vector remains poorly trained, unable to inherit semantic properties from the frequent token `"develop"`.

4.8.3 Polysemy collapse: static context invariance

Word2Vec assigns exactly **one static vector $\mathbf{v}_w \in \mathbb{R}^d$ to each word string** w, regardless of the surrounding runtime context in which it appears. This results in **polysemy collapse**.

Definition 4.5: *Polysemy collapse*

Let word w possess M distinct semantic senses $\mathcal{S}_1, \ldots, \mathcal{S}_M$ with respective corpus mixture proportions $\pi_1, \ldots, \pi_M$ ($\sum \pi_m = 1$). Because Word2Vec minimizes prediction error across all occurrences using a single vector $\mathbf{v}_w$, the optimal vector converges to the convex combination of distinct sense centroids:

$$\mathbf{v}_w^* \approx \sum_{m=1}^{M} \pi_m \mathbf{v}_{\mathcal{S}_m}. \tag{4.63}$$

For example, consider the polysemous word `"bat"`:

Sense 1 (Zoological): "A **bat** is a nocturnal flying mammal."
Sense 2 (Sports Equipment): "The baseball player swung the wooden **bat**."

In Word2Vec, $\mathbf{v}_{\text{bat}}$ is pulled toward biological animals on some iterations, and toward sports equipment on others. The resulting point converges to an intermediate centroid in $\mathbb{R}^d$ that represents neither animal nor sports gear accurately, degrading cosine similarity to both senses.

In software engineering, polysemy is ubiquitous:

- `"Python"`: A biological serpent vs. a high-level dynamic programming language.
- `"Spring"`: A season of the year vs. a popular Java enterprise application framework.
- `"rust"`: Iron oxide corrosion vs. a memory-safe systems programming language.

A static Word2Vec model maps `"Rust"` to a compromise vector that conflates oxidation chemistry with memory safety guarantees.

While contextual ambiguity requires sequence models that dynamically modulate representations based on full sentence context, the Out-of-Vocabulary barrier and morphological blindness can be solved at the token representation level. By abandoning the word as an atomic unit and descending into subword character n-grams, FastText establishes a morphologically aware embedding paradigm.

4.9 FastText: subword character n-gram embeddings

To conquer the OOV barrier and enable morphological parameter sharing, **FastText** was introduced in [40] (and covered on Slides 13–14 of the course lecture deck).

4.9.1 Character n-gram decomposition and boundary markers

Rather than treating a word as an atomic token, FastText represents each word as a **bag of character n-grams**.

1. **Boundary Enclosure**: To distinguish prefixes and suffixes from character sequences occurring in the interior of a word, each word w is bracketed by special boundary symbols `"<"` and `">"`:
$$\tilde{w} = \texttt{"<"} \circ w \circ \texttt{">"}. \tag{4.64}$$
This ensures that the prefix sequence `"<re"` (as in `"<refactor>"`) is mathematically distinct from the internal substring `"re"` (as in `"<where>"`).

2. **Subword Extraction**: All character n-grams of length $n \in [n_{\min}, n_{\max}]$ (typically $n \in [3, 6]$) are extracted, along with the special sequence containing the entire word itself $\tilde{w}$.

Course Slide Example: ⟨`where`⟩. Following Slide 13 of the course lecture deck, consider the word `"where"` with character trigrams ($n = 3$):

$$\texttt{"<where>"} \implies \mathcal{G}_{\texttt{where}} = \left\{\texttt{"<wh"}, \texttt{"whe"}, \texttt{"her"}, \texttt{"ere"}, \texttt{"re>"}, \texttt{"<where>"}\right\}. \tag{4.65}$$

4.9.2 Vector sum composition

Each subword n-gram g is assigned a latent continuous embedding vector $\mathbf{z}_g \in \mathbb{R}^d$. The final representation of word w is defined as the **sum of its constituent subword vectors**:

$$\mathbf{v}_w = \sum_{g \in \mathcal{G}_w} \mathbf{z}_g \in \mathbb{R}^d. \tag{4.66}$$

Under the Skip-Gram Negative Sampling objective, the similarity score between center word w and context word c becomes:

$$s(w, c) = \mathbf{v}_w^\top \mathbf{u}_c = \sum_{g \in \mathcal{G}_w} \mathbf{z}_g^\top \mathbf{u}_c. \tag{4.67}$$

During backpropagation, the prediction error with respect to $\mathbf{v}_w$ distributes gradients to every constituent subword vector $\mathbf{z}_g$.

4.9.3 Resolving OOV tokens dynamically and morphological sharing in code

This simple compositional formulation yields two decisive capabilities:

1. **Zero-Shot Synthesis of Out-of-Vocabulary Tokens**: When an unseen word w_{oov} appears at test time (e.g., a compound identifier like `"refactorable"` or a code typo like `"recusive"`), even if the full token was never seen during training, its constituent character n-grams (`"<re"`, `"ref"`, `"fact"`, `"able>"`) were updated across other vocabulary words. FastText synthesizes an accurate embedding vector on the fly by summing the vectors of its known subwords:
$$\mathbf{v}_{w_{\text{oov}}} = \sum_{g \in \mathcal{G}_{w_{\text{oov}}}} \mathbf{z}_g. \tag{4.68}$$
The OOV barrier is completely eliminated!

2. **Morphological Parameter Sharing**: In natural language, prefixes like `"un-"`, suffixes like `"-tion"`, and inflections like `"-ing"` accumulate shared vector representations across hundreds of distinct words. In software engineering, identifier substrings automatically share semantics:
$$\texttt{"parseXmlDocument"} \quad \text{and} \quad \texttt{"parseJsonDocument"} \tag{4.69}$$
share the character subwords for `"parse"` and `"Document"`. The model immediately recognizes that both identifiers describe parsing operations on structured documents.

Memory Bounding via the Fowler-Noll-Vo (FNV-1a) Hash Trick. Because the universe of all character n-grams ($n \in [3, 6]$) across large vocabularies exceeds tens of millions of unique strings, allocating an unbounded matrix would cause memory blowout. FastText bounds the embedding table to $K = 2 \times 10^6$ rows using the Fowler-Noll-Vo (FNV-1a) hashing function:

$$\text{index}(g) = \text{FNV-1a}(g) \pmod{K}. \tag{4.70}$$

Colliding n-grams share bucket rows; due to the high dimensionality ($d = 300$) and independent sum aggregation, hash collisions act as benign regularizing noise that does not disrupt semantic retrieval.

4.9.4 Connection to modern subword tokenizers: BPE and WordPiece

As highlighted in pedagogical breakdowns of language models [41], the subword philosophy pioneered by FastText directly inspired the tokenization mechanisms powering modern Large Language Models:

- **Byte Pair Encoding (BPE)** [12, 42]: Iteratively merges the most frequent pairs of adjacent characters or bytes to form a deterministic subword vocabulary (used in GPT-2, GPT-4, and LLaMA).
- **SentencePiece** [43]: Treats input text as a raw byte stream without whitespace pre-segmentation, training unigram language model segmentations.

FastText operates continuous character n-gram summations at the embedding layer, whereas BPE operates discrete token mergers during preprocessing. Both architectures fundamentally exist to dismantle the brittle, isolated word-level assumption and establish robust subword parameter sharing.

By decomposing words into character n-grams, FastText conquers the Out-of-Vocabulary barrier and enables morphological parameter sharing. With our embedding models fully trained, we can now step back and examine the remarkable mathematical geometry that spontaneously emerges within these dense vector spaces.

4.10 Emergent vector space geometry: clustering, metrics and relational analogies

When Word2Vec and FastText models are trained over large text corpora, the resulting continuous vector spaces exhibit profound emergent mathematical properties: high-dimensional semantic clustering, metric equivalence, and linear relational translations.

4.10.1 Semantic clustering and dimensionality reduction

In high-dimensional embedding space ($d = 300$), words with similar functional roles organize into distinct, tightly bound clusters.

Following Slide 14 of the course lecture deck, projecting 300-dimensional FastText vectors into two dimensions via Principal Component Analysis (PCA) reveals distinct semantic clusters:

- **Household Items**: `"table"`, `"chair"`, `"desk"`, `"sofa"`.
- **Mammals**: `"dog"`, `"cat"`, `"tiger"`, `"elephant"`.
- **Birds**: `"eagle"`, `"sparrow"`, `"parrot"`, `"pigeon"`.

The leading principal components preserve categorical variance: domestic entities, animal taxonomy, and biological sub-branches form linearly separable clusters in the compressed 2D projection as well as in the original 300-dimensional manifold.

4.10.2 Linear relational translations (vector analogy arithmetic)

The most celebrated emergent property of continuous word embeddings is that semantic and syntactic relationships manifest as **linear vector translation offsets**:

$$\mathbf{v}_{w_2} - \mathbf{v}_{w_1} \approx \vec{d}_{\text{relation}}. \tag{4.71}$$

Country-to-Capital Analogy (Course Slide 15). In the course lecture deck (Slide 15), subtracting the vector for `"Berlin"` from `"Germany"` produces a directional displacement vector $\vec{d}_{\text{capital_of}}$:

$$\mathbf{v}_{\text{Germany}} - \mathbf{v}_{\text{Berlin}} \approx \vec{d}_{\text{capital_of}}. \tag{4.72}$$

Applying this learned offset vector to another country predicts its capital city via simple vector addition:

$$\mathbf{v}_{\text{Spain}} + (\mathbf{v}_{\text{Germany}} - \mathbf{v}_{\text{Berlin}}) \approx \mathbf{v}_{\text{Madrid}}. \tag{4.73}$$

The Gender-Royalty Parallelogram. The canonical demonstration established in [37] is the gender-royalty parallelogram:

$$\mathbf{v}_{\text{king}} - \mathbf{v}_{\text{man}} + \mathbf{v}_{\text{woman}} \approx \mathbf{v}_{\text{queen}}. \tag{4.74}$$

Figure 4.5 illustrates this parallelogram in two dimensions.

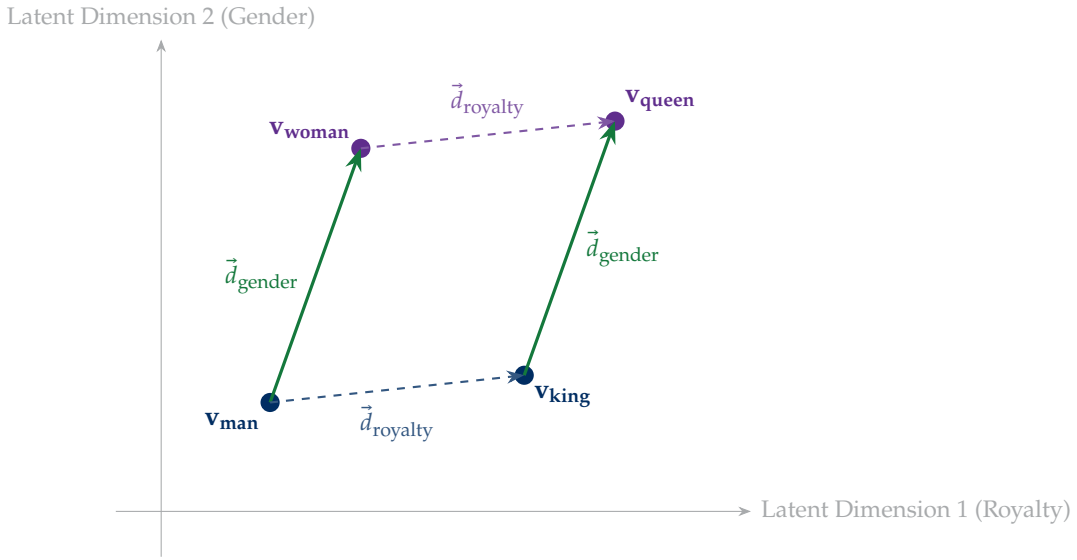


Figure 4.5: Geometric parallelogram in continuous embedding space demonstrating relational vector translations: $\mathbf{v}_{\text{King}} - \mathbf{v}_{\text{Man}} \approx \mathbf{v}_{\text{Queen}} - \mathbf{v}_{\text{Woman}}$. The directional displacement $\vec{d}_{\text{gender}}$ isolates the feminine gender transformation, while $\vec{d}_{\text{royalty}}$ isolates the monarchical status offset.

Software Engineering Analogy Equivalents. In code representation learning (e.g., CodeBERT [44], OpenAI Codex embeddings [45]), relational vector arithmetic reflects runtime architectures and language toolchains:

$$\mathbf{v}_{\text{Java}} - \mathbf{v}_{\text{JVM}} + \mathbf{v}_{\text{Python}} \approx \mathbf{v}_{\text{CPython}}, \tag{4.75}$$

$$\mathbf{v}_{\text{C++}} - \mathbf{v}_{\text{g++}} + \mathbf{v}_{\text{Rust}} \approx \mathbf{v}_{\text{rustc}}. \tag{4.76}$$

4.10.3 Metric geometry of embedding spaces: cosine vs. Euclidean distance

When querying nearest neighbors or evaluating semantic similarity between two continuous vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^d$, three metrics are commonly considered:

1. **Euclidean (L_2) Distance**:

$$D_{L_2}(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|_2 = \sqrt{\sum_{i=1}^{d} (u_i - v_i)^2}. \tag{4.77}$$

2. **Dot Product (Inner Product)**:

$$\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^\top \mathbf{v} = \sum_{i=1}^{d} u_i v_i = \|\mathbf{u}\|_2 \|\mathbf{v}\|_2 \cos\theta. \tag{4.78}$$

3. **Cosine Distance**:

$$D_{\cos}(\mathbf{u}, \mathbf{v}) = 1 - \cos(\mathbf{u}, \mathbf{v}) = 1 - \frac{\mathbf{u}^\top \mathbf{v}}{\|\mathbf{u}\|_2 \|\mathbf{v}\|_2} \in [0, 2]. \tag{4.79}$$

Theorem 4.2: *Monotonic equivalence under L_2 normalization*

Let $\mathbf{u}, \mathbf{v} \in \mathbb{R}^d$ be normalized to unit Euclidean length ($\|\mathbf{u}\|_2 = \|\mathbf{v}\|_2 = 1$). Then squared Euclidean distance is strictly proportional to Cosine distance:

$$\|\mathbf{u} - \mathbf{v}\|_2^2 = \|\mathbf{u}\|_2^2 + \|\mathbf{v}\|_2^2 - 2\mathbf{u}^\top \mathbf{v} = 1 + 1 - 2\cos(\mathbf{u}, \mathbf{v}) = 2(1 - \cos(\mathbf{u}, \mathbf{v})) = 2D_{\cos}(\mathbf{u}, \mathbf{v}). \tag{4.80}$$

Consequently, nearest neighbor rankings under Euclidean distance, Cosine distance, and Dot Product are mathematically identical on the unit hypersphere $\mathbb{S}^{d-1}$.

Why Cosine Distance Dominates Natural Language Processing. As demonstrated in educational analyses [41], when word embeddings are trained without explicit length regularization, the Euclidean norm $\|\mathbf{v}_w\|_2$ strongly correlates with unigram token frequency. Highly frequent words undergo millions of gradient pushes, systematically expanding their vector magnitude, whereas rare terms remain close to the origin. Cosine distance normalizes out vector magnitude, projecting all embeddings onto $\mathbb{S}^{d-1}$. It evaluates **pure angular orientation** (conceptual topic) rather than **vector length** (frequency artifact).

4.10.4 Dimensionality trade-offs and the Johnson-Lindenstrauss lemma

Why do word embeddings operate effectively in dimensions $d \in [100, 300]$, and modern LLMs in $d \in [768, 4096]$?

- **Under-parameterization ($d < 50$)**: Severe geometric crowding occurs. The space lacks enough orthogonal degrees of freedom to separate independent semantic concepts, forcing unrelated topics into spurious proximity.
- **Over-parameterization ($d \gg 10{,}000$)**: The space suffers from the curse of dimensionality. The volume of the unit ball concentrates entirely in a thin outer shell, distance distributions become uniform, and memory bandwidth demands explode.

The mathematical guarantee underwriting this stability is the **Johnson-Lindenstrauss Lemma**:

Theorem 4.3: *Johnson-Lindenstrauss Lemma*

Given $0 < \epsilon < 1$, any set of N points in arbitrary Euclidean space can be embedded into a Euclidean space of dimension:

$$d = O\left(\frac{\log N}{\epsilon^2}\right), \tag{4.81}$$

such that all pairwise Euclidean distances are preserved within a distortion factor of $1 \pm \epsilon$:

$$(1-\epsilon)\|\mathbf{u}-\mathbf{v}\|_2^2 \leq \|f(\mathbf{u})-f(\mathbf{v})\|_2^2 \leq (1+\epsilon)\|\mathbf{u}-\mathbf{v}\|_2^2. \tag{4.82}$$

For a vocabulary of $N = 100{,}000$ tokens and distortion tolerance $\epsilon = 0.2$, $\log(10^5)/0.04 \approx 287$, precisely justifying why $d = 300$ serves as the golden standard for dense static word vectors.

4.11 Modern subword tokenization and input pipelines for foundation models

In Sections 4.1 through 4.10, we formulated the mathematical foundations of classical distributed word representations (Word2Vec, GloVe, FastText). While continuous embeddings solved the differentiability barrier and geometric similarity, classical architectures operated under a fragmented, static paradigm: input text was naively pre-split on whitespace into word-level units, vocabulary sizes ballooned into hundreds of thousands of entries, rare or misspelled tokens collapsed into the uninformative `<UNK>` placeholder, and input sequences were treated as unordered bags or static lookup tables.

Modern Large Language Models (LLMs) such as GPT-2, GPT-3, GPT-4, LLaMA, and Claude discard this isolated design in favor of an integrated, end-to-end **data preprocessing, tokenization, and input embedding pipeline**. As formalized in *Build a Large Language Model from Scratch* [32], preparing raw text for foundation model pretraining requires an indivisible four-stage architectural pipeline:

1. **Subword Tokenization**: Segmenting raw text into atomic subword units using Byte Pair Encoding (BPE), resolving punctuation and whitespace losslessly.
2. **Token ID Mapping**: Converting discrete subwords into numerical indices via an invertible vocabulary lookup table, enriched with special operational context markers.
3. **Continuous Embedding Projection**: Retrieving dense vector representations for each token index from a learned embedding table $\mathbf{W}_{\text{tok}} \in \mathbb{R}^{V \times d}$.
4. **Composite Positional Synthesis**: Injecting sequential order into permutation-invariant neural layers by additively combining continuous token embeddings with positional embeddings: $\mathbf{X}_{\text{in}} = \mathbf{E}_{\text{tok}} + \mathbf{E}_{\text{pos}}$.

Figure 4.6 illustrates this end-to-end data pipeline from raw textual strings to the final continuous tensor ingested by Transformer architectures.

4.11.1 The tokenization spectrum: character, word, and subword trade-offs

Converting raw natural language or software code into discrete units reveals a fundamental trade-off across three representational granularities: character-level, word-level, and subword tokenization.

Why whitespace splitting fails. A naive baseline approach to text tokenization splits raw strings on whitespace characters (‘\s’). However, whitespace splitting fails catastrophically on punctuation:

```
"Hello, world!  This is a test."  ⟹ (whitespace split)  ['Hello,', 'world!', 'This', 'is',
                                                          'a', 'test.']
```

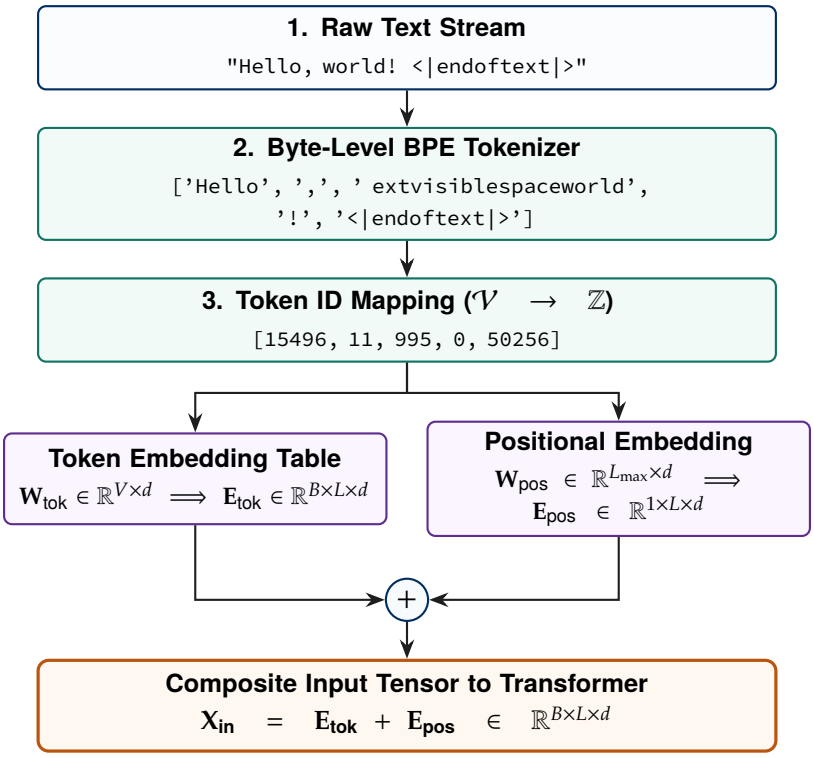


Figure 4.6: The modern foundation model tokenization and input preparation pipeline. Raw text is segmented into byte-level BPE subwords, mapped to unique integer token IDs, and converted into continuous latent vectors where token embeddings and positional embeddings combine additively.

Table 4.2: Comprehensive taxonomy and trade-off analysis of tokenization paradigms in natural language processing and modern foundation models.

Paradigm	Vocab (V)	Length (L)	OOV	Linguistic & Systems Limitations	Representative Models
Character-Level	100 – 256	Massive (4 – 6×)	Zero	High sequence length explodes computational complexity $O(L^2)$ in self-attention; lacks word-level semantic priors.	Char-RNN, Canine, ByT5
Word-Level	50k – 1M+	Minimal (1×)	Severe	Out-of-vocabulary words collapse to `<UNK>`; morphologically blind; massive embedding matrix $\mathbf{W} \in \mathbb{R}^{V \times d}$.	Word2Vec, GloVe, N-gram LMs
Subword-Level	32k – 100k	Moderate (1.2 – 1.4×)	Zero	Requires statistical offline pre-tokenization merge training; optimal Pareto frontier of length and vocabulary.	BPE (GPT-2/3/4, LLaMA), WordPiece (BERT)

The comma is fused to "Hello", and periods are fused to "world" and "test". Consequently, the model assigns distinct vocabulary indices to "world", "world!", "world,", and "world?". This fragments training counts, bloats the vocabulary with redundant punctuation variants, and destroys syntactic generalization.

Regex-based tokenization with punctuation isolation. To isolate punctuation while preserving them as distinct semantic tokens, a regular expression splitting rule can be formulated [32] that captures punctuation symbols, em-dashes, and whitespaces:

```python
import re

class SimpleTokenizerV1:
    def __init__(self, vocab):
        self.str_to_int = vocab
        self.int_to_str = {i: s for s, i in vocab.items()}

    def encode(self, text):
        # Split on whitespace, punctuation marks, and special dash patterns
        tokens = re.split(r'([,.:;?_!"()\']|--|\s)', text)
        tokens = [t.strip() for t in tokens if t.strip()]
        token_ids = [self.str_to_int[t] for t in tokens]
        return token_ids

    def decode(self, token_ids):
        text = " ".join([self.int_to_str[i] for i in token_ids])
        # Clean up spaces before punctuation marks
        text = re.sub(r'\s+([,.?!])', r'\1', text)
        return text

# Building the vocabulary from unique preprocessed tokens
text = "Hello, world. Is this-- a test?"
raw_tokens = re.split(r'([,.:;?_!"()\']|--|\s)', text)
tokens = [t.strip() for t in raw_tokens if t.strip()]
vocab = {token: idx for idx, token in enumerate(sorted(set(tokens)))}
tokenizer = SimpleTokenizerV1(vocab)
encoded = tokenizer.encode("Hello, world.")
print("Encoded IDs:", encoded)
print("Decoded text:", tokenizer.decode(encoded))
```

Listing 4.1: Basic regular expression text tokenization with punctuation isolation and vocabulary construction [32].

4.11.2 Special context tokens: document boundaries, padding, and masked batching

In practical machine learning and software engineering workflows, a vocabulary cannot consist solely of natural language words. The model requires **special context tokens** to encode structural and operational metadata:

1. **Unknown Word Token** (<|unk|>): In fixed-vocabulary systems, words unseen during training cannot be mapped to an integer index. Such tokens are mapped to <|unk|>. While this prevents code execution errors, it irreversibly destroys semantic information: a technical identifier or typo is flattened into a generic void.

2. **Document Boundary Delimiter** (<|endoftext|>): In large-scale autoregressive pretraining, billions of tokens spanning Wikipedia articles, books, and GitHub repositories are concatenated into continuous sequential streams. Without an explicit separator, the causal

attention mechanism would assume that the first sentence of an article is the causal continuation of an unrelated previous document! The token <|endoftext|> signals that all preceding context is decoupled from subsequent tokens, allowing clean document demarcation.

3. **Padding Token** (<|pad|>): Mini-batch processing on GPU hardware requires rectangular tensors $\mathbf{X} \in \mathbb{Z}^{B \times L}$. When batching sequences of varying lengths ($L_1 \neq L_2$), shorter sequences are padded with <|pad|> tokens up to maximum length L. Crucially, the model's self-attention mechanism is configured with an **attention mask**:

$$M_{ij} = \begin{cases} 0 & \text{if token } j \text{ is a genuine text token,} \\ -\infty & \text{if token } j \text{ is a <|pad|> token.} \end{cases} \tag{4.83}$$

Adding M_{ij} to attention logits before Softmax drives pad attention weights to $\frac{e^{-\infty}}{\sum e^z} = 0$, completely preventing padded positions from corrupting token representations.

4.11.3 Byte Pair Encoding (BPE): algorithmic mechanics and merge trees

To conquer the Out-of-Vocabulary barrier without blowing up sequence length, modern LLMs deploy **Byte Pair Encoding (BPE)**. Originally invented for data compression in [46], BPE was adapted to natural language processing in [42].

Algorithmic formalization. BPE constructs a subword vocabulary through an iterative bottom-up statistical merge procedure partitioned into distinct training and inference phases:

1. **Base Alphabet Initialization**: Initialize base vocabulary $\mathcal{V}_0$ with all individual characters occurring across the training corpus C, augmenting each word with an explicit end-of-word boundary marker (e.g., '</w>'):

$$\mathcal{V}_0 = \{c \mid c \in w,\ w \in C\} \cup \{\texttt{</w>}\}. \tag{4.84}$$

Every word in C is initially split into a sequence of individual character symbols.

2. **Adjacent Pair Frequency Tally**: At iteration k, construct the co-occurrence frequency histogram over all adjacent symbol pairs (p_1, p_2) across all tokenized words in C:

$$(p_1^*, p_2^*) = \arg\max_{(p_1,p_2)} \sum_{w \in C} \text{count}(w) \cdot \mathbb{I}\{(p_1, p_2) \subseteq w\}. \tag{4.85}$$

3. **Greedy Merge and Priority Rule Logging**: Merge the globally most frequent pair (p_1^*, p_2^*) into an atomic new subword unit $p_{\text{new}} = p_1^* \circ p_2^*$. Crucially, record this merge operation in an ordered priority dictionary $\mathcal{M}$:

$$\mathcal{M}[k] = ((p_1^*, p_2^*) \mapsto p_{\text{new}}). \tag{4.86}$$

4. **In-Place Corpus Contraction**: Substitute every contiguous occurrence of (p_1^*, p_2^*) across corpus C with the merged symbol p_{new}, and expand the vocabulary:

$$\mathcal{V}_k = \mathcal{V}_{k-1} \cup \{p_{\text{new}}\}. \tag{4.87}$$

5. **Termination and Inference Application**: Repeat Steps 2–4 until reaching the target vocabulary budget $|\mathcal{V}_k| = V_{\text{target}}$. At inference time, any arbitrary input word is first split into its constituent base characters, and the learned merge rules in $\mathcal{M}$ are applied sequentially according to their training rank.

A concrete clarification example: training merges and inference on unseen words. To illustrate how BPE resolves out-of-vocabulary tokens without explosion, consider a small transparent corpus consisting of four words with their occurrence frequencies:

$$C = \{\texttt{"hug"} : 10,\ \texttt{"pug"} : 5,\ \texttt{"pun"} : 12,\ \texttt{"hugs"} : 5\}. \tag{4.88}$$

The total corpus count is $10 + 5 + 12 + 5 = 32$ word instances. The initial base vocabulary consists of 7 symbols: $\mathcal{V}_0 = \{\texttt{'h'}, \texttt{'u'}, \texttt{'g'}, \texttt{'p'}, \texttt{'n'}, \texttt{'s'}, \texttt{'</w>'}\}$. The segmented corpus begins as:

```
h u g </w> :  10    p u g </w> :  5    p u n </w> :  12    h u g s </w> :  5
```

- **Iteration 1**: We evaluate the frequency of all adjacent bigrams across the corpus:
  - ('u','g') : 10(hug) + 5(pug) + 5(hugs) = 20.
  - ('p','u') : 5(pug) + 12(pun) = 17.
  - ('h','u') : 10(hug) + 5(hugs) = 15.
  - ('g','</w>') : 10(hug) + 5(pug) = 15.
  - ('u','n') : 12(pun), ('n','</w>') : 12(pun).
  - ('g','s') : 5(hugs), ('s','</w>') : 5(hugs).

  The most frequent pair is ('u','g') with frequency 20. We log Merge Rule 1: ('u','g') → 'ug' and update $\mathcal{V}_1 = \mathcal{V}_0 \cup \{\texttt{'ug'}\}$. The corpus contracts to:

```
h ug </w> :  10    p ug </w> :  5    p u n </w> :  12    h ug s </w> :  5
```

- **Iteration 2**: We recompute adjacent pair frequencies with the merged symbol 'ug':
  - ('h','ug') : 10 + 5 = 15.
  - ('ug','</w>') : 10 + 5 = 15.
  - ('p','u') : 12(pun), ('u','n') : 12(pun), ('n','</w>') : 12(pun).
  - ('p','ug') : 5(pug), ('ug','s') : 5(hugs), ('s','</w>') : 5(hugs).

  The highest frequency is 15. Selecting ('h','ug'), we log Merge Rule 2: ('h','ug') → 'hug' and update $\mathcal{V}_2 = \mathcal{V}_1 \cup \{\texttt{'hug'}\}$. The corpus becomes:

```
hug </w> :  10    p ug </w> :  5    p u n </w> :  12    hug s </w> :  5
```

Inference walkthrough on the unseen word "pugs". Now consider what happens when a model deployed in production encounters the word "pugs" (the plural of pug). Notice that "pugs" **never appeared in the training corpus as a complete word** (the training data only contained "pug" and "hugs").

1. Under a traditional word-level tokenizer, "pugs" is completely absent from vocabulary $\mathcal{V}$ and collapses into the information-destroying token <UNK>.
2. Under Byte Pair Encoding:
   - The word is initialized into base characters: ['p','u','g','s'].
   - The tokenizer queries the learned merge rules in priority order. Merge Rule 1 ('u','g') → 'ug' matches immediately! The sequence is rewritten as:

$$\texttt{['p','u','g','s']} \xrightarrow{\text{Merge 1}} \texttt{['p','ug','s']}.$$

- Merge Rule 2 (`'h'`, `'ug'`) → `'hug'` is queried next. It does not match because the preceding symbol is `'p'`, not `'h'`.
- No further merge rules match adjacent symbols.

3. The final emitted token sequence is:

$$\texttt{encode("pugs")} = [\texttt{'p'}, \texttt{'ug'}, \texttt{'s'}]. \tag{4.89}$$

The Out-of-Vocabulary barrier is overcome: the model represents the novel word via known morphological primitives—the initial consonant `'p'`, the shared stem `'ug'`, and the plural suffix `'s'`.

Figure 4.7 visualizes both the bottom-up hierarchical merge tree constructed during training and the top-down decomposition of unseen words at inference time.

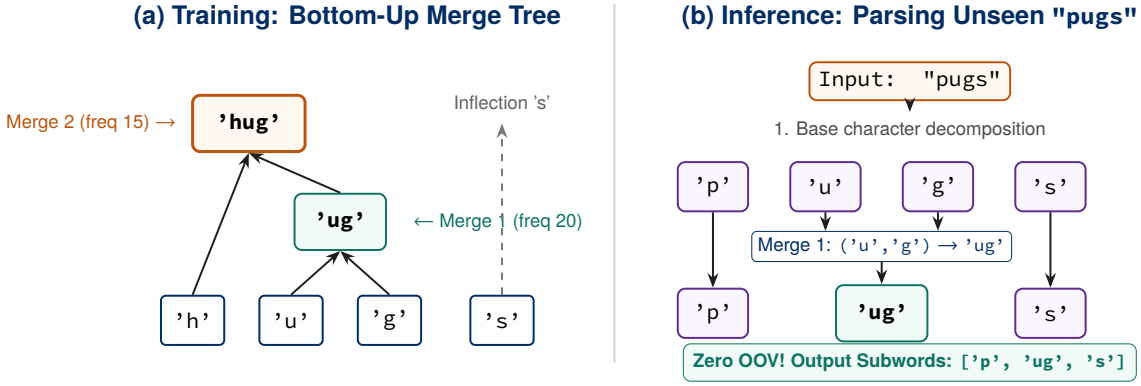


Figure 4.7: Byte Pair Encoding (BPE) mechanics. (a) During training, adjacent high-frequency symbol pairs are greedily merged into subwords. (b) At inference time, unseen words (e.g., `"pugs"`) are parsed by applying learned merge rules in priority order, resolving morphology without out-of-vocabulary collapse.

4.11.4 Byte-level BPE: the 256 UTF-8 byte foundation and elimination of unknown tokens

Despite the success of character-level BPE, applying it directly to Unicode strings faces a severe scaling barrier: the Unicode standard contains over 150,000 unique characters spanning hundreds of languages and symbol sets. Initializing $\mathcal{V}_0$ with every conceivable character creates an unmanageably large base alphabet, yet still fails when a user inputs rare historical glyphs or newly released emojis.

The 256 UTF-8 base byte breakthrough. In GPT-2, this dilemma was solved in [12] by introducing **Byte-Level Byte Pair Encoding (BBPE)**. Instead of characters, the base alphabet $\mathcal{V}_0$ is initialized strictly with the **256 distinct byte values** ($0x00$ through $0xFF$) of the UTF-8 encoding standard:

$$\mathcal{V}_0 = \{0x00, 0x01, \ldots, 0xFF\}, \quad |\mathcal{V}_0| = 256. \tag{4.90}$$

Theorem 4.4: *Total elimination of unknown tokens in byte-level BPE*

Under Byte-Level BPE with base alphabet $\mathcal{V}_0 = \{0x00, \ldots, 0xFF\}$, the probability of an Out-of-Vocabulary (OOV) token is strictly zero:

$$\mathbb{P}(w_{\text{oov}}) = 0 \quad \forall S \in \Sigma^*. \tag{4.91}$$

Proof. By definition of the Unicode transformation format (UTF-8), any arbitrary character or sequence of symbols in the Unicode code space ($U+0000$ to $U+10FFFF$) encodes deterministically into a sequence of between 1 and 4 raw bytes in $\{0x00, \ldots, 0xFF\}$. Because the base alphabet $\mathcal{V}_0$ contains all 256 possible 8-bit byte values, every byte sequence can be parsed. If an input string contains an unseen compound word, a programming syntax construct, a severe typo, or an unknown emoji, the tokenizer attempts to apply merged subword rules; if no higher-level merge rule applies, the tokenizer simply falls back to emitting individual byte tokens from $\mathcal{V}_0$. Because no input byte can ever fall outside $\{0x00, \ldots, 0xFF\}$, no token can ever be unmapped. The token `<|unk|>` is completely eliminated. ■

Lossless whitespace preservation. Standard natural language processing pipelines strip whitespace or split on spaces, making exact detokenization ambiguous. In Byte-Level BPE, whitespaces are treated as regular characters/bytes. In GPT-2 and Hugging Face tokenizers, a leading whitespace is mapped to a special character (e.g., `' extvisiblespace'` or byte $0x20$), binding the space directly to the succeeding word:

$$\texttt{"Hello world"} \overset{\text{BBPE}}{\Longrightarrow} \texttt{['Hello', ' extvisiblespaceworld']}$$

This guarantees that tokenization is **strictly lossless and fully invertible**:

$$\texttt{decode}(\texttt{encode}(S)) \equiv S \quad \forall S \in \Sigma^*. \tag{4.92}$$

Production implementations: OpenAI's `tiktoken`. In modern production LLM engineering, tokenization must process terabytes of training data without creating CPU bottlenecks. OpenAI's open-source library `tiktoken` implements byte-level BPE in pure Rust, executing tokenization orders of magnitude faster than Python regular expressions:

- **GPT-2 Tokenizer** (`gpt2`): Vocabulary size $V = 50{,}257$ (composed of 256 base byte tokens +50,000 learned BPE merges +1 special `<|endoftext|>` token).
- **GPT-4 Tokenizer** (`cl100k_base`): Vocabulary size $V = 100{,}277$. Expanding vocabulary from 50k to 100k reduces the total token count of natural language and code by $15 - 20\%$, directly reducing sequence length L, decreasing GPU memory footprint, and slashing API inference costs.

```python
import tiktoken

# Instantiate the GPT-2 tokenizer (vocab size 50,257)
tokenizer = tiktoken.get_encoding("gpt2")

text = "Hello, do you like tea? <|endoftext|> In the sunlit terraces of
    someunknownPlace."
# Disallow special tokens check or explicitly allow specific ones
integers = tokenizer.encode(text, allowed_special={"<|endoftext|>"})
print("Token IDs:", integers)

# Zero-shot handling of unknown words:
# 'someunknownPlace' is decomposed into subwords: ['some', 'unknown', 'Place']
for token_id in integers:
    print(f"{token_id:>5} -> {tokenizer.decode([token_id])!r}")

# Lossless roundtrip reconstruction
decoded_text = tokenizer.decode(integers)
assert decoded_text == text, "Decoded text must match original string identically!"
print("Lossless decoding confirmed!")
```

Listing 4.2: Byte-level BPE tokenization using OpenAI's `tiktoken` library [32].

Transition to continuous vector representations. Having established how Byte-Level BPE eliminates the out-of-vocabulary barrier and deterministically encodes raw text streams into discrete integer token IDs $\mathbf{T} = [t_0, t_1, \ldots, t_{L-1}] \in \mathbb{Z}^L$, we now confront the final representational bridge: how do discrete numerical indices enter the continuous, differentiable computational graph of modern neural networks? In the following subsections, we first establish the high-level architecture of the language model input pipeline, and then embark on an exhaustive investigation into positional representations.

4.11.5 From token IDs to composite inputs: high-level architecture of the foundation model input pipeline

Having transformed raw text into sequences of discrete token IDs $\mathbf{T} \in \mathbb{Z}^L$ (batched into mini-batch integer tensors $\mathbf{X} \in \mathbb{Z}^{B \times L}$), modern foundation models execute a unified input preparation pipeline that maps discrete token indices into continuous, differentiable vector representations equipped with sequential order prior to Transformer ingestion.

1. Delimiter and architectural control tokens. Before projecting indices into continuous embedding space, language models inject dedicated **special control tokens** into the sequence to regulate generation boundaries and downstream task execution:

- **Sequence demarcation tokens**:
  - ⟨BOS⟩ (*Beginning of Sequence*): Prepended to every sequence to prime autoregressive decoders, signaling the onset of generation.
  - ⟨EOS⟩ (*End of Sequence*): Appended to target text; during inference, emitting this token instructs the generation loop to terminate.
- **Architectural and task-specific delimiter tokens**:
  - [CLS] (*Classification token*): Introduced in bidirectional encoder architectures such as BERT. Prepend-ed at sequence index 0, its final contextual representation aggregates whole-sequence semantics for sentence classification or regression tasks.
  - [SEP] (*Separator token*): Delimits distinct segments within a single combined input sequence (for example, demarcating the boundary between a Question and an Answer, or between a code docstring and its method body), enabling the model to learn segment-specific interaction patterns.

2. The token embedding layer. In PyTorch, continuous token representations are instantiated via an embedding matrix:

```
token_embedding_layer = torch.nn.Embedding(num_embeddings=vocab_size, embedding_dim=d)
```

Passing batch tensor $\mathbf{X} \in \mathbb{Z}^{B \times L}$ retrieves the corresponding continuous embedding vectors:

$$\mathbf{E}_{\text{tok}} = \texttt{Embedding}(\mathbf{X}) \in \mathbb{R}^{B \times L \times d}, \tag{4.93}$$

where slice $(\mathbf{E}_{\text{tok}})_{b,t,:} = \mathbf{w}_{X_{b,t}} \in \mathbb{R}^d$ is the d-dimensional embedding vector for the token at position t in batch item b.

3. High-level additive synthesis of the composite Transformer input. To endow the model with sequential awareness, the input pipeline combines the lexical token representation $\mathbf{E}_{\text{tok}}$ and a dedicated positional tensor $\mathbf{E}_{\text{pos}} \in \mathbb{R}^{1\times L\times d}$ via **elementwise addition** rather than concatenation:

$$\mathbf{X}_{\text{in}} = \mathbf{E}_{\text{tok}} + \mathbf{E}_{\text{pos}} \in \mathbb{R}^{B\times L\times d}, \tag{4.94}$$

where for each batch sample $b \in \{0, \dots, B-1\}$ and temporal coordinate $pos \in \{0, \dots, L-1\}$:

$$(\mathbf{X}_{\text{in}})_{b,pos,:} = \mathbf{w}_{X_{b,pos}} + \mathbf{p}_{pos} \in \mathbb{R}^{d}. \tag{4.95}$$

Listing 4.3 demonstrates how this composite synthesis is executed concretely in PyTorch using tensor broadcasting.

```python
import torch

vocab_size = 50257     # GPT-2 BPE vocabulary size
output_dim = 256       # Latent embedding dimension d
context_length = 4     # Sequence length L

# 1. Allocate trainable token embedding layer
token_embedding_layer = torch.nn.Embedding(vocab_size, output_dim)

# 2. Allocate positional embedding layer (e.g., learned or precomputed sinusoidal)
pos_embedding_layer = torch.nn.Embedding(context_length, output_dim)

# Example batch of token IDs: [batch_size=2, context_length=4]
inputs = torch.tensor([[40, 367, 2885, 1464],
                       [1807, 3619, 402, 271]])

# Extract token embeddings: [2, 4, 256]
token_embeddings = token_embedding_layer(inputs)

# Generate position indices: tensor([0, 1, 2, 3]) -> [4, 256]
pos_indices = torch.arange(context_length)
pos_embeddings = pos_embedding_layer(pos_indices)

# 3. Additive composite synthesis with PyTorch tensor broadcasting
# token_embeddings: [2, 4, 256] + pos_embeddings: [4, 256] -> [2, 4, 256]
input_embeddings = token_embeddings + pos_embeddings

print("Token Embeddings Shape:", token_embeddings.shape)  # torch.Size([2, 4, 256])
print("Pos Embeddings Shape:  ", pos_embeddings.shape)    # torch.Size([4, 256])
print("Final Composite Shape: ", input_embeddings.shape)  # torch.Size([2, 4, 256])
```

Listing 4.3: PyTorch synthesis of token embeddings and positional embeddings [32].

Causal motivation for a deep dive into positional representations. While Equation (4.94) and Listing 4.3 summarize the high-level computational contract of the LM input pipeline, they raise foundational questions: *Why is the positional tensor* $\mathbf{E}_{pos}$ *strictly mandatory? Why do naive indexing schemes fail catastrophically? And what mathematical principles govern its construction?* We now dedicate the next subsection to an exhaustive investigation into positional representations.

4.11.6 Positional representations: intuition, mathematical formulation, and implementations

1. The permutation invariance dilemma: why positional encoding is mandatory. In classical Recurrent Neural Networks (RNNs), sequential order is implicitly baked into the architecture: the

hidden state transitions recurrently via $\mathbf{h}_t = f(\mathbf{h}_{t-1}, \mathbf{x}_t)$, guaranteeing that token $\mathbf{x}_t$ is processed strictly after token $\mathbf{x}_{t-1}$. In Convolutional Neural Networks (CNNs), sequential order is preserved through localized spatial receptive fields.

In contrast, the cornerstone mathematical operator of the Transformer—**Self-Attention**—is strictly **permutation-equivariant**:

Theorem 4.5: *Permutation equivariance of self-attention*

Let $\mathbf{X} \in \mathbb{R}^{L\times d}$ be an input sequence of L vectors, and let $\mathbf{P} \in \{0,1\}^{L\times L}$ be an arbitrary permutation matrix (a square matrix having exactly one entry 1 per row and column, satisfying $\mathbf{P}^\top = \mathbf{P}^{-1}$). Under standard dot-product self-attention:

$$\texttt{Attention}(\mathbf{X}) = \text{softmax}\left(\frac{\mathbf{X}\mathbf{W}_Q(\mathbf{X}\mathbf{W}_K)^\top}{\sqrt{d_k}}\right)\mathbf{X}\mathbf{W}_V, \tag{4.96}$$

the operator is equivariant to input permutation:

$$\texttt{Attention}(\mathbf{P}\mathbf{X}) = \mathbf{P}\,\texttt{Attention}(\mathbf{X}). \tag{4.97}$$

Proof. Let $\mathbf{X}' = \mathbf{P}\mathbf{X}$. The query and key projections evaluate to $\mathbf{Q}' = \mathbf{P}\mathbf{X}\mathbf{W}_Q$ and $\mathbf{K}' = \mathbf{P}\mathbf{X}\mathbf{W}_K$. The unscaled attention score matrix becomes:

$$\mathbf{Q}'(\mathbf{K}')^\top = (\mathbf{P}\mathbf{X}\mathbf{W}_Q)(\mathbf{P}\mathbf{X}\mathbf{W}_K)^\top = \mathbf{P}(\mathbf{X}\mathbf{W}_Q\mathbf{W}_K^\top\mathbf{X}^\top)\mathbf{P}^\top. \tag{4.98}$$

Because the Softmax function is evaluated independently along each row, applying a permutation matrix $\mathbf{P}$ on the left permutes the rows, while $\mathbf{P}^\top$ on the right permutes the columns:

$$\text{softmax}\left(\frac{\mathbf{P}(\mathbf{X}\mathbf{W}_Q\mathbf{W}_K^\top\mathbf{X}^\top)\mathbf{P}^\top}{\sqrt{d_k}}\right) = \mathbf{P}\,\text{softmax}\left(\frac{\mathbf{X}\mathbf{W}_Q\mathbf{W}_K^\top\mathbf{X}^\top}{\sqrt{d_k}}\right)\mathbf{P}^\top. \tag{4.99}$$

Finally, multiplying by the value projection $\mathbf{V}' = \mathbf{P}\mathbf{X}\mathbf{W}_V$:

$$\begin{aligned}\texttt{Attention}(\mathbf{P}\mathbf{X}) &= \left[\mathbf{P}\,\text{softmax}\left(\frac{\mathbf{X}\mathbf{W}_Q\mathbf{W}_K^\top\mathbf{X}^\top}{\sqrt{d_k}}\right)\mathbf{P}^\top\right](\mathbf{P}\mathbf{X}\mathbf{W}_V)\\ &= \mathbf{P}\,\text{softmax}\left(\frac{\mathbf{X}\mathbf{W}_Q\mathbf{W}_K^\top\mathbf{X}^\top}{\sqrt{d_k}}\right)(\mathbf{P}^\top\mathbf{P})\mathbf{X}\mathbf{W}_V\\ &= \mathbf{P}\,\texttt{Attention}(\mathbf{X}),\end{aligned} \tag{4.100}$$

where the identity $\mathbf{P}^\top\mathbf{P} = \mathbf{I}$ collapses the internal permutation. ■ □

The unordered multiset collapse. Theorem 4.5 establishes that if we permute the input tokens, the output vectors undergo the exact same permutation with *identical values*. The attention mechanism treats the input sequence as an **unordered multiset** (a "bag of vectors") with zero intrinsic knowledge of temporal order!

To understand the catastrophic consequence of this property, consider the two contrasting sentences analyzed in the course lecture:

$$\begin{aligned}&\textbf{Sentence 1}\text{: } \textit{"I ate an apple"} \quad \Longrightarrow \quad \{\mathbf{e}_{\text{I}}, \mathbf{e}_{\text{ate}}, \mathbf{e}_{\text{an}}, \mathbf{e}_{\text{apple}}\}\\ &\textbf{Sentence 2}\text{: } \textit{"An apple, I ate"} \quad \Longrightarrow \quad \{\mathbf{e}_{\text{an}}, \mathbf{e}_{\text{apple}}, \mathbf{e}_{\text{I}}, \mathbf{e}_{\text{ate}}\}\end{aligned}$$

Or similarly:

$$\begin{aligned} &\textbf{Sentence A}: \textit{“The dog bit the man”} \quad \Longrightarrow \quad \{\mathbf{e}_{\text{The}}, \mathbf{e}_{\text{dog}}, \mathbf{e}_{\text{bit}}, \mathbf{e}_{\text{the}}, \mathbf{e}_{\text{man}}\} \\ &\textbf{Sentence B}: \textit{“The man bit the dog”} \quad \Longrightarrow \quad \{\mathbf{e}_{\text{The}}, \mathbf{e}_{\text{man}}, \mathbf{e}_{\text{bit}}, \mathbf{e}_{\text{the}}, \mathbf{e}_{\text{dog}}\} \end{aligned}$$

In both pairs, the sentences express fundamentally different or opposite semantic realities. Yet, because each pair comprises the exact same multiset of token embeddings, an unaugmented Transformer evaluates identical attention weights. To the model, subject, verb, and object become entirely conflated. Sequential order must therefore be injected into the representations explicitly.

Why naive positional representations fail. Before arriving at the modern formulation, it is vital to understand why naive indexing schemes break down:

1. **Naive Integer Addition ($pos \in \{0, 1, 2, \ldots, L-1\}$)**: If we add scalar indices pos to token embeddings, early tokens receive tiny perturbations (+0, +1), while token 2048 receives +2048. Because embedding vectors are initialized with mean 0 and unit variance, large coordinates overwhelm the semantic representations completely, and models fail to generalize to sequence lengths beyond training.
2. **Normalized Coordinates ($pos/L \in [0, 1]$)**: Normalizing positions by sequence length bounds values within $[0, 1]$. However, the physical meaning of a step size becomes variable: in a 10-token sentence, an offset of $\Delta = 0.1$ corresponds to an adjacent word, whereas in a 1000-token document, $\Delta = 0.1$ spans 100 words. The model cannot learn consistent relative distances.
3. **Coordinate Concatenation ($[\mathbf{w}_t; \mathbf{p}_t]$)**: Concatenating positional coordinates expands embedding dimensionality to $d + d_{\text{pos}}$, increasing parameter counts across all subsequent projection matrices ($\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V, \mathbf{W}_O$) and preventing seamless parameter sharing.

2. The geometric clock intuition (the “Clock Way of Thought”). How can we construct an encoding that remains strictly bounded within $[-1, 1]$, assigns a unique signature to every position, and preserves consistent geometric notions of distance?

Intuition & analogy: From binary counting to continuous multi-scale clocks

Consider how a digital computer represents numbers using bounded bits $\{0, 1\}$. In a 3-bit binary counter:

- The lowest bit (Bit 0) alternates every single step: $0, 1, 0, 1, 0, 1 \ldots$ (high-frequency wave).
- The middle bit (Bit 1) alternates every two steps: $0, 0, 1, 1, 0, 0 \ldots$ (medium-frequency wave).
- The highest bit (Bit 2) alternates every four steps: $0, 0, 0, 0, 1, 1 \ldots$ (low-frequency wave).

Every integer receives an unambiguous signature, even though individual bits are bounded in $\{0, 1\}$. However, discrete step functions are non-differentiable. **Sinusoidal positional encoding is the smooth, continuous, differentiable generalization of binary counting!**
Rather than binary square waves, we assign continuous trigonometric waves oscillating at geometrically decreasing frequencies across embedding channels:

- **Fast Clocks (Seconds Hand)**: Oscillate rapidly across low dimension indices, capturing fine-grained local syntax and adjacent word order.

- **Medium Clocks (Minutes Hand)**: Oscillate at intermediate frequencies, capturing phrase and clause boundaries.
- **Slow Clocks (Hours Hand)**: Oscillate extremely slowly across high dimension indices, capturing document-level flow and paragraphs.

The clock hand problem: why sine alone fails. Why can we not simply use sine waves with varying frequencies? Consider an analog clock where you can only observe the vertical height of the hand ($Y = \sin\theta$). Because $\sin(\theta) = \sin(\pi - \theta)$, the hand is at the exact same vertical height at 2:00 and 10:00! Furthermore, $\sin(0) = \sin(\pi) = 0$. A single 1D wave suffers from severe directional ambiguity (aliasing).

To establish an unambiguous position, we need both coordinates of the clock face:

$$(X, Y) = (\cos\theta, \sin\theta). \tag{4.101}$$

Together, $(\sin\theta, \cos\theta)$ traces a point along the unit circle S^1. On a 2D circle, every angular orientation $\theta \in [0, 2\pi)$ has an unambiguous coordinate! This is why Transformer positional encodings alternate sin and cos across dimension pairs.

Figure 4.8 illustrates this multi-frequency clock mechanism and demonstrates how pairing sine and cosine eliminates 1D wave aliasing.

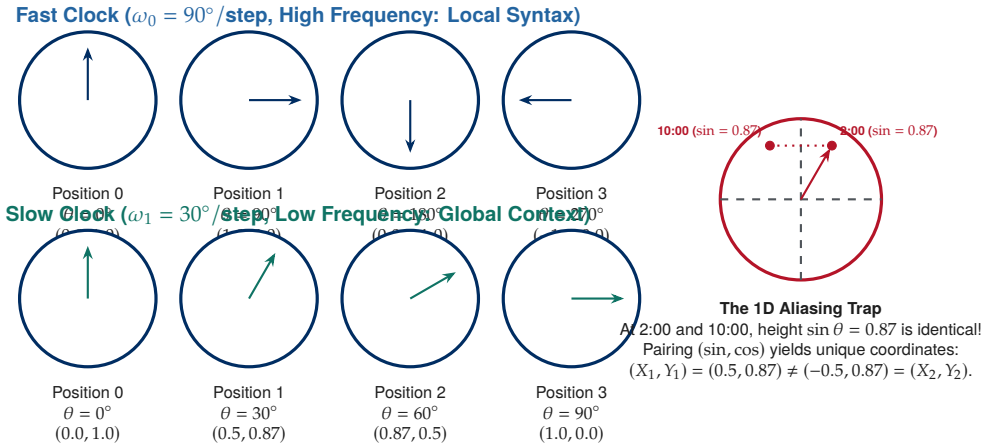


Figure 4.8: The geometric clock intuition behind positional encodings. Each dimension pair acts as an analog clock hand rotating at an angular frequency ω_i. Fast clocks disambiguate immediate neighbors (local syntax), while slow clocks preserve long-range sequence context. Pairing sin and cos prevents 1D wave aliasing by establishing a unique 2D coordinate on the unit circle S^1.

3. Taxonomy of positional representations in language models. Modern architectures implement positional awareness through three primary paradigms:

1. **Fixed Sinusoidal Positional Encodings [1]**: Fully deterministic, parameter-free functions computed from geometric progressions of trigonometric frequencies.
2. **Learned Absolute Positional Embeddings [12, 47, 48]**: Trainable parameter weights stored in an embedding matrix $\mathbf{W}_{\text{pos}} \in \mathbb{R}^{L_{\max} \times d}$ optimized via backpropagation (used in GPT-2, GPT-3, and BERT).

3. **Relative and Rotary Positional Embeddings (RoPE / ALiBi) [49–51]**: Encodings that inject positional information directly into attention scores as a function of relative distance $|i - j|$. Notably, Rotary Position Embedding (RoPE) [50] directly generalizes the 2D rotation matrix property of sinusoidal encodings into query and key representations.

4. Sinusoidal positional encodings: step-by-step mathematical derivation. In the seminal architecture introduced in [1], positional coordinates are generated deterministically:

Definition 4.6: *Sinusoidal positional encoding*

For a token at temporal sequence coordinate $pos \in \{0, 1, \ldots, L-1\}$ and channel index $i \in \{0, 1, \ldots, \frac{d}{2} - 1\}$, the components of the positional vector $\mathbf{p}_{pos} = \mathsf{PE}_{(pos,:)} \in \mathbb{R}^d$ are defined by:

$$\mathsf{PE}_{(pos,2i)} = \sin\left(\frac{pos}{10000^{2i/d}}\right), \tag{4.102}$$

$$\mathsf{PE}_{(pos,2i+1)} = \cos\left(\frac{pos}{10000^{2i/d}}\right). \tag{4.103}$$

Let us examine every term in this formulation:

- pos: The integer position of the token within the sequence $(0, 1, 2, \ldots, L-1)$.
- i: The frequency channel index $(0 \leq i < d/2)$.
- $2i$ and $2i + 1$: Even indices receive sine values, odd indices receive cosine values.
- Angular frequency ω_i: Defined as $\omega_i = \frac{1}{10000^{2i/d}} = 10000^{-2i/d}$.
- Wavelength λ_i: The wavelength $\lambda_i = \frac{2\pi}{\omega_i} = 2\pi \cdot 10000^{2i/d}$ forms a geometric progression:

$$\text{At } i = 0: \quad \lambda_0 = 2\pi \cdot 10000^0 = 2\pi \approx 6.28 \text{ tokens}, \tag{4.104}$$

$$\text{At } i = \frac{d}{2} - 1: \quad \lambda_{\max} \approx 2\pi \cdot 10000^1 \approx 62{,}832 \text{ tokens}. \tag{4.105}$$

- **Why the base 10000?** As highlighted by the professor, 10,000 guarantees that within typical context lengths (up to $\approx$ 60,000 tokens), the slowest frequency does not complete a single cycle. This guarantees that every position vector in the operational horizon is globally unique.

Mathematical properties and proofs. Sinusoidal positional encodings satisfy three pivotal theoretical properties:

1. **Positional uniqueness**: Because each channel i oscillates at a distinct frequency ω_i, no two positional vectors are identical. The vectors remain distinct for at least $2\pi \cdot 10000 \approx 62{,}832$ positions before periodic recurrence can occur.
2. **The linear relative shift property**: For any fixed spatial offset $k \in \mathbb{Z}$, the positional vector at position $pos + k$ can be computed as a direct linear transformation of the vector at position pos:

Theorem 4.6: *Linear relative transformation property*

For any frequency $\omega_i = 10000^{-2i/d}$ and spatial offset $k \in \mathbb{Z}$, there exists an orthogonal rotation matrix $\mathbf{M}_{i,k} \in \mathbb{R}^{2\times 2}$ independent of pos such that:

$$\begin{bmatrix} \mathsf{PE}_{(pos+k,2i)} \\ \mathsf{PE}_{(pos+k,2i+1)} \end{bmatrix} = \begin{bmatrix} \cos(\omega_i k) & \sin(\omega_i k) \\ -\sin(\omega_i k) & \cos(\omega_i k) \end{bmatrix} \begin{bmatrix} \mathsf{PE}_{(pos,2i)} \\ \mathsf{PE}_{(pos,2i+1)} \end{bmatrix}. \tag{4.106}$$

Proof. Applying the standard trigonometric angle addition identities:

$$\sin(\omega_i(pos+k)) = \sin(\omega_i pos)\cos(\omega_i k) + \cos(\omega_i pos)\sin(\omega_i k), \tag{4.107}$$

$$\cos(\omega_i(pos+k)) = \cos(\omega_i pos)\cos(\omega_i k) - \sin(\omega_i pos)\sin(\omega_i k). \tag{4.108}$$

Expressing this system in matrix notation yields Equation (4.106). Because the rotation matrix $\mathbf{M}_{i,k}$ depends strictly on offset k and is independent of absolute position pos, linear attention projection layers ($\mathbf{W}_Q, \mathbf{W}_K$) can easily learn to attend to relative positions. ■ □

3. **Metric similarity and distance decay**: In self-attention, query and key vectors interact via inner products. The dot product between the 2D positional components of two tokens at positions p_1 and p_2 evaluates to:

$$\mathbf{p}_{p_1}^{(i)} \cdot \mathbf{p}_{p_2}^{(i)} = \sin(\omega_i p_1)\sin(\omega_i p_2) + \cos(\omega_i p_1)\cos(\omega_i p_2) = \cos(\omega_i(p_1 - p_2)). \tag{4.109}$$

The dot product depends **strictly on the relative distance** $\Delta p = p_1 - p_2$:

- When $p_1 = p_2$, $\cos(0) = 1$ (maximum correlation along the diagonal).
- As $|p_1 - p_2|$ increases, the similarity smoothly decays.

In the course lecture, the professor demonstrated this property empirically by computing the complete pairwise cosine similarity matrix across 500 positions:

- The 2D Gram matrix exhibits a bright, luminous main diagonal (cos = 1).
- A horizontal slice along row $pos = 100$ reveals a sharp symmetric peak at index 100, followed by a smooth, monotonic decay as distance $|pos - 100|$ expands.

Concrete numerical calculation of sinusoidal coordinates. To illustrate this formulation concretely, consider the numerical configuration presented in the course lecture: sequence position $pos = 1$ (the second token, under 0-indexed coordinates) within a standard latent embedding space of dimension $d = 768$:

1. **Lowest channel pair ($i = 0$, high-frequency regime)**:

$$2i = 0 \implies \mathsf{PE}_{(1,0)} = \sin\left(\frac{1}{10000^{0/768}}\right) = \sin(1) \approx 0.8415, \tag{4.110}$$

$$2i + 1 = 1 \implies \mathsf{PE}_{(1,1)} = \cos\left(\frac{1}{10000^{0/768}}\right) = \cos(1) \approx 0.5403. \tag{4.111}$$

2. **Subsequent even channel ($i = 2$, channel index $2i = 4$)**:

$$\begin{aligned} 2i = 4 \implies \mathsf{PE}_{(1,4)} &= \sin\left(\frac{1}{10000^{4/768}}\right) = \sin\left(\frac{1}{10000^{0.005208}}\right) \\ &\approx \sin\left(\frac{1}{1.0490}\right) = \sin(0.9533) \approx 0.8153. \end{aligned} \tag{4.112}$$

These exact values (0.8415, 0.5403, and 0.8153) illustrate how trigonometric arguments shift smoothly across embedding indices, providing each positional coordinate with an unambiguous, deterministic signature.

5. Learned positional embeddings (GPT convention). An alternative approach, adopted by architectures such as GPT-2 and GPT-3 [2, 12], treats positional encodings as trainable parameter weights rather than fixed sinusoidal functions:

$$\mathbf{W}_{\text{pos}} \in \mathbb{R}^{L_{\max} \times d}. \tag{4.113}$$

Each integer coordinate $p \in \{0, 1, \ldots, L_{\max} - 1\}$ indexes a row vector $\mathbf{p}_p = (\mathbf{W}_{\text{pos}})_{p,:} \in \mathbb{R}^d$ optimized via gradient descent alongside token embeddings.

Table 4.3: Architectural comparison: sinusoidal positional encodings vs. learned positional embeddings.

Property	Sinusoidal encodings [1]	Learned embeddings [12]
Parameters	0 trainable weights (fully deterministic)	$L_{\max} \times d$ trainable parameters
Extrapolation	Can theoretically extrapolate to sequence lengths $> L_{\max}$	Strictly confined to context length $L \leq L_{\max}$
Relative shift	Exact orthogonal rotation $\mathbf{M}_{i,k}$ provable analytically	Must be approximated empirically from data
Empirical similarity	Pure smooth monotonic distance decay around diagonal	Smooth decay near diagonal with data-driven periodic peaks
Empirical performance	Comparable translation BLEU in [1]	Standard in modern autoregressive decoders

As highlighted in the lecture, computing the cosine similarity matrix of learned GPT-2 embeddings reveals high values along the main diagonal, accompanied by periodic secondary peaks at larger distances. These secondary peaks reflect statistical regularities in the training corpus (such as paragraph transitions, section breaks, or dialogue conventions).

6. Final clarification walkthrough: disambiguating opposite sentences. To crystallize all of these principles into an indelible mental picture, let us trace how a mini 2-clock system disambiguates our contrasting sentences:

Sentence 1: *"The dog bit the man."* vs. **Sentence 2**: *"The man bit the dog."*

Let $d = 4$ (consisting of 2 clocks):

- **Clock 1 (Fast Clock, $\omega_0 = 90°$/step)**: Rotates 90° clockwise per step.
- **Clock 2 (Slow Clock, $\omega_1 = 30°$/step)**: Rotates 30° clockwise per step.

Evaluating the positional vectors $\mathbf{PE}(pos) = [\sin(\omega_0 pos), \cos(\omega_0 pos), \sin(\omega_1 pos), \cos(\omega_1 pos)]$ yields:

Now consider the additive inputs:

- In **Sentence 1**: $\mathbf{x}_{\text{dog}} = \mathbf{E}_{\text{dog}} + \mathbf{p}_1$, $\mathbf{x}_{\text{bit}} = \mathbf{E}_{\text{bit}} + \mathbf{p}_2$, $\mathbf{x}_{\text{man}} = \mathbf{E}_{\text{man}} + \mathbf{p}_4$.
- In **Sentence 2**: $\mathbf{x}_{\text{man}} = \mathbf{E}_{\text{man}} + \mathbf{p}_1$, $\mathbf{x}_{\text{bit}} = \mathbf{E}_{\text{bit}} + \mathbf{p}_2$, $\mathbf{x}_{\text{dog}} = \mathbf{E}_{\text{dog}} + \mathbf{p}_4$.

Table 4.4: Exact numerical positional vectors for positions 0 through 4 under the 2-clock system.

Position (pos)	**Clock 1 ($\omega_0 = 90° \cdot pos$)**	**Clock 2 ($\omega_1 = 30° \cdot pos$)**	**Positional Vector $\mathbf{p}_{pos}$**
0	$\theta = 0° \implies [0.0, 1.0]$	$\theta = 0° \implies [0.0, 1.0]$	$[0.0, 1.0, 0.0, 1.0]$
1	$\theta = 90° \implies [1.0, 0.0]$	$\theta = 30° \implies [0.5, 0.87]$	$[1.0, 0.0, 0.5, 0.87]$
2	$\theta = 180° \implies [0.0, -1.0]$	$\theta = 60° \implies [0.87, 0.5]$	$[0.0, -1.0, 0.87, 0.5]$
3	$\theta = 270° \implies [-1.0, 0.0]$	$\theta = 90° \implies [1.0, 0.0]$	$[-1.0, 0.0, 1.0, 0.0]$
4	$\theta = 360° \implies [0.0, 1.0]$	$\theta = 120° \implies [0.87, -0.5]$	$[0.0, 1.0, 0.87, -0.5]$

How attention identifies the subject. In English syntax, the grammatical subject of the active verb "bit" resides 1 step prior (relative offset $k = -1$). At position $pos = 2$, Clock 1 points South (180°, vector $[0.0, -1.0]$). To identify the doer of the action, an attention head querying offset $k = -1$ rotates 90° counterclockwise:

$$180° - 90° = 90° \implies [1.0, 0.0]. \tag{4.114}$$

The attention query searches for the token whose Clock 1 points East ($[1.0, 0.0]$):

- In **Sentence 1**, the token at position 1 with East-pointing clock is **"dog"**. The attention mechanism routes information from "dog" to "bit", resolving the dog as the subject.
- In **Sentence 2**, the token at position 1 with East-pointing clock is **"man"**. The attention mechanism routes information from "man" to "bit", resolving the man as the subject.

The ambiguity is completely resolved: through sinusoidal clock rotations, the Transformer effortlessly deciphers who bit whom!

CHAPTER 5

The Transformer architecture and attention mechanisms

5.1 The architectural impasse of recurrent sequence models

For nearly a decade, sequence modeling was dominated by Recurrent Neural Networks (RNNs) and their gated variants, Long Short-Term Memory (LSTM) networks and Gated Recurrent Units (GRUs). As established in Chapter 3, recurrent models represent a sequence of tokens by recursively updating an internal hidden state vector:

$$\mathbf{h}_t = f(\mathbf{h}_{t-1}, \mathbf{x}_t), \quad t \in \{1, 2, \dots, L\}, \tag{5.1}$$

where $\mathbf{h}_t \in \mathbb{R}^{d_h}$ encapsulates the historical context observed up to time step t, and $\mathbf{x}_t \in \mathbb{R}^d$ is the input vector representation at time t.

While gating mechanisms and Constant Error Carousels mitigated the vanishing gradient problem over moderate temporal spans, recurrence inherently imposes two severe, insurmountable bottlenecks that halted further scaling in language modeling:

1. The sequential execution wall. The transition equation $\mathbf{h}_t = f(\mathbf{h}_{t-1}, \mathbf{x}_t)$ enforces an unbroken, strictly sequential causal dependency chain: computation of $\mathbf{h}_t$ cannot commence until the physical memory write of $\mathbf{h}_{t-1}$ is completed. Consequently, an input sequence of L tokens mandates $O(L)$ sequential computational operations. Modern deep learning accelerators (GPUs and TPUs) derive their performance from massive data parallelism, executing thousands of arithmetic matrix operations concurrently across high-bandwidth tensor cores. Under recurrent processing, hardware parallelism across the temporal dimension is impossible: the accelerator is forced to idle between consecutive time steps, creating a computational wall that restricts training across massive datasets.

2. The information compression bottleneck of Seq2Seq encoders. In classical sequence-to-sequence architectures, an encoder compresses an input sequence of arbitrary length L into a single, fixed-size vector $\mathbf{h}_L \in \mathbb{R}^{d_h}$ (or a tuple $(\mathbf{h}_L, \mathbf{c}_L)$ in LSTMs). This single vector is subsequently handed over to the decoder to initiate target sequence generation. As the input length L expands—such as in multi-turn dialogues, technical documentation, or software engineering source code files spanning thousands of tokens—compressing all semantic, syntactic, and structural relationships into a finite-dimensional vector becomes physically impossible. Information from early time steps undergoes continuous dissipation, inducing severe contextual amnesia.

The architectural objective. These limitations raised a fundamental question: *Can we design a neural architecture that completely eliminates recurrence, operates across all tokens in parallel with $O(1)$ sequential operations, and allows every token to directly communicate with every other token in the*

sequence without a fixed-size information bottleneck? In 2017, this question was answered by the introduction of the **Transformer** [1], an architecture built entirely upon multi-head attention mechanisms.

5.2 The complete Transformer architecture: encoder-decoder breakdown

The foundational Transformer proposed in [1] adopts an **Encoder-Decoder** paradigm, structured to eliminate the recurrent compression bottleneck while enabling end-to-end parallel training. Figure 5.1 reproduces the canonical architectural diagram from the landmark paper [1].

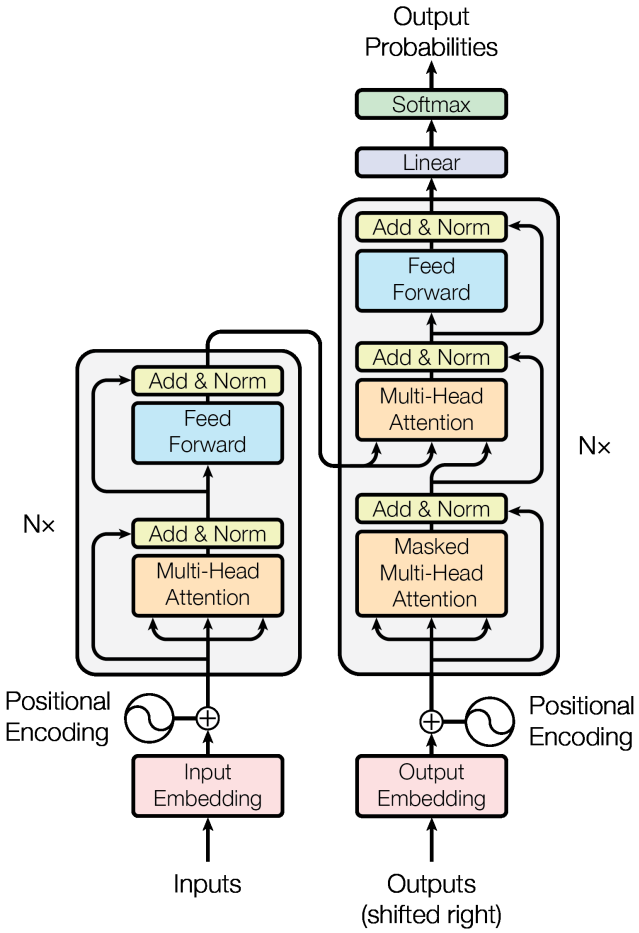


Figure 5.1: The original Transformer model architecture as published in [1]. The left stack corresponds to the Encoder ($N = 6$ identical layers), and the right stack corresponds to the Decoder ($N = 6$ identical layers), equipped with causal masked self-attention and cross-attention over the encoder representations.

1. The Encoder stack: uncompressed contextual representations. The Encoder consists of a stack of N identical layers (typically $N = 6$ in the base model). Crucially, the Encoder does *not* compress the input sequence into a single summary vector. Instead, it takes the input sequence of position-augmented embeddings $\mathbf{X} \in \mathbb{R}^{L_{\text{enc}} \times d_{\text{model}}}$ and transforms it into an equally sized sequence of contextual representations:

$$\mathbf{H}_{\text{enc}} = [\mathbf{h}_1, \mathbf{h}_2, \ldots, \mathbf{h}_{L_{\text{enc}}}] \in \mathbb{R}^{L_{\text{enc}} \times d_{\text{model}}}. \tag{5.2}$$

Every output vector $\mathbf{h}_i$ possesses a global receptive field that incorporates information from every other token in the sequence through bidirectional self-attention, preserving fine-grained token-level semantics without information loss.

2. The Decoder stack: autoregressive generation and cross-attention. The Decoder also consists of a stack of N identical layers, operating autoregressively at inference time:

1. **Autoregressive sequence initialization**: Generation is primed by feeding the special beginning-of-sequence delimiter $\langle \text{BOS} \rangle$ as the initial token.
2. **Causal step-by-step prediction**: The Decoder processes the tokens generated so far, attends to the encoder representations $\mathbf{H}_{\text{enc}}$, and emits logits over vocabulary $\mathcal{V}$. The highest-scoring token (or sampled token) y_1 is generated.
3. **Prefix feedback loop**: The newly emitted token is appended to the decoder input, forming the context $[\langle \text{BOS} \rangle, y_1]$ to predict y_2. This loop repeats until the model outputs the end-of-sequence delimiter $\langle \text{EOS} \rangle$, terminating generation.

3. The training phase: Teacher Forcing and sequence parallelization. At inference, the decoder must generate tokens sequentially. However, during training, the entire target sequence $[y_0, y_1, \ldots, y_{L_{\text{dec}}-1}]$ is already known. By employing **Teacher Forcing** combined with causal attention masking (which prevents tokens from attending to subsequent target tokens), all L_{dec} generation steps are computed simultaneously in a single forward pass. The decoder does not unroll an iterative loop during training: the loss across all output positions is evaluated concurrently on the GPU.

5.3 Attention from first principles: differentiable soft dictionary retrieval

5.3.1 Why we need the attention mechanism: from RNN bottlenecks to dynamic interaction

To understand the conceptual necessity of the attention mechanism, we must examine the architectural bottleneck that afflicted sequential models prior to the Transformer, as formalized by Sebastian Raschka [32].

When translating natural language text (e.g., from German or Italian to English) or performing code generation from formal specifications, literal word-by-word substitution is fundamentally impossible. Grammatical structures, morphological agreements, and clause ordering diverge substantially between linguistic systems. Consequently, sequence transduction requires a deep model consisting of two interacting submodules: an **Encoder** that ingests and understands the source sequence, and a **Decoder** that synthesizes the target sequence.

Prior to Transformers, the dominant paradigm for sequence-to-sequence modeling was the recurrent neural network (RNN, Chapter 3). In an encoder-decoder RNN:

1. The encoder ingests source tokens $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_L$ sequentially, updating its recurrent hidden state at each time step via $\mathbf{h}_t = f(\mathbf{h}_{t-1}, \mathbf{x}_t)$.
2. At the conclusion of the source sequence, the encoder attempts to compress the entire semantic and syntactic content of the input into its final hidden state vector $\mathbf{h}_L \in \mathbb{R}^{d_h}$.
3. This single vector $\mathbf{h}_L$ is handed over to the decoder as its initial context vector $\mathbf{c} = \mathbf{h}_L$, from which the decoder must generate the translated output token by token.

This architecture suffers from a catastrophic mathematical limitation: the **information compression bottleneck** [32]. Forcing an arbitrary-length sentence—spanning tens, hundreds, or thousands of tokens—into a single fixed-dimensional embedding vector creates an information-theoretic bottleneck. As sequence length expands:

- The decoder possesses *zero direct access* to intermediate representations $\mathbf{h}_1, \mathbf{h}_2, \ldots, \mathbf{h}_{L-1}$ from earlier encoder steps.
- Gradients backpropagating through time undergo exponential decay or explosion across recurrent chain products $\prod_{k=t+1}^{L} \frac{\partial \mathbf{h}_k}{\partial \mathbf{h}_{k-1}}$.
- Information from early tokens undergoes continuous dissipative dilution, causing severe context amnesia on long sequences.

To solve this dilemma, Bahdanau et al. [31] introduced the attention mechanism for RNNs, providing the decoder with dynamic, direct connections to *all* intermediate encoder states $\mathbf{h}_1, \ldots, \mathbf{h}_L$ at every generation step. Three years later, Vaswani et al. [1] realized that recurrence itself was entirely superfluous. By introducing **Self-Attention**, the model allows every position within a sequence to directly attend to and dynamically interact with every other position in the sequence across an $O(1)$ sequential path length, computing rich contextual embeddings where each word's representation is dynamically updated based on the relevant context of the entire sentence [32].

5.3.2 The associative memory analogy: queries, keys, and values

To understand how attention operates mathematically from first principles, it is instructive to draw an analogy to associative retrieval structures in computer science [5, 52].

1. Classical dictionary lookup (discrete hash table). In computer systems, a dictionary or associative array stores information as a collection of m discrete key-value pairs:

$$\mathcal{D} = \{(\kappa_1, v_1), (\kappa_2, v_2), \ldots, (\kappa_m, v_m)\}. \tag{5.3}$$

When a discrete search query q is issued, the dictionary performs an exact binary match against the keys:

$$\text{Lookup}(q, \mathcal{D}) = \begin{cases} v_k & \text{if } \exists\, k \text{ such that } q = \kappa_k, \\ \text{None} & \text{otherwise.} \end{cases} \tag{5.4}$$

In this classical paradigm, retrieval is binary (0 or 1) and non-differentiable: if query q is slightly perturbed, the lookup either matches completely or fails catastrophically.

2. Soft, differentiable key-value retrieval. Attention generalizes dictionary lookup to continuous metric vector spaces $\mathbb{R}^d$:

- A **Query** vector $\mathbf{q} \in \mathbb{R}^{d_k}$ represents the information need: *"What linguistic, syntactic, or semantic attributes am I searching for?"*
- A set of m **Key** vectors $\mathbf{k}_j \in \mathbb{R}^{d_k}$ represent content identifiers or addresses: *"What attributes do I offer?"*
- A set of m **Value** vectors $\mathbf{v}_j \in \mathbb{R}^{d_v}$ represent the informative content or payload: *"What information do I communicate if selected?"*

Instead of checking exact equality $q = \kappa_k$, we evaluate a continuous compatibility score $s(\mathbf{q}, \mathbf{k}_j) \in \mathbb{R}$ using vector inner products:

$$s_j = \mathbf{q}^\top \mathbf{k}_j. \tag{5.5}$$

The raw alignment scores $[s_1, s_2, \ldots, s_m]$ are mapped onto the probability simplex via the softmax operator, yielding non-negative attention weights that sum to unity:

$$\alpha_j = \frac{\exp(s_j)}{\sum_{l=1}^{m} \exp(s_l)}, \quad \sum_{j=1}^{m} \alpha_j = 1, \quad \alpha_j \geq 0. \tag{5.6}$$

Finally, the retrieved representation is computed as the **convex combination** (weighted sum) of all value vectors:

$$\mathbf{o} = \sum_{j=1}^{m} \alpha_j \mathbf{v}_j \in \mathbb{R}^{d_v}. \tag{5.7}$$

Because inner products, exponentials, and summations are smooth operations, the entire lookup pipeline is fully differentiable with well-defined gradients $\frac{\partial \mathbf{o}}{\partial \mathbf{q}}$, $\frac{\partial \mathbf{o}}{\partial \mathbf{k}_j}$, and $\frac{\partial \mathbf{o}}{\partial \mathbf{v}_j}$.

Illustrative example: contextual displacement of a fluffy blue creature. To understand how soft key-value retrieval transforms static representations into rich contextual embeddings in high-dimensional vector spaces, consider the visual intuition formalized by Grant Sanderson [52] and illustrated in Figure 5.2:

1. **The static embedding**: Initially, each word in the phrase *"A fluffy blue creature roamed the verdant forest"* begins as an isolated unigram embedding vector $\mathbf{e} \in \mathbb{R}^{d_{\text{model}}}$. The static vector for *"creature"* merely occupies a generic semantic coordinate indicating an animal or organism, completely devoid of context regarding its visual appearance.

2. **Queries as informational requests**: When passed into an attention block, linear projection $\mathbf{W}_Q$ maps the vector for *"creature"* into a Query vector $\mathbf{q}_{\text{creature}}$. Geometrically, this query acts as a directional probe asking: *"What descriptive adjectives precede me?"*

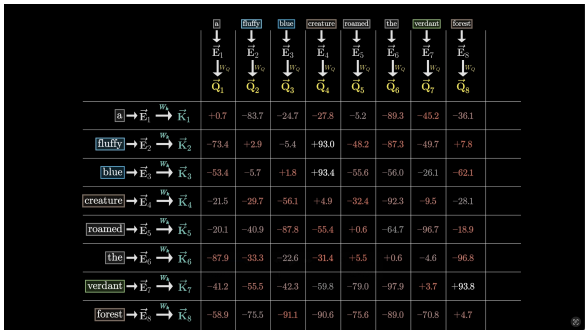
3. **Keys as feature advertisements**: Concurrently, projection matrix $\mathbf{W}_K$ maps all tokens into Key vectors. The preceding adjectives *"fluffy"* and *"blue"* produce keys $\mathbf{k}_{\text{fluffy}}$ and $\mathbf{k}_{\text{blue}}$ advertising: *"I am a tactile descriptor"* and *"I am a visual color descriptor"*.

4. **Inner-product compatibility**: Computing inner products $\mathbf{q}_{\text{creature}}^\top \mathbf{k}_j$ evaluates geometric alignment (Figure 5.2a). Because the query explicitly seeks adjective descriptors, it achieves high positive inner products with $\mathbf{k}_{\text{fluffy}}$ (+93.0) and $\mathbf{k}_{\text{blue}}$ (+93.4), but low or negative alignment with irrelevant tokens like *"roamed"* or *"forest"*.

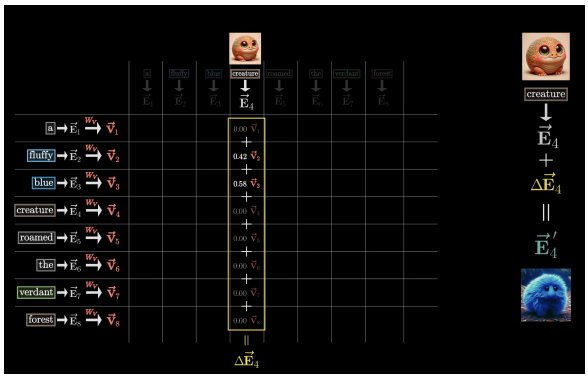
5. **Value routing and displacement**: Linear projection $\mathbf{W}_V$ extracts the informative payload: Value vectors $\mathbf{v}_{\text{fluffy}}$ and $\mathbf{v}_{\text{blue}}$ encode specific sensory features. Taking their weighted sum produces a contextual displacement vector (Figure 5.2b):

$$\Delta \mathbf{e}_{\text{creature}} = \alpha_{\text{fluffy}} \mathbf{v}_{\text{fluffy}} + \alpha_{\text{blue}} \mathbf{v}_{\text{blue}}. \tag{5.8}$$

   When added to the initial representation via residual connection, $\mathbf{e}_{\text{creature}} + \Delta \mathbf{e}_{\text{creature}}$ physically moves the point in embedding space into the specialized manifold region corresponding to a *fluffy, blue creature*!



(a) Query-Key compatibility matrix ($\mathbf{QK}^\top$)



(b) Value matrix weighting and vector update ($\Delta\mathbf{e}$)

Figure 5.2: Geometric attention dynamics and matrix transformations in 3Blue1Brown's pedagogical breakdown [52]. (a) Pairwise Query-Key dot products evaluate compatibility across all token pairs in *"A fluffy blue creature roamed the verdant forest"*; the Query for *"creature"* strongly matches the Keys for descriptive modifiers *"fluffy"* (+93.0) and *"blue"* (+93.4). (b) The attention table routes the corresponding Value vectors: after softmax normalization (0.42 and 0.58), the weighted sum produces a displacement vector $\Delta\mathbf{E}_4 = 0.42\mathbf{V}_2 + 0.58\mathbf{V}_3$. Adding $\Delta\mathbf{E}_4$ to the generic creature vector $\mathbf{E}_4$ displaces the representation into the specialized manifold coordinates of a fluffy blue creature.

5.4 Scaled dot-product attention and variance stabilization

5.4.1 Definition of the scaled attention mechanism

When computing alignment scores between queries and keys, standard dot product $s = \mathbf{q}^\top\mathbf{k}$ suffers from a critical numerical failure mode as vector dimensionality increases. To resolve this, the landmark formulation in [1] introduced **Scaled Dot-Product Attention**.

Definition 5.1: *Scaled dot-product attention*

Let $\mathbf{Q} \in \mathbb{R}^{L_Q \times d_k}$, $\mathbf{K} \in \mathbb{R}^{L_K \times d_k}$, and $\mathbf{V} \in \mathbb{R}^{L_K \times d_v}$ denote Query, Key, and Value matrices. The Scaled Dot-Product Attention operator evaluates:

$$\texttt{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{QK}^\top}{\sqrt{d_k}} + \mathbf{M}\right)\mathbf{V}, \tag{5.9}$$

where $\mathbf{M} \in \{0, -\infty\}^{L_Q \times L_K}$ is an optional masking matrix, and softmax is applied row-wise.

5.4.2 Why linear projections are mandatory

In self-attention, input tokens start as embeddings $\mathbf{x}_i \in \mathbb{R}^{d_{\text{model}}}$. If we computed dot products directly between raw token embeddings $\mathbf{x}_i^\top\mathbf{x}_j$, two critical limitations would cripple model expressivity:

1. **Symmetry trap**: Because the dot product is commutative ($\mathbf{x}_i^\top\mathbf{x}_j = \mathbf{x}_j^\top\mathbf{x}_i$), the unprojected affinity between token i and token j would be strictly symmetric. A token could not express an asymmetric relationship, such as a verb actively seeking its direct object without the direct object reciprocating with identical strength.

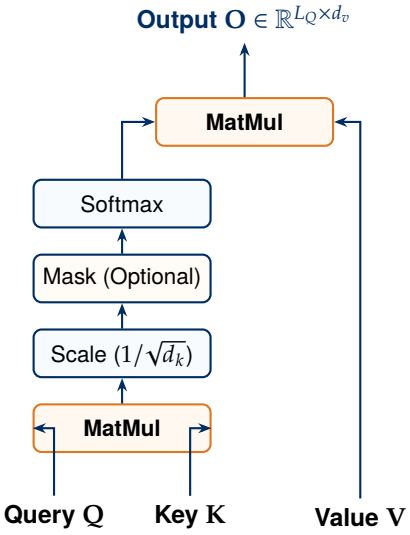


Figure 5.3: Computational dataflow of Scaled Dot-Product Attention [1]. The dot-product matrix is scaled by $1/\sqrt{d_k}$ to prevent gradient saturation before row-wise softmax normalization and value multiplication.

2. **Role collapse**: A token vector would be forced to play three fundamentally distinct roles simultaneously: formulating a search request (Query), advertising its semantic contents (Key), and providing the contextual payload (Value).

To break this symmetry and enable functional specialization, the model projects each input vector into three distinct latent subspaces using learned weight matrices:

$$\mathbf{Q} = \mathbf{X}\mathbf{W}_Q, \quad \mathbf{K} = \mathbf{X}\mathbf{W}_K, \quad \mathbf{V} = \mathbf{X}\mathbf{W}_V, \tag{5.10}$$

where $\mathbf{W}_Q \in \mathbb{R}^{d_{\text{model}} \times d_k}$, $\mathbf{W}_K \in \mathbb{R}^{d_{\text{model}} \times d_k}$, and $\mathbf{W}_V \in \mathbb{R}^{d_{\text{model}} \times d_v}$.

5.4.3 Complete numerical walkthrough: reproducing the lecture calculations

To illustrate these matrix projections and attention operations concretely, we walk through step-by-step the complete numerical worked example, tracking intermediate values down to four decimal places.

1. Initial setup. Consider an input sequence of $L = 3$ tokens in a 2-dimensional latent embedding space ($d_{\text{model}} = 2$), with projection dimensions $d_k = d_v = 2$:

$$\mathbf{X} = \begin{bmatrix} 0.1 & 0.5 \\ 0.1 & 0.2 \\ 0.9 & 0.9 \end{bmatrix} \in \mathbb{R}^{3\times 2}, \quad \mathbf{W}_Q = \begin{bmatrix} 0.9 & 0.9 \\ 0.5 & 0.3 \end{bmatrix}, \quad \mathbf{W}_K = \begin{bmatrix} 1.0 & 0.9 \\ 1.0 & 0.6 \end{bmatrix}, \quad \mathbf{W}_V = \begin{bmatrix} 0.8 & 0.9 \\ 0.9 & 1.0 \end{bmatrix}. \tag{5.11}$$

2. Linear projection into Queries, Keys, and Values. Multiplying $\mathbf{X}$ by each projection matrix yields:

$$\mathbf{Q} = \mathbf{X}\mathbf{W}_Q = \begin{bmatrix} 0.1(0.9) + 0.5(0.5) & 0.1(0.9) + 0.5(0.3) \\ 0.1(0.9) + 0.2(0.5) & 0.1(0.9) + 0.2(0.3) \\ 0.9(0.9) + 0.9(0.5) & 0.9(0.9) + 0.9(0.3) \end{bmatrix} = \begin{bmatrix} 0.34 & 0.24 \\ 0.19 & 0.15 \\ 1.26 & 1.08 \end{bmatrix}, \tag{5.12}$$

$$\mathbf{K} = \mathbf{X}\mathbf{W}_K = \begin{bmatrix} 0.1(1.0) + 0.5(1.0) & 0.1(0.9) + 0.5(0.6) \\ 0.1(1.0) + 0.2(1.0) & 0.1(0.9) + 0.2(0.6) \\ 0.9(1.0) + 0.9(1.0) & 0.9(0.9) + 0.9(0.6) \end{bmatrix} = \begin{bmatrix} 0.60 & 0.39 \\ 0.30 & 0.21 \\ 1.80 & 1.35 \end{bmatrix}, \tag{5.13}$$

$$\mathbf{V} = \mathbf{X}\mathbf{W}_V = \begin{bmatrix} 0.1(0.8) + 0.5(0.9) & 0.1(0.9) + 0.5(1.0) \\ 0.1(0.8) + 0.2(0.9) & 0.1(0.9) + 0.2(1.0) \\ 0.9(0.8) + 0.9(0.9) & 0.9(0.9) + 0.9(1.0) \end{bmatrix} = \begin{bmatrix} 0.53 & 0.59 \\ 0.26 & 0.29 \\ 1.53 & 1.71 \end{bmatrix}. \tag{5.14}$$

3. Directly computing raw dot products $\mathbf{QK}^\top$ and scaled logits $\mathbf{S} = \frac{\mathbf{QK}^\top}{\sqrt{d_k}}$. Evaluating the full matrix multiplication $\mathbf{QK}^\top \in \mathbb{R}^{3\times 3}$ directly:

$$\mathbf{QK}^\top = \begin{bmatrix} 0.34 & 0.24 \\ 0.19 & 0.15 \\ 1.26 & 1.08 \end{bmatrix} \begin{bmatrix} 0.60 & 0.30 & 1.80 \\ 0.39 & 0.21 & 1.35 \end{bmatrix} = \begin{bmatrix} 0.2976 & 0.1524 & 0.9360 \\ 0.1725 & 0.0885 & 0.5445 \\ 1.1772 & 0.6048 & 3.7260 \end{bmatrix}. \tag{5.15}$$

Alongside this matrix computation, examining row 1 reveals the individual scalar dot-product compatibility scores for the first token ($\mathbf{q}_1 = [0.34, 0.24]$):

$$s_{1,1} = \mathbf{q}_1^\top \mathbf{k}_1 = 0.34(0.60) + 0.24(0.39) = 0.2040 + 0.0936 = 0.2976, \tag{5.16}$$
$$s_{1,2} = \mathbf{q}_1^\top \mathbf{k}_2 = 0.34(0.30) + 0.24(0.21) = 0.1020 + 0.0504 = 0.1524, \tag{5.17}$$
$$s_{1,3} = \mathbf{q}_1^\top \mathbf{k}_3 = 0.34(1.80) + 0.24(1.35) = 0.6120 + 0.3240 = 0.9360. \tag{5.18}$$

Next, we scale the complete matrix by $\frac{1}{\sqrt{d_k}} = \frac{1}{\sqrt{2}} \approx 0.7071$:

$$\mathbf{S} = \frac{\mathbf{QK}^\top}{\sqrt{2}} = \begin{bmatrix} \frac{0.2976}{\sqrt{2}} & \frac{0.1524}{\sqrt{2}} & \frac{0.9360}{\sqrt{2}} \\ \frac{0.1725}{\sqrt{2}} & \frac{0.0885}{\sqrt{2}} & \frac{0.5445}{\sqrt{2}} \\ \frac{1.1772}{\sqrt{2}} & \frac{0.6048}{\sqrt{2}} & \frac{3.7260}{\sqrt{2}} \end{bmatrix} \approx \begin{bmatrix} 0.2104 & 0.1078 & 0.6619 \\ 0.1220 & 0.0626 & 0.3850 \\ 0.8324 & 0.4277 & 2.6347 \end{bmatrix}. \tag{5.19}$$

4. Matrix softmax normalization onto the probability simplex. Applying the row-wise softmax operator $\mathbf{A} = \mathrm{softmax}(\mathbf{S})$ exponentiates each row and normalizes by its partition sum. For row 1:

$$\exp(0.2104) \approx 1.2342, \quad \exp(0.1078) \approx 1.1138, \quad \exp(0.6619) \approx 1.9385, \tag{5.20}$$

$$\sum_{j=1}^{3} \exp(\tilde{s}_{1,j}) = 1.2342 + 1.1138 + 1.9385 = 4.2865, \tag{5.21}$$

$$\alpha_{1,1} = \frac{1.2342}{4.2865} \approx 0.2879, \quad \alpha_{1,2} = \frac{1.1138}{4.2865} \approx 0.2598, \quad \alpha_{1,3} = \frac{1.9385}{4.2865} \approx 0.4522. \tag{5.22}$$

Evaluating rows 2 and 3 identically yields the full row-stochastic attention weight matrix:

$$\mathbf{A} = \mathrm{softmax}\left(\frac{\mathbf{QK}^\top}{\sqrt{d_k}}\right) = \begin{bmatrix} 0.2879 & 0.2598 & 0.4522 \\ 0.3083 & 0.2906 & 0.4011 \\ 0.1294 & 0.0863 & 0.7843 \end{bmatrix}, \tag{5.23}$$

where each row sums precisely to 1.0000.

5. Matrix value aggregation: computing the full output matrix $\mathbf{O} = \mathbf{AV}$. The full sequence output matrix is obtained by matrix multiplication $\mathbf{O} = \mathbf{AV} \in \mathbb{R}^{3\times 2}$:

$$\mathbf{O} = \mathbf{AV} = \begin{bmatrix} 0.2879 & 0.2598 & 0.4522 \\ 0.3083 & 0.2906 & 0.4011 \\ 0.1294 & 0.0863 & 0.7843 \end{bmatrix} \begin{bmatrix} 0.53 & 0.59 \\ 0.26 & 0.29 \\ 1.53 & 1.71 \end{bmatrix} \approx \begin{bmatrix} 0.91 & 1.02 \\ 0.85 & 0.95 \\ 1.29 & 1.44 \end{bmatrix}. \tag{5.24}$$

For token 1, this matrix product directly computes the linear combination of value rows:

$$\mathbf{o}_1 = 0.2879 \begin{bmatrix} 0.53 \\ 0.59 \end{bmatrix} + 0.2598 \begin{bmatrix} 0.26 \\ 0.29 \end{bmatrix} + 0.4522 \begin{bmatrix} 1.53 \\ 1.71 \end{bmatrix} \approx \begin{bmatrix} 0.91 \\ 1.02 \end{bmatrix}. \tag{5.25}$$

This reproduces identically the exact tensor results displayed in the calculations.

5.4.4 Dimensional constraints across projection matrices

The dimensional compatibility of the projection matrices is governed by three fundamental algebraic constraints:

- **Queries and Keys must share identical dimensionality (d_k)**: The dot product $\mathbf{q}^\top \mathbf{k}$ is an inner product between two vectors. An inner product is mathematically undefined unless both vectors live in the identical Euclidean vector space $\mathbb{R}^{d_k}$. Therefore, $\mathbf{W}_Q$ and $\mathbf{W}_K$ must always share identical column dimension d_k.
- **Values can possess independent dimensionality (d_v)**: Value vectors are never multiplied with queries or keys. They are only scaled by scalar attention weights $\alpha_j \in [0, 1]$ and summed together. Consequently, d_v could mathematically differ from d_k. In standard architectures (including [1]), practitioners choose $d_k = d_v = d_{\text{model}}/h$ purely for architectural symmetry and implementation simplicity.
- **Sequence lengths in self-attention vs. cross-attention**: In self-attention, $L_Q = L_K = L_V = L$. In cross-attention, queries stem from the decoder ($L_Q = L_{\text{dec}}$), while keys and values stem from the encoder ($L_K = L_V = L_{\text{enc}}$). The dot-product matrix $\mathbf{QK}^\top$ has shape $L_{\text{dec}} \times L_{\text{enc}}$, and multiplying by $\mathbf{V} \in \mathbb{R}^{L_{\text{enc}} \times d_v}$ yields the required decoder output shape $L_{\text{dec}} \times d_v$.

5.4.5 Mathematical derivation: theoretical necessity of the $\frac{1}{\sqrt{d_k}}$ scaling factor

Why must the dot product $\mathbf{q}^\top \mathbf{k}$ be scaled specifically by $\frac{1}{\sqrt{d_k}}$? To understand the mathematical necessity of this normalization, consider the variance behavior of unscaled dot products as dimensionality expands.

Proof of dot-product variance expansion. Assume that the components of query vector $\mathbf{q} \in \mathbb{R}^{d_k}$ and key vector $\mathbf{k} \in \mathbb{R}^{d_k}$ are independent, identically distributed (i.i.d.) random variables with zero mean and unit variance:

$$\mathbb{E}[q_i] = 0, \quad \mathbb{V}\text{ar}(q_i) = 1, \quad \mathbb{E}[k_i] = 0, \quad \mathbb{V}\text{ar}(k_i) = 1, \quad \forall i \in \{1, 2, \ldots, d_k\}. \tag{5.26}$$

The unscaled dot product is the random variable $z = \mathbf{q}^\top \mathbf{k} = \sum_{i=1}^{d_k} q_i k_i$. By linearity of expectation and independence:

$$\mathbb{E}[z] = \mathbb{E}\left[\sum_{i=1}^{d_k} q_i k_i\right] = \sum_{i=1}^{d_k} \mathbb{E}[q_i]\,\mathbb{E}[k_i] = \sum_{i=1}^{d_k} (0 \cdot 0) = 0. \tag{5.27}$$

To compute the variance of z, we expand the definition $\mathbb{V}\text{ar}(z) = \sum_{i=1}^{d_k} \mathbb{V}\text{ar}(q_i k_i)$. Using identity $\mathbb{V}\text{ar}(XY) = \mathbb{E}[X^2]\,\mathbb{E}[Y^2] - (\mathbb{E}[X]\,\mathbb{E}[Y])^2$:

$$\begin{aligned} \mathbb{V}\text{ar}(q_i k_i) &= \mathbb{E}[q_i^2]\,\mathbb{E}[k_i^2] - (\mathbb{E}[q_i]\,\mathbb{E}[k_i])^2 \\ &= \left(\mathbb{V}\text{ar}(q_i) + (\mathbb{E}[q_i])^2\right)\left(\mathbb{V}\text{ar}(k_i) + (\mathbb{E}[k_i])^2\right) - 0 \\ &= (1 + 0)(1 + 0) = 1. \end{aligned} \tag{5.28}$$

Summing over all d_k independent dimensions:

$$\mathbb{V}\text{ar}(z) = \sum_{i=1}^{d_k} \mathbb{V}\text{ar}(q_i k_i) = \sum_{i=1}^{d_k} 1 = d_k. \tag{5.29}$$

Thus, the standard deviation of the dot product is $\sigma_z = \sqrt{d_k}$.

The gradient saturation consequence. For large dimensions (e.g., $d_k = 64$ or 128), the variance d_k is substantial, causing dot products to take values with magnitude $|z| \gg 1$. Recall that the derivative of the softmax function with respect to logit z_i is given by:

$$\frac{\partial \,\mathrm{softmax}(z)_i}{\partial z_j} = \mathrm{softmax}(z)_i \left(\delta_{ij} - \mathrm{softmax}(z)_j\right). \tag{5.30}$$

When logits z have large absolute variance, the exponential $\exp(z_m)$ of the maximal logit completely dominates the sum. The softmax output degenerates into a nearly one-hot indicator distribution ($\alpha_m \approx 1$, and $\alpha_{j\neq m} \approx 0$). In this regime:

$$\frac{\partial \,\mathrm{softmax}(z)_i}{\partial z_j} \approx 0, \quad \forall i, j. \tag{5.31}$$

The gradients vanish, arresting backpropagation through the attention module. By scaling z by scalar factor $\frac{1}{\sqrt{d_k}}$, we renormalize the variance back to unity:

$$\mathbb{V}\mathrm{ar}\left(\frac{z}{\sqrt{d_k}}\right) = \frac{1}{(\sqrt{d_k})^2}\,\mathbb{V}\mathrm{ar}(z) = \frac{1}{d_k} \cdot d_k = 1. \tag{5.32}$$

Renormalizing the variance preserves healthy gradient dynamics across arbitrarily deep networks.

5.5 The triad of attention mechanisms

Within the complete Transformer architecture, attention is deployed across three distinct operational settings [1], summarized in Figure 5.4.

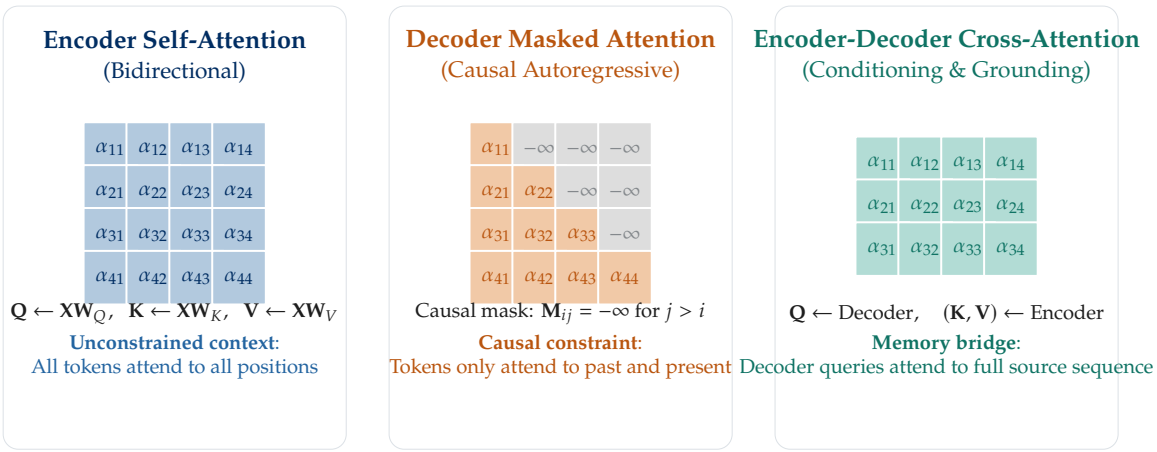


Figure 5.4: The triad of attention mechanisms across the Transformer architecture. (a) Encoder self-attention processes inputs bidirectionally with an unconstrained attention matrix. (b) Decoder masked self-attention applies a lower-triangular causal mask ($\mathbf{M}_{ij} = -\infty$ for $j > i$) to enforce autoregressive generation without future information leakage. (c) Encoder-decoder cross-attention allows decoder queries to attend across all encoder key-value pairs, bridging representations across different sequence lengths.

5.5.1 Encoder self-attention: unconstrained bidirectional contextualization

Why unconstrained bidirectional visibility is mandatory. In natural language understanding, the meaning of a word is profoundly shaped by tokens that appear both *before* and *after* it. For

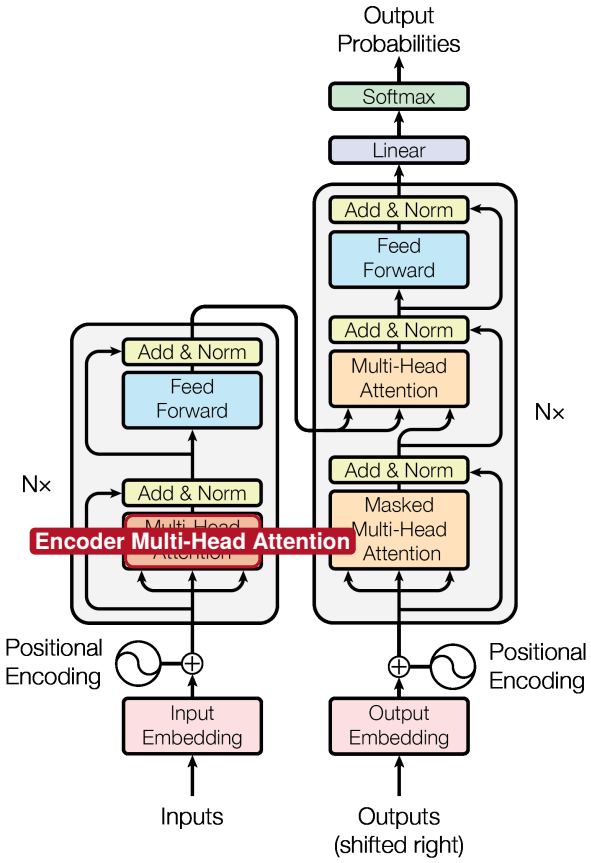


Figure 5.5: Location of Encoder Self-Attention within the canonical Transformer architecture [1]. The module resides in the left encoder stack, processing input token representations with full unconstrained bidirectional visibility.

example, consider syntactic ambiguity, pronoun reference, or polysemy: in the sentence *"The bank of the river overflowed"*, the polysemous noun *"bank"* cannot be resolved into a geological landform rather than a financial institution without inspecting the downstream word *"river"*. Because the Encoder's sole objective is representation learning rather than autoregressive word generation, there is no risk of label leakage. Therefore, the Encoder must have access to the entire sentence simultaneously: the attention matrix $\mathbf{A} \in \mathbb{R}^{L_{\text{enc}} \times L_{\text{enc}}}$ is completely unconstrained ($\mathbf{M} = \mathbf{0}$), granting every token an instantaneous $O(1)$ global receptive field across all L_{enc} positions.

In the Encoder, queries, keys, and values are all linearly projected from the preceding encoder layer's representations:

$$\mathbf{Q} = \mathbf{X}\mathbf{W}_Q, \quad \mathbf{K} = \mathbf{X}\mathbf{W}_K, \quad \mathbf{V} = \mathbf{X}\mathbf{W}_V, \tag{5.33}$$

where $\mathbf{X} \in \mathbb{R}^{L_{\text{enc}} \times d_{\text{model}}}$, $\mathbf{W}_Q, \mathbf{W}_K \in \mathbb{R}^{d_{\text{model}} \times d_k}$, and $\mathbf{W}_V \in \mathbb{R}^{d_{\text{model}} \times d_v}$. No masking matrix is applied ($\mathbf{M} = \mathbf{0}$). Every token i can attend freely to all other tokens $j \in \{1, 2, \ldots, L_{\text{enc}}\}$, producing unconstrained bidirectional representations.

Step-by-step mathematical example: bidirectional polysemy disambiguation. Consider an English input sentence of $L_{\text{enc}} = 3$ tokens:

$$X = [\text{“bank”}_1, \text{“of”}_2, \text{“river”}_3]. \tag{5.34}$$

The token *"bank"* is polysemous. Without inspecting the future token *"river"*, disambiguation is impossible. Let $d_k = d_v = 2$. Suppose the linear projections $\mathbf{X}\mathbf{W}_Q$, $\mathbf{X}\mathbf{W}_K$, and $\mathbf{X}\mathbf{W}_V$ yield:

$$\mathbf{Q} = \begin{bmatrix} 1.0 & 0.5 \\ 0.2 & 0.8 \\ 0.9 & 1.2 \end{bmatrix}, \quad \mathbf{K} = \begin{bmatrix} 0.8 & 0.4 \\ 0.1 & 0.9 \\ 1.0 & 1.1 \end{bmatrix}, \quad \mathbf{V} = \begin{bmatrix} 1.5 & 0.2 \\ 0.1 & 0.8 \\ 0.4 & 2.0 \end{bmatrix}. \tag{5.35}$$

We execute the attention mechanism step by step:

1. **Raw dot-product computation**: Evaluating unscaled inner products $\mathbf{Q}\mathbf{K}^\top \in \mathbb{R}^{3\times 3}$:

$$\mathbf{Q}\mathbf{K}^\top = \begin{bmatrix} (1.0)(0.8) + (0.5)(0.4) & (1.0)(0.1) + (0.5)(0.9) & (1.0)(1.0) + (0.5)(1.1) \\ (0.2)(0.8) + (0.8)(0.4) & (0.2)(0.1) + (0.8)(0.9) & (0.2)(1.0) + (0.8)(1.1) \\ (0.9)(0.8) + (1.2)(0.4) & (0.9)(0.1) + (1.2)(0.9) & (0.9)(1.0) + (1.2)(1.1) \end{bmatrix}. \tag{5.36}$$

   Evaluating each scalar entry yields:

$$\mathbf{Q}\mathbf{K}^\top = \begin{bmatrix} 1.00 & 0.55 & 1.55 \\ 0.48 & 0.74 & 1.08 \\ 1.20 & 1.17 & 2.22 \end{bmatrix}. \tag{5.37}$$

2. **Scaling by $\sqrt{d_k} = \sqrt{2} \approx 1.414$**:

$$\mathbf{S} = \frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{2}} \approx \begin{bmatrix} 0.707 & 0.389 & 1.096 \\ 0.339 & 0.523 & 0.764 \\ 0.849 & 0.827 & 1.570 \end{bmatrix}. \tag{5.38}$$

3. **Bidirectional softmax normalization**: Since no mask is present ($\mathbf{M} = \mathbf{0}$), we compute row-wise softmax directly across all columns: For row 1 (corresponding to *"bank"*):

$$\exp(0.707) \approx 2.028, \quad \exp(0.389) \approx 1.475, \quad \exp(1.096) \approx 2.992, \tag{5.39}$$

$$Z_1 = 2.028 + 1.475 + 2.992 = 6.495, \tag{5.40}$$

$$\alpha_{11} = \frac{2.028}{6.495} \approx 0.312, \quad \alpha_{12} = \frac{1.475}{6.495} \approx 0.227, \quad \alpha_{13} = \frac{2.992}{6.495} \approx 0.461. \tag{5.41}$$

Applying this across all rows yields the dense attention matrix:

$$\mathbf{A}_{\text{enc}} = \begin{bmatrix} 0.312 & 0.227 & 0.461 \\ 0.268 & 0.322 & 0.410 \\ 0.248 & 0.243 & 0.510 \end{bmatrix}. \tag{5.42}$$

4. **Contextual value aggregation**: Multiplying $\mathbf{A}_{\text{enc}}$ by $\mathbf{V} \in \mathbb{R}^{3\times 2}$ yields the updated contextual representations:

$$\mathbf{O}_{\text{enc}} = \mathbf{A}_{\text{enc}}\mathbf{V} = \begin{bmatrix} 0.312\mathbf{v}_1 + 0.227\mathbf{v}_2 + 0.461\mathbf{v}_3 \\ 0.268\mathbf{v}_1 + 0.322\mathbf{v}_2 + 0.410\mathbf{v}_3 \\ 0.248\mathbf{v}_1 + 0.243\mathbf{v}_2 + 0.510\mathbf{v}_3 \end{bmatrix} \approx \begin{bmatrix} 0.675 & 1.165 \\ 0.598 & 1.131 \\ 0.600 & 1.263 \end{bmatrix}. \tag{5.43}$$

Key architectural takeaway. Notice that token 1 (*"bank"*) allocates its largest attention weight ($\alpha_{13} = 0.461$) to token 3 (*"river"*), looking *ahead* into future context. This unconstrained bidirectional information exchange allows *"bank"* to instantly absorb the geological semantics of *"river"* into its contextual output $\mathbf{o}_1$.

5.5.2 Decoder masked self-attention: causal autoregressive constraint

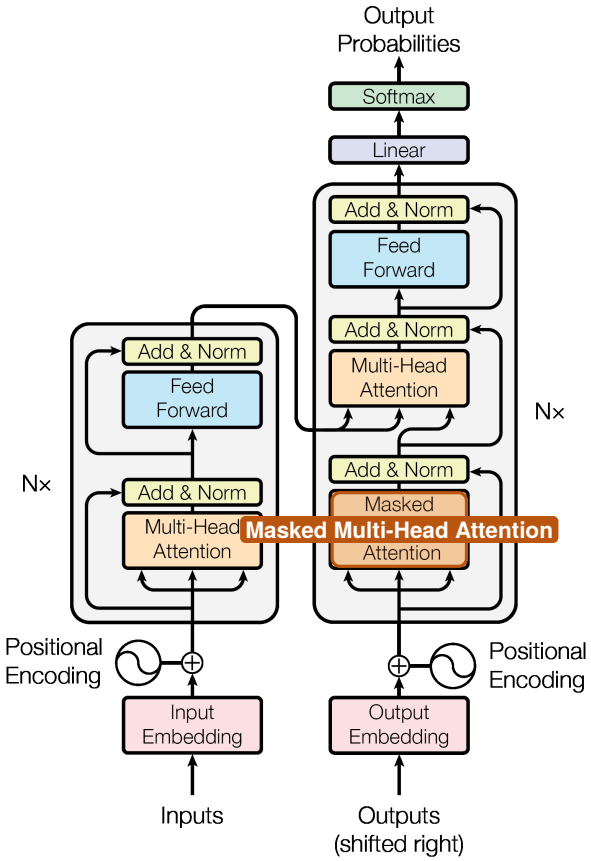


Figure 5.6: Location of Masked Multi-Head Attention within the canonical Transformer architecture [1]. The module resides at the base of the right decoder stack, applying causal masking to enforce strictly autoregressive sequence generation.

Why the causal mask is mandatory for autoregressive decoding. Unlike the Encoder, the Decoder is tasked with sequential language generation: producing the next token y_t conditioned strictly on the prefix of tokens emitted so far $[y_1, y_2, \ldots, y_{t-1}]$. At test time, future tokens $y_{t+1}, y_{t+2}, \ldots$ do not exist. During training, however, the entire target sequence is supplied to the model simultaneously to enable parallel execution via Teacher Forcing. If unconstrained self-attention were permitted in the decoder, a token at position t could directly attend to position $t+1$ and cheat by copying the ground-truth answer. The model would suffer from catastrophic information leakage, failing to learn any predictive distribution over future tokens. To enforce the arrow of time while preserving parallel tensor execution on hardware, a **causal mask M** is added to the attention logits prior to softmax, strictly setting all future attention weights to zero.

To preserve causality while retaining parallel training under Teacher Forcing, a **causal mask** $\mathbf{M} \in \mathbb{R}^{L_{\text{dec}} \times L_{\text{dec}}}$ is added to the scaled dot products prior to the softmax operator:

$$M_{i,j} = \begin{cases} 0 & \text{if } j \le i, \\ -\infty & \text{if } j > i. \end{cases} \tag{5.44}$$

In matrix form for a sequence of length 3:

$$\mathbf{M} = \begin{bmatrix} 0 & -\infty & -\infty \\ 0 & 0 & -\infty \\ 0 & 0 & 0 \end{bmatrix}. \tag{5.45}$$

Because $\exp(-\infty) = 0$, evaluating the row-wise softmax guarantees:

$$\alpha_{i,j} = 0 \quad \forall j > i. \tag{5.46}$$

From a hardware efficiency standpoint, computing the full dense GEMM operation $\mathbf{QK}^\top$ followed by a tensor addition with $\mathbf{M}$ fully saturates GPU tensor cores and memory buses, outperforming jagged triangular loops.

Step-by-step mathematical example: autoregressive generation under causal masking. Now consider the Decoder generating the Italian translation prefix of length $L_{\text{dec}} = 3$:

$$Y = [y_1 = \langle \text{BOS} \rangle,\ y_2 = \text{“riva”},\ y_3 = \text{“del”}]. \tag{5.47}$$

All queries, keys, and values are projected from the decoder's target token embeddings ($d_k = d_v = 2$):

$$\mathbf{Q} = \begin{bmatrix} 0.5 & 0.5 \\ 1.2 & 0.3 \\ 0.8 & 1.0 \end{bmatrix}, \quad \mathbf{K} = \begin{bmatrix} 0.6 & 0.4 \\ 1.0 & 0.2 \\ 0.7 & 0.9 \end{bmatrix}, \quad \mathbf{V} = \begin{bmatrix} 0.2 & 0.5 \\ 1.8 & 0.4 \\ 0.6 & 1.4 \end{bmatrix}. \tag{5.48}$$

Let us trace the computation step by step:

1. **Raw scaled dot products (unmasked)**: Computing $\mathbf{S} = \frac{\mathbf{QK}^\top}{\sqrt{2}}$:

$$\mathbf{S} = \frac{1}{\sqrt{2}} \begin{bmatrix} 0.50 & 0.60 & 0.80 \\ 0.84 & 1.26 & 1.11 \\ 0.88 & 1.00 & 1.46 \end{bmatrix} \approx \begin{bmatrix} 0.354 & 0.424 & 0.566 \\ 0.594 & 0.891 & 0.785 \\ 0.622 & 0.707 & 1.032 \end{bmatrix}. \tag{5.49}$$

   Notice that unmasked score $S_{13} = 0.566$ is the highest in the first row: without masking, $\langle \text{BOS} \rangle$ would strongly attend to future token *“del”*!

2. **Applying the causal mask M**: We add $\mathbf{M}$ to $\mathbf{S}$:

$$\mathbf{S}_{\text{masked}} = \mathbf{S} + \mathbf{M} = \begin{bmatrix} 0.354 & -\infty & -\infty \\ 0.594 & 0.891 & -\infty \\ 0.622 & 0.707 & 1.032 \end{bmatrix}. \tag{5.50}$$

3. **Row-wise softmax under masking**:

   - **Row 1 ($y_1 = \langle$BOS$\rangle$)**: $\exp(0.354) \approx 1.425$, while $\exp(-\infty) = 0$:

   $$\alpha_{11} = \frac{1.425}{1.425 + 0 + 0} = 1.000, \quad \alpha_{12} = 0.000, \quad \alpha_{13} = 0.000. \tag{5.51}$$

   Position 1 can *only* attend to itself!

   - **Row 2 (y_2 = "riva")**: $\exp(0.594) \approx 1.811$, $\exp(0.891) \approx 2.438$, $\exp(-\infty) = 0$:

   $$Z_2 = 1.811 + 2.438 = 4.249, \tag{5.52}$$

   $$\alpha_{21} = \frac{1.811}{4.249} \approx 0.426, \quad \alpha_{22} = \frac{2.438}{4.249} \approx 0.574, \quad \alpha_{23} = 0.000. \tag{5.53}$$

   Position 2 attends strictly to the prefix $\{y_1, y_2\}$; future token y_3 is zeroed out.

   - **Row 3 (y_3 = "del")**: $\exp(0.622) \approx 1.863$, $\exp(0.707) \approx 2.028$, $\exp(1.032) \approx 2.807$:

   $$Z_3 = 1.863 + 2.028 + 2.807 = 6.698 \implies \alpha_{31} \approx 0.278, \quad \alpha_{32} \approx 0.303, \quad \alpha_{33} \approx 0.419. \tag{5.54}$$

   This produces the strictly lower-triangular attention matrix:

   $$\mathbf{A}_{\text{dec}} = \begin{bmatrix} 1.000 & 0.000 & 0.000 \\ 0.426 & 0.574 & 0.000 \\ 0.278 & 0.303 & 0.419 \end{bmatrix}. \tag{5.55}$$

4. **Causal value aggregation**: Multiplying by $\mathbf{V} \in \mathbb{R}^{3\times 2}$:

$$\mathbf{O}_{\text{dec}} = \mathbf{A}_{\text{dec}}\mathbf{V} = \begin{bmatrix} 1.000\mathbf{v}_1 \\ 0.426\mathbf{v}_1 + 0.574\mathbf{v}_2 \\ 0.278\mathbf{v}_1 + 0.303\mathbf{v}_2 + 0.419\mathbf{v}_3 \end{bmatrix} \approx \begin{bmatrix} 0.200 & 0.500 \\ 1.118 & 0.443 \\ 0.852 & 0.847 \end{bmatrix}. \tag{5.56}$$

Key architectural takeaway. Contrast this directly with Encoder self-attention: despite $S_{13} = 0.566$ having high compatibility, the causal mask completely extinguishes its influence ($\alpha_{13} = 0$). Output $\mathbf{o}_1$ is computed strictly from $\mathbf{v}_1$, preserving the causality required for autoregressive sequence modeling.

5.5.3 Encoder-decoder cross-attention: dynamic source grounding

Why queries originate from the decoder while keys and values originate from the encoder. In sequence-to-sequence translation, the decoder is generating tokens in the target language, while the encoder holds the semantic context of the source sentence. Recall the fundamental roles of the attention triad:

- A **Query** asks: *"What information am I currently searching for to generate my next word?"* Because the entity generating the next word is the decoder, the queries $\mathbf{Q}$ must be projected from the decoder's current representation.
- A **Key** advertises: *"What features do I describe in the source sentence?"*
- A **Value** provides: *"What source meaning should be retrieved?"*

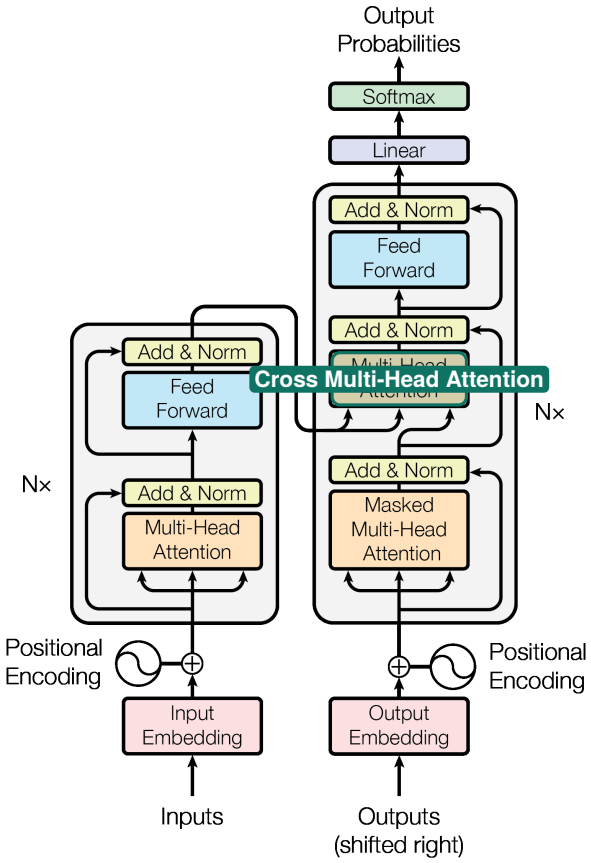


Figure 5.7: Location of Encoder-Decoder Cross-Attention within the canonical Transformer architecture [1]. The module bridges the two stacks: queries $\mathbf{Q}$ are generated by the decoder, while keys $\mathbf{K}$ and values $\mathbf{V}$ are supplied directly from the output of the encoder stack.

Because the source content is held exclusively in the encoder, the keys $\mathbf{K}$ and values $\mathbf{V}$ must originate from the encoder's output representations. If queries were taken from the encoder instead, the source sentence would be searching the target words, which completely contradicts the directional objective of translating source meaning into target text. By projecting queries from the decoder and keys/values from the encoder, cross-attention acts as a dynamic search engine allowing each generated target token to query the entire uncompressed source sentence.

In cross-attention, the model bridges the target sequence with the source sequence. The queries originate from the decoder's intermediate representations $\mathbf{H}_{\text{dec}} \in \mathbb{R}^{L_{\text{dec}} \times d_{\text{model}}}$, while keys and values originate from the uncompressed output of the encoder stack $\mathbf{H}_{\text{enc}} \in \mathbb{R}^{L_{\text{enc}} \times d_{\text{model}}}$:

$$\mathbf{Q} = \mathbf{H}_{\text{dec}}\mathbf{W}_Q^{\text{cross}} \in \mathbb{R}^{L_{\text{dec}} \times d_k}, \quad \mathbf{K} = \mathbf{H}_{\text{enc}}\mathbf{W}_K^{\text{cross}} \in \mathbb{R}^{L_{\text{enc}} \times d_k}, \quad \mathbf{V} = \mathbf{H}_{\text{enc}}\mathbf{W}_V^{\text{cross}} \in \mathbb{R}^{L_{\text{enc}} \times d_v}. \tag{5.57}$$

Crucially:

- **Asymmetric sequence lengths**: The number of queries L_{dec} does not need to equal the number of keys/values L_{enc}. In translation, sentences in different languages rarely have identical token counts.
- **No causal mask on the source**: The source sentence was fully processed by the encoder before decoding began. Thus, every decoder query position can inspect *all* L_{enc} source tokens ($\mathbf{M} = \mathbf{0}$).

Step-by-step mathematical example: cross-lingual alignment with asymmetric lengths. Suppose our decoder has generated a target prefix of length $L_{\text{dec}} = 2$ ([⟨BOS⟩, "riva"]), and cross-attends to our earlier English source sentence of length $L_{\text{enc}} = 3$ (["bank", "of", "river"]). The decoder provides $L_{\text{dec}} = 2$ queries $\mathbf{Q} \in \mathbb{R}^{2\times 2}$ (taken from the first two rows of (5.48)), while the encoder provides $L_{\text{enc}} = 3$ keys and values $\mathbf{K}, \mathbf{V} \in \mathbb{R}^{3\times 2}$ (from (5.35)):

$$\mathbf{Q} = \begin{bmatrix} 0.5 & 0.5 \\ 1.2 & 0.3 \end{bmatrix}, \quad \mathbf{K} = \begin{bmatrix} 0.8 & 0.4 \\ 0.1 & 0.9 \\ 1.0 & 1.1 \end{bmatrix}, \quad \mathbf{V} = \begin{bmatrix} 1.5 & 0.2 \\ 0.1 & 0.8 \\ 0.4 & 2.0 \end{bmatrix}. \tag{5.58}$$

Let us trace the rectangular cross-attention computation step by step:

1. **Rectangular dot-product matrix ($L_{\text{dec}} \times L_{\text{enc}} = 2 \times 3$)**:

$$\mathbf{Q}\mathbf{K}^\top = \begin{bmatrix} (0.5)(0.8) + (0.5)(0.4) & (0.5)(0.1) + (0.5)(0.9) & (0.5)(1.0) + (0.5)(1.1) \\ (1.2)(0.8) + (0.3)(0.4) & (1.2)(0.1) + (0.3)(0.9) & (1.2)(1.0) + (0.3)(1.1) \end{bmatrix}. \tag{5.59}$$

   Evaluating the pairwise inner products yields:

$$\mathbf{Q}\mathbf{K}^\top = \begin{bmatrix} 0.60 & 0.50 & 1.05 \\ 1.08 & 0.39 & 1.53 \end{bmatrix}. \tag{5.60}$$

2. **Scaling by $\sqrt{d_k} = \sqrt{2}$**:

$$\mathbf{S}_{\text{cross}} = \frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{2}} \approx \begin{bmatrix} 0.424 & 0.354 & 0.742 \\ 0.764 & 0.276 & 1.082 \end{bmatrix} \in \mathbb{R}^{2\times 3}. \tag{5.61}$$

3. **Row-wise softmax normalization (over $L_{\text{enc}} = 3$ source keys)**:
   - **Row 1 ($\mathbf{q}_{\langle \text{BOS} \rangle}$)**: $\exp(0.424) \approx 1.528$, $\exp(0.354) \approx 1.425$, $\exp(0.742) \approx 2.100$:

$$Z_1 = 1.528 + 1.425 + 2.100 = 5.053 \implies \mathbf{A}_{1,:} \approx [0.302,\ 0.282,\ 0.416]. \tag{5.62}$$

- **Row 2 ($\mathbf{q}_{\text{“riva”}}$)**: $\exp(0.764) \approx 2.147$, $\exp(0.276) \approx 1.318$, $\exp(1.082) \approx 2.951$:

$$Z_2 = 2.147 + 1.318 + 2.951 = 6.416 \implies \mathbf{A}_{2,:} \approx [0.335,\ 0.205,\ 0.460]. \tag{5.63}$$

This yields the rectangular cross-attention matrix:

$$\mathbf{A}_{\text{cross}} = \begin{bmatrix} 0.302 & 0.282 & 0.416 \\ 0.335 & 0.205 & 0.460 \end{bmatrix} \in \mathbb{R}^{2\times 3}. \tag{5.64}$$

4. **Cross-attention context retrieval**: Multiplying $\mathbf{A}_{\text{cross}} \in \mathbb{R}^{2\times 3}$ by encoder values $\mathbf{V} \in \mathbb{R}^{3\times 2}$ yields the grounded representations $\mathbf{O}_{\text{cross}} \in \mathbb{R}^{2\times 2}$:

$$\mathbf{O}_{\text{cross}} = \mathbf{A}_{\text{cross}}\mathbf{V} = \begin{bmatrix} 0.302\mathbf{v}_1 + 0.282\mathbf{v}_2 + 0.416\mathbf{v}_3 \\ 0.335\mathbf{v}_1 + 0.205\mathbf{v}_2 + 0.460\mathbf{v}_3 \end{bmatrix} \approx \begin{bmatrix} 0.648 & 1.117 \\ 0.706 & 1.151 \end{bmatrix}. \tag{5.65}$$

Key architectural takeaway. Query $\mathbf{q}_{\text{“riva”}}$ strongly aligns with source key $\mathbf{k}_{\text{“river”}}$ ($\alpha_{23} = 0.460$) and source key $\mathbf{k}_{\text{“bank”}}$ ($\alpha_{21} = 0.335$). The decoder directly pulls the exact semantic coordinates of the geological shore from the source representation without compressing the source into a single bottleneck vector!

5.6 Multi-head attention (MHA)

The limitation of single-head attention. In single-head attention, the model computes a single set of attention weights $\alpha_{i,j}$, averaging all values into a single weighted sum. However, a single token often participates in multiple simultaneous linguistic relationships:

- A verb must track both its grammatical subject (often located earlier in the sentence) and its direct object (often located downstream).
- In programming languages, an identifier must link to its type declaration, its scope boundary, and its mutating assignments.

A single attention head cannot focus simultaneously on disparate positions with different semantic intents: the softmax distribution is forced to average across them, diluting specific relational signals.

Mathematical formulation. Multi-Head Attention (MHA) resolves this by linearly projecting queries, keys, and values h times with different, learned parameter matrices:

Definition 5.2: *Multi-head attention*

Let h denote the number of attention heads. Multi-Head Attention projects $\mathbf{Q}, \mathbf{K}, \mathbf{V}$ into h distinct representation subspaces of dimension $d_k = d_{\text{model}}/h$ and $d_v = d_{\text{model}}/h$:

$$\texttt{MultiHead}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = [\text{head}_1 \,\|\, \text{head}_2 \,\|\, \cdots \,\|\, \text{head}_h]\,\mathbf{W}^O, \tag{5.66}$$

where each individual head is evaluated via:

$$\text{head}_i = \texttt{Attention}(\mathbf{Q}\mathbf{W}_i^Q, \mathbf{K}\mathbf{W}_i^K, \mathbf{V}\mathbf{W}_i^V), \quad i \in \{1, 2, \ldots, h\}, \tag{5.67}$$

with parameter projection matrices $\mathbf{W}_i^Q \in \mathbb{R}^{d_{\text{model}}\times d_k}$, $\mathbf{W}_i^K \in \mathbb{R}^{d_{\text{model}}\times d_k}$, $\mathbf{W}_i^V \in \mathbb{R}^{d_{\text{model}}\times d_v}$, and output projection matrix $\mathbf{W}^O \in \mathbb{R}^{hd_v\times d_{\text{model}}}$.

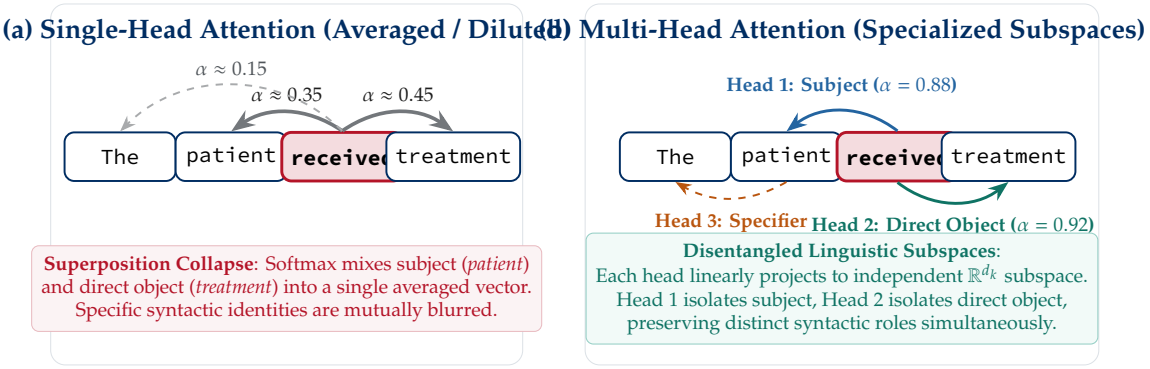


Figure 5.8: Linguistic relationship modeling: Single-Head vs. Multi-Head Attention. In Single-Head Attention (a), a token involved in multiple syntactic dependencies must compromise its single softmax distribution, averaging disparate features and diluting specific relational roles. In Multi-Head Attention (b), projection into h distinct representation subspaces enables separate heads to concurrently isolate orthogonal linguistic relationships (such as grammatical subject tracking and direct object binding) without mutual interference.

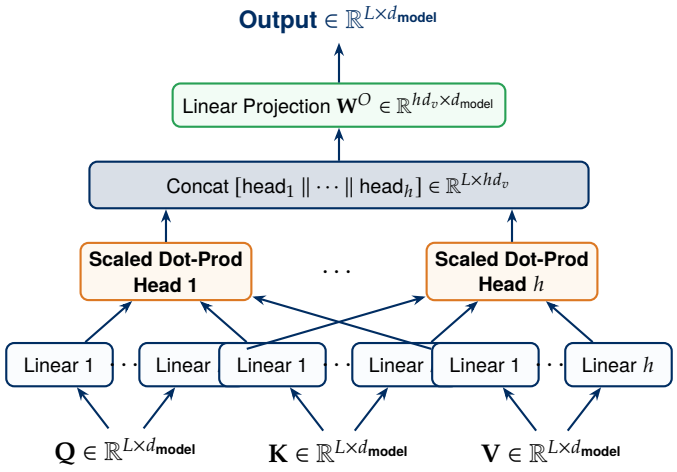


Figure 5.9: Architecture of Multi-Head Attention (MHA) [1]. Projections map inputs into h parallel representation subspaces of reduced dimension $d_k = d_v = d_{\text{model}}/h$, whose outputs are concatenated and linearly projected back to d_{model}.

Parameter equivalence and computational efficiency. Due to the reduced dimension of each head ($d_k = d_v = d_{\text{model}}/h$), the total computational cost of multi-head attention is identical to that of single-head attention with full dimensionality d_{model}:

$$h \times \left(d_{\text{model}} \times \frac{d_{\text{model}}}{h} \right) = d_{\text{model}}^2 \text{ operations for projections.} \tag{5.68}$$

In standard configurations (e.g., $d_{\text{model}} = 512, h = 8 \implies d_k = d_v = 64$, or $d_{\text{model}} = 768, h = 12 \implies d_k = d_v = 64$), parallel heads learn specialized sub-behaviors: some heads attend to local syntactic structure, others to long-range dependencies, and others to delimiter tokens.

5.7 Sublayer mechanics: residual connections and Layer Normalization

In deep Transformer architectures comprising dozens of layers, stacking attention blocks and non-linear feed-forward networks directly would lead to optimization failure due to exploding or vanishing gradients. To ensure stable gradient propagation, each Transformer sublayer is wrapped in a residual connection followed by Layer Normalization:

$$\mathbf{y} = \texttt{LayerNorm}(\mathbf{x} + \texttt{Sublayer}(\mathbf{x})). \tag{5.69}$$

1. Residual skip connections (gradient superhighways). First introduced in [53] for deep residual networks (ResNets), identity shortcut connections pass the unmodified input $\mathbf{x}$ around the sublayer (Figure 5.10):

$$\mathbf{y} = \mathbf{x} + \mathcal{F}(\mathbf{x}), \tag{5.70}$$

where $\mathcal{F}(\mathbf{x})$ represents the multi-head attention or feed-forward transformation.

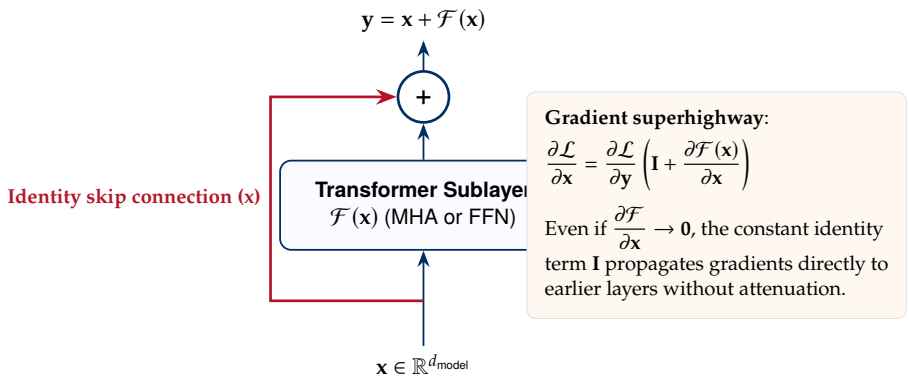


Figure 5.10: Residual skip connection and gradient propagation dynamics. The input $\mathbf{x}$ bypasses the non-linear sublayer $\mathcal{F}(\mathbf{x})$ via an identity shortcut. During backpropagation, the derivative $\frac{\partial \mathbf{y}}{\partial \mathbf{x}} = \mathbf{I} + \frac{\partial \mathcal{F}}{\partial \mathbf{x}}$ ensures that the gradient $\frac{\partial \mathcal{L}}{\partial \mathbf{y}}$ flows unimpeded across arbitrary layer depths via the identity term $\mathbf{I}$.

During backpropagation, the gradient of the loss $\mathcal{L}$ with respect to input $\mathbf{x}$ expands via the chain rule to:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{x}} = \frac{\partial \mathcal{L}}{\partial \mathbf{y}} \frac{\partial \mathbf{y}}{\partial \mathbf{x}} = \frac{\partial \mathcal{L}}{\partial \mathbf{y}} \left(\mathbf{I} + \frac{\partial \mathcal{F}(\mathbf{x})}{\partial \mathbf{x}} \right) = \frac{\partial \mathcal{L}}{\partial \mathbf{y}} + \frac{\partial \mathcal{L}}{\partial \mathbf{y}} \frac{\partial \mathcal{F}(\mathbf{x})}{\partial \mathbf{x}}. \tag{5.71}$$

The presence of the identity matrix $\mathbf{I}$ guarantees an unobstructed **gradient superhighway**: even if the Jacobian $\frac{\partial \mathcal{F}}{\partial \mathbf{x}}$ vanishes (for instance, when weights are initialized small or activations saturate), term $\frac{\partial \mathcal{L}}{\partial \mathbf{y}}$ flows directly back to earlier layers in a single step, completely preventing gradient extinction.

2. Layer Normalization. Unlike Batch Normalization (which computes statistics across the mini-batch dimension, making it sensitive to batch size and variable sequence lengths), **Layer Normalization** [54] computes normalization statistics across the feature dimensions for each individual token independently (Figure 5.11):

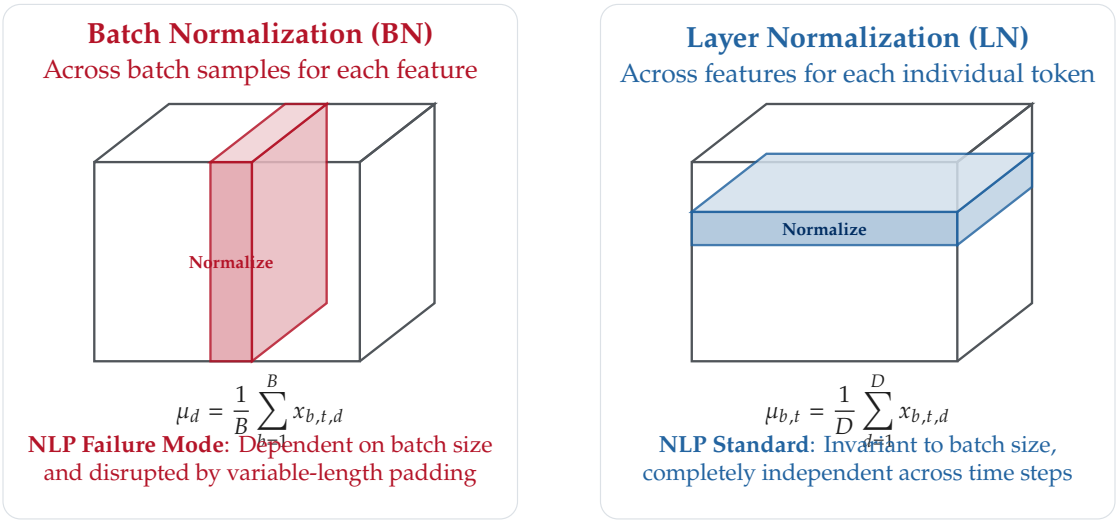


Figure 5.11: Comparison of normalization geometries in 3D activation tensor space $\mathcal{X} \in \mathbb{R}^{B\times T\times D}$. (Left) Batch Normalization computes statistics across all batch instances B for each feature channel d, creating inter-sample dependencies and breaking down under variable sequence lengths and sequence padding. (Right) Layer Normalization computes statistics across all feature channels D for each token vector $\mathbf{x}_{b,t}$ independently, ensuring strict batch-size invariance and identical execution during training and autoregressive inference.

Definition 5.3: *Layer Normalization*

For a token representation $\mathbf{x} = [x_1, x_2, \ldots, x_d]^\top \in \mathbb{R}^d$, Layer Normalization evaluates:

$$\mu = \frac{1}{d}\sum_{j=1}^{d} x_j, \quad \sigma^2 = \frac{1}{d}\sum_{j=1}^{d}(x_j - \mu)^2, \tag{5.72}$$

$$\texttt{LayerNorm}(\mathbf{x}) = \frac{\mathbf{x} - \mu}{\sqrt{\sigma^2 + \epsilon}} \odot \boldsymbol{\gamma} + \boldsymbol{\beta}, \tag{5.73}$$

where $\epsilon > 0$ is a small numerical stabilization constant, and $\boldsymbol{\gamma}, \boldsymbol{\beta} \in \mathbb{R}^d$ are learnable affine gain and bias parameters initialized to $\boldsymbol{\gamma} = \mathbf{1}$ and $\boldsymbol{\beta} = \mathbf{0}$.

Because Layer Normalization operates on a per-token basis without batch-level pooling, its behavior is identical during training and evaluation, providing exceptional stability across variable-length sequences.

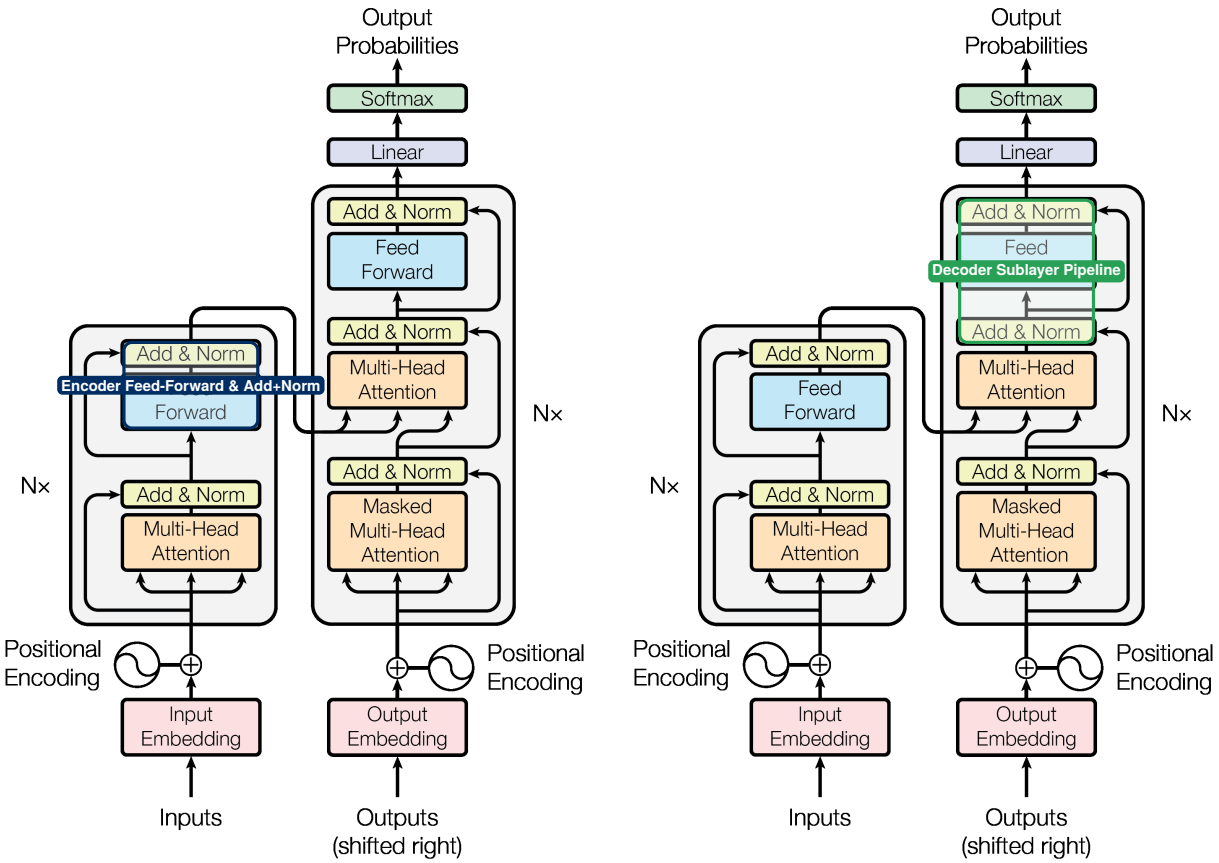
3. Position-wise Feed-Forward Networks (FFN). In addition to attention sublayers, each layer in the encoder and decoder contains a fully connected **Feed-Forward Network** applied to each position separately and identically:

$$\texttt{FFN}(\mathbf{x}) = \max(0, \mathbf{x}\mathbf{W}_1 + \mathbf{b}_1)\,\mathbf{W}_2 + \mathbf{b}_2, \tag{5.74}$$

where $\mathbf{W}_1 \in \mathbb{R}^{d_{\text{model}} \times d_{\text{ff}}}$, $\mathbf{b}_1 \in \mathbb{R}^{d_{\text{ff}}}$, $\mathbf{W}_2 \in \mathbb{R}^{d_{\text{ff}} \times d_{\text{model}}}$, and $\mathbf{b}_2 \in \mathbb{R}^{d_{\text{model}}}$. Typically, the inner hidden dimension is expanded by a factor of 4 ($d_{\text{ff}} = 4d_{\text{model}}$, e.g., $d_{\text{ff}} = 2048$ for $d_{\text{model}} = 512$). While attention routes information across different temporal positions, the FFN provides non-linear feature transformation within each individual position.

5.7.1 End-to-end sublayer integration: the complete mathematical pipeline

Having analyzed Multi-Head Attention, residual skip connections, Layer Normalization, and Feed-Forward Networks individually, we now formalize how these mathematical operators chain together within complete Transformer layers, tracing their exact locations in the canonical architecture (Figure 5.12).



(a) Encoder layer stack (b) Decoder layer stack

Figure 5.12: End-to-end sublayer connectivity within the canonical Transformer architecture [1]. Every attention and feed-forward block is coupled with an identity residual shortcut and an "Add & Norm" operator.

1. Complete Encoder layer pipeline. An encoder layer $l \in \{1, \ldots, N\}$ receives tensor $\mathbf{H}_{\text{enc}}^{(l-1)} \in \mathbb{R}^{L_{\text{enc}} \times d_{\text{model}}}$ (with $\mathbf{H}_{\text{enc}}^{(0)} = \mathbf{X}_{\text{enc}}$) and applies two composite sublayers:

1. **Sublayer 1 (Bidirectional Multi-Head Self-Attention)**:

$$\mathbf{Z}^{(l)} = \texttt{MultiHead}\left(\mathbf{H}_{\text{enc}}^{(l-1)}, \mathbf{H}_{\text{enc}}^{(l-1)}, \mathbf{H}_{\text{enc}}^{(l-1)}\right), \tag{5.75}$$

$$\tilde{\mathbf{H}}_{\text{enc}}^{(l)} = \texttt{LayerNorm}\left(\mathbf{H}_{\text{enc}}^{(l-1)} + \mathbf{Z}^{(l)}\right). \tag{5.76}$$

2. **Sublayer 2 (Position-wise Feed-Forward Network)**:

$$\mathbf{F}^{(l)} = \texttt{FFN}\left(\tilde{\mathbf{H}}^{(l)}_{\text{enc}}\right), \tag{5.77}$$

$$\mathbf{H}^{(l)}_{\text{enc}} = \texttt{LayerNorm}\left(\tilde{\mathbf{H}}^{(l)}_{\text{enc}} + \mathbf{F}^{(l)}\right). \tag{5.78}$$

2. Complete Decoder layer pipeline. A decoder layer $l \in \{1, \ldots, N\}$ receives target representation $\mathbf{H}^{(l-1)}_{\text{dec}} \in \mathbb{R}^{L_{\text{dec}} \times d_{\text{model}}}$ along with the terminal encoder output $\mathbf{H}^{(N)}_{\text{enc}} \in \mathbb{R}^{L_{\text{enc}} \times d_{\text{model}}}$, executing three consecutive sublayers:

1. **Sublayer 1 (Causal Masked Multi-Head Self-Attention)**:

$$\mathbf{Z}^{(l)}_{\text{self}} = \texttt{MaskedMultiHead}\left(\mathbf{H}^{(l-1)}_{\text{dec}}, \mathbf{H}^{(l-1)}_{\text{dec}}, \mathbf{H}^{(l-1)}_{\text{dec}}\right), \tag{5.79}$$

$$\mathbf{H}^{(l)}_{\text{dec},1} = \texttt{LayerNorm}\left(\mathbf{H}^{(l-1)}_{\text{dec}} + \mathbf{Z}^{(l)}_{\text{self}}\right). \tag{5.80}$$

2. **Sublayer 2 (Encoder-Decoder Multi-Head Cross-Attention)**:

$$\mathbf{Z}^{(l)}_{\text{cross}} = \texttt{MultiHead}\left(\mathbf{H}^{(l)}_{\text{dec},1}, \mathbf{H}^{(N)}_{\text{enc}}, \mathbf{H}^{(N)}_{\text{enc}}\right), \tag{5.81}$$

$$\mathbf{H}^{(l)}_{\text{dec},2} = \texttt{LayerNorm}\left(\mathbf{H}^{(l)}_{\text{dec},1} + \mathbf{Z}^{(l)}_{\text{cross}}\right). \tag{5.82}$$

   Here, queries are projected from $\mathbf{H}^{(l)}_{\text{dec},1}$, while keys and values are projected from $\mathbf{H}^{(N)}_{\text{enc}}$.

3. **Sublayer 3 (Position-wise Feed-Forward Network)**:

$$\mathbf{F}^{(l)}_{\text{dec}} = \texttt{FFN}\left(\mathbf{H}^{(l)}_{\text{dec},2}\right), \tag{5.83}$$

$$\mathbf{H}^{(l)}_{\text{dec}} = \texttt{LayerNorm}\left(\mathbf{H}^{(l)}_{\text{dec},2} + \mathbf{F}^{(l)}_{\text{dec}}\right). \tag{5.84}$$

Post-LN vs. Pre-LN parameter stabilization. The formulation above represents the original **Post-LN** architecture of [1], where normalization is placed on the residual sum: $\mathbf{y} = \texttt{LN}(\mathbf{x} + \mathcal{F}(\mathbf{x}))$. While mathematically elegant, deep Post-LN models require delicate learning rate warm-up schedules because gradients passing through the unnormalized residual additions can destabilize early training. Modern implementations (such as GPT-2, LLaMA, and Mistral) predominantly adopt the **Pre-LN** formulation:

$$\mathbf{y} = \mathbf{x} + \mathcal{F}\left(\texttt{LayerNorm}(\mathbf{x})\right), \tag{5.85}$$

which preserves an unbroken, completely unnormalized identity shortcut throughout the entire depth of the network, enabling stable optimization without learning rate warm-up.

5.8 Relative positional representations

In Chapter 3, we introduced absolute positional encodings—both fixed sinusoidal functions [1] and learned absolute embedding vectors. Under absolute encoding, each index in the sequence ($pos = 0, 1, 2, \ldots$) is assigned a dedicated vector added directly to the token embedding before entering the Transformer.

Motivation for relative representations. In natural language and software engineering source code, the precise absolute coordinate of a token is frequently less important than its **relative displacement** with respect to other tokens. For example, whether a verb appears at index 12 or index 84, its grammatical subject is typically found 2 or 3 tokens earlier. To incorporate this inductive bias directly into the attention mechanism, Shaw et al. [49] formulated **Relative Positional Representations**.

Mathematical formulation. Instead of adding absolute vectors to the input, relative positional embeddings inject pairwise distance information directly into the scaled dot-product attention logits :

$$\texttt{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top + \mathbf{S}_{\text{rel}}}{\sqrt{d_k}}\right)\mathbf{V}, \tag{5.86}$$

where $\mathbf{S}_{\text{rel}} \in \mathbb{R}^{L\times L}$ is a relative position bias matrix. For query position i and key position j, the entry is defined by:

$$S_{\text{rel}}[i, j] = \mathbf{q}_i^\top \mathbf{a}^K_{\text{clip}(i-j,k)}, \tag{5.87}$$

where $\mathbf{a}^K$ is a learned embedding representing the relative offset $i - j$, and $\text{clip}(x, k) = \max(-k, \min(k, x))$ truncates relative distances beyond a maximum clipping horizon k.

Asymmetry of relative positional embeddings: self-attention vs. cross-attention. Relative positional embeddings make sense exclusively for self-attention, and cannot be naturally defined for cross-attention:

- **Self-attention**: Both queries and keys originate from the same sequence. The scalar distance $i - j$ represents a physical, well-defined offset within that sequence (e.g., token j occurs 2 steps before token i).
- **Cross-attention**: Queries stem from the target sequence (e.g., generated English words in machine translation), while keys stem from the source sequence (e.g., input Italian words). Because source and target sentences possess different lengths, word orders, and grammatical alignments, there is no shared coordinate frame: asking whether an English target word is "three positions ahead" of an Italian source word is mathematically meaningless. Consequently, cross-attention cannot leverage relative positional biases.

5.9 Parallelization advantages and computational complexity

To appreciate why the Transformer superseded Recurrent Neural Networks, we analyze three foundational computational metrics: computational complexity per layer, sequential operations, and maximum signal path length [1], summarized in Table 5.1.

Table 5.1: Computational complexity and operational parallelism across sequence modeling layers [1].

Layer type	Complexity per layer	Sequential operations	Maximum path length
Self-Attention	$O(L^2 \cdot d)$	$O(1)$	$O(1)$
Recurrent (RNN/LSTM)	$O(L \cdot d^2)$	$O(L)$	$O(L)$
Convolutional	$O(k \cdot L \cdot d^2)$	$O(1)$	$O(\log_k(L))$

1. Sequential operations: shattering the recurrence barrier. In an RNN, computing the representation at step L requires traversing an unbroken chain of L sequential state updates ($O(L)$ sequential operations). In Self-Attention, computing all pairwise dot products $\mathbf{QK}^\top$ and value weighted sums involves dense matrix multiplications that execute in a single parallel step ($O(1)$ sequential operations) across GPU tensor cores.

2. Maximum path length: constant-time gradient propagation. In recurrent networks, information and backpropagation gradients traveling between token 1 and token L must traverse L intermediate state transitions. As proven by the Jacobian product $\prod_{k=t+1}^{T} \frac{\partial \mathbf{h}_k}{\partial \mathbf{h}_{k-1}}$ in Chapter 3, long path lengths cause exponential gradient decay. In Self-Attention, the distance between any arbitrary pair of tokens (i, j) is strictly $O(1)$: the attention weight $\alpha_{i,j}$ directly links token i to token j in a single operation. This constant path length completely eliminates topological distance as an obstacle to learning long-range dependencies.

3. Asymptotic trade-off: sequence length vs. latent dimension. Self-attention incurs a quadratic time and memory complexity $O(L^2 \cdot d)$ with respect to sequence length L. When sequence length L is smaller than embedding dimension d (as was typical in early machine translation benchmarks where $L \approx 50$ and $d = 512$), $O(L^2 \cdot d)$ is strictly faster than the recurrent cost $O(L \cdot d^2)$. While long context windows ($L > 8192$) eventually encounter the $O(L^2)$ memory wall, the massive parallel speedup during training outweighs this cost for standard operational contexts.

5.10 The full Encoder-Decoder paradigm: T5

Following the introduction of the unified Encoder-Decoder Transformer in [1], the machine learning community recognized that many tasks do not require both components simultaneously. Maintaining both stacks incurs unnecessary parameter bloat and memory overhead when the target problem involves either pure sequence transduction with distinct input and output vocabularies, or pure feature representation and bidirectional sequence understanding. Consequently, the architecture bifurcated into specialized paradigms: full Encoder-Decoder architectures and pure Encoder-Only architectures. (We will address pure autoregressive Decoder-Only architectures in the subsequent chapter dedicated to large-scale autoregressive pre-training and scaling laws).

5.10.1 The unified text-to-text framework

The full Encoder-Decoder architecture was revitalized by Raffel et al. [55] through the **Text-to-Text Transfer Transformer (T5)** (Figure 5.13).

1. The unified text-to-text framing. Rather than designing separate network architectures for different NLP tasks (e.g., classification heads for sentiment analysis, sequence taggers for named entity recognition, and autoregressive decoders for summarization), T5 casts **every** problem into an identical format: text-to-text sequence generation. The input string encodes both the task directive and the input data via a prepended **task prefix**:

- **Machine translation**: `"translate English to German: That is good."` ⟶ Output: `"Das ist gut."`
- **Summarization**: `"summarize: State authorities dispatched emergency crews..."` ⟶ Output: `"Crews responded to flood alerts."`

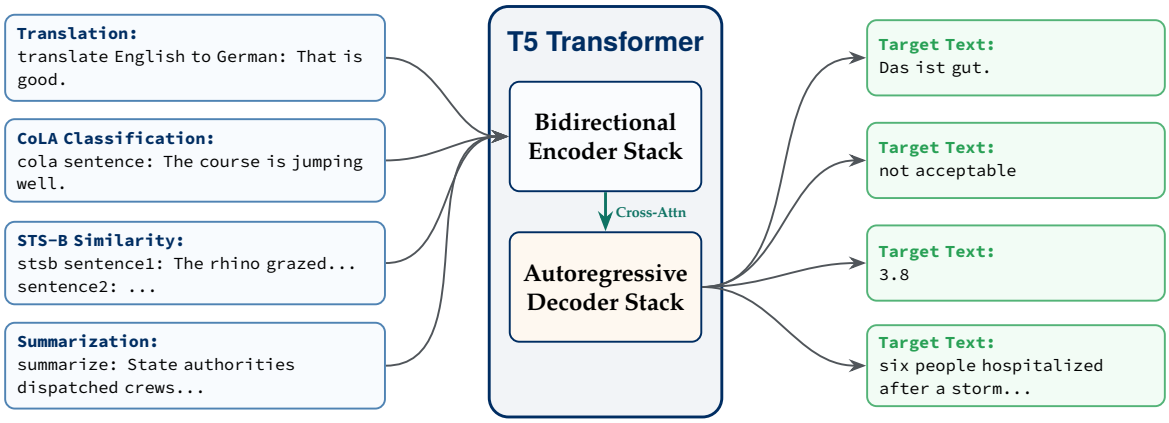


Figure 5.13: The T5 unified text-to-text framework [55]. Every natural language problem—including machine translation, linguistic acceptability (CoLA), sentence similarity scoring (STS-B), and document summarization—is framed as receiving a task-conditioned text prompt with a fixed prefix and autoregressively generating a target text string.

- **Linguistic acceptability (CoLA)**: `"cola sentence: The course is great."` ⟶ Output: `"acceptable"`

2. Critical distinction: task conditioning vs. instruction tuning. While T5's prefix-based behavior superficially resembles modern conversational assistants, **T5 is not instruction-tuned**. T5 was trained strictly on fixed, rigid task prefixes. The model learns to recognize these specific keywords and route its internal representations accordingly. However, it exhibits no zero-shot generalization to unseen instructions or altered formulations:

- A slightly verbose variation, such as `"Can you translate English to German the following sentence: That is good"`, causes the model to break down and emit ungrammatical gibberish.
- Prompts requesting novel reasoning, such as `"Compute 2 + 2"`, fail completely because arithmetic was not included in its pre-training task templates.

5.10.2 Trade-offs of full encoder-decoder architectures

While the unified encoder-decoder architecture provides exceptional flexibility for arbitrary sequence transduction, maintaining both an encoder and an autoregressive decoder introduces significant computational and representational overhead when the downstream task does not involve sequence generation. In sequence classification, token tagging, or representation retrieval, running an autoregressive decoder step-by-step is completely wasteful. This motivates isolating the encoder stack into a standalone, deeply bidirectional architecture.

5.11 The Encoder-Only paradigm: BERT (Bidirectional Encoder Representations from Transformers)

When the objective is sequence understanding, document classification, or semantic code search rather than autoregressive text generation, the decoder stack is completely superfluous. Devlin et al. [48] isolated the bidirectional encoder stack to create **BERT** (Figure 5.14).

To master BERT from first principles, it is essential to systematically decompose and analyze each of its core architectural and training components:

1. **Input representation**: Why we must analyze the tripartite embedding construction (Token, Segment, Position) and special tokens (`[CLS]`, `[SEP]`), which enables a single unified Transformer stack to ingest either isolated sentences or structured sentence pairs without architectural modifications.
2. **Masked Language Model (MLM)**: Why we must analyze bidirectional Cloze pre-training, resolving the fundamental paradox that standard autoregressive next-token prediction causes trivial label leakage in bidirectional self-attention networks.
3. **Next Sentence Prediction (NSP)**: Why intra-sentence bidirectional token conditioning alone is insufficient for discourse-level relational reasoning, requiring explicit cross-sentence relationship pre-training.
4. **Attention map anatomy and emergent linguistic patterns**: Why we must inspect the empirical behavior across the 144 attention heads, discovering how self-attention spontaneously organizes into syntactic dependency trackers, coreference resolvers, and delimiter registers without supervised parse trees.
5. **Downstream fine-tuning mechanisms**: Why we must formalize the adaptation mechanics (sequence-level pooling vs. token-level classification heads), demonstrating how pre-trained bidirectional representations transfer to diverse task topologies with minimal task-specific parameters.

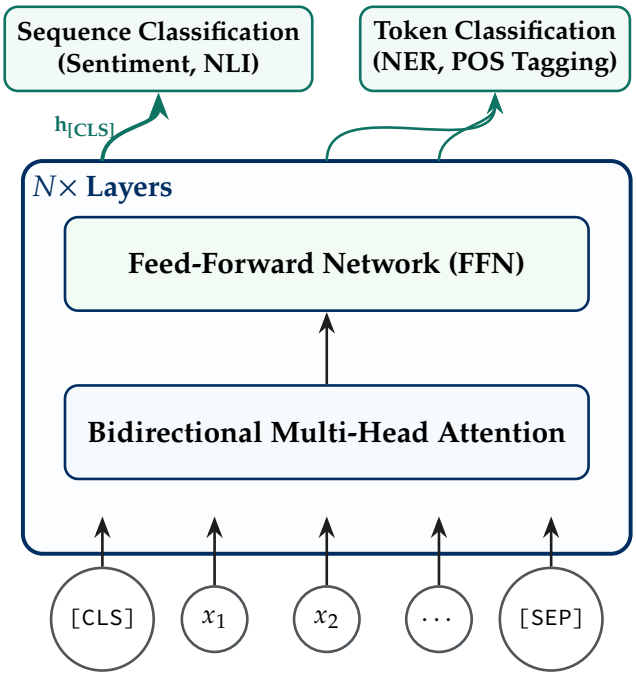


Figure 5.14: The Encoder-only architecture (BERT) [48]. The autoregressive decoder is discarded. Stacked bidirectional Transformer encoder layers process the full context simultaneously. Downstream tasks branch directly off the contextual representations: sequence-level tasks utilize the pooled $\mathbf{h}_{[\text{CLS}]}$ representation, while token-level tasks attach classification heads to individual token outputs $\mathbf{h}_i$.

5.11.1 Input representation and special tokens

Motivation for analyzing input representation. A central challenge in designing a universal sequence understanding backbone is handling diverse downstream input topologies: single

text sequences (e.g., sentiment analysis, spam detection), paired sequences with asymmetric roles (e.g., Question Answering premise-hypothesis pairs), and span-level entities. Rather than designing disparate input pipelines for each task type, BERT unifies all inputs into a single continuous stream. Analyzing this construction reveals how lexical identity, sentence membership, and ordinal sequence position are simultaneously communicated to permutation-invariant self-attention layers.

BERT supports pairs of sentences (A, B) packed into a single sequence (Figure 5.15). Every input token representation is constructed by summing three distinct embedding vectors:

$$\mathbf{x}_i = \texttt{TokenEmbedding}(w_i) + \texttt{SegmentEmbedding}(s_i) + \texttt{PositionEmbedding}(i), \tag{5.88}$$

where:

- **Token embeddings**: Subword tokens derived from WordPiece tokenization (with vocabulary cardinality $|\mathcal{V}| = 30{,}522$).
- **Segment embeddings**: Distinguishes whether the token belongs to Sentence A (E_A) or Sentence B (E_B). This allows self-attention queries to identify cross-sentence relationships and apply sentence-specific relational transformations.
- **Position embeddings**: Learned absolute positional vectors $(E_0, \ldots, E_{511})$ that inject spatial sequence ordering.

BERT introduces two canonical special tokens:

- `[CLS]` (**Classification**): Always prepended as the very first token ($i = 0$). Because self-attention is bidirectional, the final hidden state $\mathbf{h}_{[\text{CLS}]} \in \mathbb{R}^{d_{\text{model}}}$ aggregates context from the entire sequence and serves as the pooled representation for sentence-level classification.
- `[SEP]` (**Separator**): Appended to mark the boundary between Sentence A and Sentence B, as well as the termination of the sequence.

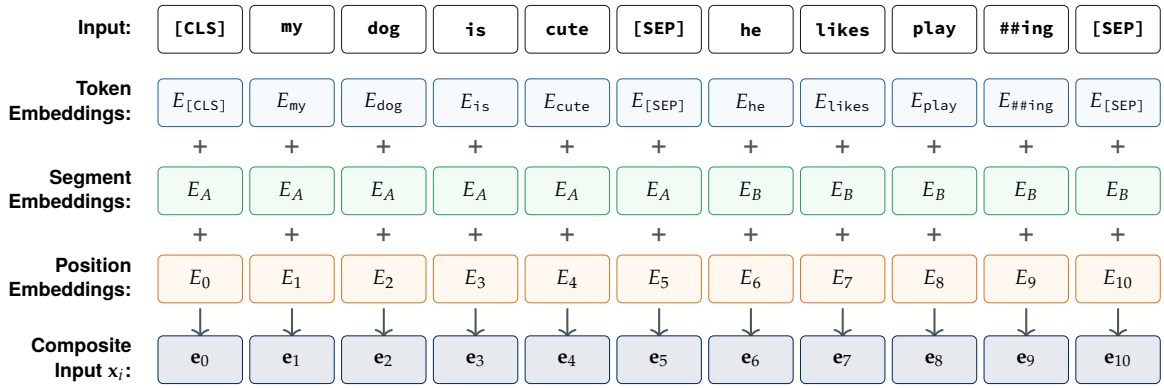


Figure 5.15: BERT input representation structure [48]. Each input vector $\mathbf{e}_i$ is constructed by the element-wise sum of its Token Embedding (WordPiece vocabulary), Segment Embedding (E_A for Sentence A, E_B for Sentence B), and Position Embedding $(E_0, \ldots, E_{10})$. Sequences begin with `[CLS]` and separate sentences with `[SEP]`. Distinct learned segment embeddings allow the self-attention mechanism to discriminate between paired sentences.

5.11.2 Pre-training task 1: Masked Language Model (MLM)

Motivation for analyzing the MLM objective. Analyzing the MLM objective is crucial because bidirectional representations cannot be pre-trained using standard autoregressive next-token

prediction. In a bidirectional self-attention model, token i can attend to all tokens $j > i$. If tasked with predicting x_{i+1}, the network would simply route attention to the next position and trivial identity-copy the answer, collapsing gradient learning. The MLM task formulates a non-trivial, label-leakage-free self-supervised objective inspired by the Cloze procedure [48].

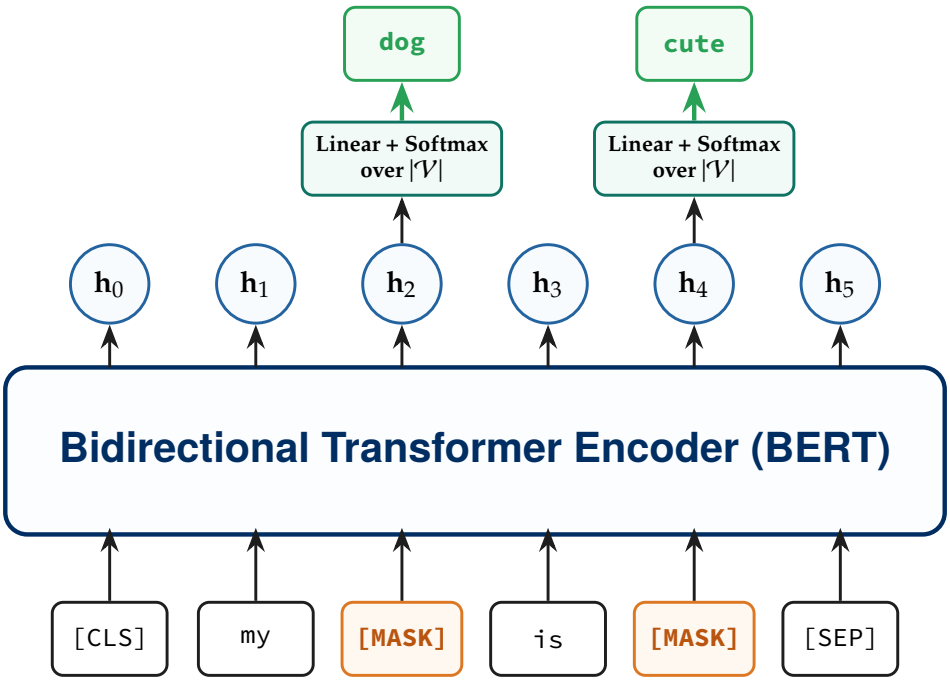


Figure 5.16: The Masked Language Model (MLM) pre-training task in BERT. 15% of input tokens are corrupted using the 80/10/10 masking rule. The model passes corrupted sequences through bidirectional self-attention and trains linear classification heads over the full vocabulary $\mathcal{V}$ specifically on hidden representations $\mathbf{h}_i$ at masked positions to reconstruct the original ground-truth tokens.

In each input sequence, a subset $\mathcal{M} \subset \{1, 2, \ldots, L\}$ representing approximately 15% of all input tokens is uniformly selected to provide pre-training supervision. However, how should these selected positions be presented to the network?

The general motivation: resolving the pre-training vs. fine-tuning mismatch. The naive Cloze procedure would simply replace every selected token in $\mathcal{M}$ with the artificial marker `[MASK]`. While conceptually straightforward, this naive strategy introduces a fatal structural flaw known as the **pre-training vs. fine-tuning discrepancy** (or domain shift):

1. **Absence of `[MASK]` in downstream tasks**: When BERT is subsequently fine-tuned on real-world applications (e.g., sequence classification, sentiment analysis, named entity recognition, or question answering), the input text consists exclusively of genuine natural language tokens. The artificial delimiter `[MASK]` *never appears during inference*.

2. **Pathological representation specialization**: If the encoder were exposed to `[MASK]` whenever a prediction was required, its internal self-attention heads would learn a degenerative heuristic: treat all regular words as unquestioned, static context, and only activate deep predictive feature extraction when encountering the explicit `[MASK]` token. Consequently, during fine-tuning, the network would fail to generate optimal contextual representations for normal tokens.

3. **Neglect of the observed token**: If target positions are always overwritten with a mask

symbol, the model never learns to evaluate or preserve the identity of the token physically occupying position i.

To force the model to build high-quality contextual representations for *all* tokens—while keeping the model uncertain about which positions are being graded—Devlin et al. [48] formulated the **80/10/10 corruption policy**. Once the 15% target positions $\mathcal{M}$ are chosen, they are perturbed according to three distinct operations:

1. **The 80% replacement with `[MASK]` (Cloze semantic reconstruction)**: In 80% of the selected positions (12% of total sequence tokens), the original token is replaced with `[MASK]`:

$$\texttt{"my dog is cute"} \longrightarrow \texttt{"my [MASK] is cute.}$$

   Motivation: This forms the core Cloze engine. Because the ground-truth token is physically removed from the input stream, the model cannot copy it. It is compelled to synthesize bidirectional semantic and syntactic clues from the entire left and right contexts (e.g., using `"my"` and `"is cute"` to infer a domestic pet like `"dog"`).

2. **The 10% replacement with a random token (Denoising and contextual validation)**: In 10% of the selected positions (1.5% of total sequence tokens), the target token is replaced with an arbitrary token drawn uniformly from the vocabulary $w_{\text{rand}} \sim \mathcal{V}$:

$$\texttt{"my dog is cute"} \longrightarrow \texttt{"my apple is cute.}$$

   Motivation: If the model only ever saw `[MASK]` or correct words, it would assume that every non-masked word in the sentence is guaranteed to be 100% truthful and contextually valid. By injecting random corrupted words, self-attention heads are forced to act as an active **anomaly detector** and **denoiser**. When the encoder encounters `"apple"` framed by `"my"` and `"is cute"`, it must detect that `"apple"` creates a semantic contradiction, discount the corrupted surface form, and reconstruct the true underlying concept (`"dog"`). This prevents the network from blindly trusting unmasked tokens.

3. **The 10% retention of the original token (Grounding and representation anchoring)**: In the remaining 10% of selected positions (1.5% of total sequence tokens), the token is kept strictly unchanged:

$$\texttt{"my dog is cute"} \longrightarrow \texttt{"my dog is cute.}$$

   Motivation: If target positions were *always* altered (either masked or replaced with random noise), the encoder would learn that the input symbol at target position i is never the right answer, incentivizing it to completely discard the token's own embedding. By leaving the correct word intact 10% of the time, the model is taught that the observed word may well be genuine. This forces the final hidden representation $\mathbf{h}_i$ to remain faithfully anchored to the actual lexical token observed at that position, directly bridging the gap to downstream fine-tuning where all input tokens are real.

The global awareness effect. Because of this tri-fold distribution, when BERT computes self-attention across any token x_i, it operates under total uncertainty:

- The network does not know whether token x_i is one of the supervised targets being evaluated by cross-entropy loss.
- If x_i is a target, the network does not know whether it represents an uncorrupted word, an anomalous impostor, or an explicit mask.

This deliberate ambiguity eliminates any possibility of lazy shortcut heuristics, compelling the Transformer encoder to maintain an expressive, contextually verified representation $\mathbf{h}_i$ for **every single token** across the entire sequence.

For each selected position $i \in \mathcal{M}$, the topmost encoder output $\mathbf{h}_i \in \mathbb{R}^{d_{\text{model}}}$ is projected onto the full vocabulary $\mathcal{V}$ through a linear classification layer parameterized by $\mathbf{W}_{\text{MLM}} \in \mathbb{R}^{|\mathcal{V}| \times d_{\text{model}}}$ (whose weights are tied with the input token embedding matrix $\mathbf{E}_{\text{tok}}$) and bias $\mathbf{b}_{\text{MLM}} \in \mathbb{R}^{|\mathcal{V}|}$:

$$P(w \mid \tilde{\mathbf{X}}; \theta) = \frac{\exp(\mathbf{w}_w^\top \mathbf{h}_i + b_w)}{\sum_{v \in \mathcal{V}} \exp(\mathbf{w}_v^\top \mathbf{h}_i + b_v)}, \quad \forall w \in \mathcal{V}, \tag{5.89}$$

where $\tilde{\mathbf{X}}$ denotes the corrupted input sequence. The MLM pre-training objective minimizes the cross-entropy loss aggregated across all masked positions:

$$\mathcal{L}_{\text{MLM}}(\theta) = -\sum_{i \in \mathcal{M}} \log P(w_i^* \mid \tilde{\mathbf{X}}; \theta), \tag{5.90}$$

where w_i^* is the original ground-truth token at position i.

5.11.3 Pre-training task 2: Next Sentence Prediction (NSP)

Motivation for analyzing the NSP objective. While the Masked Language Model objective forces the encoder to capture deep intra-sentence lexical and syntactic relationships, many critical downstream tasks—such as Question Answering (QA), Natural Language Inference (NLI), and code-comment retrieval—depend fundamentally on understanding logical coherence across *pairs* of sentences. Analyzing NSP explains how the network develops cross-sentence relational awareness, and motivates why subsequent architectures (e.g., RoBERTa) later audited whether NSP was truly necessary or suboptimal.

During corpus pre-training, input sequences are constructed by sampling pairs of sentences (A, B) concatenated with segment markers (Figure 5.17):

- **50% of the time (Label $y = 1$, IsNext)**: Sentence B is the actual contiguous sentence immediately succeeding Sentence A in the source document.
- **50% of the time (Label $y = 0$, NotNext)**: Sentence B is a random sentence sampled independently from an entirely unrelated document in the pre-training corpus.

To classify the sentence pair without introducing an ad-hoc pooling mechanism, BERT leverages the special leading token representation $\mathbf{h}_{[\text{CLS}]} \in \mathbb{R}^{d_{\text{model}}}$. Because self-attention is bidirectional, $\mathbf{h}_{[\text{CLS}]}$ computes attention over every token in both Sentence A and Sentence B. This pooled vector is projected through a non-linear dense transformation:

$$\mathbf{h}_{\text{pool}} = \tanh(\mathbf{W}_{\text{pool}} \mathbf{h}_{[\text{CLS}]} + \mathbf{b}_{\text{pool}}), \quad \mathbf{W}_{\text{pool}} \in \mathbb{R}^{d_{\text{model}} \times d_{\text{model}}}, \ \mathbf{b}_{\text{pool}} \in \mathbb{R}^{d_{\text{model}}}. \tag{5.91}$$

The binary prediction probability is emitted by a logistic regression head:

$$\hat{y}_{\text{NSP}} = \sigma(\mathbf{w}_{\text{NSP}}^\top \mathbf{h}_{\text{pool}} + b_{\text{NSP}}), \quad \mathbf{w}_{\text{NSP}} \in \mathbb{R}^{d_{\text{model}}}, \ b_{\text{NSP}} \in \mathbb{R}, \tag{5.92}$$

and evaluated under the binary cross-entropy loss:

$$\mathcal{L}_{\text{NSP}}(\theta) = -y \log \hat{y}_{\text{NSP}} - (1 - y) \log(1 - \hat{y}_{\text{NSP}}). \tag{5.93}$$

The unified pre-training objective jointly optimizes the composite loss across all sampled sequences:

$$\mathcal{L}_{\text{BERT}}(\theta) = \mathcal{L}_{\text{MLM}}(\theta) + \mathcal{L}_{\text{NSP}}(\theta). \tag{5.94}$$

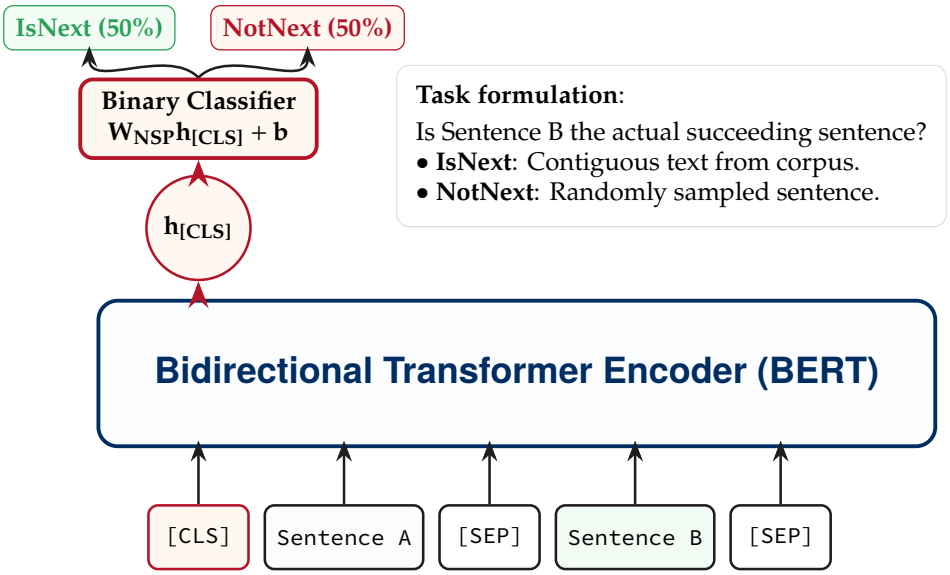


Figure 5.17: The Next Sentence Prediction (NSP) pre-training task in BERT. Two sentences are packed with delimiters `[CLS]` Sentence A `[SEP]` Sentence B `[SEP]`. The pooled final representation $\mathbf{h}_{[\text{CLS}]}$ is projected through a binary classification head $\mathbf{W}_{\text{NSP}}$ to predict whether Sentence B logically follows Sentence A.

5.11.4 Attention map anatomy and emergent linguistic structures

Motivation for analyzing attention maps. Analyzing internal attention maps demystifies the "black box" of deep Transformer encoders. It allows researchers to verify whether the statistical optimizations of MLM and NSP genuinely induce human-like linguistic and syntactic competencies, or whether the model merely relies on shallow statistical surface correlations. By visualizing the attention matrices $\mathbf{A}^{(l,h)}$, we observe how individual attention heads autonomously specialize into interpretable linguistic operators.

A standard BERT-Base model comprises $L = 12$ stacked encoder layers with $h = 12$ attention heads per layer, operating across a hidden dimension of $d_{\text{model}} = 768$ ($d_k = d_{\text{model}}/h = 64$). Consequently, for any input sequence, the network computes:

$$12 \times 12 = 144 \text{ distinct attention heads (and corresponding attention maps).} \tag{5.95}$$

For an input sequence of length N, each attention head generates an $N \times N$ row-stochastic attention weight matrix $\mathbf{A}^{(l,h)} \in [0,1]^{N \times N}$ satisfying $\sum_{j=1}^{N} A_{ij}^{(l,h)} = 1$.

Consider the canonical linguistic probe evaluated in [56]:

`[CLS] The dog ate the food because it was hungry [SEP]` ($N = 11$ tokens).

Across the 144 attention maps, empirical probing reveals that self-attention layers autonomously partition into highly specialized linguistic registers without explicit syntactic supervision (Figure 5.18):

1. **Coreference and anaphora resolution (intermediate layers, e.g., Layer 5, Head 4)**: Specific attention heads in the middle layers specialize in resolving pronominal anaphora across clausal boundaries. In the running example, when computing attention from the pronoun query token `"it"`, the head routes over 82% of its total attention mass directly to the antecedent noun key `"dog"` ($\alpha = 0.82$). This allows the contextual representation of `"it"` to absorb the semantic properties of the entity it replaces.

2. **Syntactic dependency tracking (early-to-intermediate layers)**: Distinct heads capture grammatical relations: verb-object heads route strong attention from transitive verbs to

their direct objects (e.g., "ate" → "food" with $\alpha = 0.74$), determiner heads link articles to head nouns ("The" → "dog", "the" → "food"), and predicate heads connect adjectives to sentence subjects ("hungry" → "dog").

3. **Surface sequential tracking (early layers)**: In early layers, several heads attend predominantly to adjacent tokens ($i \to i-1$ or $i \to i+1$), preserving localized n-gram ordering within the permutation-invariant self-attention matrix.

4. **The delimiter and neutral register phenomenon (deep layers, e.g., Layer 10, Head 8)**: In deeper layers, Clark et al. [56] and Michel et al. [57] observed that multiple attention heads allocate between 50% and 90% of their total attention weight directly to the special delimiter token [SEP]. This behavior arises from a fundamental mathematical constraint: the softmax operator strictly enforces $\sum_{j=1}^{N} A_{ij} = 1$. When a specialized attention head (such as a coreference head) processes a token that does not participate in that specific linguistic relation (e.g., the conjunction "because" or punctuation tokens), the head must still distribute a probability mass of 1.0. Distributing this mass across arbitrary semantic words would inject irrelevant noise and distort their representations. Therefore, the network designates [SEP] as a **neutral default register / "no-op" target**: unassigned attention is dumped onto [SEP], leaving the contextual representation of the source token virtually unaltered ($\Delta \mathbf{h} \approx \mathbf{0}$).

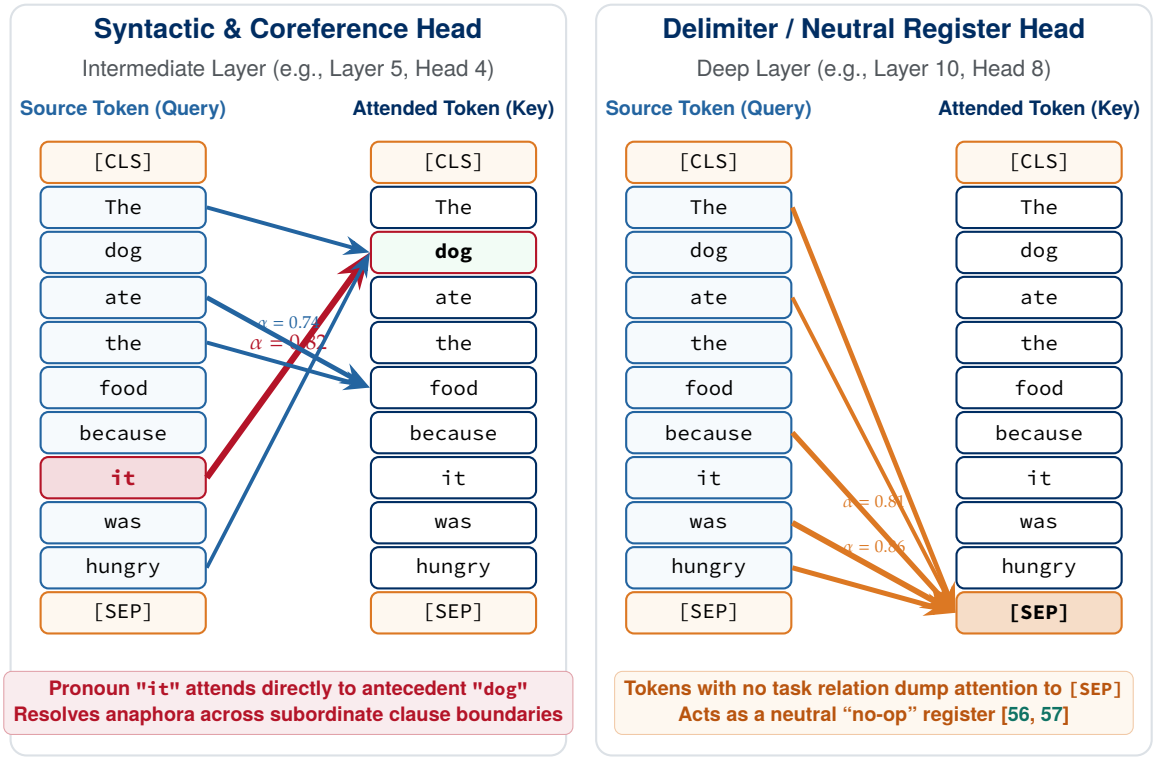


Figure 5.18: Emergent linguistic specialization and delimiter behavior across BERT attention heads for the input sequence "[CLS] The dog ate the food because it was hungry [SEP]" (11 tokens, evaluated across $12 \times 12 = 144$ attention maps). Left panel: An intermediate-layer head (Layer 5, Head 4) resolves anaphoric coreference by routing strong attention from pronoun "it" to antecedent "dog" ($\alpha = 0.82$), alongside verb-object dependencies ("ate" → "food"). Right panel: A deep-layer head (Layer 10, Head 8) exhibits delimiter pooling where unassigned tokens dump over 80% of their attention mass onto [SEP], functioning as a neutral no-op register that leaves vector representations undisturbed [56, 57].

Empirical layer-wise emergence: from surface adjacency to syntax and delimiter registers. To deeply understand how representations transform across the encoder depth, we inspect the actual attention heatmaps across the entire network hierarchy (Figures 5.19 and 5.21). This reveals a pronounced, progressive stratification of linguistic functions across depth:

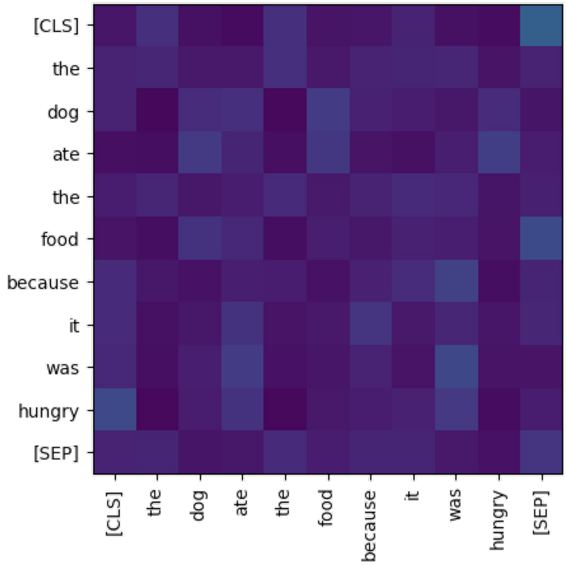


Figure 5.19: Attention map for Layer 1, Head 1 on "[CLS] The dog ate the food because it was hungry [SEP]". Early layer heads attend primarily to immediate neighboring tokens and local syntax (e.g., verb "ate" focusing attention on object "food"). Each row sums to 1.0.

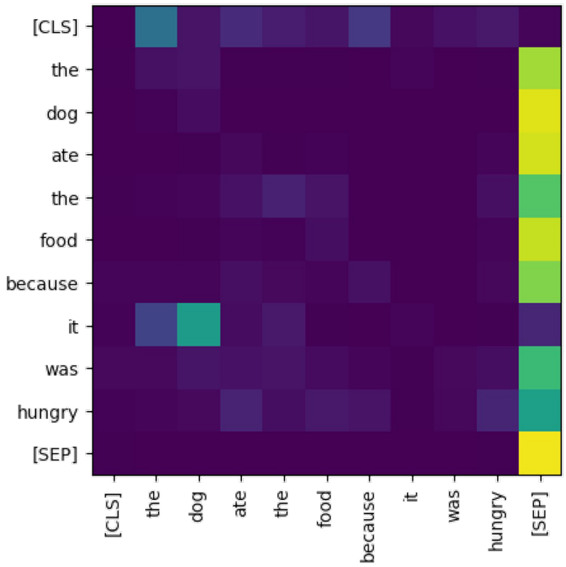


Figure 5.20: Intermediate-layer attention head resolving ambiguous pronominal coreference. Query token "it" allocates massive attention mass to antecedent "dog", allowing BERT to accurately bind pronominal entities across clausal boundaries.

The emergent patterns across the layers decompose into three canonical phases:

1. **Early layers (Layers 1–3: Surface and Local N-gram Geometry)**: As shown in Figure 5.19, attention weights are concentrated heavily along the main diagonal ($i \to i$) and adjacent off-diagonals ($i \to i-1$ and $i \to i+1$). The model first reconstructs sequential positional adjacency, functioning much like convolutional filters with small receptive fields. Furthermore, basic lexical dependencies begin forming (e.g., "ate" linking with "food").

2. **Middle layers (Layers 4–8: Syntactic Trees and Anaphora Resolution)**: In the intermediate layers, attention patterns become markedly sparse and structured (Figure 5.20). The heads no longer attend to adjacent words; instead, they bridge long-range syntactic gaps. Dedicated heads track verb-object dependencies, noun-modifier bindings, and pronoun coreferences across subordinate conjunctions ("it" $\to$ "dog"). This proves that the multi-head mechanism divides labor: different heads track different syntactic relations concurrently.

3. **Deep layers (Layers 9–12: High-Level Task Abstraction and Delimiter Pooling)**: In the final layers, attention distributions shift again. Many heads show broad, diffuse patterns, while a significant fraction dump 80%–90% of their attention mass onto the special token [SEP]. Because the final layers synthesize task-level features, tokens that have already aggregated their required syntactic context utilize [SEP] as an inert sink. This preserves their accumulated representation vectors without absorbing unwanted semantic drift in the topmost layers before classification or MLM loss computation.

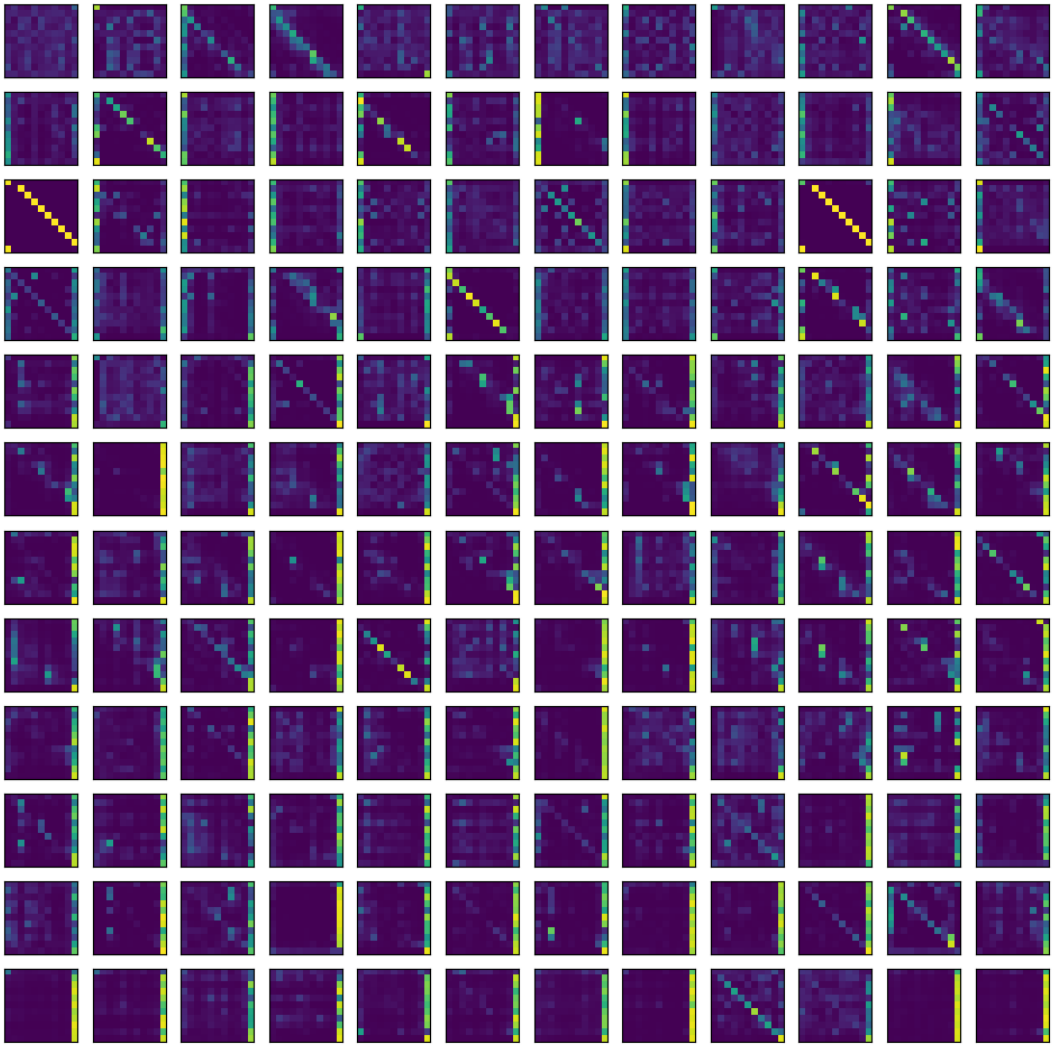


Figure 5.21: The complete landscape of $12 \times 12 = 144$ attention maps across BERT-Base for the 11-token sequence. Vertical axis denotes layer index (Layers 1–12 from top to bottom); horizontal axis denotes head index (Heads 1–12 from left to right). The grid visually confirms the progression: early layers track adjacent tokens and surface patterns; middle layers exhibit sparse, directed syntactic and coreference links; final layers dump attention mass onto the delimiter `[SEP]` as a neutral no-op register [56, 57].

5.11.5 Downstream fine-tuning mechanisms

Motivation for analyzing fine-tuning mechanisms. A pre-trained foundation model is only as valuable as its adaptability to real-world tasks. Analyzing BERT's downstream adaptation demonstrates the paradigm shift from task-specific feature engineering to task-agnostic representation fine-tuning. By modifying only the topmost prediction layer while allowing gradients to backpropagate through all 110 million parameters, BERT achieves state-of-the-art performance across diverse natural language tasks with minimal training time.

Following pre-training on MLM and NSP, the generic pre-training classification layers are discarded. The pre-trained Transformer weights are retained, and minimal task-specific projection heads are appended to adapt the model to downstream NLP applications with end-to-end gradient updates:

1. **Sequence-level classification** (e.g., sentiment analysis, code defect prediction, natural language inference): The pooled sentence representation $\mathbf{h}_{\text{pool}} \in \mathbb{R}^{d_{\text{model}}}$ from Equation (5.91) is projected through a task-specific linear head:

$$\hat{\mathbf{y}} = \text{softmax}(\mathbf{W}_{\text{seq}}\mathbf{h}_{\text{pool}} + \mathbf{b}_{\text{seq}}), \quad \mathbf{W}_{\text{seq}} \in \mathbb{R}^{C \times d_{\text{model}}}, \ \mathbf{b}_{\text{seq}} \in \mathbb{R}^{C}, \tag{5.96}$$

   where C denotes the number of target classes. All parameters (both $\mathbf{W}_{\text{seq}}$ and the encoder backbone) are fine-tuned via cross-entropy loss.

2. **Token-level sequence labeling** (e.g., Named Entity Recognition NER, Part-of-Speech tagging, code token vulnerability tagging): Instead of pooling, the final contextual representation $\mathbf{h}_i \in \mathbb{R}^{d_{\text{model}}}$ of each individual token $i \in \{1, 2, \ldots, L\}$ is mapped to a discrete label distribution:

$$\hat{\mathbf{y}}_i = \text{softmax}(\mathbf{W}_{\text{tok}}\mathbf{h}_i + \mathbf{b}_{\text{tok}}), \quad \mathbf{W}_{\text{tok}} \in \mathbb{R}^{K \times d_{\text{model}}}, \ \mathbf{b}_{\text{tok}} \in \mathbb{R}^{K}, \tag{5.97}$$

   where K is the number of entity or syntactic tags.

3. **Extractive Question Answering** (e.g., SQuAD): Conditioned on a passage and a question concatenated as Sentence A and Sentence B, the model identifies the continuous answer span by learning a start vector $\mathbf{w}_{\text{start}} \in \mathbb{R}^{d_{\text{model}}}$ and an end vector $\mathbf{w}_{\text{end}} \in \mathbb{R}^{d_{\text{model}}}$. The probability that token i is the start of the answer span is:

$$P_{\text{start}}(i) = \frac{\exp(\mathbf{w}_{\text{start}}^{\top}\mathbf{h}_i)}{\sum_{j=1}^{L} \exp(\mathbf{w}_{\text{start}}^{\top}\mathbf{h}_j)}, \tag{5.98}$$

   with an identical formulation for $P_{\text{end}}(i)$.

Architectural transition: from bidirectional representations to autoregressive generation. While the Encoder-Decoder paradigm addresses sequence transduction and the Encoder-Only paradigm excels at bidirectional representation learning and sequence classification, pure autoregressive generative tasks (such as open-ended text synthesis and conversational reasoning) rely on Decoder-Only architectures. We defer the comprehensive architectural analysis of large-scale Decoder-Only models to Chapter 6, where scaling laws and autoregressive pre-training are developed in full detail.

However, regardless of whether generation occurs within the decoder stack of an Encoder-Decoder model or a dedicated autoregressive language decoder, the fundamental problem of selecting discrete tokens from continuous output distributions remains mathematically invariant. In the subsequent section, we formalize the probabilistic mechanics governing autoregressive decoding.

5.12 Autoregressive decoding and sampling strategies

At each step t of autoregressive sequence generation, an autoregressive language decoder projects its topmost contextual hidden state $\mathbf{h}_t \in \mathbb{R}^{d_{\text{model}}}$ through an unembedding linear transformation $\mathbf{W}_U \in \mathbb{R}^{|\mathcal{V}| \times d_{\text{model}}}$ parameterized with vocabulary bias $\mathbf{b}_U \in \mathbb{R}^{|\mathcal{V}|}$. This maps the continuous latent manifold into raw unnormalized affinities over discrete token space:

$$\mathbf{z}_t = \mathbf{W}_U \mathbf{h}_t + \mathbf{b}_U \in \mathbb{R}^{|\mathcal{V}|}, \tag{5.99}$$

where $\mathbf{z}_t = [z_{t,1}, z_{t,2}, \ldots, z_{t,|\mathcal{V}|}]^\top \in \mathbb{R}^{|\mathcal{V}|}$ denotes the unnormalized log-affinity (logit) vector across the vocabulary $\mathcal{V} = \{v_1, v_2, \ldots, v_{|\mathcal{V}|}\}$. Passing $\mathbf{z}_t$ through the canonical softmax operator $\text{softmax} : \mathbb{R}^{|\mathcal{V}|} \to \Delta^{|\mathcal{V}|-1}$ maps the continuous unbounded logits onto the standard $(|\mathcal{V}| - 1)$-dimensional probability simplex $\Delta^{|\mathcal{V}|-1} = \{\mathbf{p} \in \mathbb{R}^{|\mathcal{V}|} : p_i \geq 0, \sum_{i=1}^{|\mathcal{V}|} p_i = 1\}$.

This defines the discrete categorical conditional probability mass function governing the next-token random variable $X_t \sim \text{Categorical}(\mathbf{p}_t)$:

$$\begin{aligned} P(X_t = v_i \mid w_{<t}) &= [\text{softmax}(\mathbf{z}_t)]_i \\ &= \frac{\exp(z_{t,i})}{\sum_{j=1}^{|\mathcal{V}|} \exp(z_{t,j})} \\ &= \frac{\exp\left(\mathbf{w}_{U,i}^\top \mathbf{h}_t + b_{U,i}\right)}{\sum_{j=1}^{|\mathcal{V}|} \exp\left(\mathbf{w}_{U,j}^\top \mathbf{h}_t + b_{U,j}\right)}, \quad \forall v_i \in \mathcal{V}, \end{aligned} \tag{5.100}$$

where $\mathbf{w}_{U,i}^\top \in \mathbb{R}^{1 \times d_{\text{model}}}$ denotes the i-th row vector of the unembedding matrix $\mathbf{W}_U$ (often tied with the input embedding matrix $\mathbf{E}_{\text{tok}}^\top$ in weight-tied models), and $b_{U,i} \in \mathbb{R}$ represents the corresponding vocabulary bias term. By construction, the probabilities satisfy the Kolmogorov axioms:

$$P(X_t = v_i \mid w_{<t}) \in (0, 1) \quad \text{and} \quad \sum_{i=1}^{|\mathcal{V}|} P(X_t = v_i \mid w_{<t}) = 1. \tag{5.101}$$

In modern language models, the vocabulary cardinality $|\mathcal{V}|$ spans tens of thousands of subword units (for example, $|\mathcal{V}| = 30{,}522$ in BERT's WordPiece, $|\mathcal{V}| = 50{,}257$ in GPT-2's Byte-Pair Encoding). Consequently, $\mathbf{p}_t$ represents a massive, highly non-uniform probability distribution over tens of thousands of discrete outcomes.

Once this dense probability distribution $P(w_i \mid w_{<t})$ is obtained, the fundamental challenge of autoregressive inference arises: **how do we select the discrete token $\hat{w}_t$ to append to the generated sequence?** The choice of decoding policy governs the trade-off between deterministic precision and stochastic diversity, directly deciding whether the model produces factual truth, creative narrative, generic clichés, or ungrammatical gibberish.

To establish an identical, controlled empirical testbed across all decoding policies studied in this section, we condition each strategy on the opening prompt drawn from Arthur Conan Doyle's *The Red-Headed League* (as examined in lecture):

> *"I had called upon my friend, Mr. Sherlock Holmes, one day in the autumn of last year and found him in deep conversation with a very stout, florid-faced, elderly gentleman with fiery red hair. With an apology for my intrusion, I was about to withdraw when Holmes pulled me abruptly into the room and closed the door behind me."*

In the following subsections, we analyze the six canonical decoding paradigms in strict linear progression: from deterministic greedy selection and beam optimization to stochastic sampling, entropy modulation, and dynamic truncation filters.

5.12.1 Greedy decoding (argmax) and the degeneration trap

Mathematical formulation. The simplest and computationally most direct decoding policy is **Greedy Search**. At each temporal step t, the algorithm deterministically selects the single token assigned the highest instantaneous conditional probability:

$$\hat{w}_t = \text{argmax}_{w \in \mathcal{V}} P(w \mid w_{<t}). \tag{5.102}$$

Greedy decoding requires zero hyperparameters and incurs minimal computational overhead: it performs a single $O(|\mathcal{V}|)$ reduction across the logit vector at each time step.

Concrete numerical example. Suppose after processing the prompt, the model emits the normalized probabilities across a representative candidate subset shown in Table 5.2.

Table 5.2: Worked probability distribution for next-token selection at step $t = 1$.

Token Candidate w	`"I`	`"He`	`"The`	`"Mr.`	`"What`	... [Remaining 50,252 tokens]
Logit z	14.20	13.50	12.80	11.90	10.10	< 8.00
Softmax $P(w \mid w_{<t})$	**0.521**	0.260	0.130	0.053	0.009	$\sum = 0.027$

Under greedy search, the algorithm evaluates $\text{argmax}_w P(w \mid w_{<t})$ and deterministically emits token $\hat{w}_t =$ `"I`. All remaining hypotheses are discarded permanently.

The failure mode: mathematical myopia and cyclic repetition. Despite its superficial appeal, greedy decoding exhibits two catastrophic failure modes in open-ended language generation:

1. **Mathematical Myopia (Local Greediness ≠ Global Sequence Optimality)**: Sequence likelihood factorizes via the probabilistic chain rule as $P(w_{1:T}) = \prod_{t=1}^{T} P(w_t \mid w_{<t})$. A greedy local maximum at step t frequently leads into a barren probability subspace at subsequent steps:

$$\text{argmax}_{w_1,\ldots,w_T} \prod_{t=1}^{T} P(w_t \mid w_{<t}) \neq \prod_{t=1}^{T} \text{argmax}_{w_t} P(w_t \mid w_{<t}). \tag{5.103}$$

   As illustrated in Figure 5.22, committing to token A because $P(A) = 0.60 > P(B) = 0.40$ traps the generator if A can only be followed by low-probability tokens ($P(C \mid A) = 0.10$, yielding path likelihood 0.06), whereas selecting the locally sub-optimal token B uncovers a highly fluent sequence ($P(D \mid B) = 0.80$, yielding path likelihood 0.32).

2. **Neural Text Degeneration**: Holtzman et al. [58] demonstrated that greedy decoding on neural language models triggers self-reinforcing attractor loops. Common syntactic markers (e.g., pronouns, auxiliary verbs) dominate local argmax selections. Once a repetitive n-gram is appended to the context, self-attention attending back to those identical tokens artificially elevates their conditional probabilities even higher at subsequent steps, locking the model into an inescapable loop of cyclic degeneration:

```
"I am sorry, Mr. Holmes," I said. "I am sorry, Mr. Holmes, for having
been so rude to you. I am sorry, Mr. Holmes, for having been so rude
to you. I am sorry, Mr. Holmes, for having been so rude to you..."
```

Causal bridge to Beam Search. Because greedy search suffers from irrecoverable path pruning, we require an algorithm capable of looking ahead and exploring multiple alternative sequences concurrently. This directly motivates **Beam Search**.

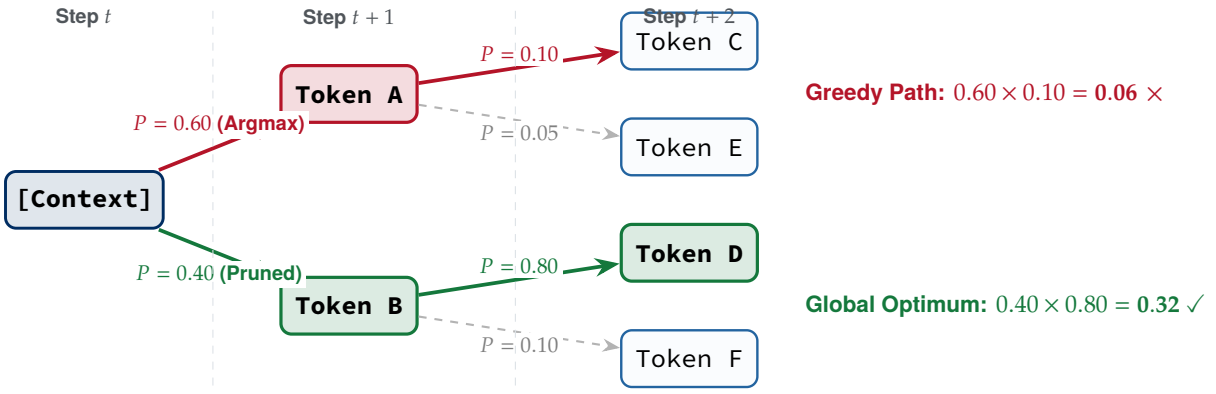


Figure 5.22: The failure mechanism of greedy decoding. By greedily selecting `Token A` ($P = 0.60$) over `Token B` ($P = 0.40$) at step $t+1$, the search space collapses into low-probability continuations ($P = 0.06$). The global optimum sequence through `Token B` ($P = 0.32$) is permanently pruned.

5.12.2 Beam search: exploring parallel hypotheses

Mathematical formulation. **Beam Search** avoids the strict myopia of greedy decoding by maintaining a fixed set of B top candidate sequences (termed the *beam width* B, or k in literature) at each time step.

Let $\mathcal{Y}_t = \{\mathbf{y}_t^{(1)}, \mathbf{y}_t^{(2)}, \ldots, \mathbf{y}_t^{(B)}\}$ denote the set of active partial hypotheses of length t. At generation step $t+1$:

1. Every hypothesis $\mathbf{y}_t^{(b)} \in \mathcal{Y}_t$ is expanded across all $|\mathcal{V}|$ vocabulary candidates, forming $B \times |\mathcal{V}|$ child candidate sequences $\mathbf{y}_{t+1} = (\mathbf{y}_t^{(b)}, w)$.

2. The cumulative score of each sequence is evaluated in logarithmic space to prevent numerical underflow:
$$\text{Score}(\mathbf{y}_{1:t+1}) = \sum_{i=1}^{t+1} \log P(y_i \mid \mathbf{y}_{<i}). \tag{5.104}$$

3. The expanded set of $B \cdot |\mathcal{V}|$ candidates is sorted, and only the top-B candidates with the highest cumulative log-probabilities are preserved:
$$\mathcal{Y}_{t+1} = \text{argtop-B}_{\mathbf{y} \in \mathcal{Y}_t \times \mathcal{V}} \text{Score}(\mathbf{y}). \tag{5.105}$$
All other $B \cdot (|\mathcal{V}| - 1)$ hypotheses are permanently pruned.

Length penalty normalization. Because $\log P(y_i \mid \mathbf{y}_{<i}) \leq 0$ for all valid probabilities, the objective in Equation (5.104) is strictly non-increasing with sequence length. Consequently, unconstrained beam search inherently penalizes longer generations, favoring unnaturally short, truncated outputs. To counteract this bias, standard implementations incorporate a length normalization heuristic:

$$\text{Score}_{\text{norm}}(Y) = \frac{\sum_{i=1}^{|Y|} \log P(y_i \mid \mathbf{y}_{<i})}{\text{LP}(Y)}, \quad \text{where} \quad \text{LP}(Y) = \frac{(5 + |Y|)^\alpha}{(5+1)^\alpha}, \tag{5.106}$$

with length penalty parameter $\alpha \in [0.6, 1.0]$. When $\alpha = 0$, no length normalization occurs; when $\alpha = 1.0$, the score is normalized by length.

Worked numerical example. Consider beam search with beam width $B = 2$ expanding from the root token `"The"`, exactly replicating the lecture derivation:

1. **Step** $t = 1$: From `"The"`, candidate probabilities are:

$$P(\texttt{"nice"} \mid \texttt{"The"}) = 0.50, \quad P(\texttt{"dog"} \mid \texttt{"The"}) = 0.40, \quad P(\texttt{"car"} \mid \texttt{"The"}) = 0.10. \quad (5.107)$$

   The top-2 beams retained are $\mathcal{Y}_1 = \{(\texttt{"The"}, \texttt{"nice"}), (\texttt{"The"}, \texttt{"dog"})\}$. The branch `"car"` is pruned.

2. **Step** $t = 2$: Each of the 2 active beams expands its candidates:

   - From `"The dog"`: `"has"` ($P = 0.90 \implies$ joint $0.40 \times 0.90 = 0.36$), `"and"` ($P = 0.05 \implies$ joint 0.02), `"run"` ($P = 0.05 \implies$ joint 0.02).
   - From `"The nice"`: `"woman"` ($P = 0.40 \implies$ joint $0.50 \times 0.40 = 0.20$), `"house"` ($P = 0.30 \implies$ joint 0.15), `"guy"` ($P = 0.30 \implies$ joint 0.15).

   Across all 6 expanded paths, the top-2 global sequences are `"The dog has"` ($P = 0.36$) and `"The nice woman"` ($P = 0.20$). All other 4 paths are pruned.

Figure 5.23 illustrates this exact operational trellis.

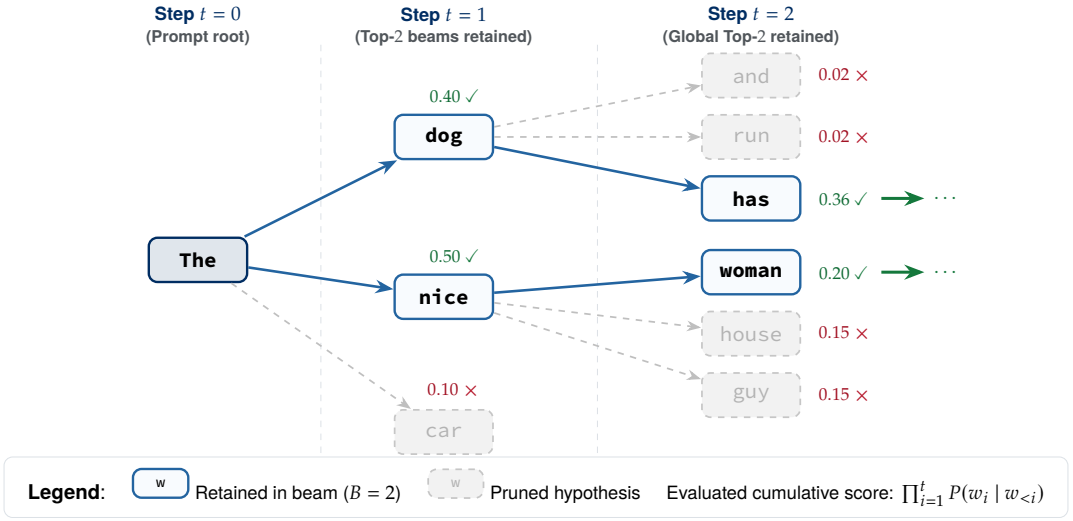


Figure 5.23: Beam search decoding with beam width $B = 2$. At each generation step, all B candidate sequences are expanded across vocabulary candidates. The algorithm prunes lower-probability branches, maintaining only the top-B cumulative probability hypotheses (here, `"The dog has"` with $P = 0.36$ and `"The nice woman"` with $P = 0.20$).

The failure mode of beam search in open-ended generation. While beam search is the gold standard for closed, goal-oriented tasks with narrow target spaces (such as neural machine translation and formal code generation), it fails strikingly in open-ended creative text generation:

- **Deterministic Blandness and Clichés**: Human communication does not optimize for maximum likelihood at every step. Natural human language contains an alternating rhythm of predictable syntactic connectors (low information entropy) and surprising, domain-specific words (high information entropy). Maximizing cumulative likelihood forces the model into high-probability clichés and generic sentences.
- **Repetitive Stagnation**: Holtzman et al. [58] demonstrated that increasing the beam width B in open-ended language generation actually *degrades* perceived quality: as $B \to \infty$, the search uncovers pathological repeating loops that accumulate high probability mass over extended horizons.

Causal bridge to Random Sampling. To escape deterministic clichés and inject human-like entropy, we must transition from deterministic maximization to **stochastic generation**. This leads directly to **Pure Random Sampling**.

5.12.3 Pure random sampling and the catastrophic tail risk

Mathematical formulation. Instead of deterministically tracking top-ranking candidates, **Pure Random Sampling** (Multinomial Sampling) draws the next token directly from the categorical probability distribution generated by the softmax unembedding:

$$w_t \sim \text{Categorical}\left(P(w_1 \mid w_{<t}), P(w_2 \mid w_{<t}), \ldots, P(w_{|\mathcal{V}|} \mid w_{<t})\right). \tag{5.108}$$

Sampling is implemented computationally via Inverse Transform Sampling: given a uniform pseudo-random draw $u \sim \mathcal{U}[0,1)$, select index $\hat{i}$ satisfying:

$$\sum_{j=1}^{\hat{i}-1} P(w_j \mid w_{<t}) \leq u < \sum_{j=1}^{\hat{i}} P(w_j \mid w_{<t}). \tag{5.109}$$

Worked numerical example. Returning to the distribution from Table 5.2:

- Under greedy search, `"I` is selected with 100% probability.
- Under random sampling:

$$P(w_t = \texttt{"I"}) = 52.1\%, \quad P(w_t = \texttt{"He"}) = 26.0\%, \quad P(w_t = \texttt{"The"}) = 13.0\%, \quad \ldots \tag{5.110}$$

  If the pseudo-random variable evaluates to $u = 0.650$, since $0.521 \leq 0.650 < (0.521 + 0.260 = 0.781)$, the generator selects token $\hat{w}_t =$ `"He"`. The generated continuation gains immediate narrative variety.

The failure mode: the unreliable long tail and hallucination cascades. While random sampling completely eliminates repetitive loops, it introduces a severe vulnerability: **the catastrophic tail hazard**. In language models with $|\mathcal{V}| \approx 50{,}000$, although individual tokens in the tail have infinitesimal probabilities (e.g., $P(w) < 10^{-4}$), the *cumulative aggregate mass* of the thousands of tail tokens commonly sums to 15% to 30% of the entire probability simplex (Figure 5.24).

Consequently, when generating a multi-token sequence, the probability of sampling *at least one* erratic tail token over N steps is $1 - (1 - P_{\text{tail}})^N \approx 1$. Once an aberrant token is sampled into the context, self-attention attends to it, irrevocably corrupting the topic representation and causing severe semantic collapse:

```
"...and my amazement owled wild-it seems an impossibility-and while it
was gone in an stupor and good hearing from me I pretended to know his
name, assuming some personal information which I would..."
```

Causal bridge to Temperature Scaling. Before resorting to hard truncation of the vocabulary, can we smoothly reshape the probability landscape itself, suppressing the tail while accentuating dominant modes? This leads to **Temperature Scaling**.

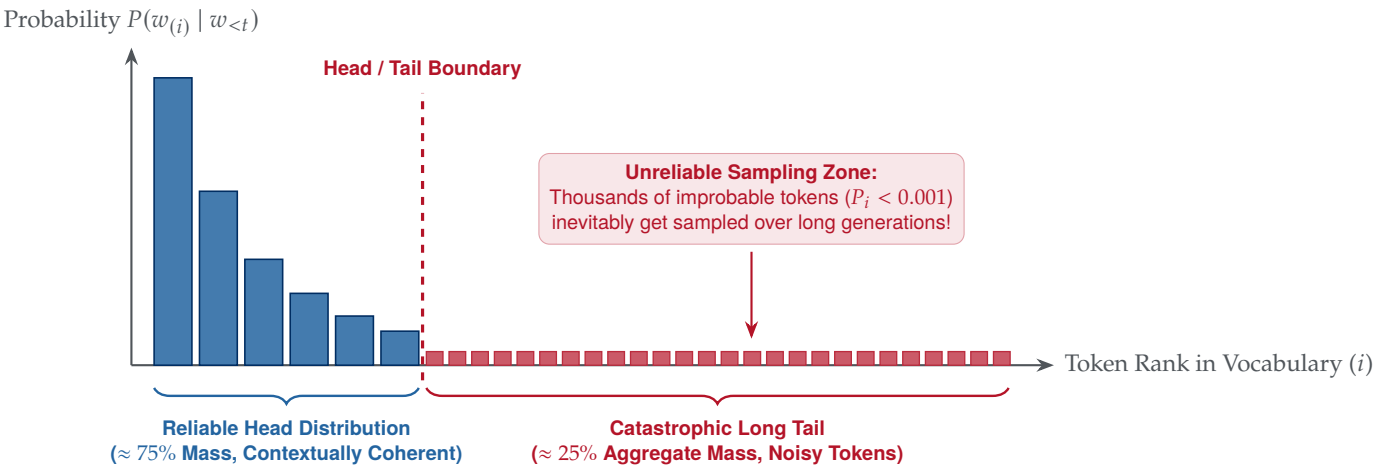


Figure 5.24: The long-tail vulnerability of pure multinomial random sampling. Although individual low-probability tokens have negligible probabilities, their cumulative probability mass (≈ 25%) guarantees that over a horizon of dozens of tokens, the generator will sample an erratic, nonsensical token from the tail, corrupting the subsequent context history.

5.12.4 Temperature scaling: thermodynamic entropy modulation

Mathematical formulation. **Temperature Scaling** modifies the sharpness of the categorical probability distribution by introducing a continuous thermodynamic parameter $T > 0$ that scales raw logits prior to exponentiation:

$$P(w_i; T) = \frac{\exp(z_i/T)}{\sum_{j=1}^{|\mathcal{V}|} \exp(z_j/T)}, \quad \forall w_i \in \mathcal{V}. \tag{5.111}$$

The parameter T acts as a continuous entropy dial:

- **Low Temperature ($T < 1.0$)**: Amplifies relative logit differences $z_i - z_j$, concentrating probability mass on the highest-scoring tokens (reducing Shannon entropy).
- **Standard Temperature ($T = 1.0$)**: Preserves the original, calibrated categorical distribution learned during maximum-likelihood pre-training.
- **High Temperature ($T > 1.0$)**: Dampens logit disparities, flattening the distribution toward maximum-entropy uniformity.

Concrete numerical example. Consider a model evaluating three candidate tokens with raw logits $\mathbf{z} = [10.0, 8.0, 5.0]^\top$. Table 5.3 computes the resulting probabilities across four temperature regimes.

Table 5.3: Behavior of probabilities under temperature scaling for logits $\mathbf{z} = [10.0, 8.0, 5.0]^\top$.

Regime	Scaled Logits $\mathbf{z}/T$	$\exp(\mathbf{z}/T)$	Probabilities $P(w;T)$	Shannon Ent
$T = 0.5$ (Low / Sharp)	$[20.0, 16.0, 10.0]$	$[4.85 \cdot 10^8, 8.89 \cdot 10^6, 2.20 \cdot 10^4]$	$[\mathbf{0.9820}, 0.0180, 0.00004]$	0.092 na
$T = 1.0$ (Default)	$[10.0, 8.0, 5.0]$	$[22026, 2981, 148]$	$[\mathbf{0.8756}, 0.1185, 0.0059]$	0.388 na
$T = 2.0$ (High / Soft)	$[5.0, 4.0, 2.5]$	$[148.4, 54.6, 12.2]$	$[\mathbf{0.6896}, 0.2537, 0.0567]$	0.784 na
$T = 5.0$ (Extreme)	$[2.0, 1.6, 1.0]$	$[7.39, 4.95, 2.72]$	$[\mathbf{0.4907}, 0.3287, 0.1806]$	1.042 na

As demonstrated, decreasing T from 1.0 to 0.5 drives the top token's probability from 87.6% to 98.2%, virtually suppressing all tail tokens. Conversely, increasing T to 5.0 flattens the distribution toward equal weighting ($1/3 \approx 0.333$).

Figure 5.25 visualizes this transformation across the vocabulary spectrum.

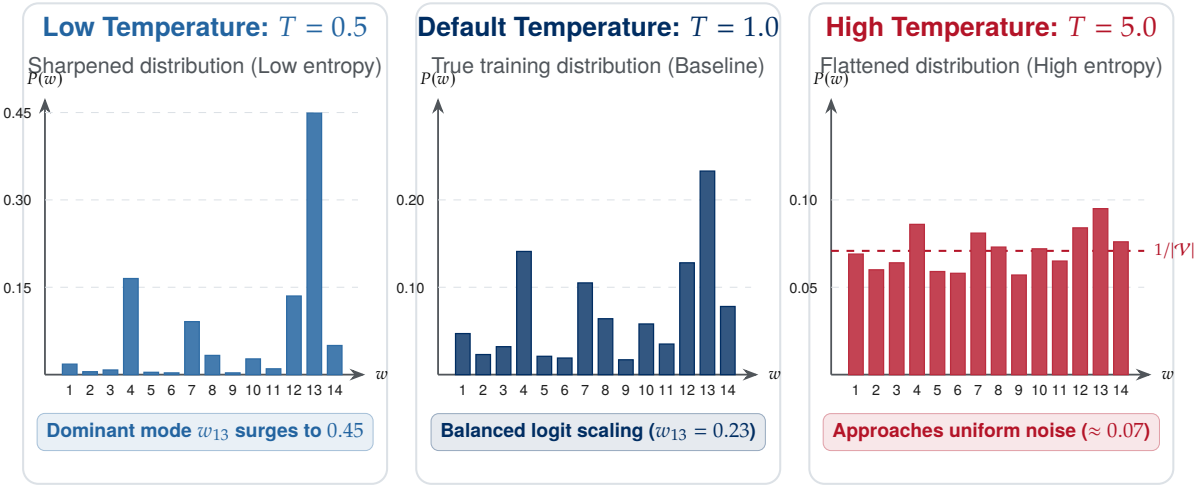


Figure 5.25: Temperature scaling of the softmax distribution. Scaling logits by temperature T modulates distribution entropy: $T < 1.0$ (left) exponentially sharpens differences toward the dominant mode (w_{13}), approaching deterministic greedy decoding as $T \to 0$; $T = 1.0$ (center) preserves the model's true training logits; $T > 1.0$ (right) flattens logit disparities toward a uniform distribution, producing erratic and ungrammatical token sequences.

Analytical limits and mathematical proofs. The behavior of temperature scaling is bounded by two rigorous analytical limits:

1. **Low-Temperature Limit ($T \to 0^+$, Greedy Convergence)**: Let $z^* = \max_j z_j$ denote the unique maximal logit. Factoring out $\exp(z^*/T)$ from the numerator and denominator of Equation (5.111):

$$\lim_{T\to 0^+} P(w_i;T) = \lim_{T\to 0^+} \frac{\exp((z_i - z^*)/T)}{\sum_{j=1}^{|\mathcal{V}|} \exp((z_j - z^*)/T)}. \tag{5.112}$$

For the maximal token where $z_i = z^*$, $(z_i - z^*)/T = 0$, so $\exp(0) = 1$. For any sub-optimal token k where $z_k < z^*$, the numerator satisfies $z_k - z^* = -\delta < 0$ ($\delta > 0$). As $T \to 0^+$, $-\delta/T \to -\infty$, yielding $\exp(-\delta/T) \to 0$. Therefore:

$$\lim_{T\to 0^+} P(w_i;T) = \begin{cases} 1 & \text{if } z_i = \max_j z_j, \\ 0 & \text{if } z_i < \max_j z_j. \end{cases} \tag{5.113}$$

Temperature sampling converges continuously to deterministic **Greedy decoding**.

2. **High-Temperature Limit ($T \to \infty$, Uniform Collapse)**: As $T \to \infty$, for every finite logit z_i, the argument $z_i/T \to 0$, implying $\exp(z_i/T) \to 1$. Consequently:

$$\lim_{T\to\infty} P(w_i;T) = \frac{1}{\sum_{j=1}^{|\mathcal{V}|} 1} = \frac{1}{|\mathcal{V}|}. \tag{5.114}$$

The distribution collapses into a maximal-entropy uniform distribution across all $|\mathcal{V}|$ vocabulary tokens.

3. **Monotonicity of Shannon Entropy**: The Shannon entropy of the distribution is defined as $H(P;T) = -\sum_{i=1}^{|\mathcal{V}|} P_i(T) \log P_i(T)$. Differentiating with respect to T yields:

$$\frac{\partial H(P;T)}{\partial T} = \frac{1}{T^3} \mathrm{Var}_P(z) = \frac{1}{T^3} \left(\sum_{i=1}^{|\mathcal{V}|} P_i(T) z_i^2 - \left(\sum_{i=1}^{|\mathcal{V}|} P_i(T) z_i \right)^2 \right) \geq 0. \tag{5.115}$$

Because variance is strictly non-negative ($\mathrm{Var}_P(z) \geq 0$), entropy increases strictly monotonically with T whenever logits are non-identical.

Limitation of temperature scaling. Although temperature scaling compresses or flattens the distribution, **it never zeros out the probability of any token**: every token in the vocabulary retains $P(w_i; T) > 0$. In low-entropy contexts where only a single token is correct, even a lowered temperature leaves non-zero probability mass in the tail, still leaving room for improbable hallucinations. Conversely, lowering temperature globally to combat tail noise also suppresses legitimate synonyms in creative passages.

Causal bridge to Top-k Sampling. To guarantee that low-probability tokens cannot be sampled under any circumstances, we must apply explicit **truncation** to the support of the distribution. This motivates **Top-k Sampling**.

5.12.5 Top-k sampling: hard cardinality truncation

Mathematical formulation. Introduced by Fan et al. [59], **Top-k Sampling** bounds tail risk by strictly restricting the sampling candidate set to the k most probable tokens at each generation step.

Definition 5.4: *Top-k sampling*

Let $\mathcal{V}^{(k)} \subset \mathcal{V}$ denote the subset of size $|\mathcal{V}^{(k)}| = k$ containing the tokens with the highest probabilities under $P(w \mid w_{<t})$:

$$\sum_{w \in \mathcal{V}^{(k)}} P(w \mid w_{<t}) \quad \text{is maximized subject to } |\mathcal{V}^{(k)}| = k. \tag{5.116}$$

All tokens outside $\mathcal{V}^{(k)}$ have their probabilities truncated to zero. The remaining k probabilities are renormalized by dividing by their cumulative mass $S_k = \sum_{w_j \in \mathcal{V}^{(k)}} P(w_j)$:

$$P'(w_i) = \begin{cases} \frac{P(w_i)}{\sum_{w_j \in \mathcal{V}^{(k)}} P(w_j)} & \text{if } w_i \in \mathcal{V}^{(k)}, \\ 0 & \text{otherwise.} \end{cases} \tag{5.117}$$

The next token is sampled from this truncated categorical distribution: $w_t \sim \text{Categorical}(P')$.

Worked numerical example. Suppose a model evaluates the token distribution in Table 5.4 under a top-k policy with $k = 3$.

Table 5.4: Worked example of Top-k truncation ($k = 3$) and probability renormalization.

Token Candidate w	`"dog"`	`"cat"`	`"bird"`	`"fish"`	`"horse"`	`...` [Tail]
Original $P(w)$	0.40	0.30	0.15	0.08	0.04	$\sum = 0.03$
In Top-3 Set $\mathcal{V}^{(k)}$**?**	✓	✓	✓	×	×	×
Truncated Mass	0.40	0.30	0.15	0.00	0.00	0.00
Renormalized $P'(w)$	$\frac{0.40}{0.85} = \mathbf{0.471}$	$\frac{0.30}{0.85} = \mathbf{0.353}$	$\frac{0.15}{0.85} = \mathbf{0.176}$	0.000	0.000	0.000

The cumulative mass of the top-3 candidates is $S_3 = 0.40 + 0.30 + 0.15 = 0.85$. Dividing each retained candidate by 0.85 rescales the probabilities so that $P'(\texttt{"dog"}) + P'(\texttt{"cat"}) + P'(\texttt{"bird"}) =$

1.000. All tail candidates (accounting for 15% of original probability mass) are strictly excluded.

Figure 5.26 illustrates this truncation and renormalization geometry.

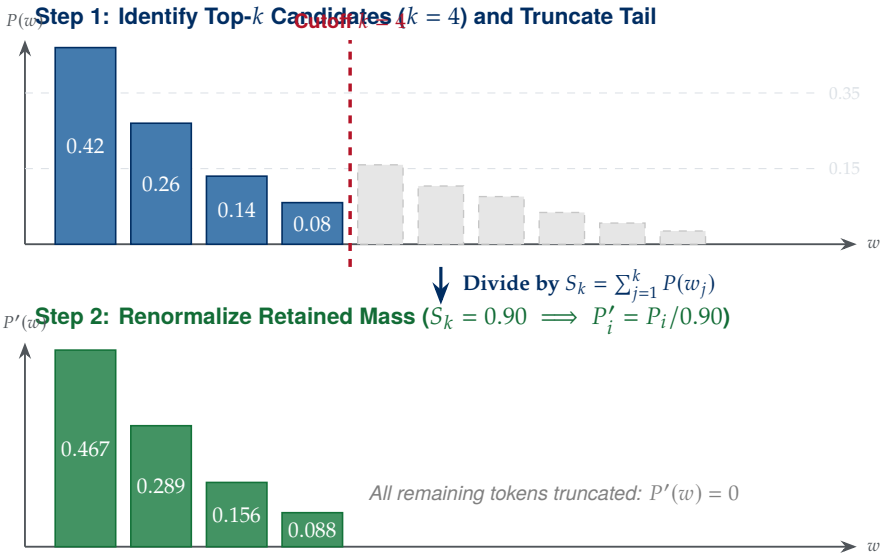


Figure 5.26: The mechanics of Top-k sampling ($k = 4$). The algorithm isolates the top-k tokens, strictly zeroing out the remaining probability mass. Renormalizing by the sum of retained mass rescales the top-k probabilities so they sum to 1.0, shielding the generator from tail errors.

Evaluating top-k sampling with $k = 50$ on the Sherlock Holmes prompt eliminates bizarre tail tokens and produces coherent dialogue:

```
"'You will think, Doctor, that a man of my age would be so much of a menace
to me, if I were to refuse a pass for him.  I should rather get a few things
away from him than any which might injure his reputation...'"
```

The failure mode: the structural rigidity of a static k. Despite its widespread adoption, top-k sampling suffers from an inherent architectural rigidity: **a static threshold k cannot adapt to varying distribution entropy across different contexts**:

1. **Low-Entropy (Peaked / Factual Context)**: Suppose the prompt is `"The capital of France is"`. The model assigns $P(\texttt{"Paris"}) = 0.98$, while all other tokens are factual errors. If $k = 50$, top-k sampling is forced to retain 49 improbable, factually erroneous tokens. Because their collective mass is renormalized, the chance of sampling an error is artificially elevated.

2. **High-Entropy (Flat / Creative Context)**: Suppose the prompt is `"She looked out into the vast horizon and saw"`. Hundreds of valid English nouns and verbs (e.g., `"mountains"`, `"ships"`, `"stars"`, `"glaciers"`) share roughly equal probabilities (~ 0.005). If $k = 10$ or $k = 20$, top-k prematurely truncates dozens of perfectly fluent, highly creative candidate tokens.

Causal bridge to Top-p (Nucleus) Sampling. To overcome the inflexibility of a fixed cardinality k, the candidate pool size must become **dynamic**, expanding in high-entropy situations and contracting in low-entropy situations. This motivates **Top-p (Nucleus) Sampling**.

5.12.6 Top-p (nucleus) sampling: dynamic cumulative mass truncation

Mathematical formulation. To eliminate the rigid cardinality constraint of top-k, Holtzman et al. [58] proposed **Top-p (Nucleus) Sampling**. Instead of fixing the number of tokens k, nucleus sampling fixes the **cumulative probability mass threshold** $p \in (0, 1]$.

Definition 5.5: *Top-p (Nucleus) sampling*

Let $p \in (0, 1]$ be a cumulative probability threshold. Let $\mathcal{V}^{(p)} \subset \mathcal{V}$ denote the smallest subset of vocabulary tokens whose cumulative probability mass exceeds threshold p:

$$\mathcal{V}^{(p)} = \text{argmin}_{V' \subseteq \mathcal{V}} |V'| \quad \text{subject to} \quad \sum_{w \in V'} P(w \mid w_{<t}) \geq p. \tag{5.118}$$

Equivalently, sorting tokens in descending order of probability $P(w_{(1)}) \geq P(w_{(2)}) \geq \cdots \geq P(w_{(|\mathcal{V}|)})$, the nucleus cutoff index $k^*(p)$ is:

$$k^*(p) = \min \left\{ k \in \{1, \ldots, |\mathcal{V}|\} \;\middle|\; \sum_{i=1}^{k} P(w_{(i)} \mid w_{<t}) \geq p \right\}. \tag{5.119}$$

All tokens with rank $i > k^*(p)$ are zeroed out, and the probabilities within $\mathcal{V}^{(p)}$ are renormalized:

$$P'(w_i) = \begin{cases} \frac{P(w_i)}{\sum_{w_j \in \mathcal{V}^{(p)}} P(w_j)} & \text{if } w_i \in \mathcal{V}^{(p)}, \\ 0 & \text{otherwise.} \end{cases} \tag{5.120}$$

The next token is sampled from P'.

Worked numerical example: adaptive candidate cardinality. To demonstrate the dynamic adaptivity of nucleus sampling, consider setting threshold $p = 0.90$ across two different contexts:

- **Context A (Peaked / Low Entropy)**: Prompt: `"The capital of France is"`.

$$P(\texttt{"Paris"}) = 0.92, \quad P(\texttt{"the"}) = 0.03, \quad P(\texttt{"a"}) = 0.01, \quad \ldots$$
$$\text{Cumulative sum}: \quad P(\texttt{"Paris"}) = 0.92 \geq 0.90.$$

  The nucleus contains **only 1 token**: $\mathcal{V}^{(p)} = \{\texttt{"Paris"}\}$. The candidate pool automatically collapses to greedy behavior, preventing hallucinations.

- **Context B (Flat / High Entropy)**: Prompt: `"She looked through the telescope and saw"`. Suppose tokens $w_{(1)}$ through $w_{(15)}$ each carry $P \approx 0.06$. To accumulate $\sum P \geq 0.90$, the nucleus retains $k^*(0.90) = 15$ tokens. The candidate pool dynamically expands, allowing rich lexical variation.

Figure 5.27 demonstrates this dynamic expansion and contraction across high-entropy and low-entropy distributions.

Evaluating nucleus sampling with $p = 0.90$ on the Sherlock Holmes prompt produces fluent, highly natural prose that maintains the atmospheric literary tone of Arthur Conan Doyle without degenerate loops or tail gibberish:

```
"Mr.  Holmes had been talking for awhile and seemed very engrossed in the
conversation.  He wished to conduct his last spoke through the eyes of
another, and by that he meant to tell me the whole of the publick story
that had been passed from one of them into the other..."
```

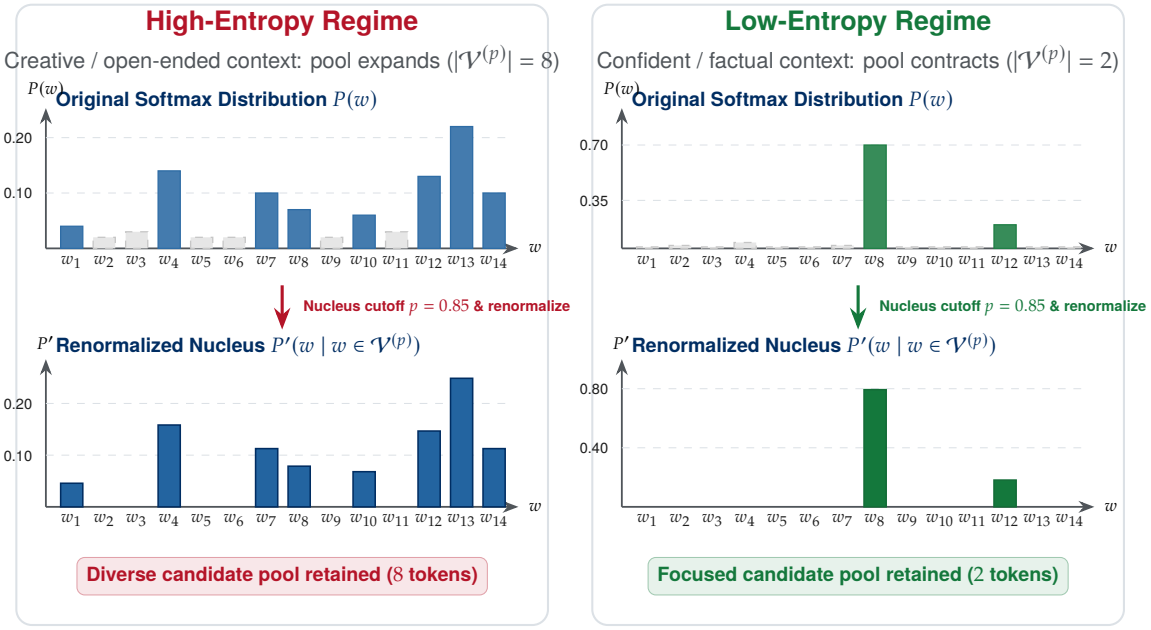


Figure 5.27: Top-p (Nucleus) sampling mechanism. Instead of a fixed k, the candidate pool dynamically expands or contracts to cover cumulative probability mass p. In high-entropy regimes (left), the pool expands to preserve lexical diversity (8 tokens); in low-entropy regimes (right), the pool automatically contracts to dominant tokens (2 tokens), preventing tail hallucinations [58].

5.12.7 Empirical synthesis: the Sherlock Holmes spectrum and decoding selection

The Sherlock Holmes generation spectrum. To directly observe how temperature interacts with sampling mechanics, Table 5.5 summarizes the empirical outputs obtained across temperature and truncation regimes conditioned on the identical Sherlock Holmes prefix.

Algorithmic selection guide for software engineering and NLP tasks. In practical production deployments, no single decoding policy dominates all use cases. The optimal strategy depends strictly on whether the downstream task prioritizes **invariance to randomness** (factual precision, exact syntax) or **stochastic diversity** (creative exploration, synthetic data generation):

1. **Deterministic Tasks (Code Generation, Unit Test Synthesis, Mathematical Reasoning, Machine Translation)**: Require strict reproducibility and syntactic validity. Standard practice employs **Greedy Decoding** ($T = 0$) or **Beam Search** with small beam width ($B \in [2, 5]$) and length penalty $\alpha \in [0.6, 0.8]$.

2. **Open-Ended Creative Tasks (Conversational Agents, Story Generation, Synthetic Documentation)**: Require human-like cadence and vocabulary richness. Standard practice employs **Top-p (Nucleus) Sampling** ($p \in [0.90, 0.95]$) coupled with moderate temperature ($T \in [0.6, 0.8]$).

3. **Hybrid Pipeline: Combining Temperature, Top-k, and Top-p**: Modern inference engines (e.g., vLLM, Hugging Face 'transformers') execute decoding as a multi-stage sequential filter pipeline:

$$\mathbf{z} \xrightarrow{\text{Temperature } T} \frac{\mathbf{z}}{T} \xrightarrow{\text{Top-}k \text{ Truncation}} \mathcal{V}^{(k)} \xrightarrow{\text{Top-}p \text{ Nucleus}} \mathcal{V}^{(p)} \xrightarrow{\text{Softmax \& Categorical Draw}} \hat{w}_t. \tag{5.121}$$

Table 5.5: Empirical generation spectrum across decoding policies on the Sherlock Holmes testbed.

Decoding Policy	Configuration	Observed Output Behavior
Greedy (Argmax)	Deterministic ($T \to 0$)	Collapses into cyclic repetition: *"I am sorry, Mr. Holmes... I am sorry, Mr. Holmes..."*
Beam Search	Width $B = 4$, $\alpha = 0.8$	Grammatically pristine but highly generic and formulaic; lacks human stylistic variance.
Pure Random	Multinomial ($T = 1.0$)	Catastrophic tail contamination: emits ungrammatical fragments (*"owled wild–it seems..."*).
Temperature	$T = 0.1$ (Near Greedy)	Repetition loop persists identically to greedy decoding.
Temperature	$T = 0.5$ (Balanced Low)	Highly coherent, narrative continuity: *"It was a very sad moment, but a true one..."*
Temperature	$T = 1.1$ (High Entropy)	Stylistic drift; bizarre metaphorical imagery (*"grim gulags..."*).
Temperature	$T = 1.5$ (Extreme Noise)	Complete entropy collapse into gibberish subwords (*"IMLEG it off... doctorm Hiraghbene"*).
Top-k	$k = 50$, $T = 0.8$	Fluent dialogue, eliminates bizarre tails, but risks hallucinating if context is strictly factual.
Top-p (Nucleus)	$p = 0.90$, $T = 0.7$	Optimal human-like balance; dynamic candidate pool preserves narrative voice while preventing collapse.

Scaling logits by T first modulates relative entropy; filtering by Top-k removes extreme vocabulary outliers (e.g., $k = 50$); filtering by Top-p dynamically adapts candidate size for the remaining distribution; and sampling from the renormalized subset provides controlled, diverse outputs.

5.13 Comparative paradigm synthesis and bridge to scaling laws

Table 5.6 synthesizes the architectural trade-offs across the three Transformer families studied in this chapter.

The causal bridge to Scaling Laws. While encoder-only models dominated semantic representation, syntax parsing, and discriminative classification, decoder-only models introduced a profound architectural unification: their pre-training objective (autoregressive next-token prediction) is identical to their inference task. By pairing decoder-only transformers with stochastic sampling strategies—such as nucleus sampling ($p \in [0.9, 0.95]$) and temperature scaling ($T \in [0.6, 0.8]$)—researchers uncovered an extraordinary emergent capability: as model parameter count, training token volume, and computational budget scale upward, language models exhibit few-shot and zero-shot reasoning without task-specific parameter fine-tuning. In the subsequent chapter, we formalize the empirical principles and mathematical boundaries governing this regime: **Scaling Laws and Large-Scale Pre-training**.

Table 5.6: Comprehensive comparison of Transformer architectural paradigms.

Feature	Encoder-Decoder (e.g., T5)	Encoder-Only (e.g., BERT)	Decoder-Only (Preview: Chapter 6)
Attention types	Self-Attention, Masked Self-Attention, Cross-Attention	Bidirectional Self-Attention only	Causal Masked Self-Attention only
Pre-training loss	Sequence-to-Sequence span corruption	Masked LM (15%) + Next Sentence Prediction	Autoregressive Next-Token Prediction
Receptive field	Bidirectional (Enc) + Causal (Dec)	Fully Bidirectional across all tokens	Strictly Causal (preceding tokens only)
Best suited for	Translation, summarization, structured text-to-text	Sentence classification, NER, semantic code search	Open-ended text generation, dialogue, few-shot prompting
Inference style	Autoregressive decoding conditioned on encoder	Single forward pass (non-autoregressive)	Autoregressive generation loop with sampling

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